Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-16 22:18:28
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.2
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

model_path............................. checkpoints_64x128/best_model.pt
input_data_path........................ data/processed_64x128/sst_final.nc
run_validation......................... True
run_test............................... True
forecast_lead_steps.................... 90
chain_stride_steps..................... 90
output_dir............................. results_64x128/
val_predictions_file................... predictions_val.nc
test_predictions_file.................. predictions_test.nc

-- Concat Chains --
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  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_64x128/std_params.nc
  trend_params_path..... data/processed_64x128/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
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METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

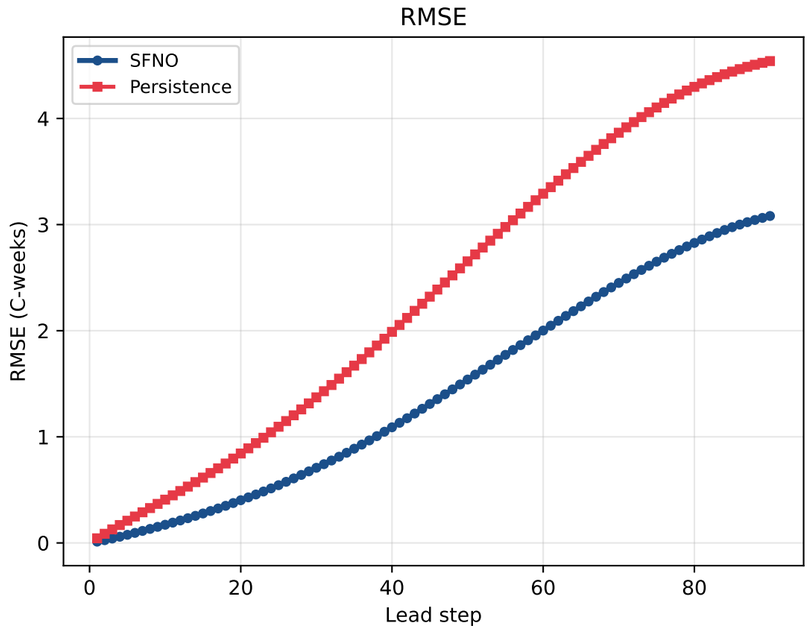
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

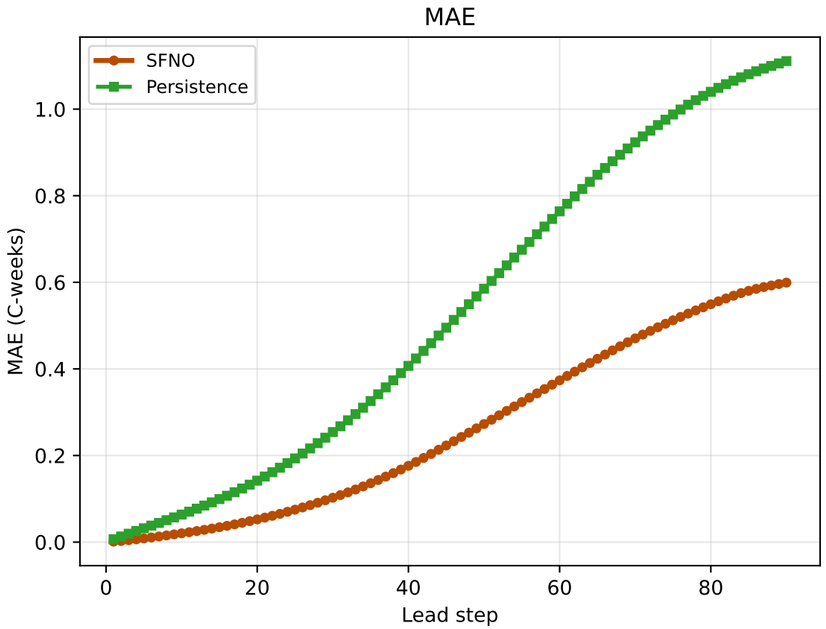
DATASETS
  Test          19 chains x 90 leads x 64 lat x 128 lon
  Val           16 chains x 90 leads x 64 lat x 128 lon
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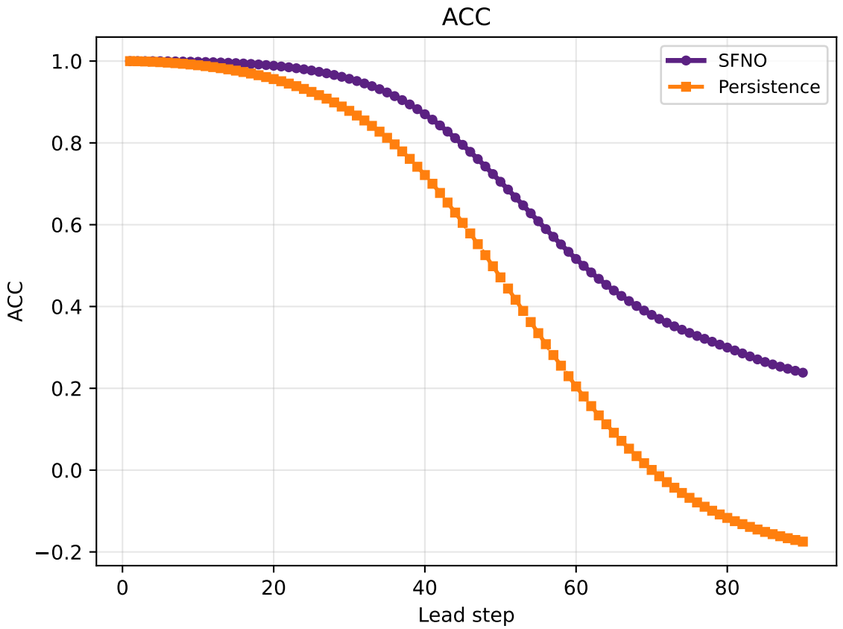
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

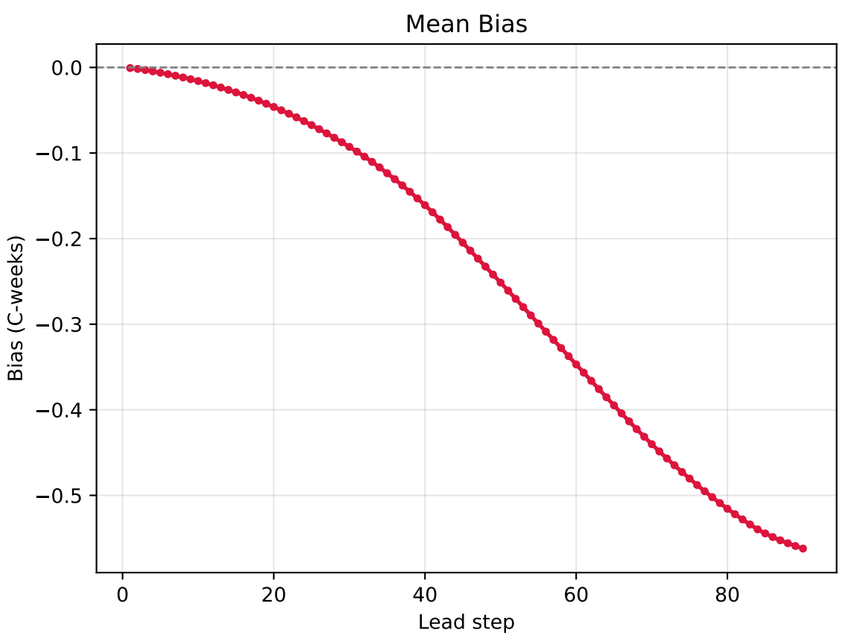
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0124	0.0437	71.7%	0.0011	0.0065	-0.0007	1.0000	0.9999	1.0000
2	0.0260	0.0862	69.9%	0.0026	0.0130	-0.0017	1.0000	0.9996	0.9999
3	0.0416	0.1278	67.5%	0.0044	0.0193	-0.0031	0.9999	0.9990	0.9998
4	0.0586	0.1687	65.3%	0.0064	0.0257	-0.0046	0.9998	0.9983	0.9996
5	0.0766	0.2091	63.4%	0.0085	0.0319	-0.0062	0.9996	0.9973	0.9994
6	0.0947	0.2491	62.0%	0.0107	0.0381	-0.0078	0.9995	0.9962	0.9991
7	0.1133	0.2888	60.8%	0.0130	0.0443	-0.0097	0.9992	0.9949	0.9987
8	0.1323	0.3285	59.7%	0.0154	0.0507	-0.0117	0.9989	0.9934	0.9982
9	0.1516	0.3680	58.8%	0.0178	0.0571	-0.0137	0.9986	0.9917	0.9976
10	0.1712	0.4079	58.0%	0.0203	0.0635	-0.0159	0.9982	0.9898	0.9969
11	0.1914	0.4484	57.3%	0.0229	0.0702	-0.0183	0.9977	0.9877	0.9962
12	0.2118	0.4893	56.7%	0.0255	0.0772	-0.0208	0.9972	0.9853	0.9952
13	0.2331	0.5310	56.1%	0.0283	0.0844	-0.0234	0.9966	0.9827	0.9942
14	0.2553	0.5733	55.5%	0.0313	0.0918	-0.0263	0.9958	0.9798	0.9930
15	0.2776	0.6158	54.9%	0.0344	0.0993	-0.0292	0.9950	0.9767	0.9916
16	0.3008	0.6590	54.4%	0.0375	0.1071	-0.0322	0.9941	0.9733	0.9900
17	0.3249	0.7031	53.8%	0.0409	0.1152	-0.0354	0.9930	0.9695	0.9882
18	0.3499	0.7484	53.3%	0.0446	0.1237	-0.0388	0.9917	0.9654	0.9862
19	0.3759	0.7951	52.7%	0.0484	0.1326	-0.0424	0.9903	0.9609	0.9838
20	0.4026	0.8429	52.2%	0.0524	0.1419	-0.0462	0.9886	0.9560	0.9812
21	0.4294	0.8913	51.8%	0.0565	0.1515	-0.0500	0.9868	0.9507	0.9783
22	0.4565	0.9406	51.5%	0.0608	0.1614	-0.0541	0.9848	0.9450	0.9751
23	0.4847	0.9912	51.1%	0.0653	0.1717	-0.0583	0.9825	0.9387	0.9715
24	0.5140	1.0427	50.7%	0.0700	0.1824	-0.0628	0.9799	0.9319	0.9674
25	0.5443	1.0951	50.3%	0.0749	0.1933	-0.0674	0.9770	0.9246	0.9629
26	0.5760	1.1486	49.8%	0.0801	0.2046	-0.0721	0.9737	0.9167	0.9578
27	0.6082	1.2029	49.4%	0.0854	0.2163	-0.0770	0.9701	0.9081	0.9522
28	0.6408	1.2585	49.1%	0.0909	0.2285	-0.0821	0.9661	0.8989	0.9460
29	0.6744	1.3152	48.7%	0.0966	0.2411	-0.0874	0.9615	0.8889	0.9392
30	0.7081	1.3721	48.4%	0.1025	0.2541	-0.0927	0.9566	0.8783	0.9319
31	0.7421	1.4298	48.1%	0.1085	0.2675	-0.0984	0.9512	0.8669	0.9239
32	0.7768	1.4883	47.8%	0.1149	0.2813	-0.1042	0.9453	0.8547	0.9152
33	0.8125	1.5482	47.5%	0.1215	0.2956	-0.1103	0.9387	0.8416	0.9057
34	0.8496	1.6090	47.2%	0.1285	0.3104	-0.1168	0.9314	0.8274	0.8952
35	0.8880	1.6707	46.8%	0.1359	0.3257	-0.1236	0.9232	0.8123	0.8838
36	0.9267	1.7328	46.5%	0.1434	0.3414	-0.1306	0.9143	0.7963	0.8714
37	0.9662	1.7960	46.2%	0.1512	0.3573	-0.1378	0.9046	0.7792	0.8581
38	1.0065	1.8599	45.9%	0.1592	0.3737	-0.1452	0.8940	0.7609	0.8438
39	1.0479	1.9244	45.5%	0.1676	0.3902	-0.1530	0.8824	0.7417	0.8284
40	1.0897	1.9890	45.2%	0.1761	0.4070	-0.1609	0.8700	0.7214	0.8122
41	1.1321	2.0544	44.9%	0.1850	0.4241	-0.1692	0.8568	0.7001	0.7951
42	1.1752	2.1205	44.6%	0.1942	0.4416	-0.1778	0.8426	0.6776	0.7773
43	1.2194	2.1870	44.2%	0.2036	0.4592	-0.1865	0.8276	0.6541	0.7586
44	1.2643	2.2540	43.9%	0.2133	0.4771	-0.1956	0.8117	0.6297	0.7393
45	1.3095	2.3209	43.6%	0.2230	0.4951	-0.2048	0.7952	0.6045	0.7195
46	1.3549	2.3878	43.3%	0.2329	0.5132	-0.2140	0.7781	0.5787	0.6993
47	1.4008	2.4546	42.9%	0.2428	0.5313	-0.2233	0.7605	0.5524	0.6789
48	1.4470	2.5213	42.6%	0.2527	0.5494	-0.2327	0.7423	0.5255	0.6582
49	1.4934	2.5876	42.3%	0.2627	0.5673	-0.2420	0.7238	0.4985	0.6374
50	1.5398	2.6535	42.0%	0.2726	0.5852	-0.2514	0.7050	0.4712	0.6165
51	1.5860	2.7188	41.7%	0.2827	0.6031	-0.2608	0.6860	0.4439	0.5959
52	1.6327	2.7840	41.4%	0.2928	0.6211	-0.2704	0.6667	0.4166	0.5754
53	1.6793	2.8487	41.0%	0.3031	0.6391	-0.2800	0.6473	0.3892	0.5550
54	1.7259	2.9131	40.8%	0.3132	0.6572	-0.2896	0.6278	0.3620	0.5349
55	1.7723	2.9772	40.5%	0.3234	0.6752	-0.2992	0.6084	0.3348	0.5152
56	1.8183	3.0409	40.2%	0.3334	0.6931	-0.3088	0.5893	0.3079	0.4960
57	1.8642	3.1042	39.9%	0.3434	0.7110	-0.3183	0.5704	0.2813	0.4773
58	1.9100	3.1668	39.7%	0.3535	0.7287	-0.3278	0.5518	0.2552	0.4591
59	1.9557	3.2292	39.4%	0.3635	0.7463	-0.3374	0.5337	0.2296	0.4415
60	2.0014	3.2912	39.2%	0.3736	0.7638	-0.3470	0.5162	0.2045	0.4247
61	2.0473	3.3528	38.9%	0.3836	0.7813	-0.3566	0.4994	0.1801	0.4086
62	2.0933	3.4139	38.7%	0.3936	0.7985	-0.3662	0.4832	0.1565	0.3931
63	2.1391	3.4741	38.4%	0.4036	0.8155	-0.3758	0.4677	0.1338	0.3784
64	2.1845	3.5329	38.2%	0.4135	0.8320	-0.3853	0.4530	0.1121	0.3646
65	2.2299	3.5910	37.9%	0.4234	0.8483	-0.3948	0.4391	0.0913	0.3514
66	2.2751	3.6481	37.6%	0.4331	0.8641	-0.4042	0.4258	0.0715	0.3390
67	2.3199	3.7045	37.4%	0.4427	0.8795	-0.4135	0.4133	0.0524	0.3272
68	2.3643	3.7596	37.1%	0.4522	0.8945	-0.4226	0.4013	0.0343	0.3160
69	2.4078	3.8136	36.9%	0.4615	0.9091	-0.4315	0.3901	0.0169	0.3054
70	2.4502	3.8659	36.6%	0.4705	0.9233	-0.4402	0.3795	0.0005	0.2955
71	2.4919	3.9166	36.4%	0.4793	0.9370	-0.4487	0.3696	-0.0149	0.2863
72	2.5327	3.9656	36.1%	0.4878	0.9503	-0.4569	0.3603	-0.0294	0.2776
73	2.5728	4.0132	35.9%	0.4961	0.9631	-0.4649	0.3516	-0.0431	0.2694
74	2.6119	4.0591	35.7%	0.5043	0.9755	-0.4727	0.3432	-0.0558	0.2616
75	2.6506	4.1036	35.4%	0.5123	0.9875	-0.4804	0.3354	-0.0678	0.2543
76	2.6882	4.1466	35.2%	0.5201	0.9991	-0.4879	0.3280	-0.0790	0.2473
77	2.7245	4.1876	34.9%	0.5277	1.0101	-0.4952	0.3210	-0.0895	0.2407
78	2.7596	4.2264	34.7%	0.5351	1.0206	-0.5022	0.3139	-0.0993	0.2340
79	2.7938	4.2630	34.5%	0.5423	1.0305	-0.5090	0.3067	-0.1084	0.2274
80	2.8271	4.2978	34.2%	0.5493	1.0400	-0.5156	0.2997	-0.1168	0.2209
81	2.8591	4.3305	34.0%	0.5561	1.0491	-0.5220	0.2925	-0.1248	0.2143
82	2.8902	4.3612	33.7%	0.5626	1.0576	-0.5281	0.2853	-0.1321	0.2078
83	2.9202	4.3898	33.5%	0.5690	1.0657	-0.5340	0.2780	-0.1388	0.2011
84	2.9491	4.4167	33.2%	0.5752	1.0733	-0.5397	0.2707	-0.1452	0.1946
85	2.9757	4.4418	33.0%	0.5803	1.0805	-0.5445	0.2643	-0.1511	0.1888
86	3.0002	4.4649	32.8%	0.5848	1.0873	-0.5487	0.2584	-0.1565	0.1835
87	3.0228	4.4865	32.6%	0.5889	1.0937	-0.5525	0.2529	-0.1616	0.1786
88	3.0436	4.5062	32.5%	0.5926	1.0998	-0.5559	0.2478	-0.1664	0.1741
89	3.0631	4.5244	32.3%	0.5961	1.1055	-0.5592	0.2429	-0.1708	0.1698
90	3.0811	4.5408	32.1%	0.5993	1.1109	-0.5622	0.2382	-0.1748	0.1658

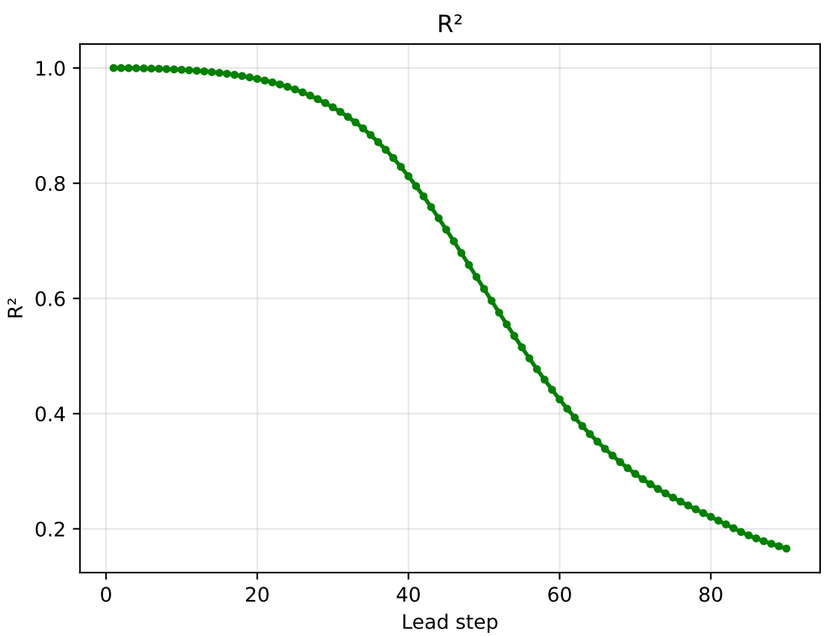
DHW Forecast Skill vs Lead Step - Test

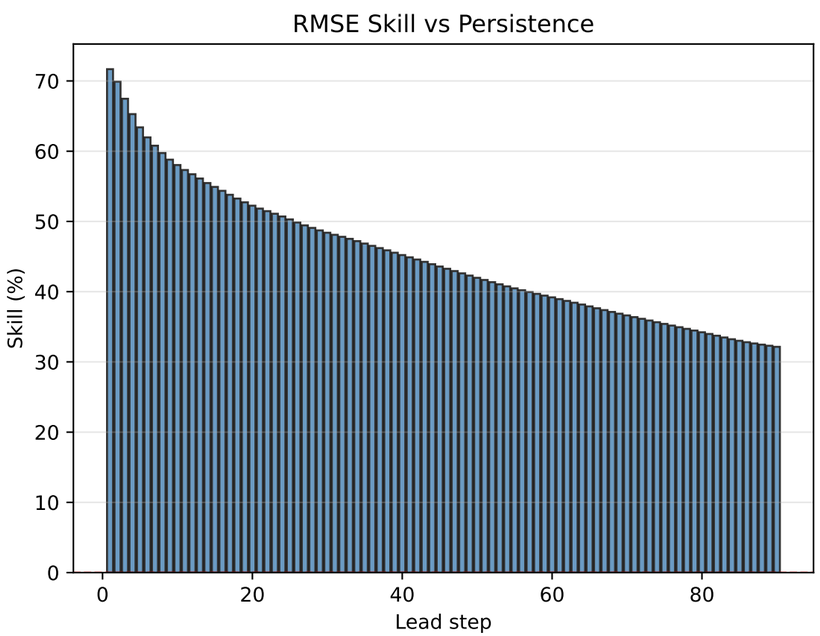


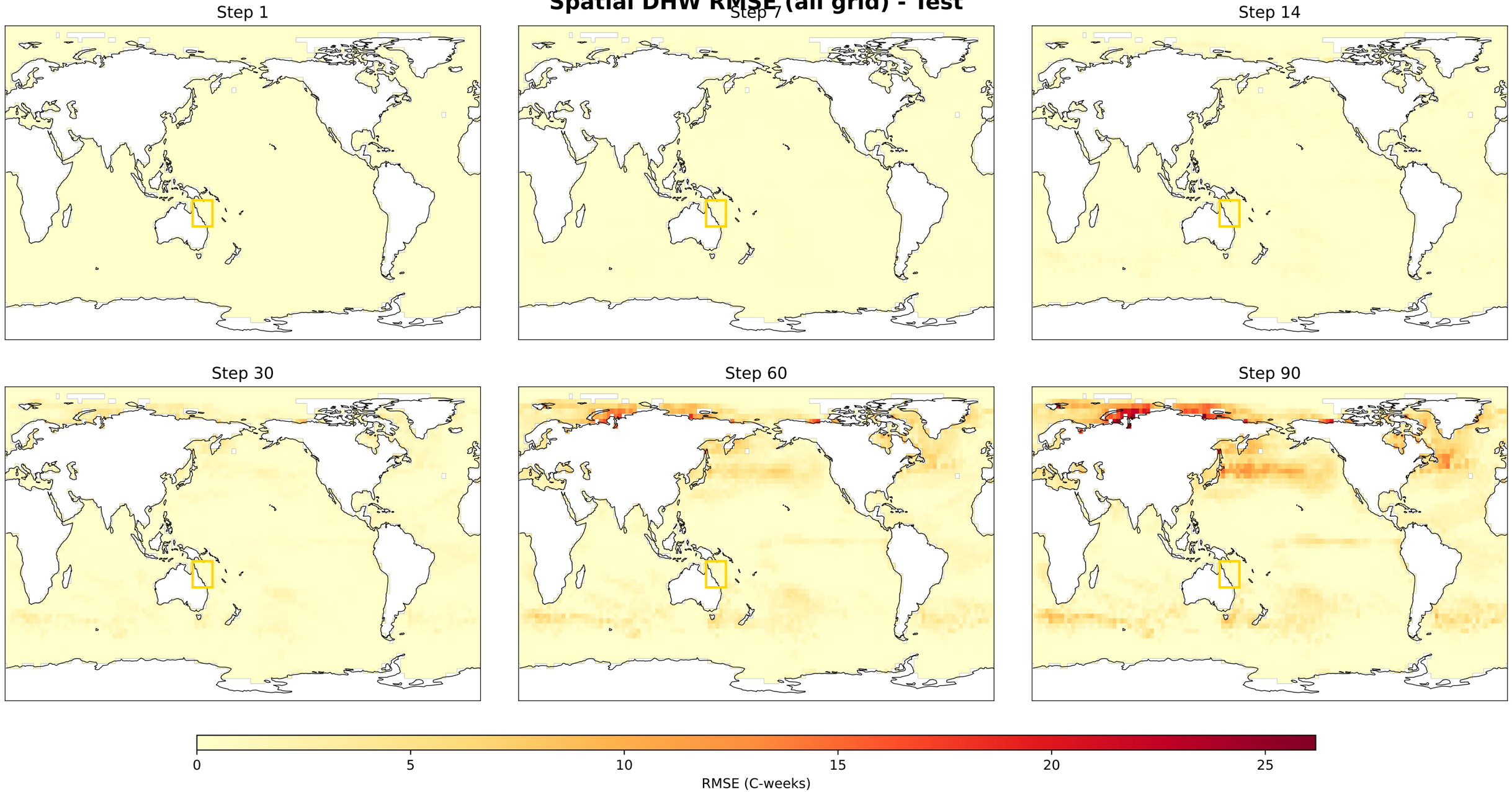


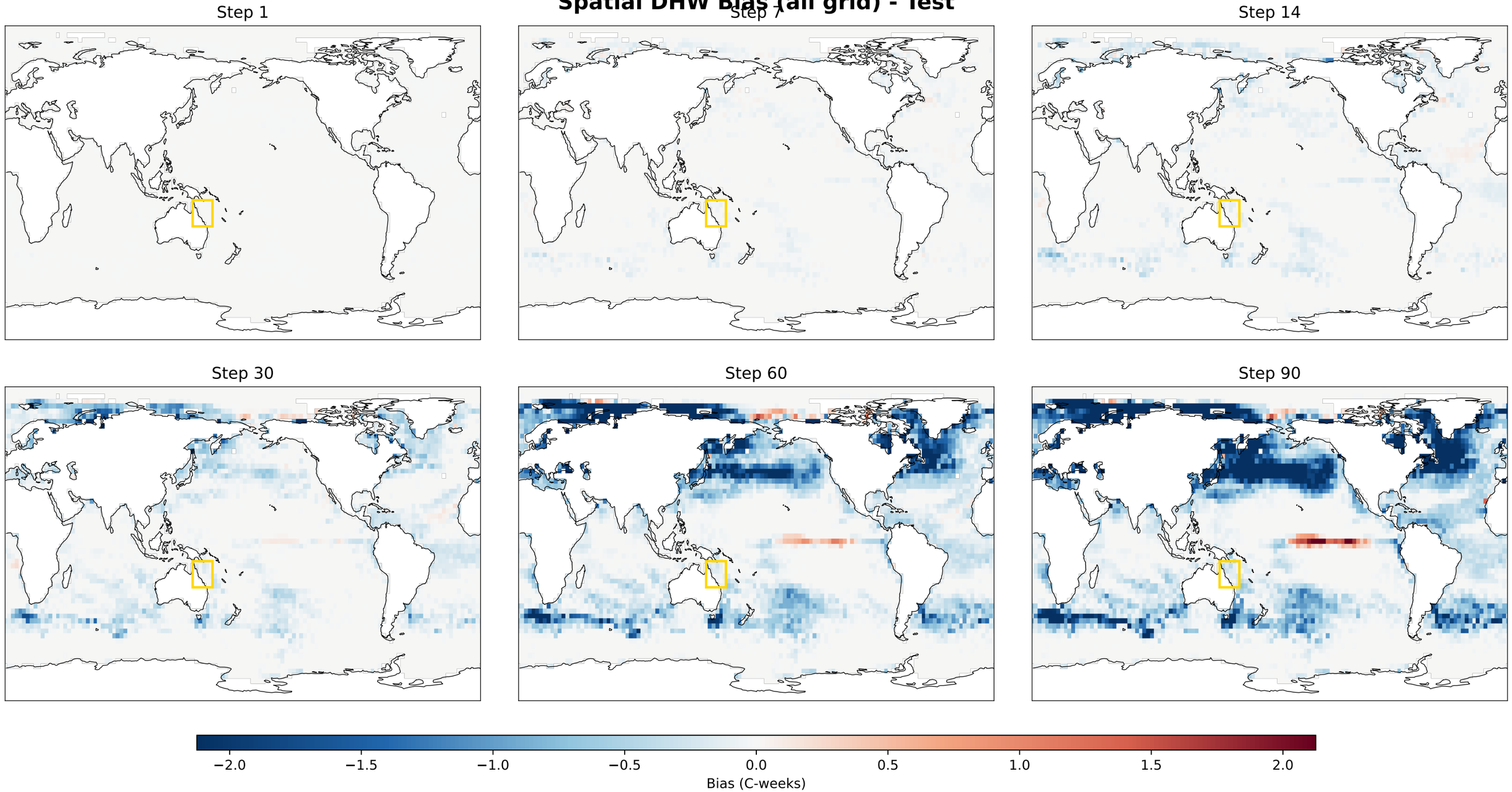


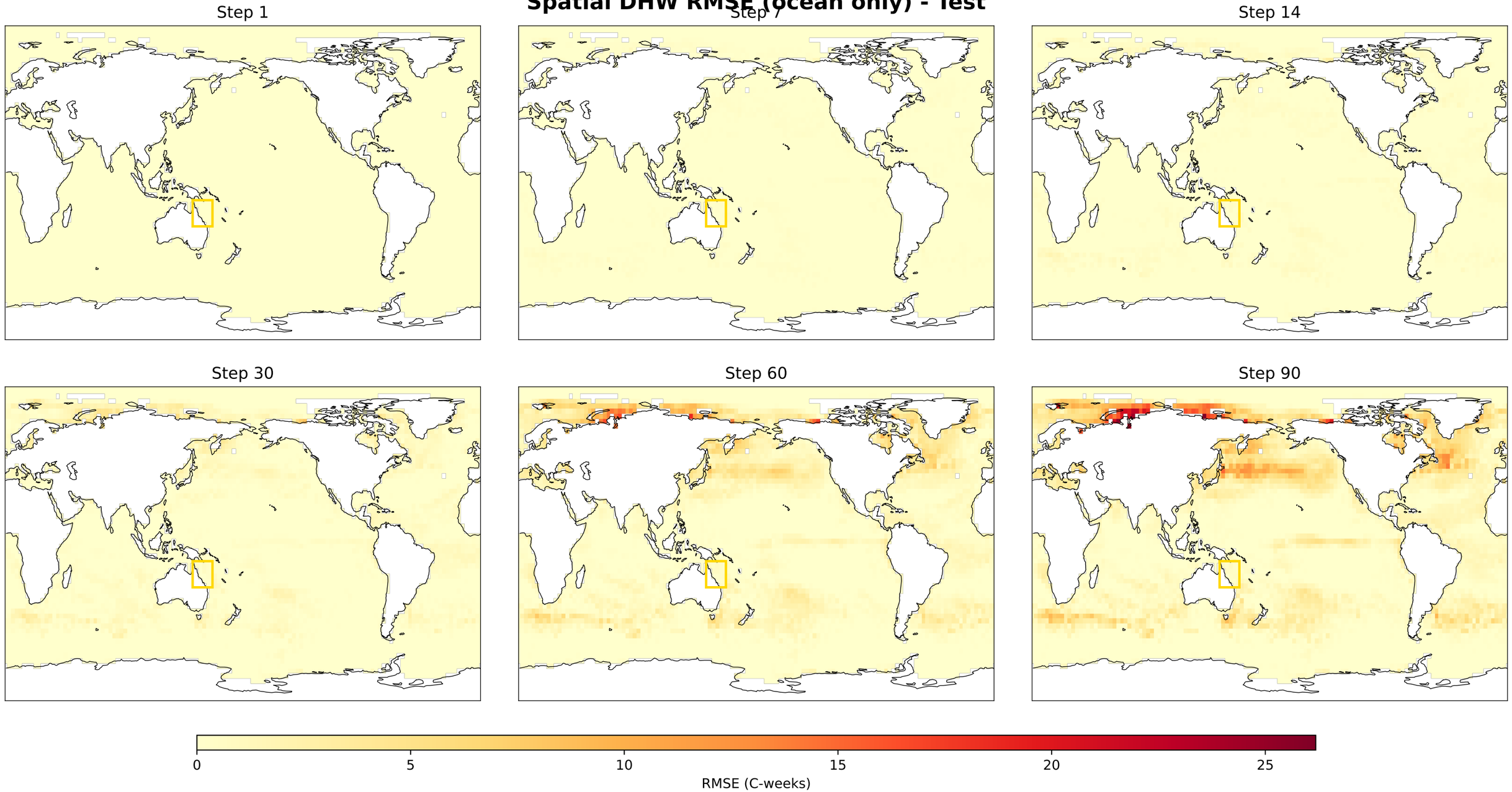


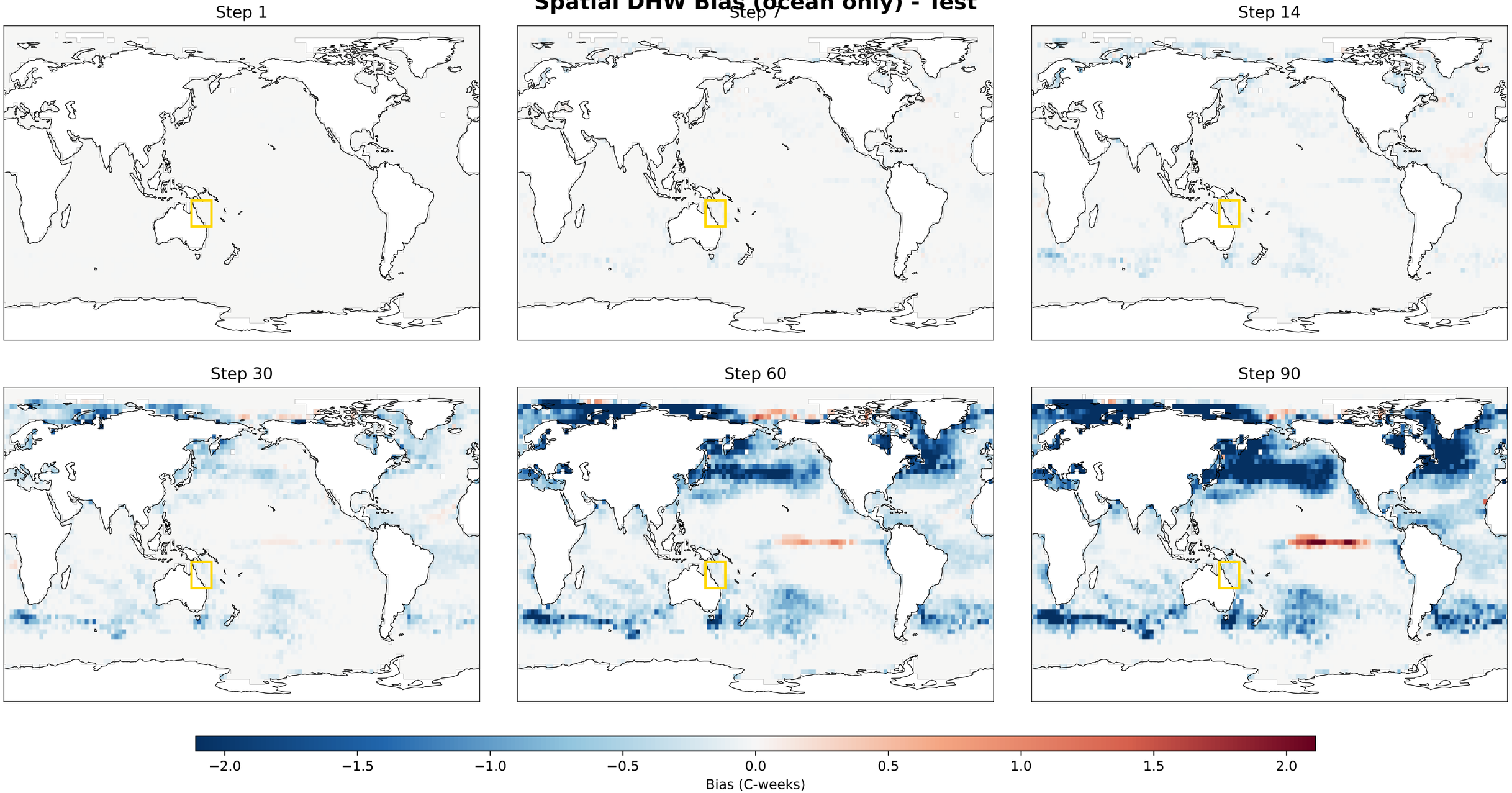





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

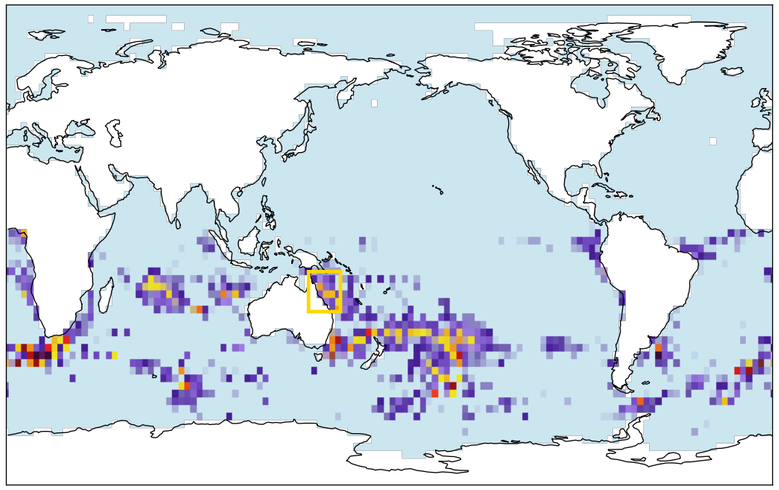

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

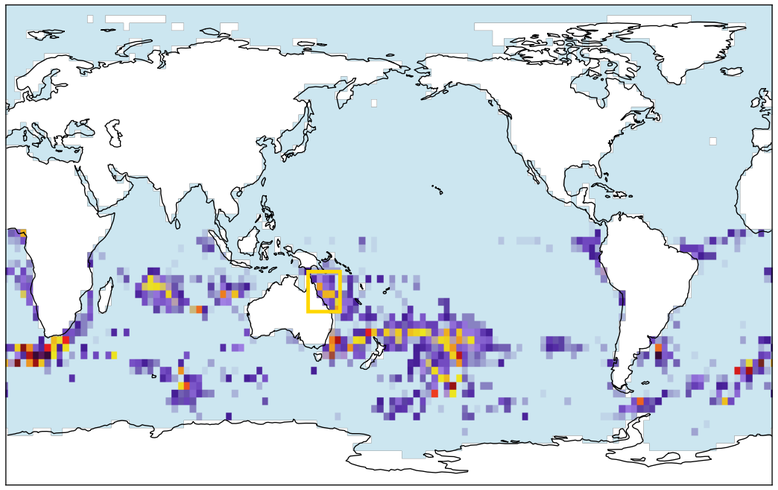

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

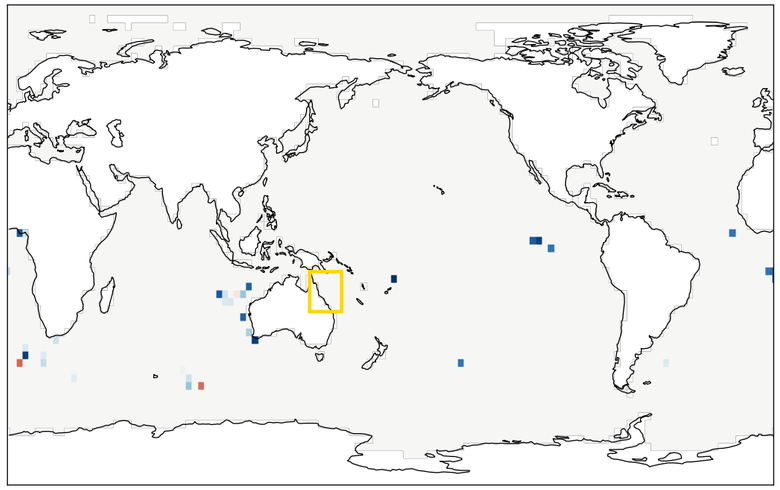
Actual DHW - Step 1



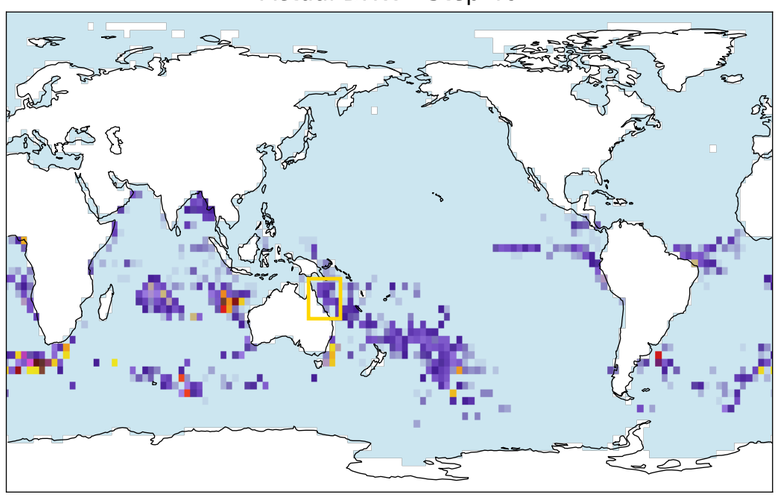
Predicted DHW - Step 1



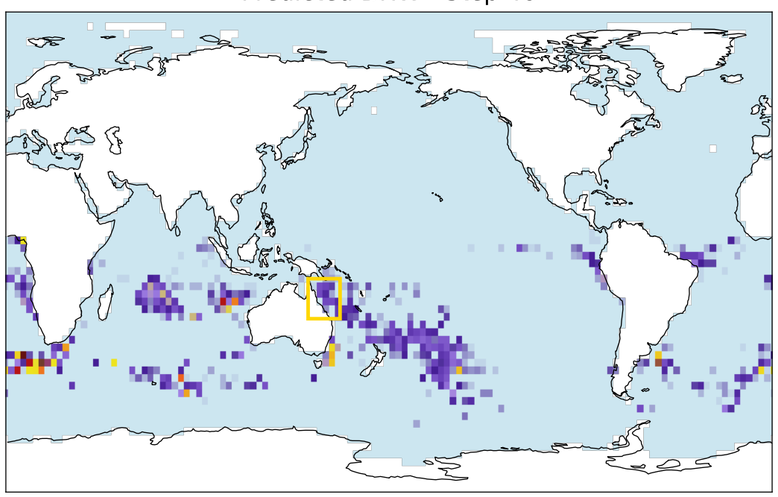
Error (Pred - Actual) - Step 1 [scale ±0.24]



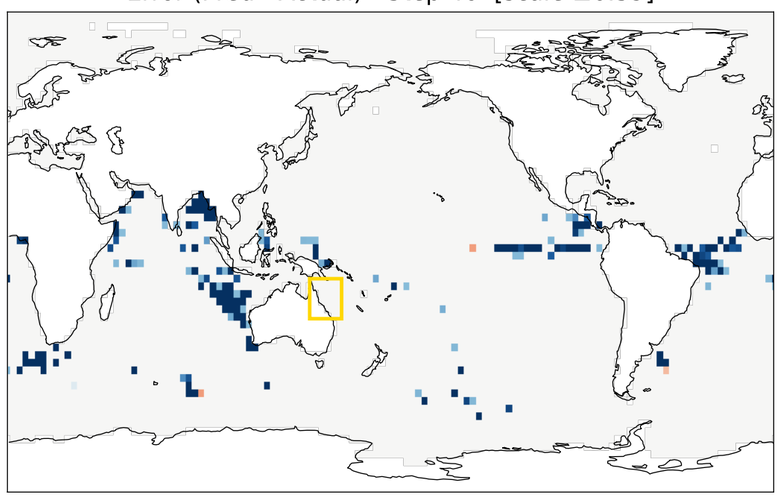
Actual DHW - Step 46



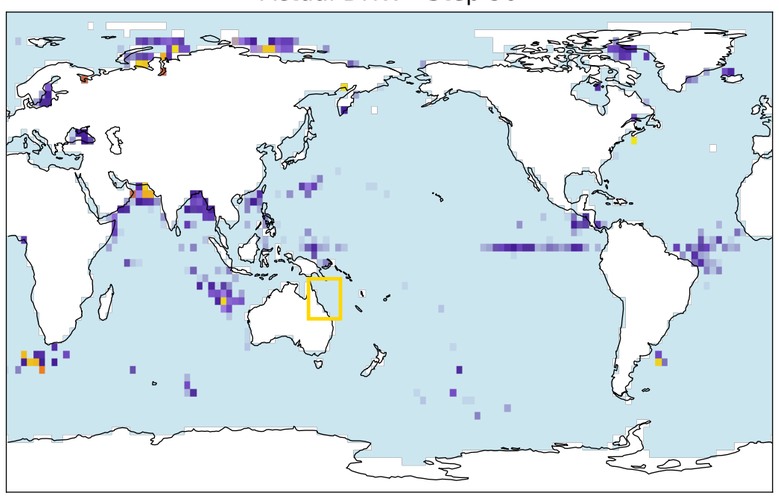
Predicted DHW - Step 46



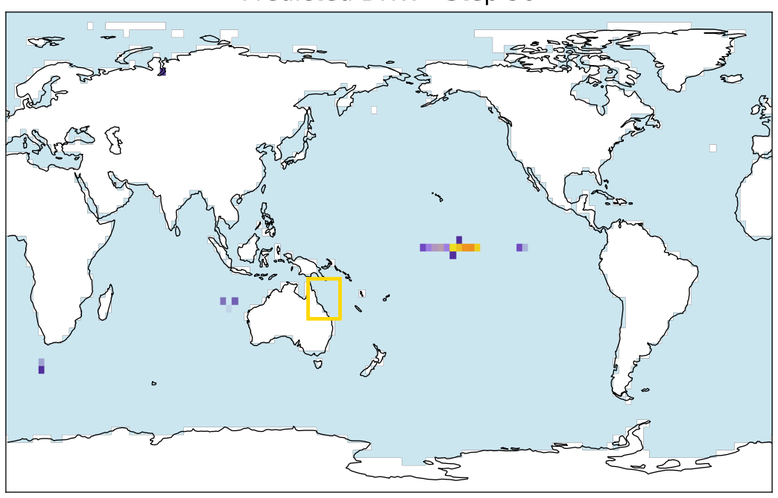
Error (Pred - Actual) - Step 46 [scale ±0.39]



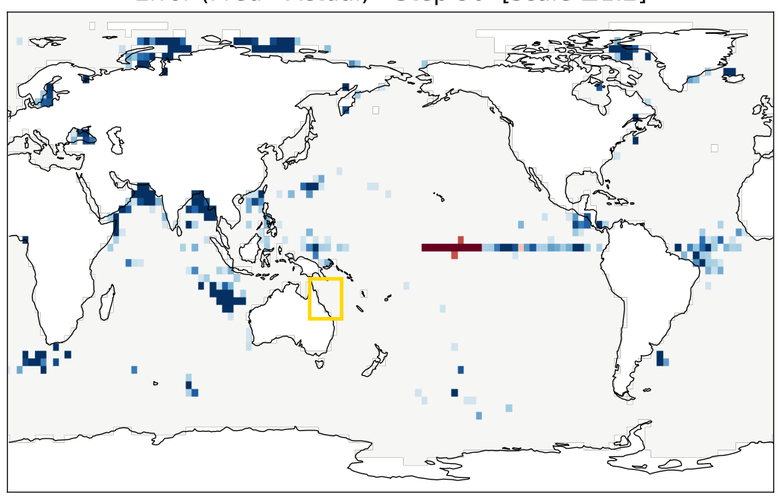
Actual DHW - Step 90

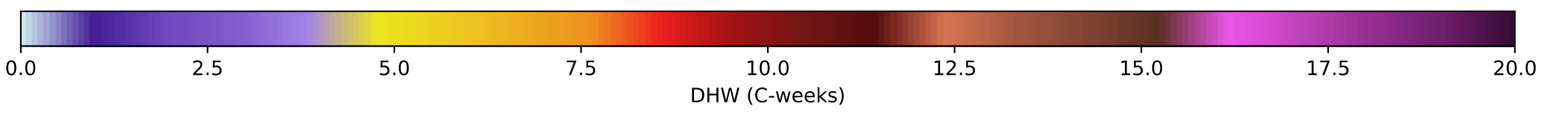


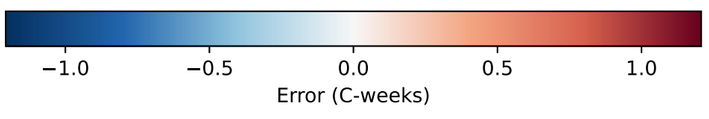
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.2]

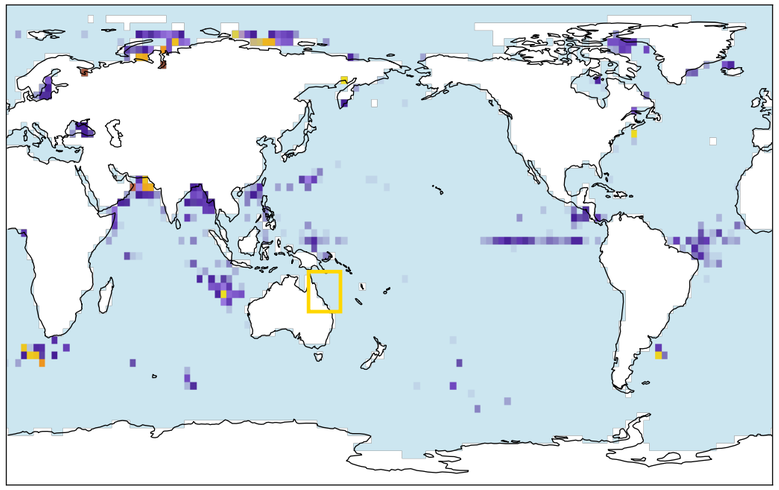




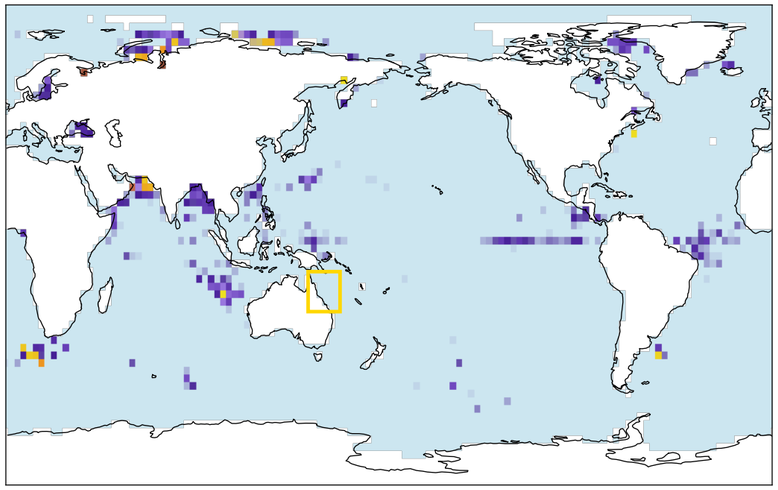


DHW Chain init: 2020-07-09 - Test (C-weeks)

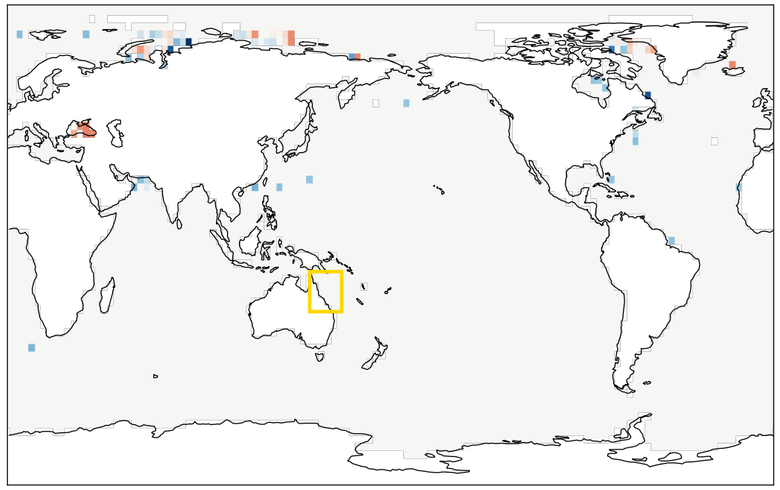
Actual DHW - Step 1



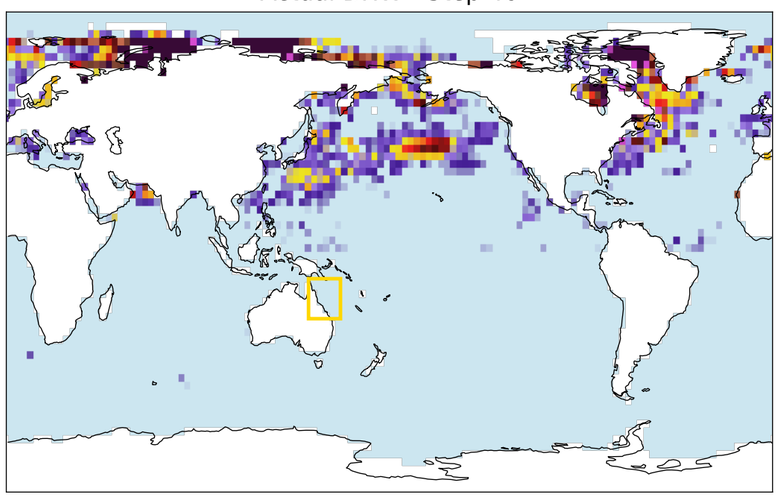
Predicted DHW - Step 1



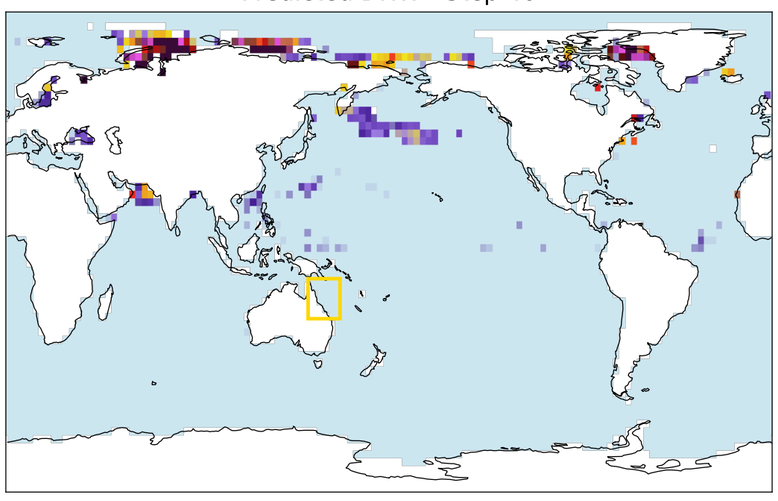
Error (Pred - Actual) - Step 1 [scale ±0.47]



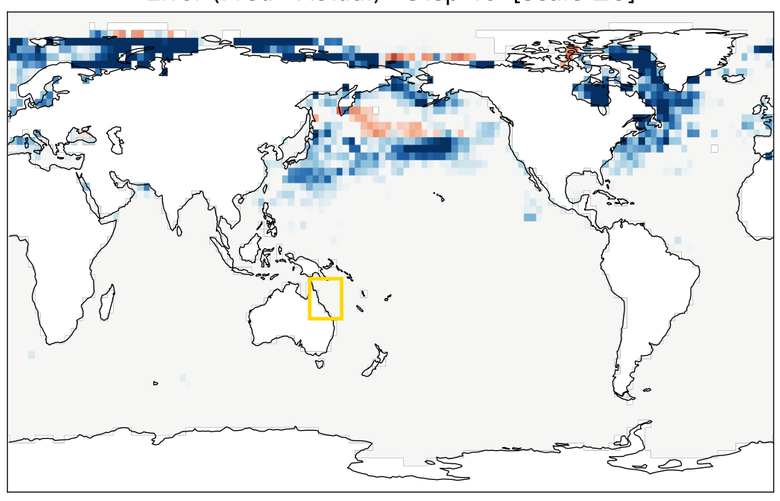
Actual DHW - Step 46



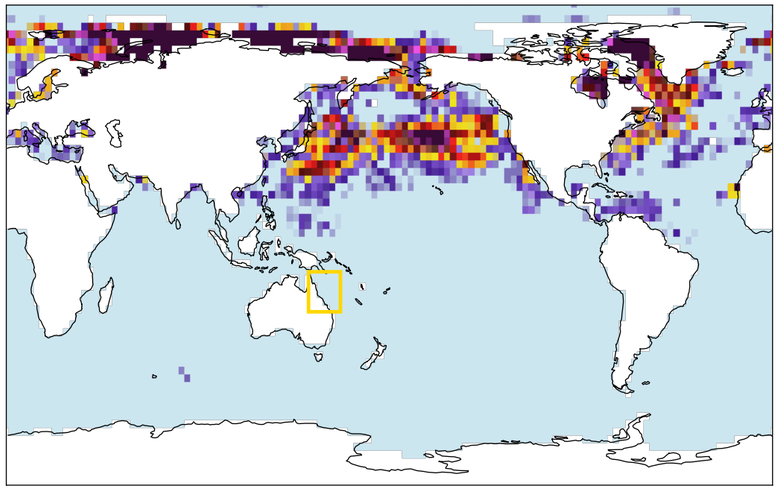
Predicted DHW - Step 46



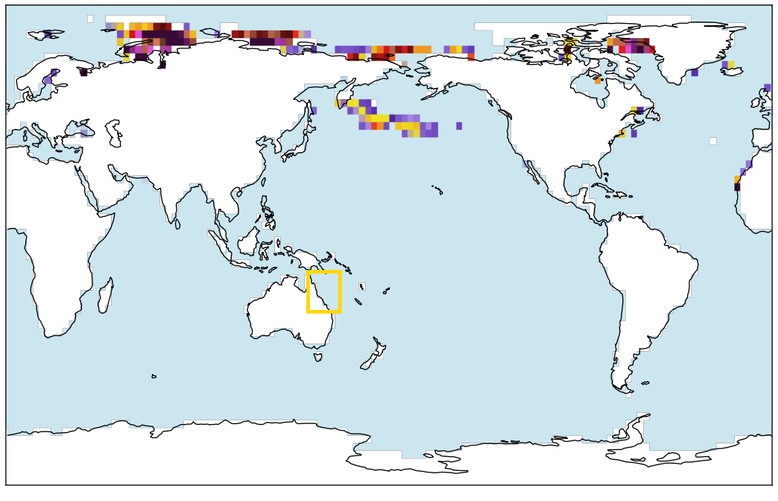
Error (Pred - Actual) - Step 46 [scale ±8]



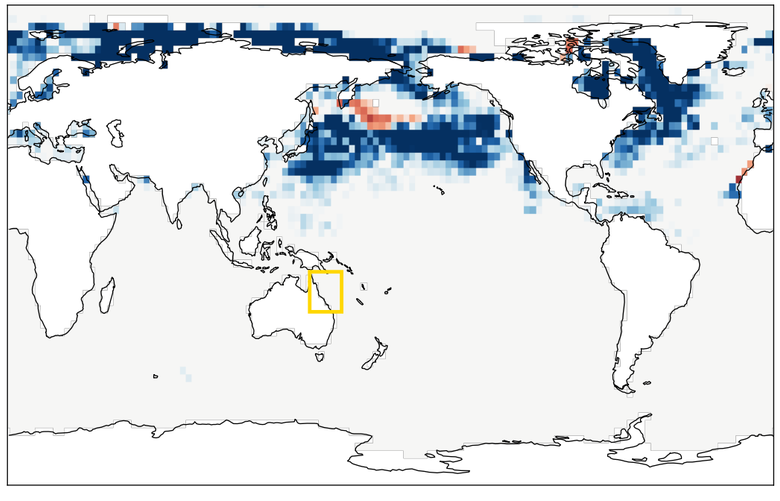
Actual DHW - Step 90

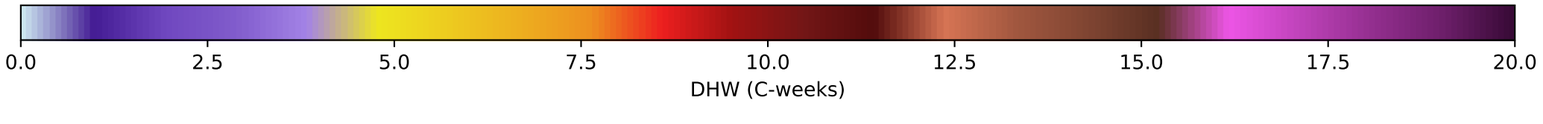


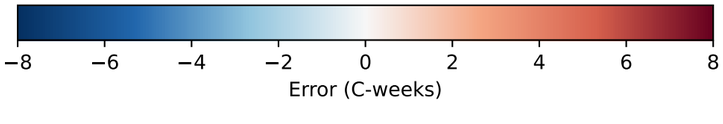
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

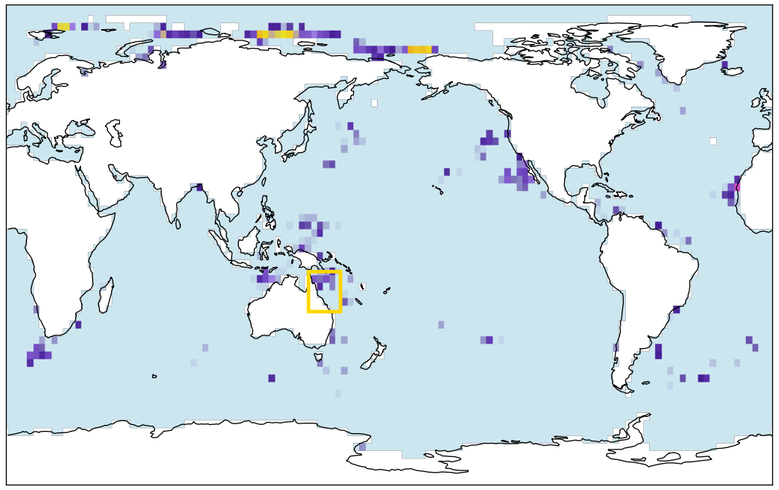




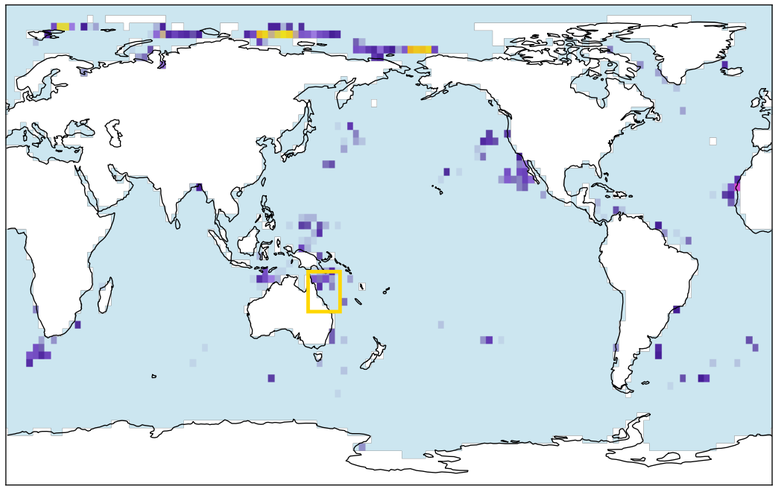


DHW Chain init: 2021-01-06 - Test (C-weeks)

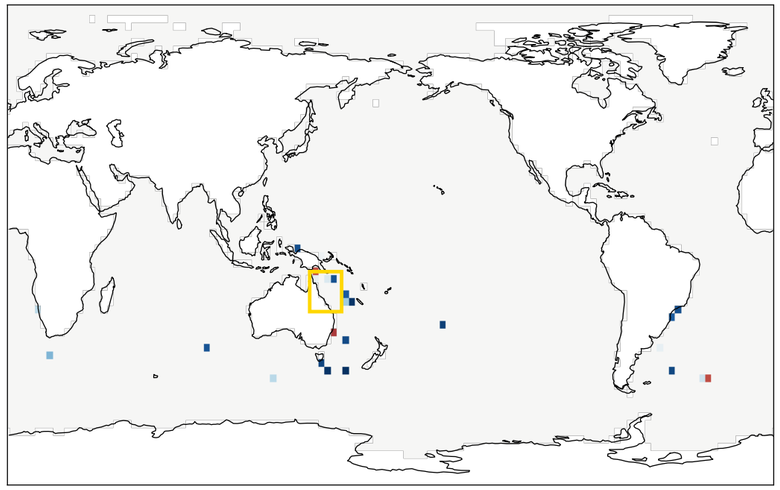
Actual DHW - Step 1



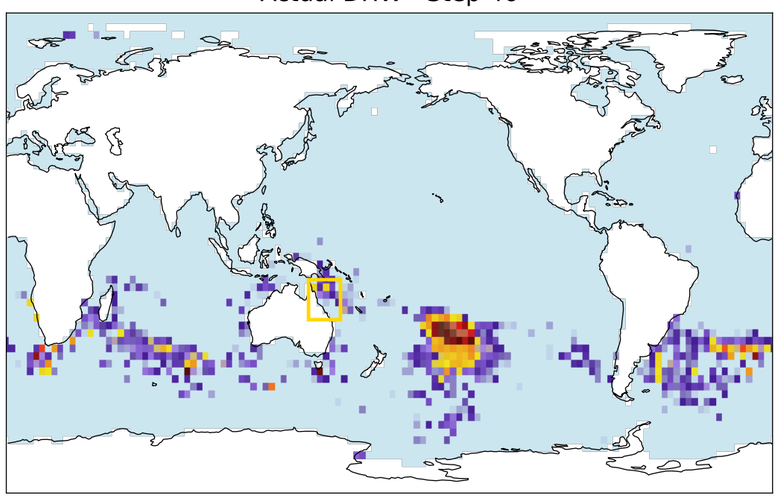
Predicted DHW - Step 1



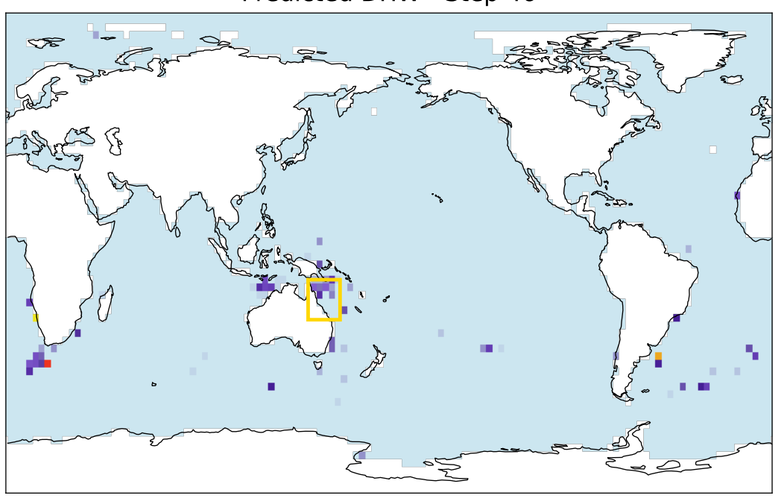
Error (Pred - Actual) - Step 1 [scale ±0.2]



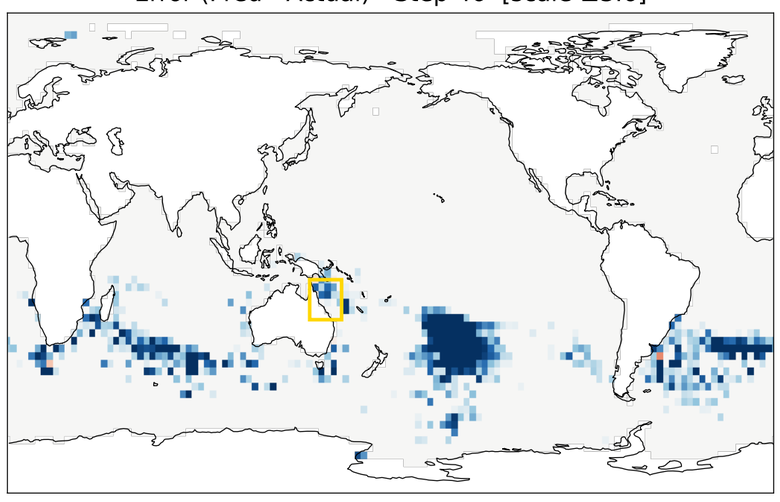
Actual DHW - Step 46



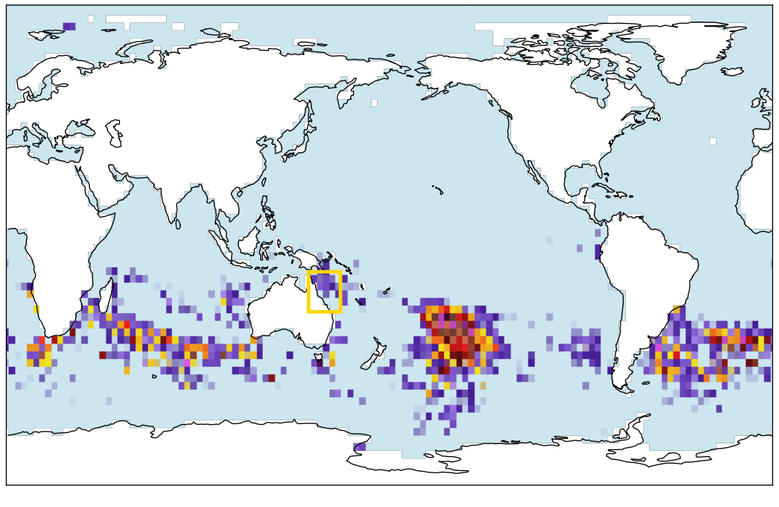
Predicted DHW - Step 46



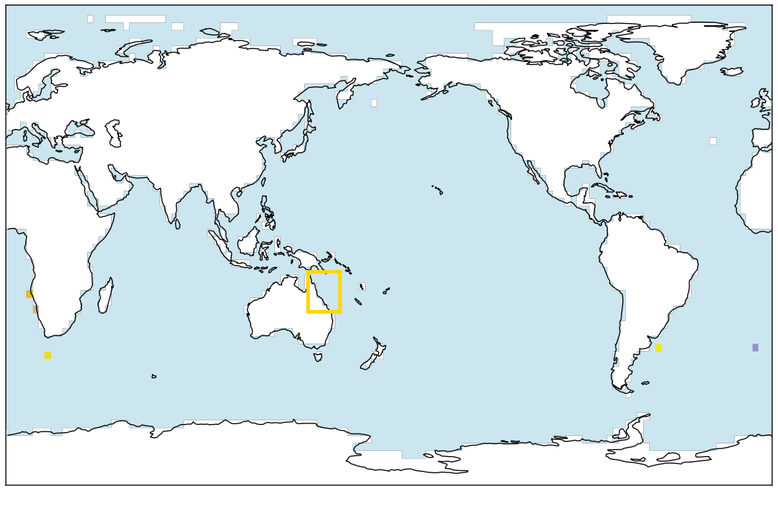
Error (Pred - Actual) - Step 46 [scale ±3.6]



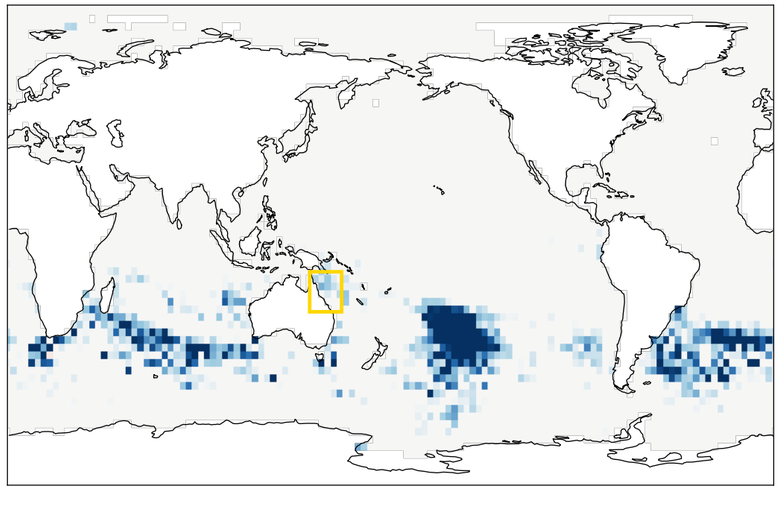
Actual DHW - Step 90

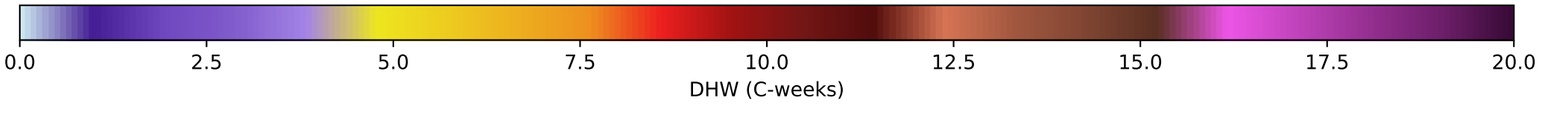


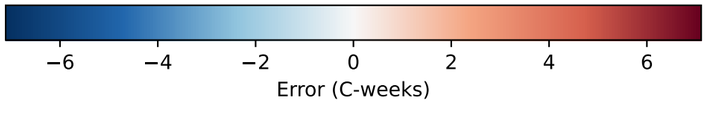
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±7.1]

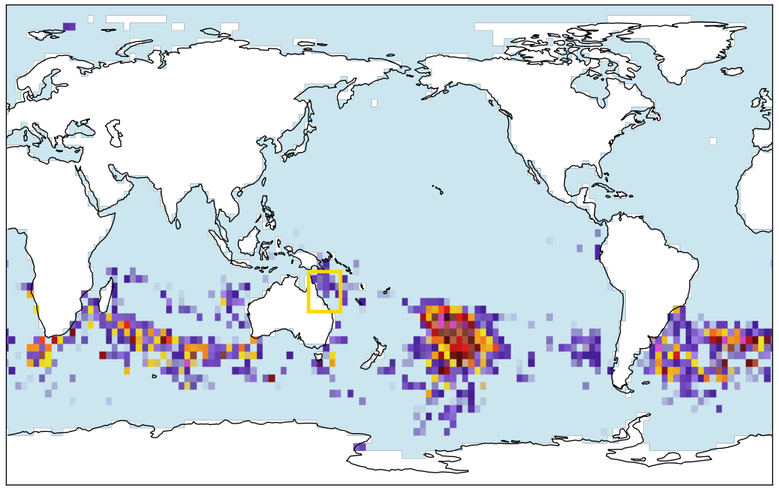




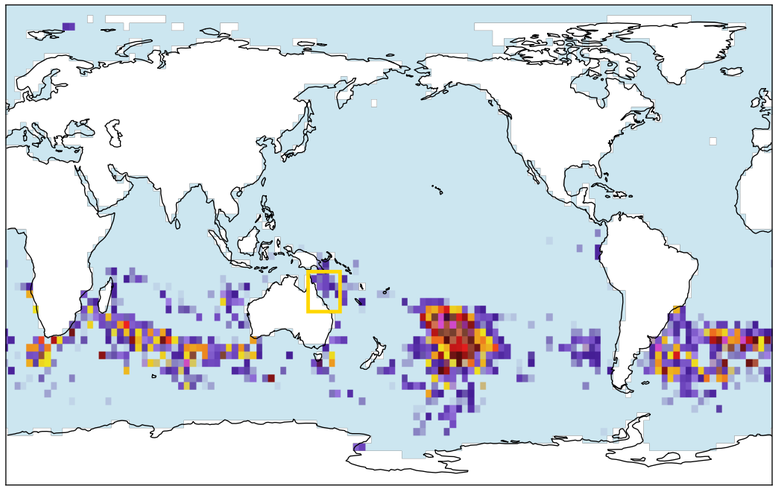


DHW Chain init: 2021-04-06 - Test (C-weeks)

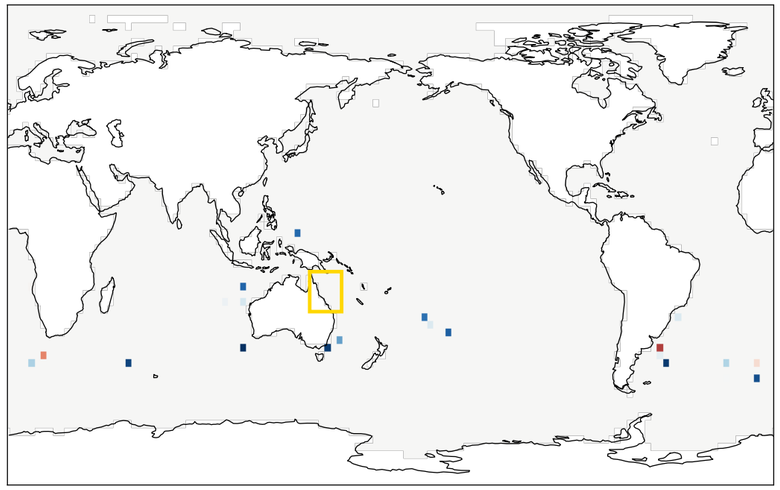
Actual DHW - Step 1



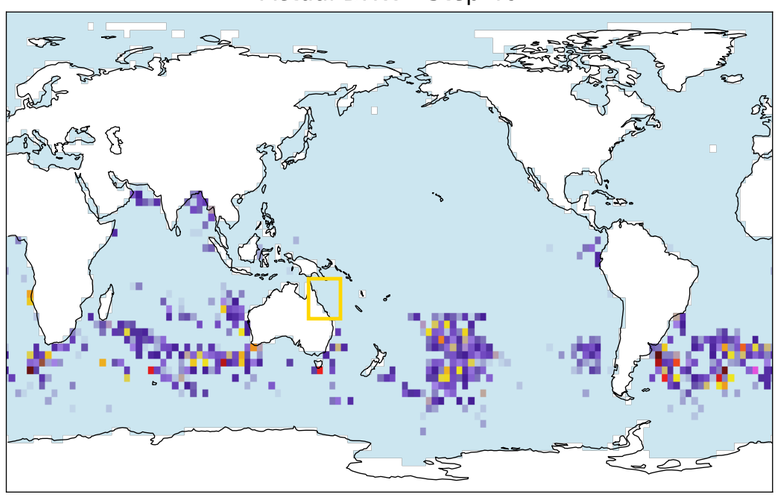
Predicted DHW - Step 1



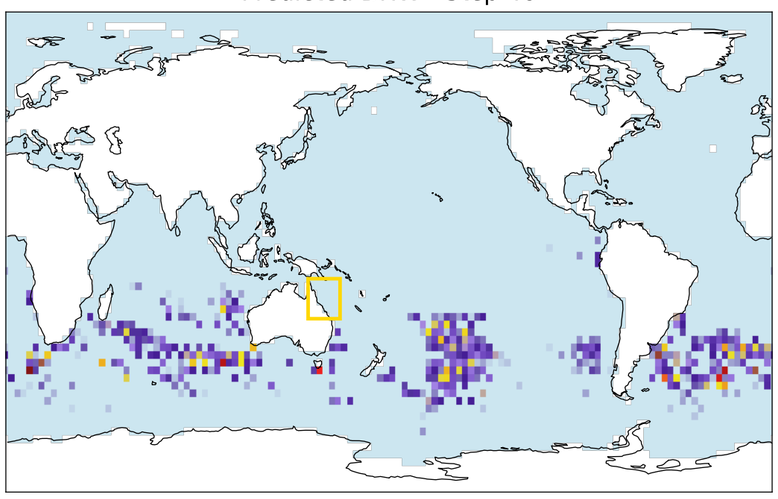
Error (Pred - Actual) - Step 1 [scale ±0.23]



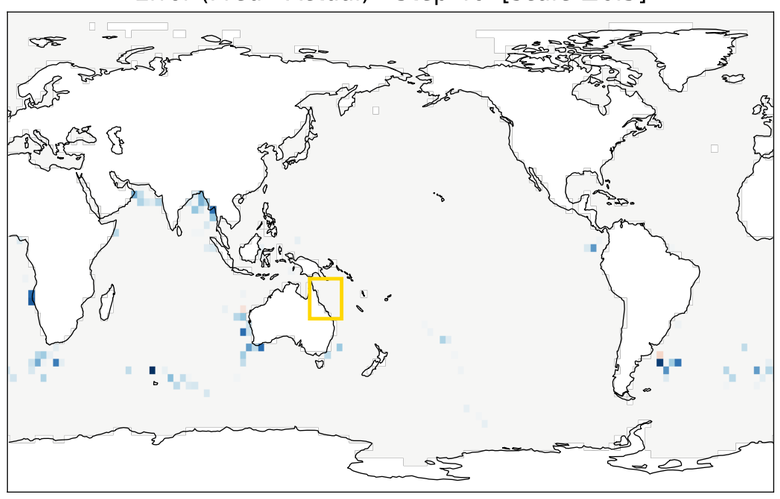
Actual DHW - Step 46



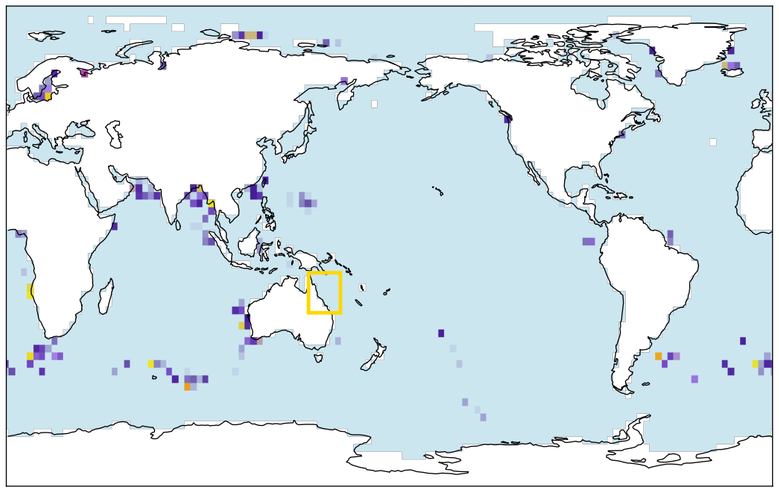
Predicted DHW - Step 46



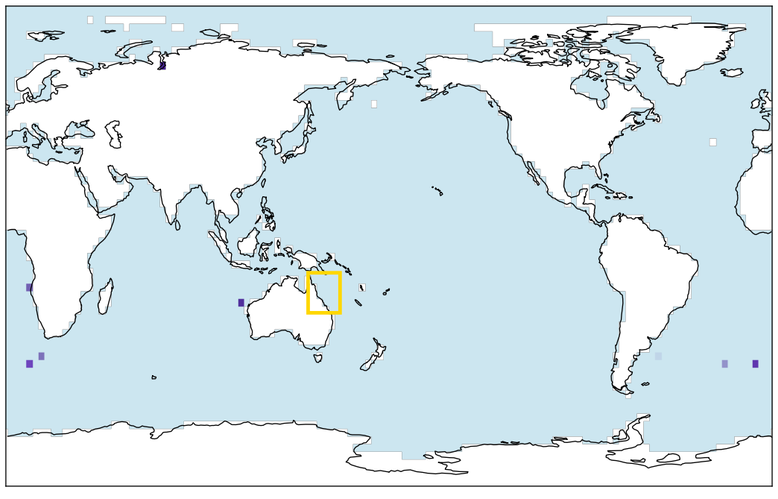
Error (Pred - Actual) - Step 46 [scale ±6.5]



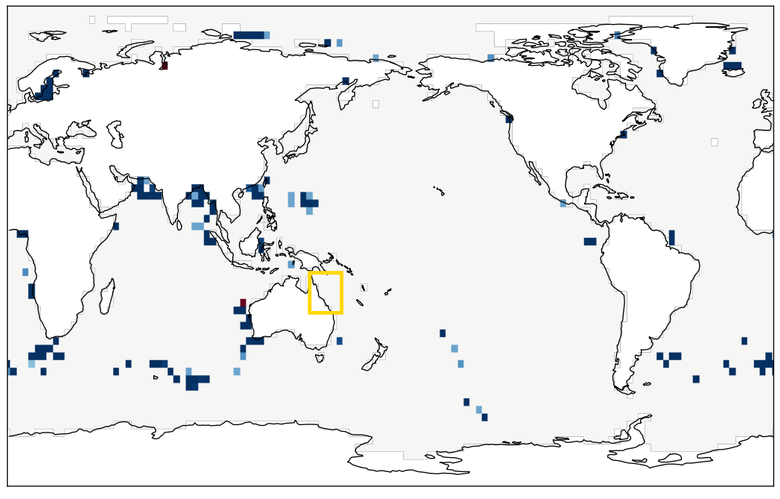
Actual DHW - Step 90

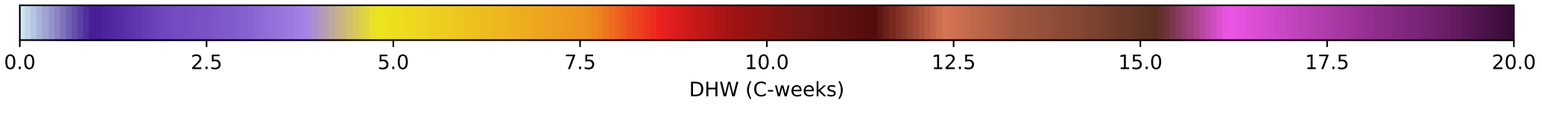


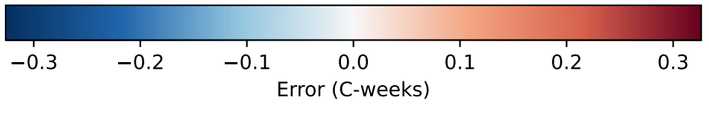
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

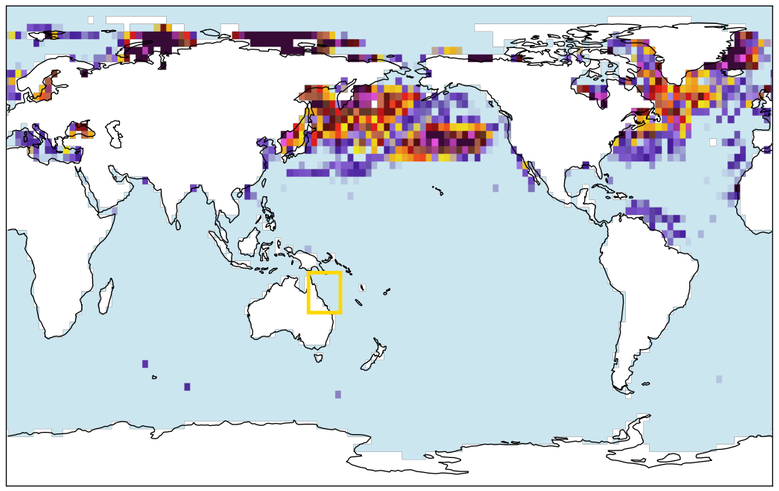




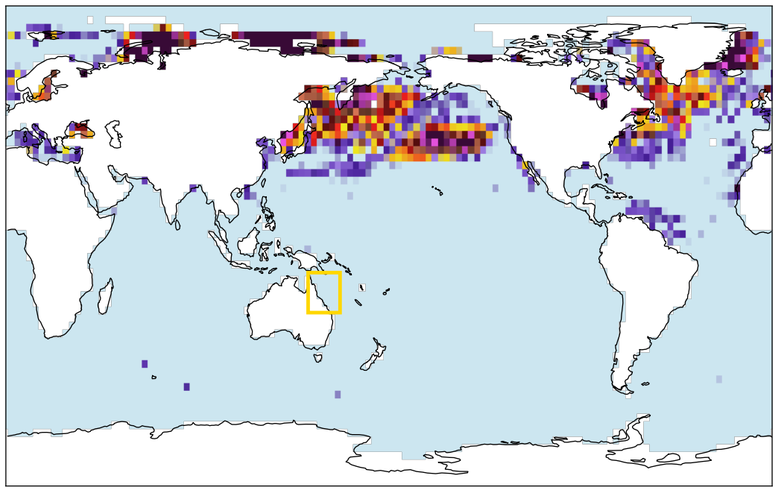


DHW Chain init: 2021-10-03 - Test (C-weeks)

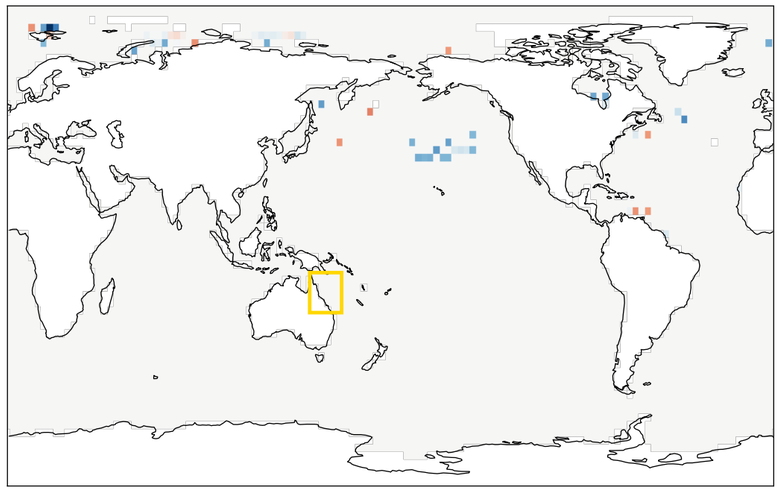
Actual DHW - Step 1



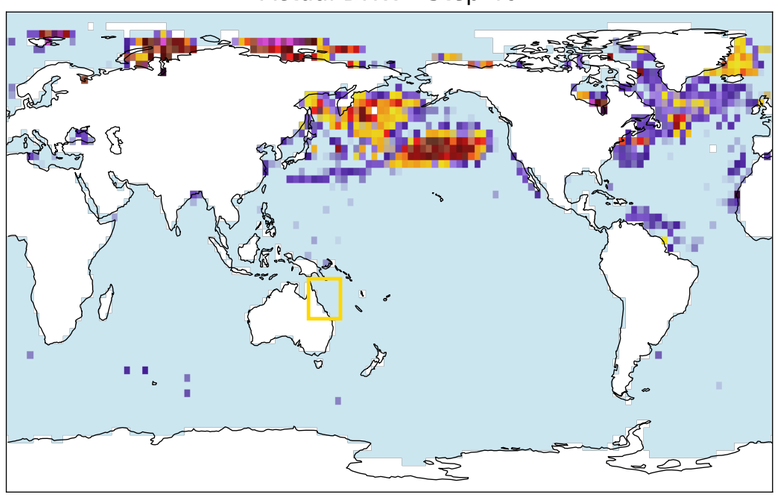
Predicted DHW - Step 1



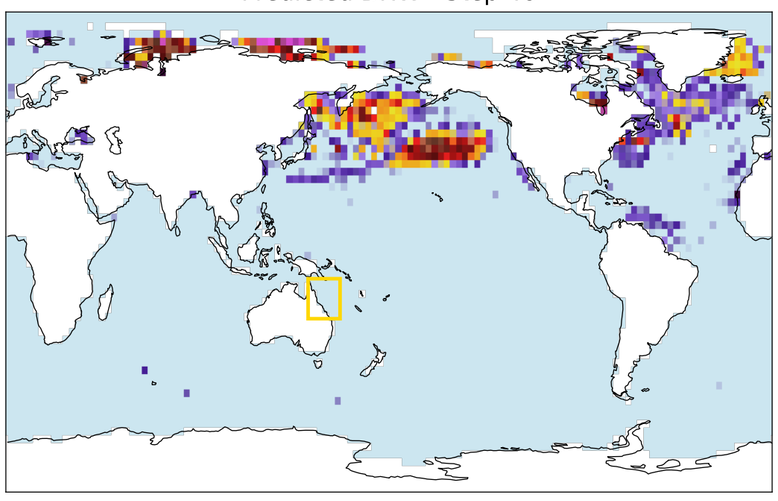
Error (Pred - Actual) - Step 1 [scale ±0.38]



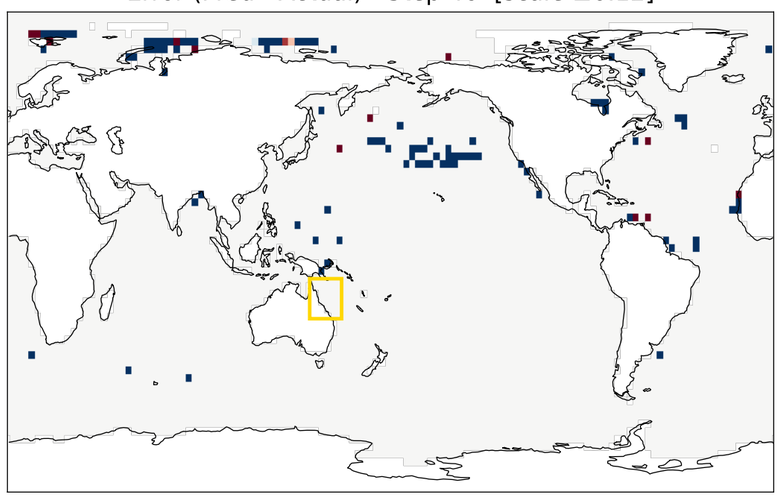
Actual DHW - Step 46



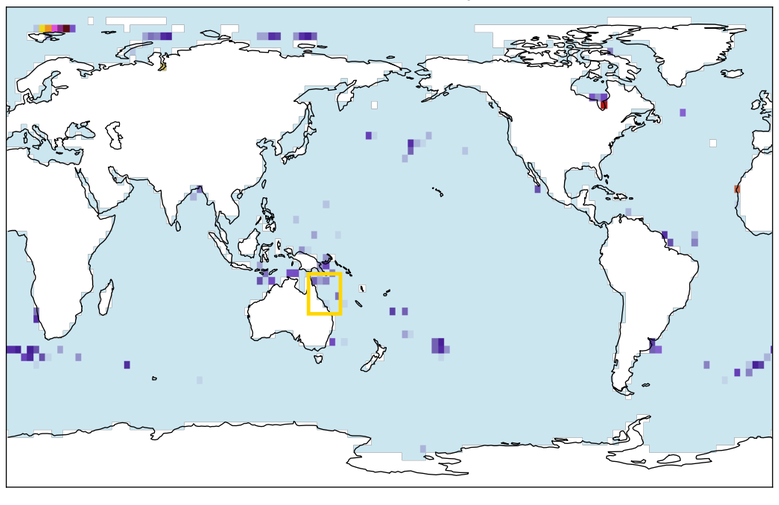
Predicted DHW - Step 46



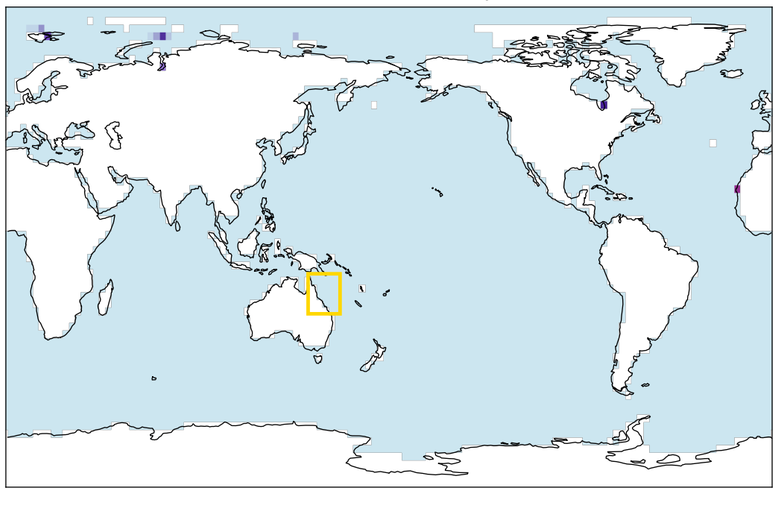
Error (Pred - Actual) - Step 46 [scale ±0.12]



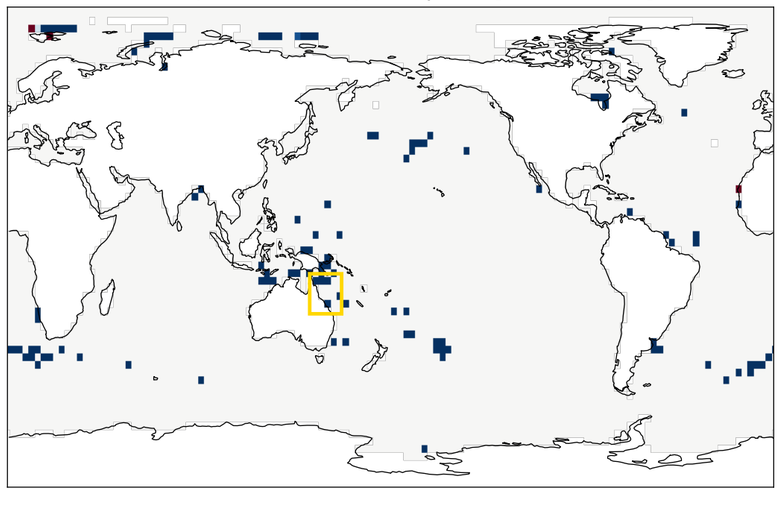
Actual DHW - Step 90

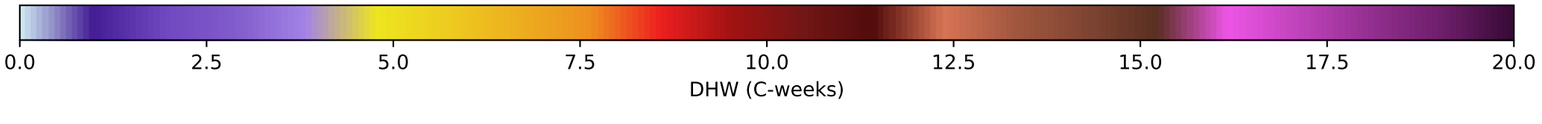


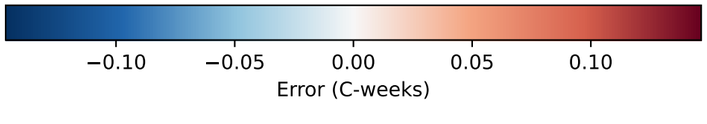
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.15]

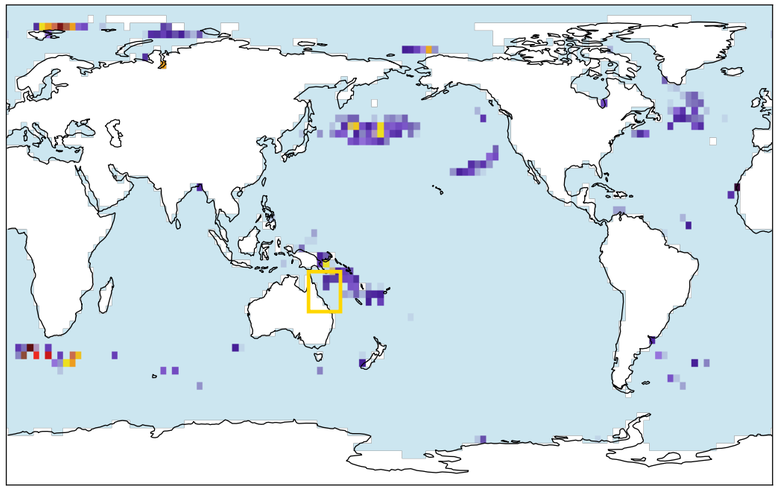




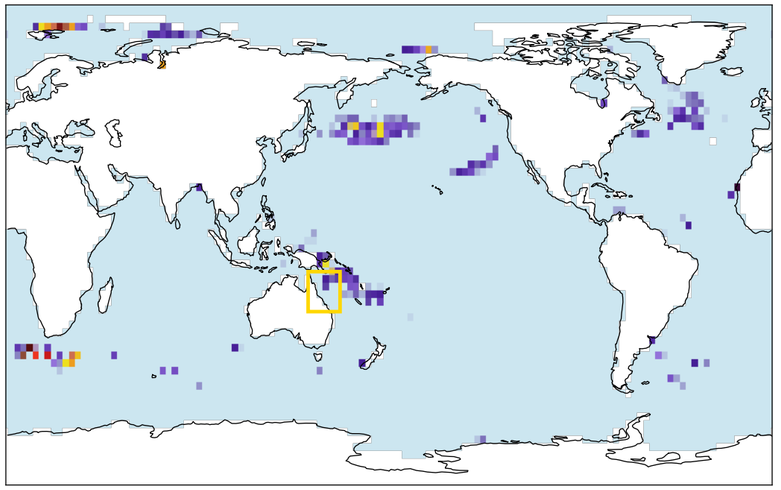


DHW Chain init: 2022-12-27 - Test (C-weeks)

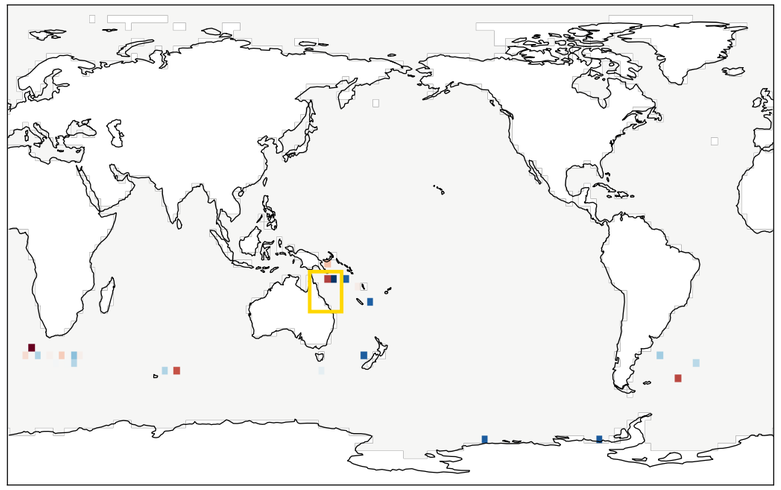
Actual DHW - Step 1



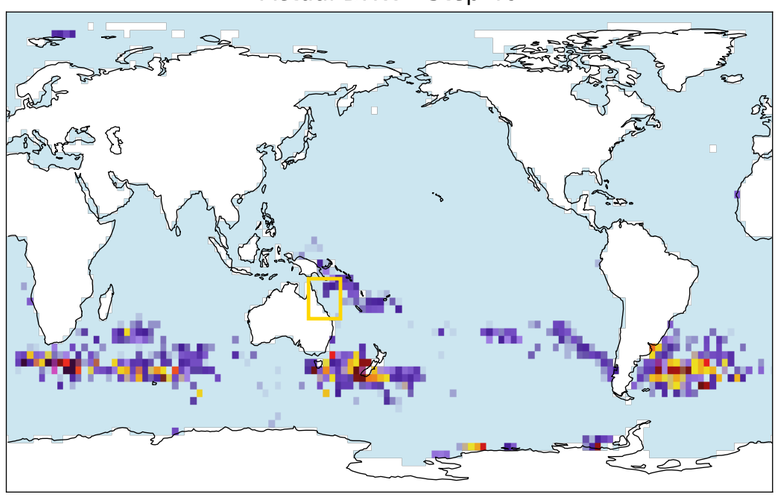
Predicted DHW - Step 1



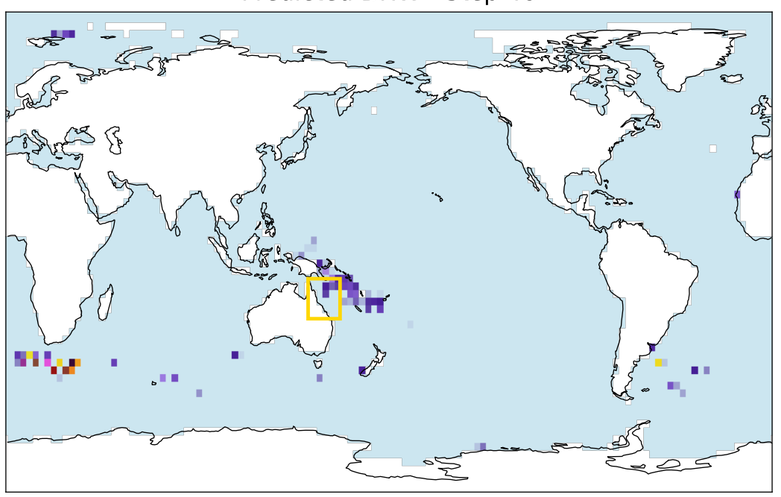
Error (Pred - Actual) - Step 1 [scale ±0.2]



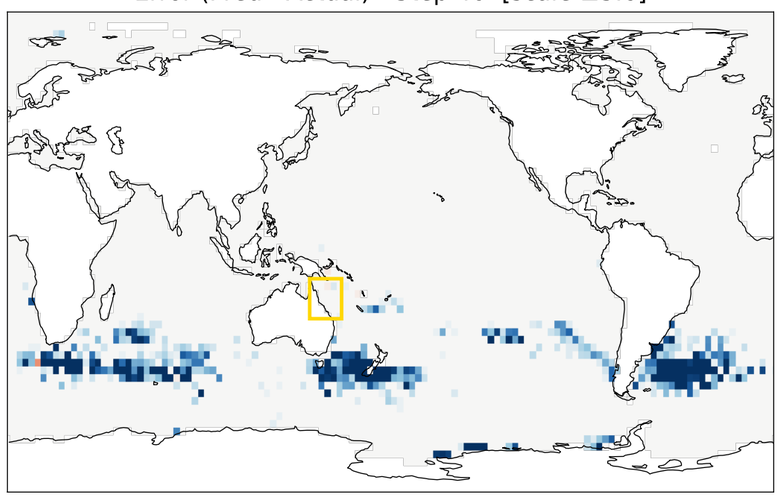
Actual DHW - Step 46



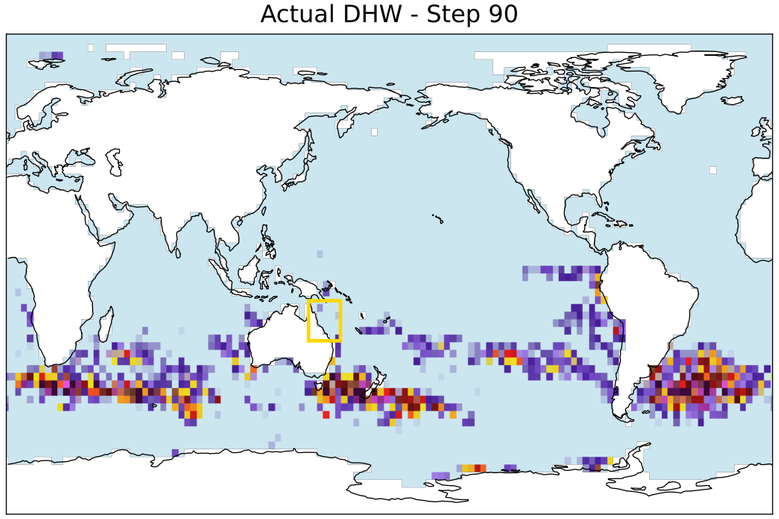
Predicted DHW - Step 46



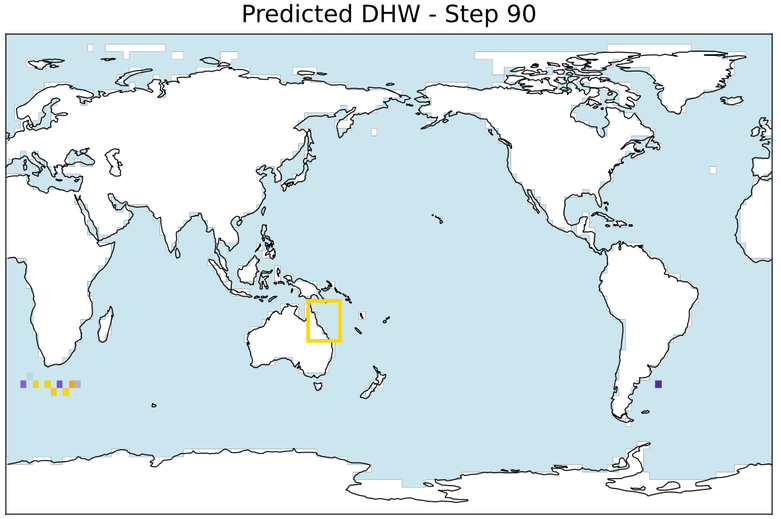
Error (Pred - Actual) - Step 46 [scale ±3.6]



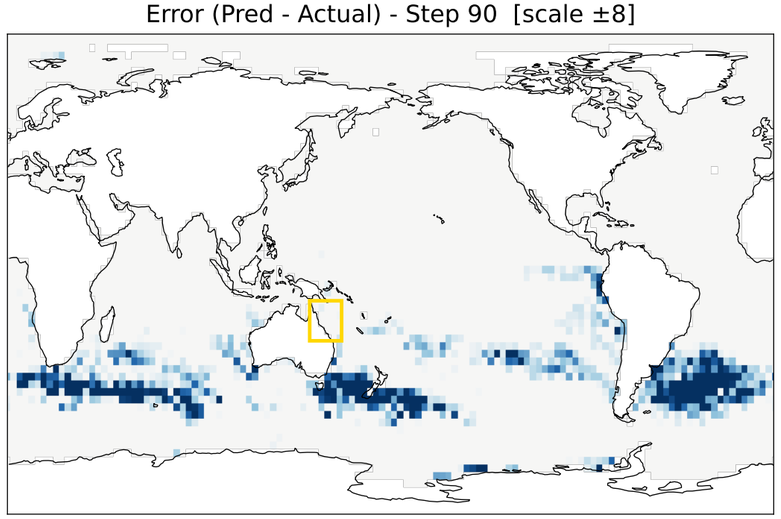
Actual DHW - Step 90

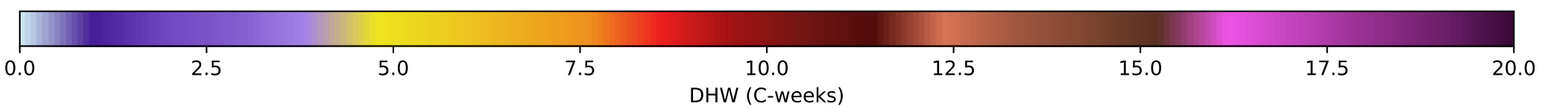


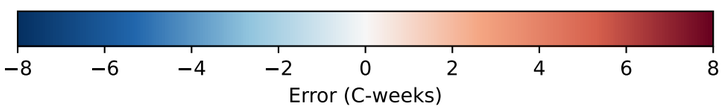
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

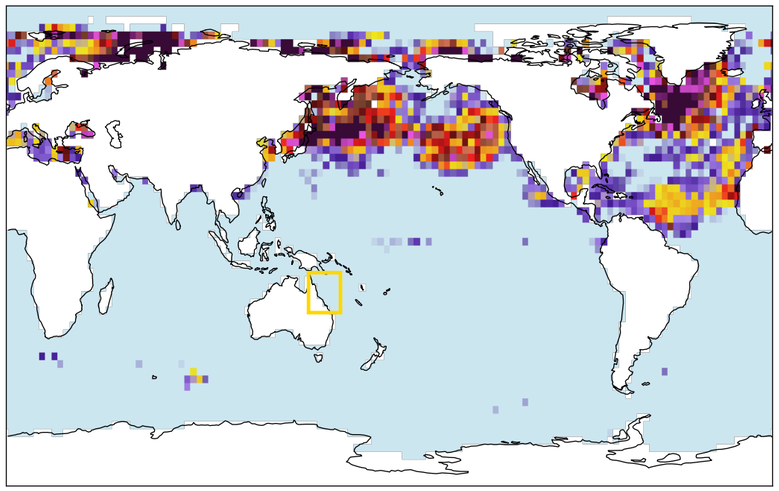




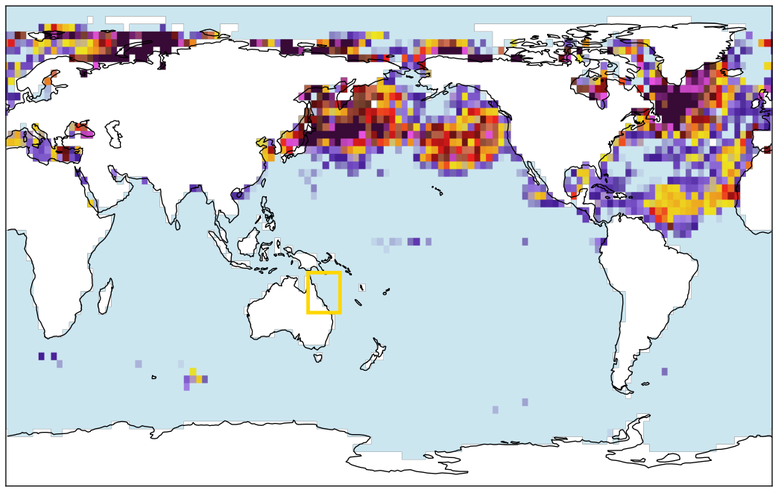


DHW Chain init: 2023-09-23 - Test (C-weeks)

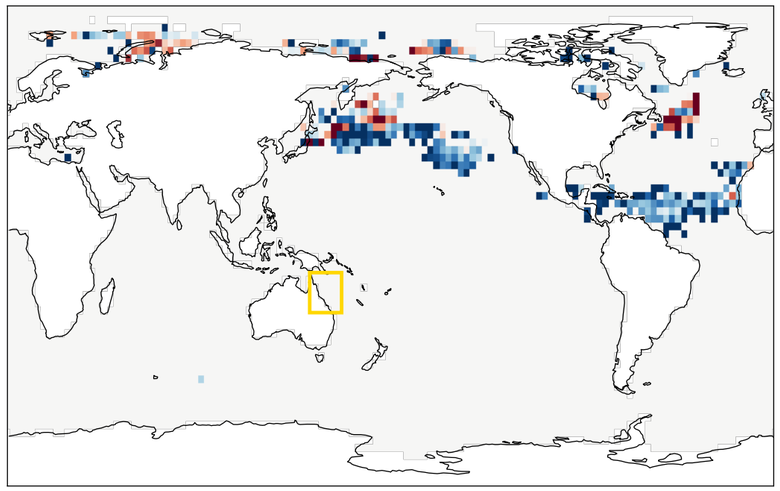
Actual DHW - Step 1



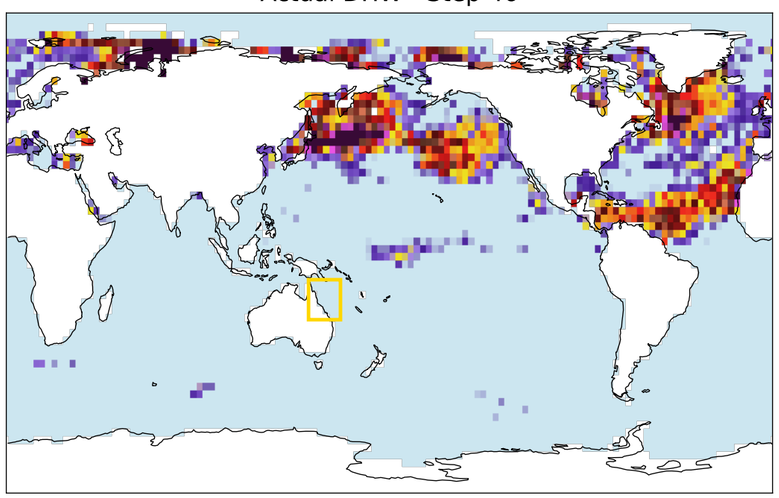
Predicted DHW - Step 1



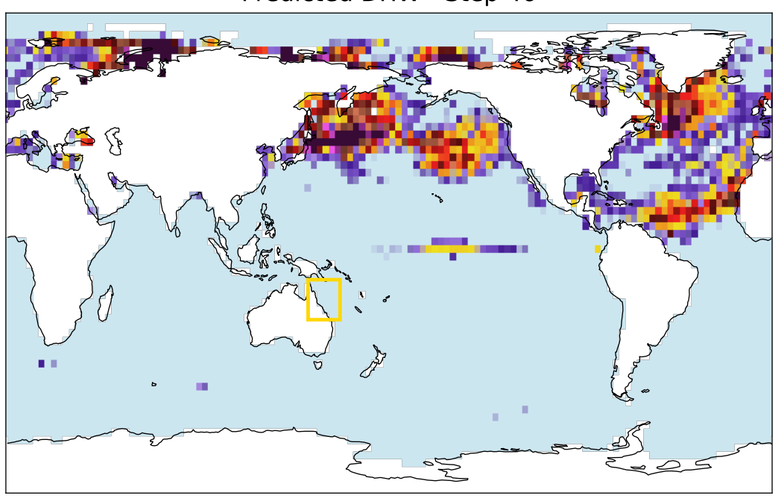
Error (Pred - Actual) - Step 1 [scale ±0.042]



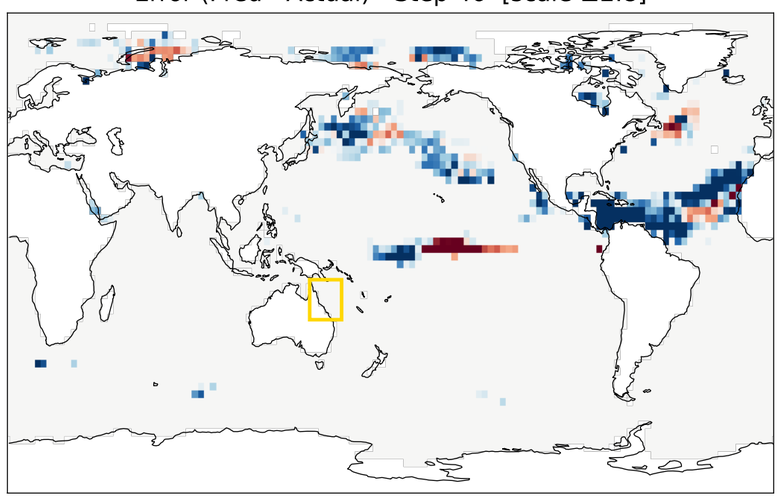
Actual DHW - Step 46



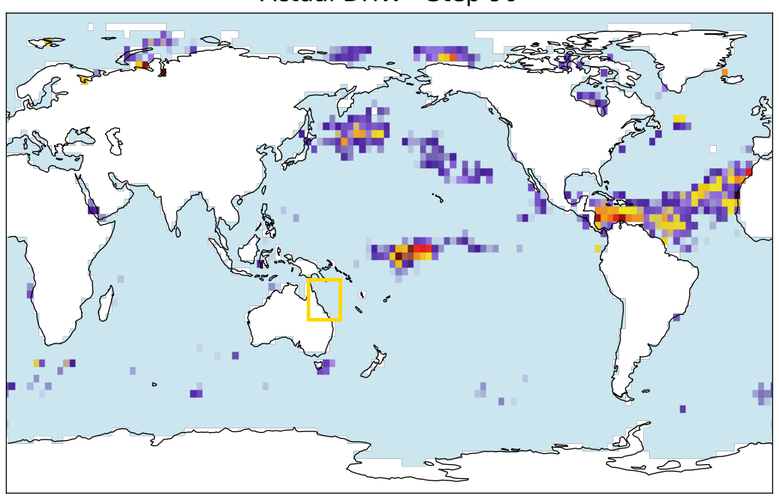
Predicted DHW - Step 46



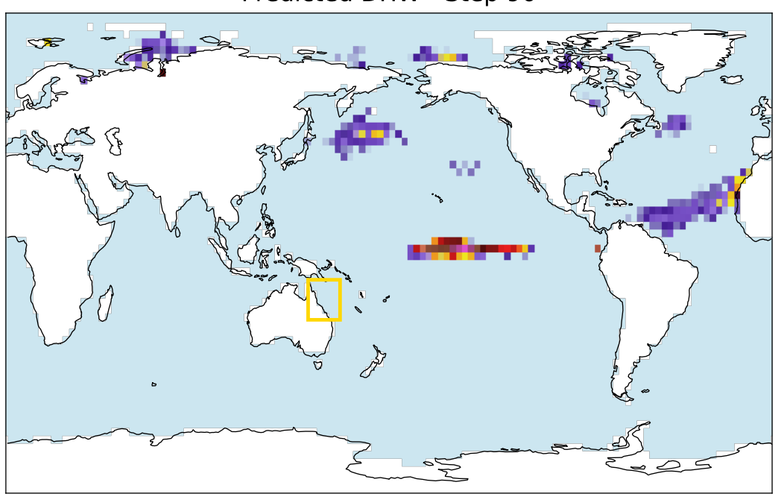
Error (Pred - Actual) - Step 46 [scale ±2.8]



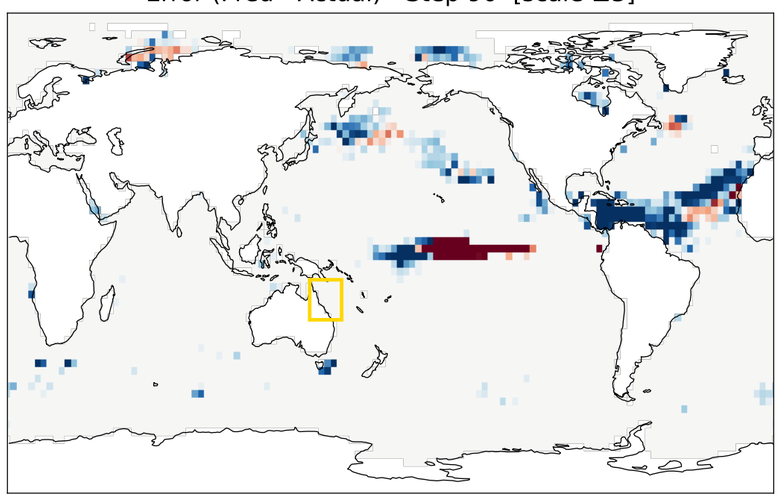
Actual DHW - Step 90

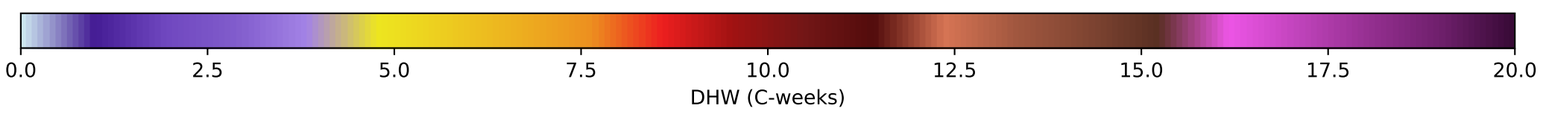


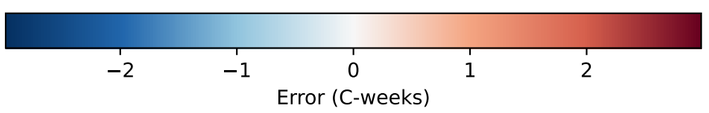
Predicted DHW - Step 90



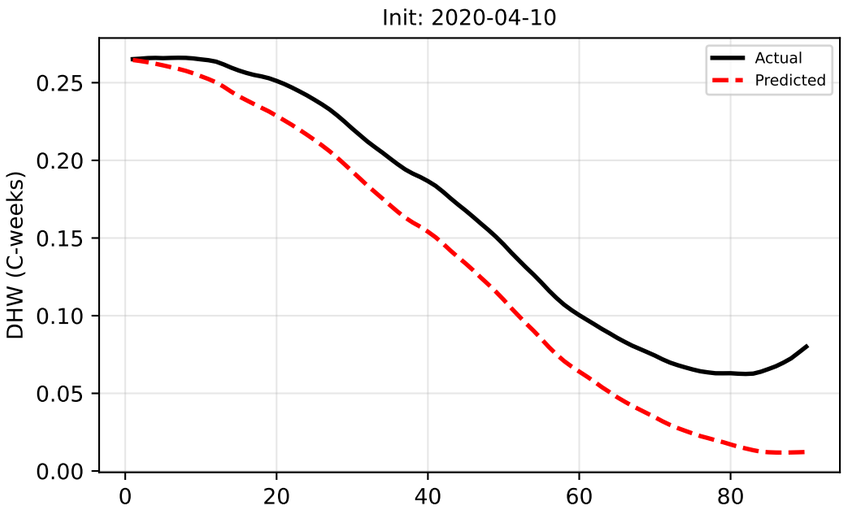
Error (Pred - Actual) - Step 90 [scale ±3]

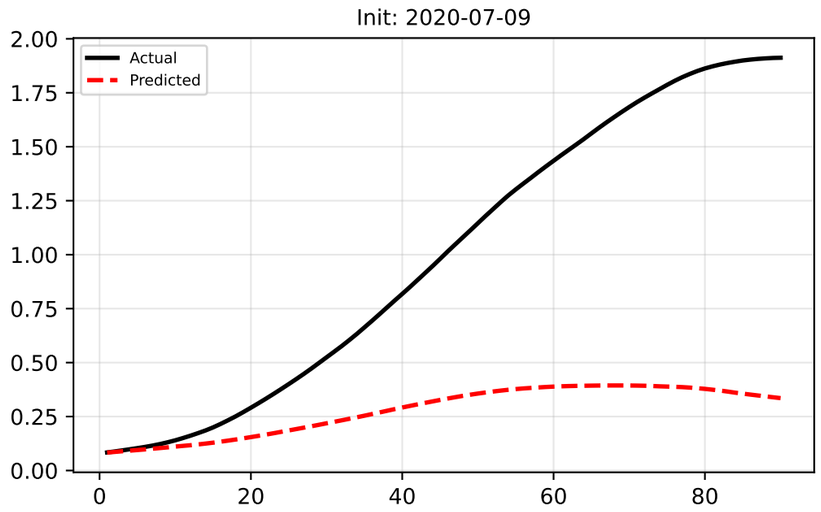


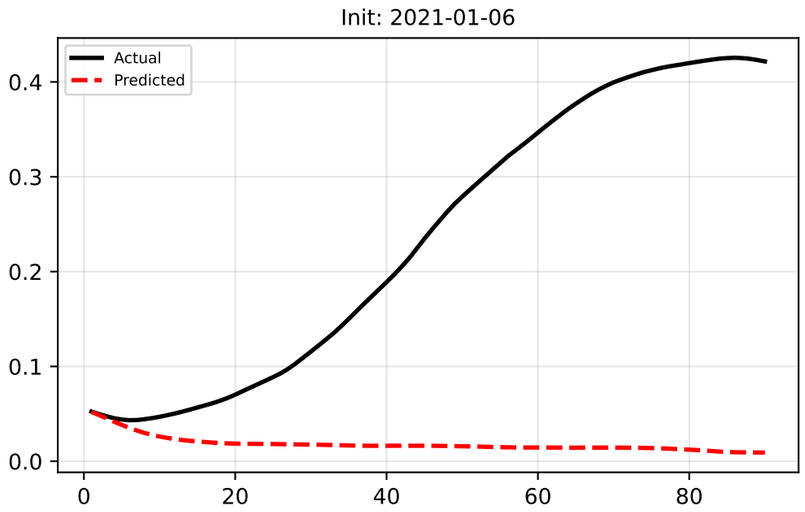


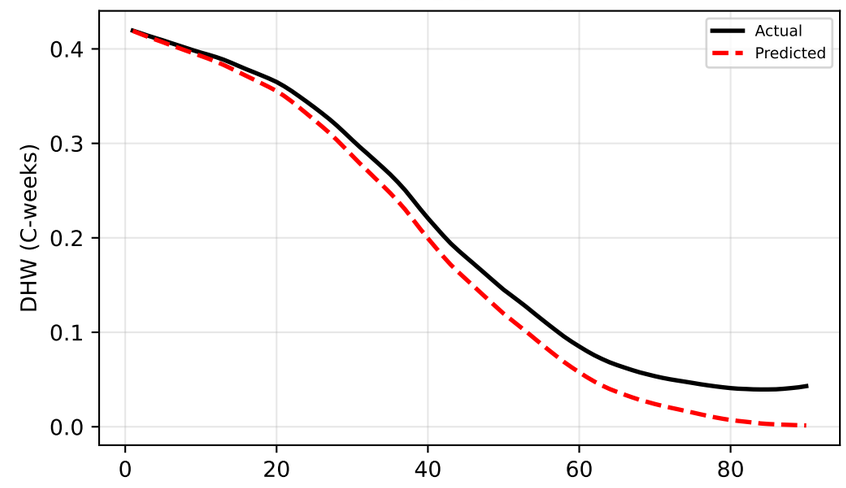


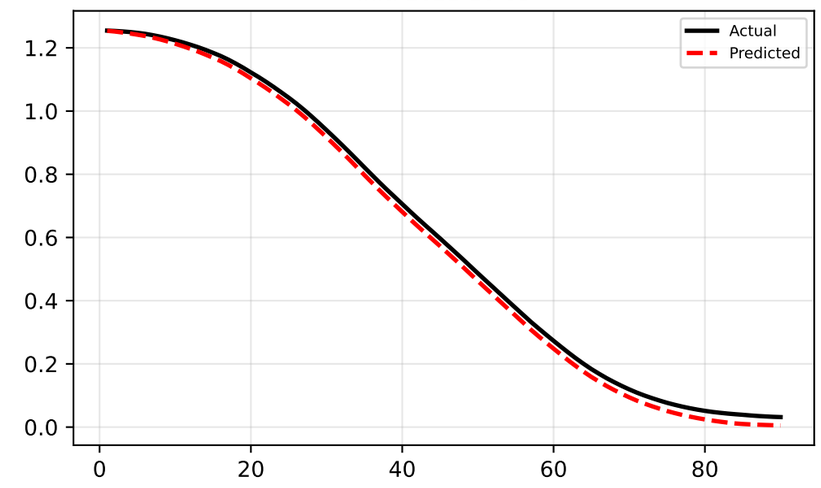
Global-Mean DHW - Test (C-weeks)

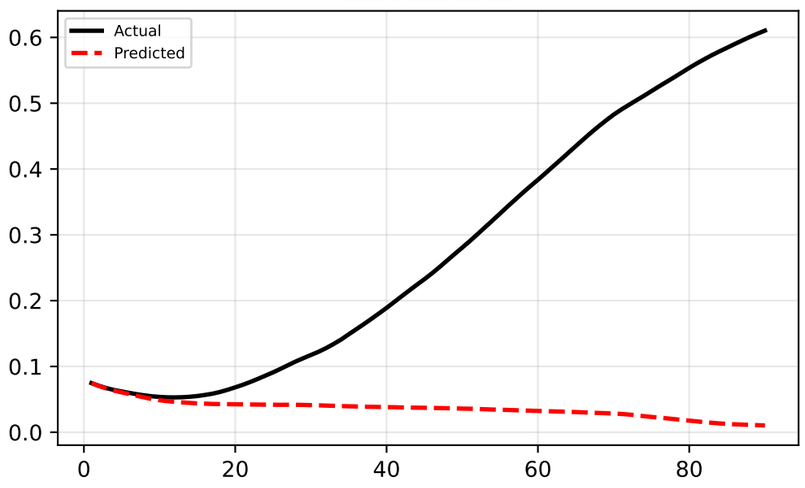


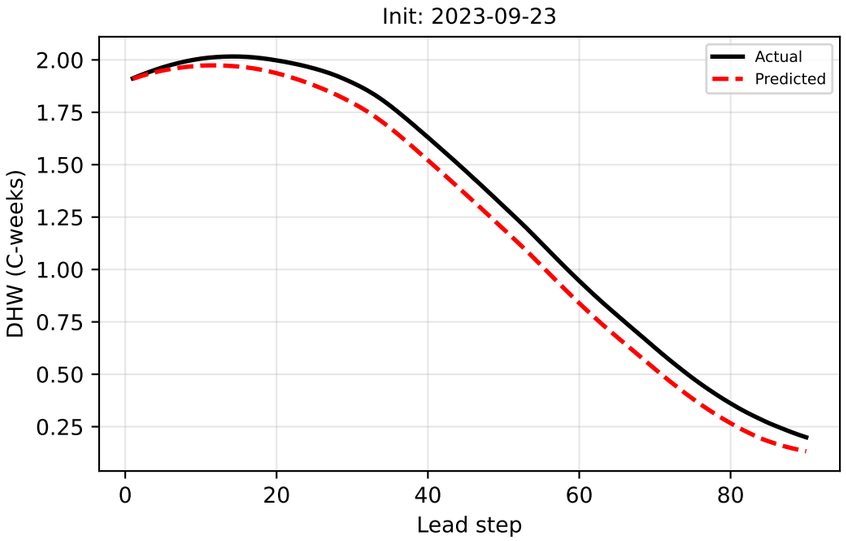




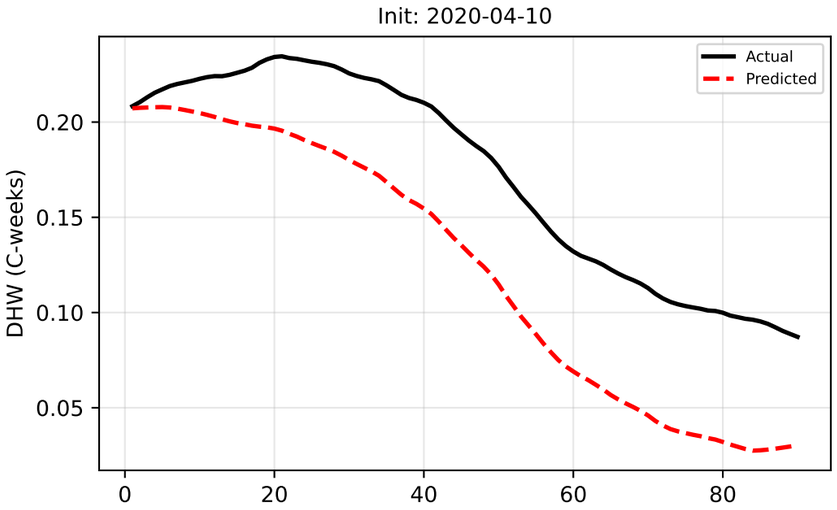


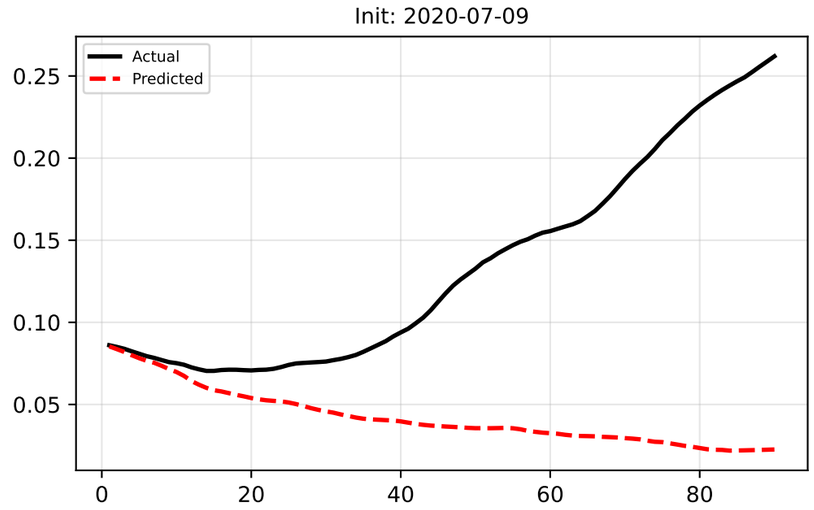


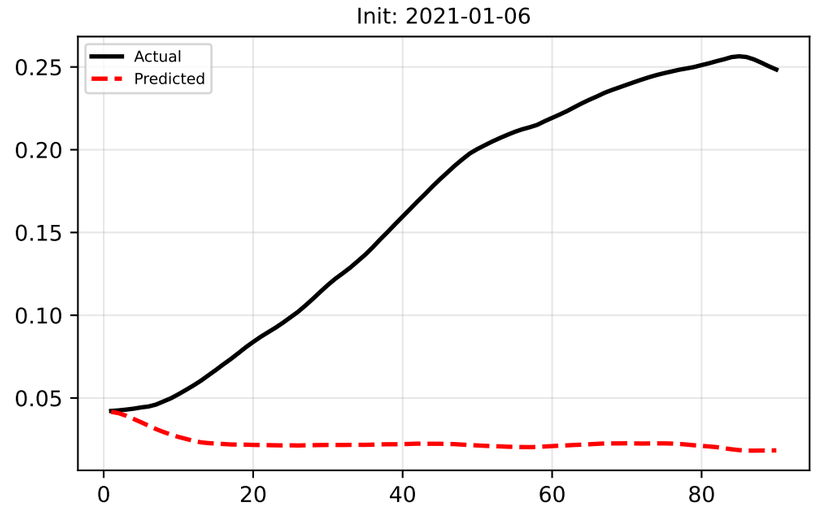


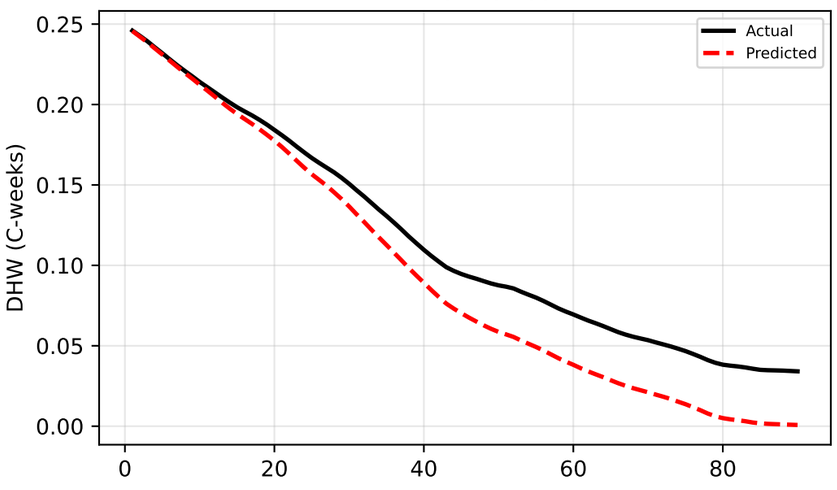


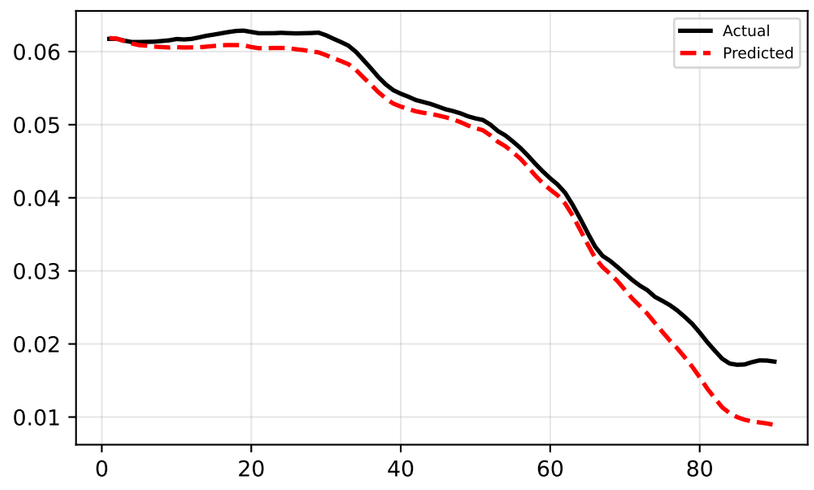
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

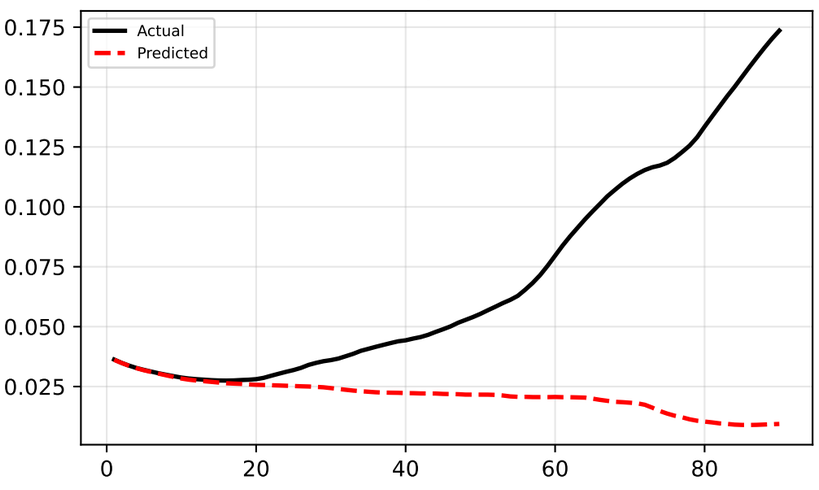


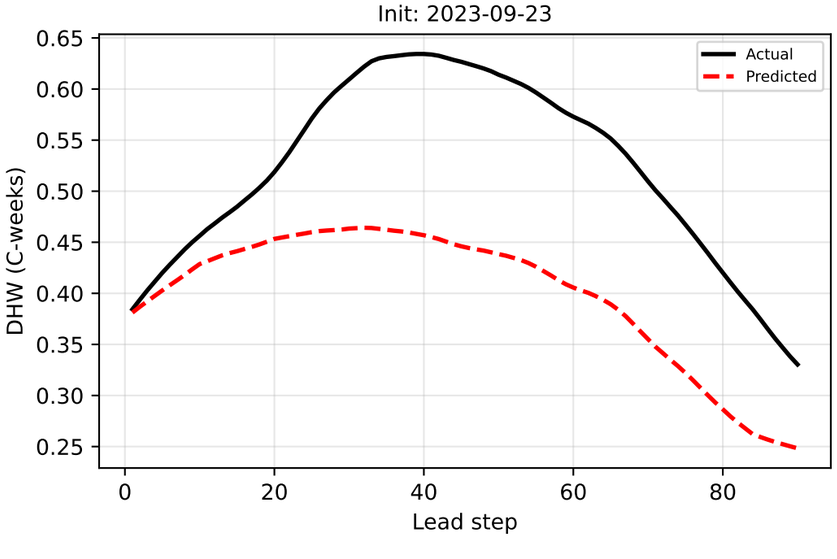


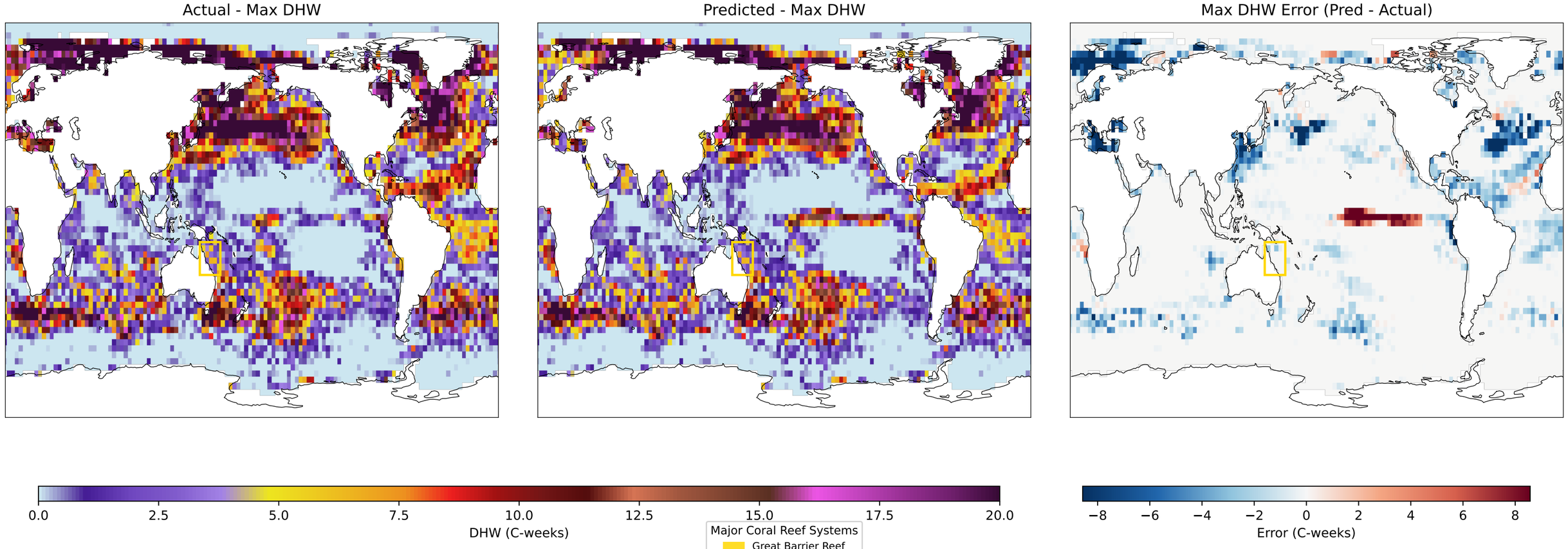


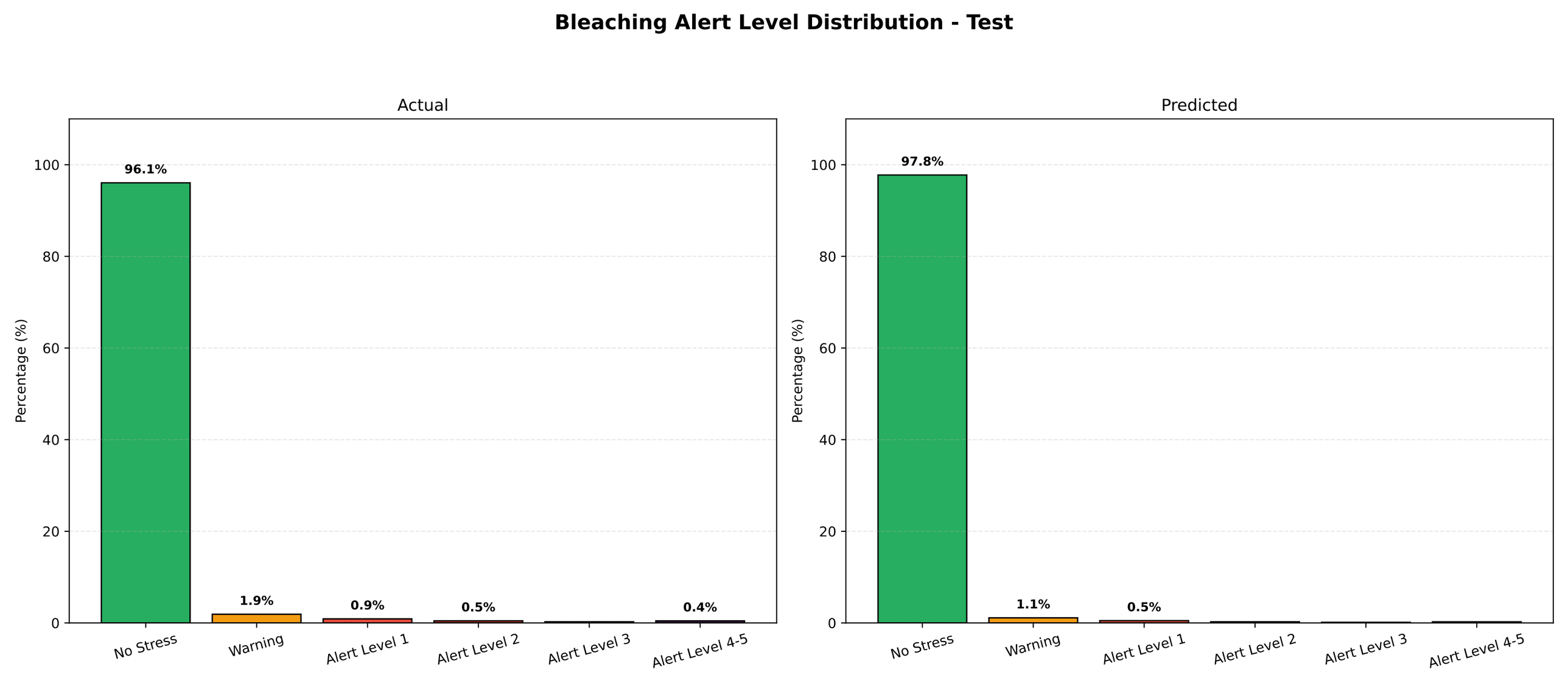






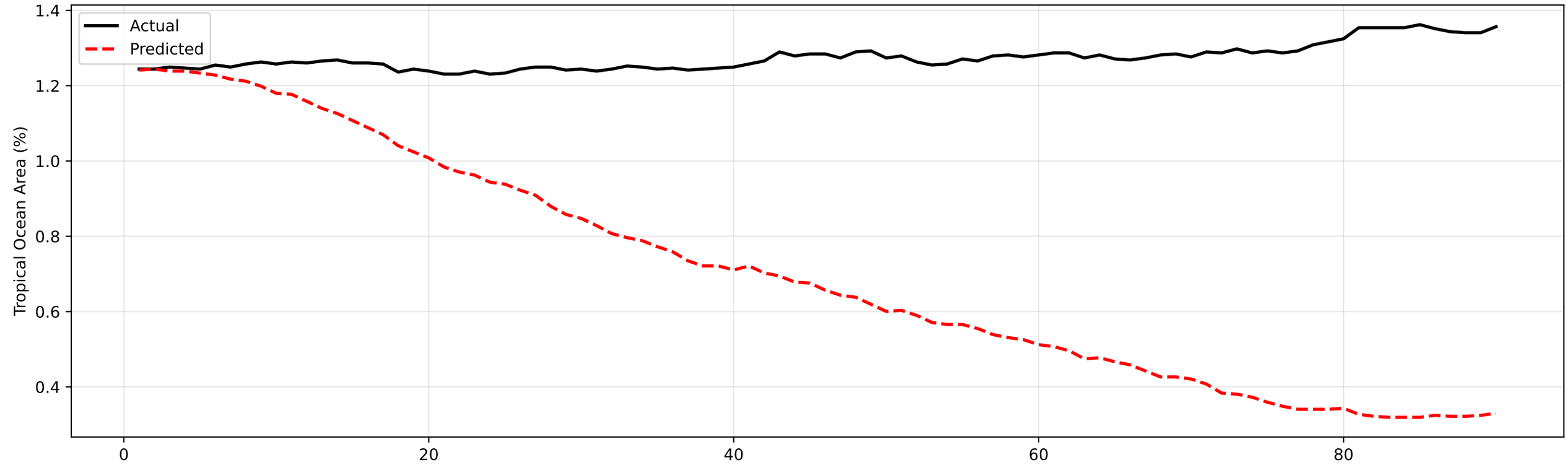



Peak DHW (max over all chains & leads) - Test (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)

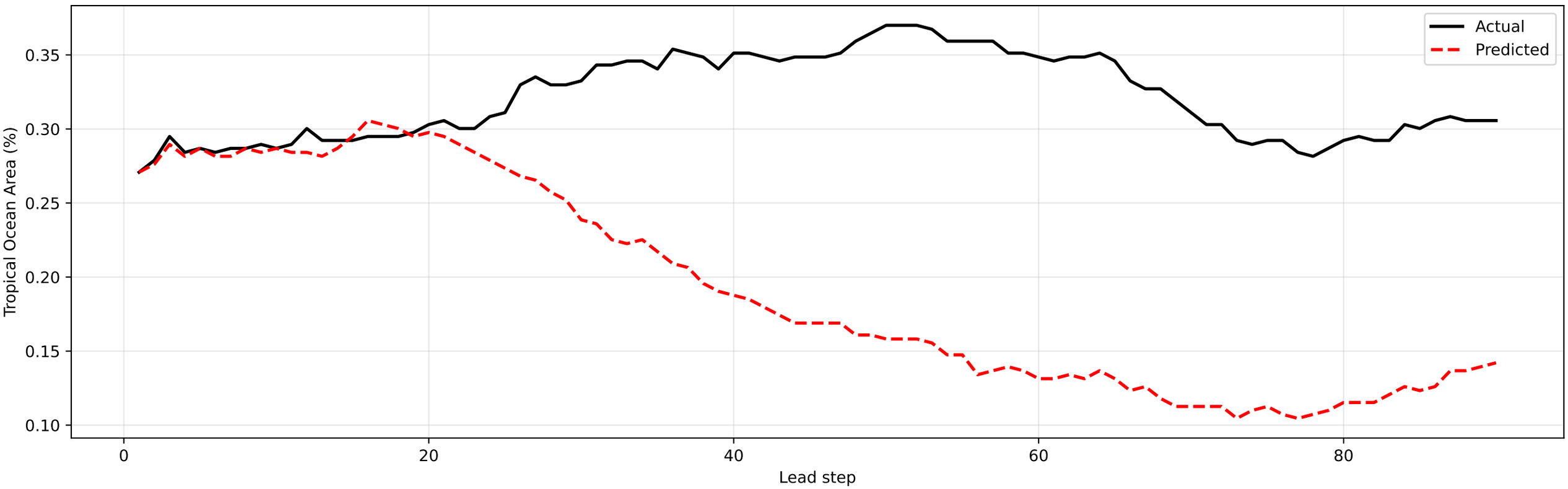

Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.1%
1.9%
0.9%
0.5%
0.4%
97.8%
1.1%
0.5%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



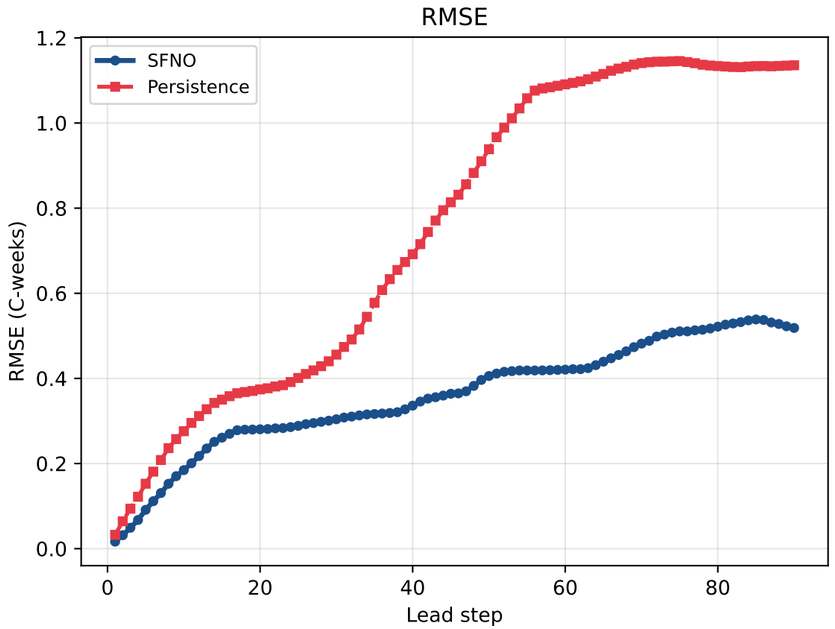
Great Barrier Reef Analysis - Test

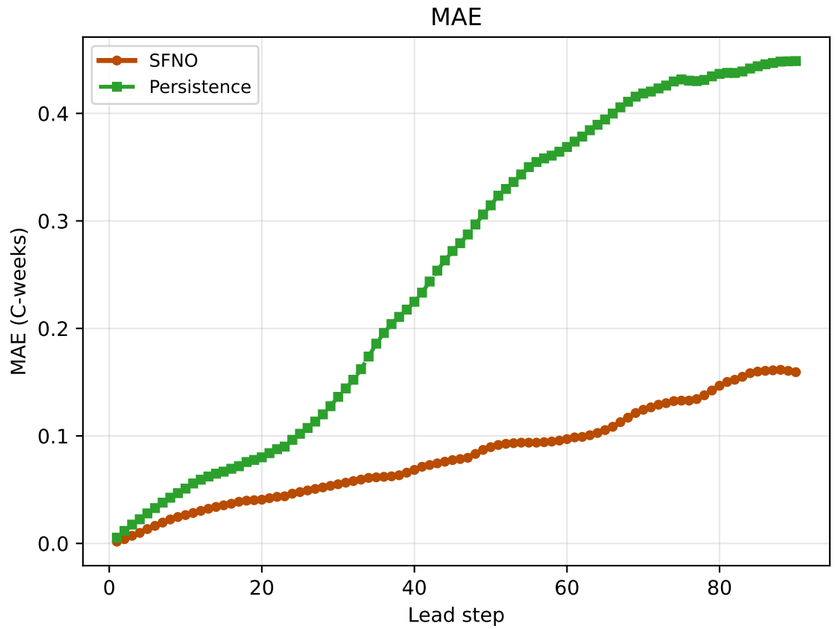
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                      Grid: 5 lat x 5 lon points
```

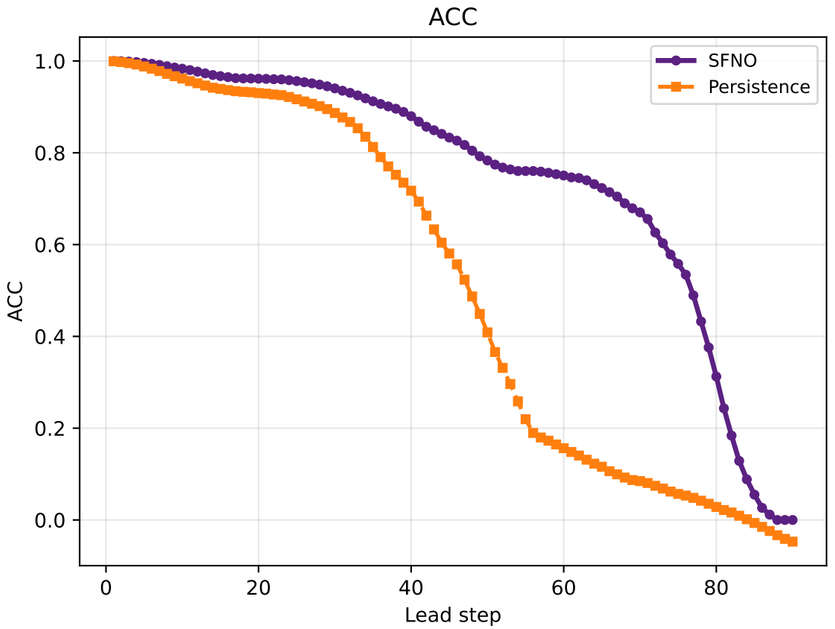
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

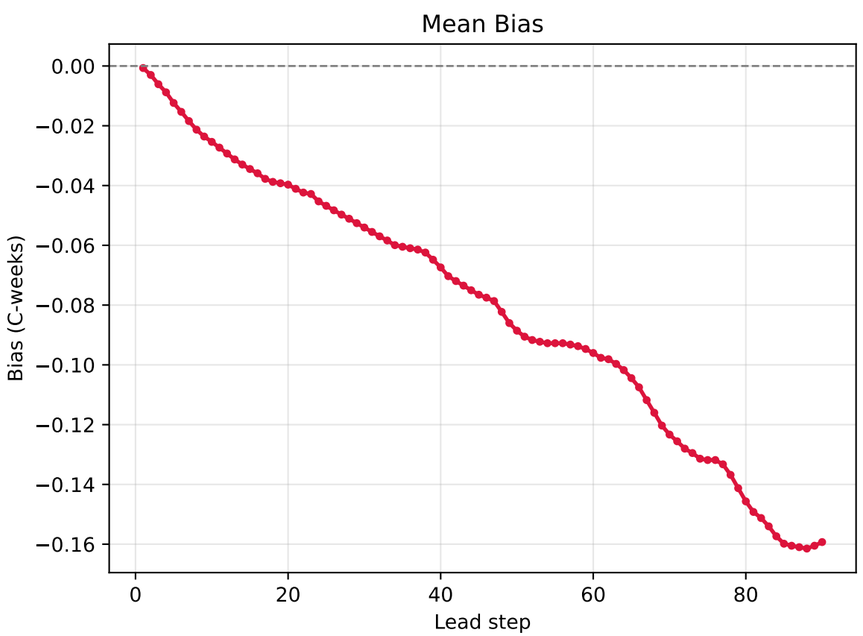
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0165	0.0325	49.1%	0.0017	0.0054	-0.0007	0.9999	0.9995	0.9997
2	0.0317	0.0643	50.8%	0.0040	0.0116	-0.0030	0.9995	0.9979	0.9990
3	0.0491	0.0939	47.7%	0.0071	0.0175	-0.0061	0.9988	0.9955	0.9976
4	0.0674	0.1219	44.7%	0.0098	0.0224	-0.0088	0.9977	0.9924	0.9955
5	0.0912	0.1525	40.2%	0.0133	0.0279	-0.0124	0.9958	0.9881	0.9919
6	0.1115	0.1810	38.4%	0.0163	0.0329	-0.0153	0.9938	0.9834	0.9879
7	0.1307	0.2083	37.2%	0.0194	0.0379	-0.0184	0.9915	0.9781	0.9835
8	0.1525	0.2362	35.5%	0.0223	0.0425	-0.0213	0.9886	0.9721	0.9778
9	0.1703	0.2574	33.8%	0.0246	0.0468	-0.0236	0.9858	0.9669	0.9725
10	0.1845	0.2759	33.1%	0.0264	0.0510	-0.0254	0.9833	0.9619	0.9676
11	0.2005	0.2955	32.1%	0.0283	0.0558	-0.0273	0.9803	0.9562	0.9617
12	0.2176	0.3116	30.2%	0.0303	0.0593	-0.0293	0.9769	0.9515	0.9552
13	0.2353	0.3277	28.2%	0.0322	0.0622	-0.0313	0.9731	0.9466	0.9479
14	0.2510	0.3424	26.7%	0.0339	0.0649	-0.0330	0.9695	0.9419	0.9411
15	0.2608	0.3502	25.5%	0.0354	0.0668	-0.0345	0.9671	0.9393	0.9365
16	0.2698	0.3582	24.7%	0.0369	0.0694	-0.0359	0.9649	0.9366	0.9322
17	0.2783	0.3652	23.8%	0.0387	0.0718	-0.0378	0.9627	0.9343	0.9281
18	0.2795	0.3680	24.0%	0.0397	0.0757	-0.0388	0.9621	0.9329	0.9270
19	0.2799	0.3705	24.5%	0.0402	0.0777	-0.0392	0.9619	0.9319	0.9267
20	0.2805	0.3742	25.0%	0.0407	0.0801	-0.0397	0.9615	0.9302	0.9258
21	0.2811	0.3768	25.4%	0.0421	0.0840	-0.0411	0.9611	0.9289	0.9251
22	0.2827	0.3810	25.8%	0.0433	0.0876	-0.0423	0.9604	0.9271	0.9239
23	0.2833	0.3841	26.2%	0.0438	0.0901	-0.0428	0.9600	0.9256	0.9232
24	0.2856	0.3914	27.0%	0.0463	0.0963	-0.0453	0.9583	0.9217	0.9201
25	0.2885	0.4011	28.1%	0.0478	0.1020	-0.0468	0.9563	0.9167	0.9163
26	0.2926	0.4104	28.7%	0.0493	0.1074	-0.0483	0.9539	0.9118	0.9119
27	0.2951	0.4189	29.6%	0.0507	0.1133	-0.0497	0.9514	0.9070	0.9074
28	0.2980	0.4287	30.5%	0.0521	0.1201	-0.0511	0.9486	0.9017	0.9023
29	0.3007	0.4402	31.7%	0.0536	0.1277	-0.0526	0.9449	0.8952	0.8957
30	0.3040	0.4558	33.3%	0.0550	0.1363	-0.0541	0.9403	0.8868	0.8875
31	0.3081	0.4739	35.0%	0.0565	0.1442	-0.0555	0.9353	0.8770	0.8785
32	0.3104	0.4915	36.9%	0.0580	0.1522	-0.0570	0.9306	0.8672	0.8702
33	0.3131	0.5150	39.2%	0.0594	0.1621	-0.0584	0.9251	0.8535	0.8605
34	0.3154	0.5448	42.1%	0.0609	0.1738	-0.0599	0.9187	0.8349	0.8492
35	0.3164	0.5777	45.2%	0.0615	0.1858	-0.0605	0.9123	0.8127	0.8379
36	0.3175	0.6076	47.7%	0.0619	0.1958	-0.0610	0.9064	0.7908	0.8277
37	0.3188	0.6333	49.7%	0.0624	0.2040	-0.0614	0.9011	0.7703	0.8186
38	0.3206	0.6548	51.0%	0.0634	0.2107	-0.0624	0.8960	0.7521	0.8100
39	0.3273	0.6735	51.4%	0.0658	0.2175	-0.0648	0.8890	0.7349	0.7983
40	0.3360	0.6918	51.4%	0.0684	0.2248	-0.0674	0.8798	0.7175	0.7827
41	0.3456	0.7155	51.7%	0.0713	0.2332	-0.0703	0.8679	0.6935	0.7627
42	0.3525	0.7441	52.6%	0.0729	0.2436	-0.0720	0.8568	0.6632	0.7439
43	0.3558	0.7709	53.8%	0.0745	0.2537	-0.0735	0.8492	0.6331	0.7303
44	0.3596	0.7952	54.8%	0.0760	0.2632	-0.0750	0.8412	0.6042	0.7159
45	0.3639	0.8139	55.3%	0.0775	0.2720	-0.0765	0.8331	0.5806	0.7009
46	0.3648	0.8315	56.1%	0.0785	0.2793	-0.0775	0.8265	0.5570	0.6884
47	0.3698	0.8560	56.8%	0.0796	0.2874	-0.0786	0.8171	0.5236	0.6707
48	0.3825	0.8827	56.7%	0.0832	0.2967	-0.0823	0.8046	0.4870	0.6479
49	0.3965	0.9103	56.4%	0.0870	0.3059	-0.0860	0.7925	0.4490	0.6256
50	0.4057	0.9384	56.8%	0.0895	0.3145	-0.0886	0.7833	0.4087	0.6080
51	0.4115	0.9668	57.4%	0.0915	0.3234	-0.0906	0.7743	0.3658	0.5921
52	0.4155	0.9892	58.0%	0.0927	0.3297	-0.0917	0.7678	0.3313	0.5812
53	0.4171	1.0111	58.7%	0.0932	0.3362	-0.0923	0.7638	0.2963	0.5732
54	0.4186	1.0344	59.5%	0.0937	0.3432	-0.0928	0.7603	0.2584	0.5657
55	0.4186	1.0582	60.4%	0.0937	0.3501	-0.0928	0.7602	0.2195	0.5631
56	0.4186	1.0761	61.1%	0.0937	0.3548	-0.0928	0.7603	0.1896	0.5612
57	0.4189	1.0812	61.3%	0.0942	0.3581	-0.0932	0.7590	0.1797	0.5567
58	0.4194	1.0841	61.3%	0.0947	0.3607	-0.0937	0.7564	0.1728	0.5508
59	0.4200	1.0875	61.4%	0.0957	0.3643	-0.0947	0.7534	0.1645	0.5438
60	0.4206	1.0909	61.4%	0.0970	0.3688	-0.0960	0.7506	0.1564	0.5371
61	0.4217	1.0943	61.5%	0.0986	0.3738	-0.0976	0.7465	0.1481	0.5274
62	0.4218	1.0980	61.6%	0.0991	0.3785	-0.0981	0.7450	0.1404	0.5211
63	0.4242	1.1030	61.5%	0.1007	0.3843	-0.0997	0.7402	0.1313	0.5132
64	0.4312	1.1092	61.1%	0.1027	0.3894	-0.1017	0.7318	0.1227	0.5023
65	0.4393	1.1152	60.6%	0.1054	0.3943	-0.1044	0.7232	0.1160	0.4925
66	0.4473	1.1228	60.2%	0.1085	0.3999	-0.1075	0.7142	0.1063	0.4822
67	0.4551	1.1280	59.7%	0.1128	0.4056	-0.1118	0.7046	0.0994	0.4718
68	0.4637	1.1321	59.0%	0.1170	0.4108	-0.1160	0.6899	0.0924	0.4549
69	0.4734	1.1374	58.4%	0.1213	0.4156	-0.1203	0.6791	0.0870	0.4440
70	0.4818	1.1409	57.8%	0.1243	0.4186	-0.1233	0.6703	0.0847	0.4351
71	0.4883	1.1428	57.3%	0.1265	0.4203	-0.1256	0.6558	0.0805	0.4193
72	0.4984	1.1442	56.4%	0.1290	0.4232	-0.1280	0.6263	0.0744	0.3878
73	0.5030	1.1443	56.0%	0.1305	0.4259	-0.1295	0.6029	0.0684	0.3612
74	0.5077	1.1451	55.7%	0.1324	0.4297	-0.1314	0.5783	0.0619	0.3364
75	0.5106	1.1456	55.4%	0.1328	0.4316	-0.1318	0.5581	0.0567	0.3161
76	0.5106	1.1433	55.3%	0.1328	0.4305	-0.1318	0.5343	0.0532	0.2952
77	0.5130	1.1405	55.0%	0.1343	0.4300	-0.1333	0.4893	0.0482	0.2589
78	0.5144	1.1370	54.8%	0.1378	0.4312	-0.1368	0.4325	0.0419	0.2168
79	0.5173	1.1351	54.4%	0.1422	0.4344	-0.1413	0.3760	0.0353	0.1775
80	0.5216	1.1339	54.0%	0.1467	0.4368	-0.1457	0.3126	0.0284	0.1339
81	0.5264	1.1328	53.5%	0.1502	0.4377	-0.1492	0.2432	0.0216	0.0864
82	0.5291	1.1317	53.2%	0.1522	0.4376	-0.1512	0.1838	0.0162	0.0449
83	0.5323	1.1313	53.0%	0.1550	0.4390	-0.1540	0.1286	0.0092	0.0050
84	0.5364	1.1328	52.6%	0.1583	0.4417	-0.1574	0.0885	0.0015	-0.0250
85	0.5385	1.1337	52.5%	0.1598	0.4438	-0.1598	0.0552	-0.0065	-0.0518
86	0.5373	1.1340	52.6%	0.1605	0.4457	-0.1605	0.0264	-0.0150	-0.0758
87	0.5314	1.1333	53.1%	0.1610	0.4469	-0.1610	0.0116	-0.0240	-0.0907
88	0.5279	1.1341	53.5%	0.1615	0.4482	-0.1615	0.0000	-0.0332	-0.1032
89	0.5225	1.1349	54.0%	0.1605	0.4484	-0.1605	-0.0000	-0.0412	-0.1042
90	0.5186	1.1357	54.3%	0.1593	0.4487	-0.1593	0.0000	-0.0472	-0.1042

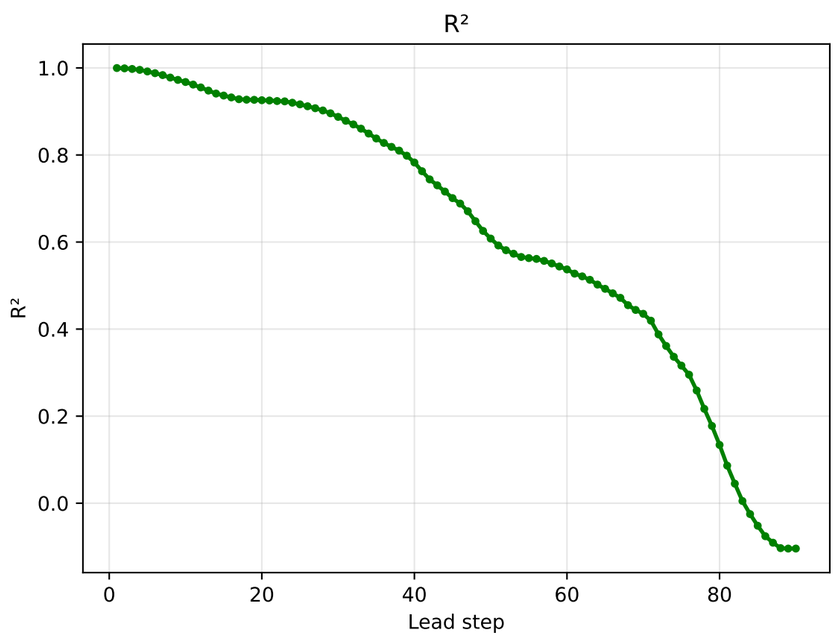
DHW Forecast Skill vs Lead Step - Test - GBR

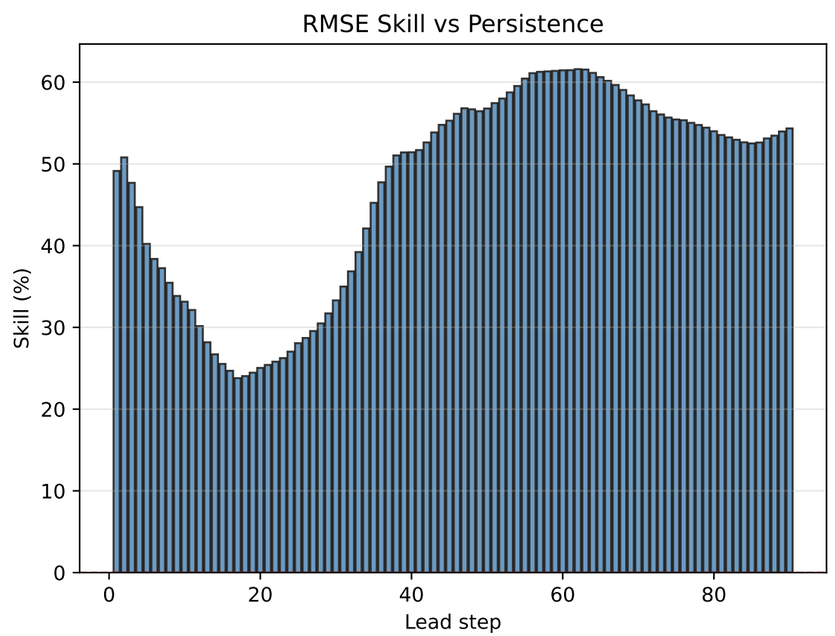


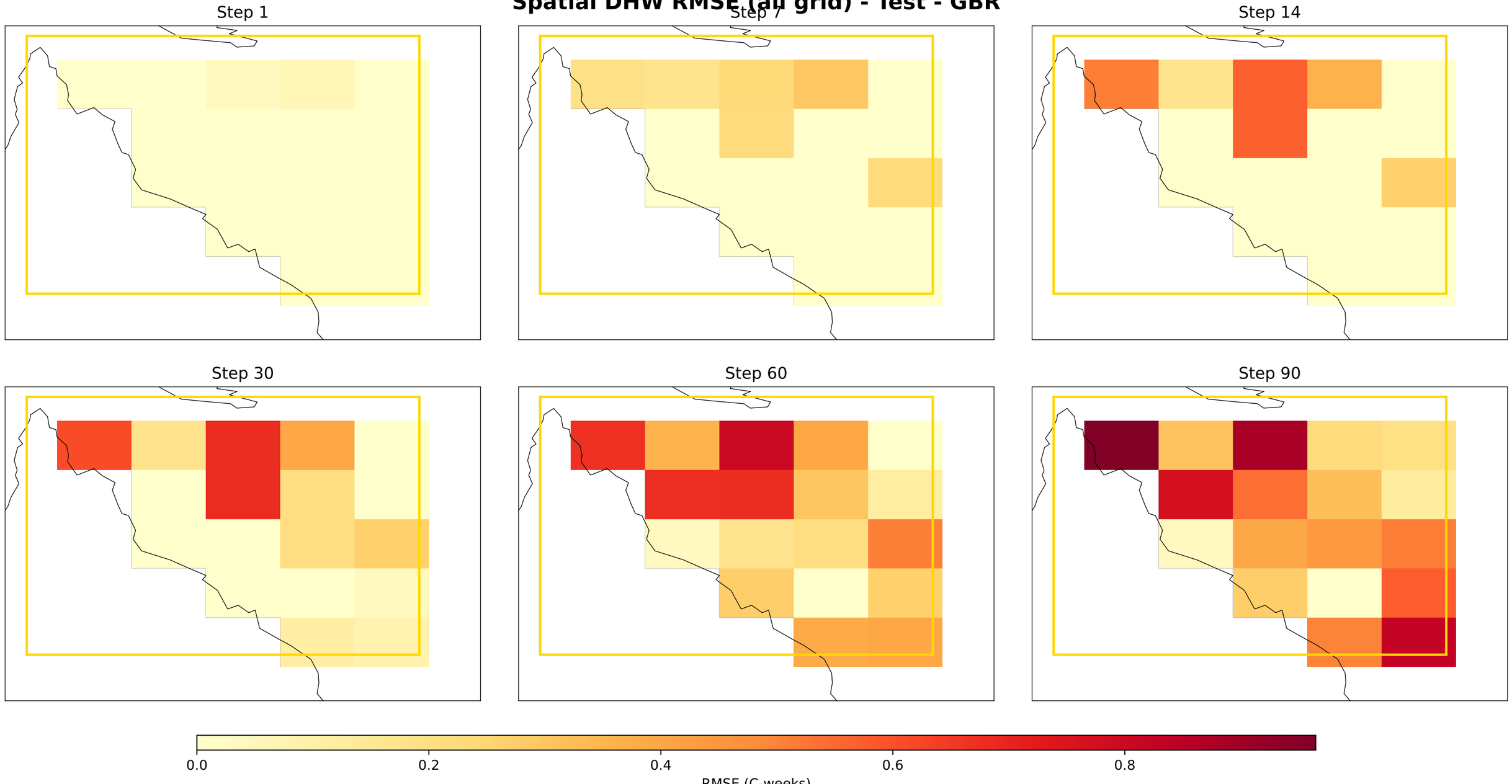


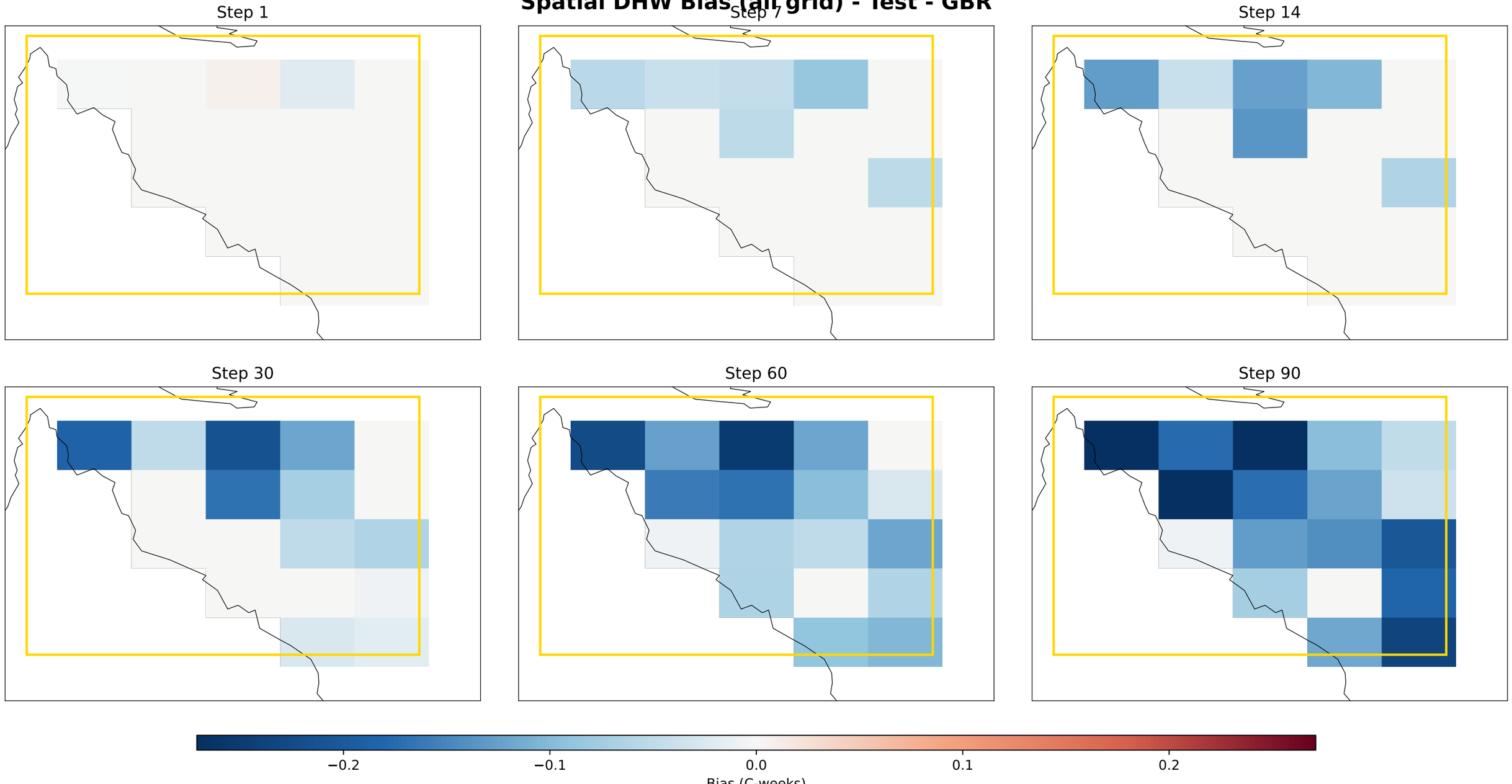


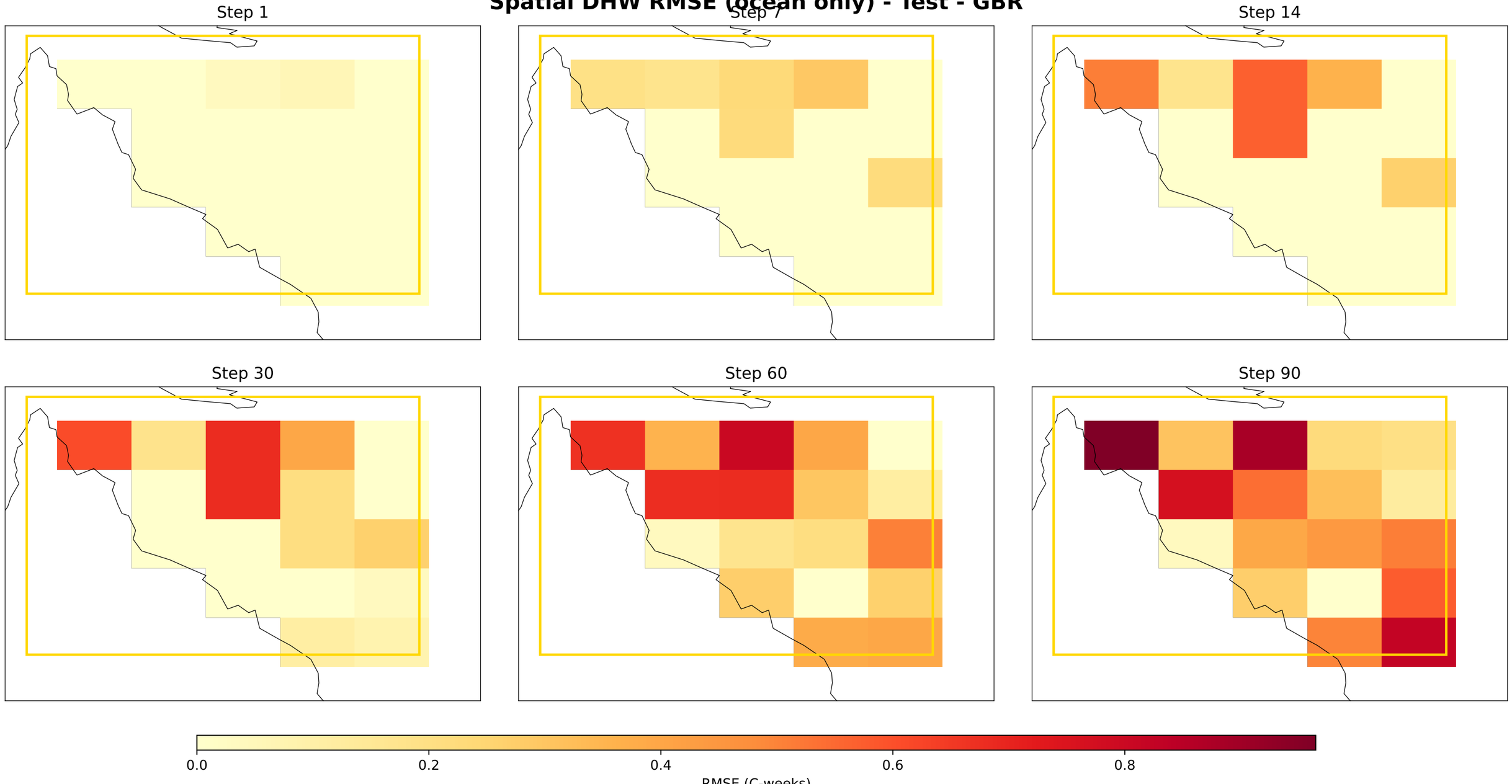


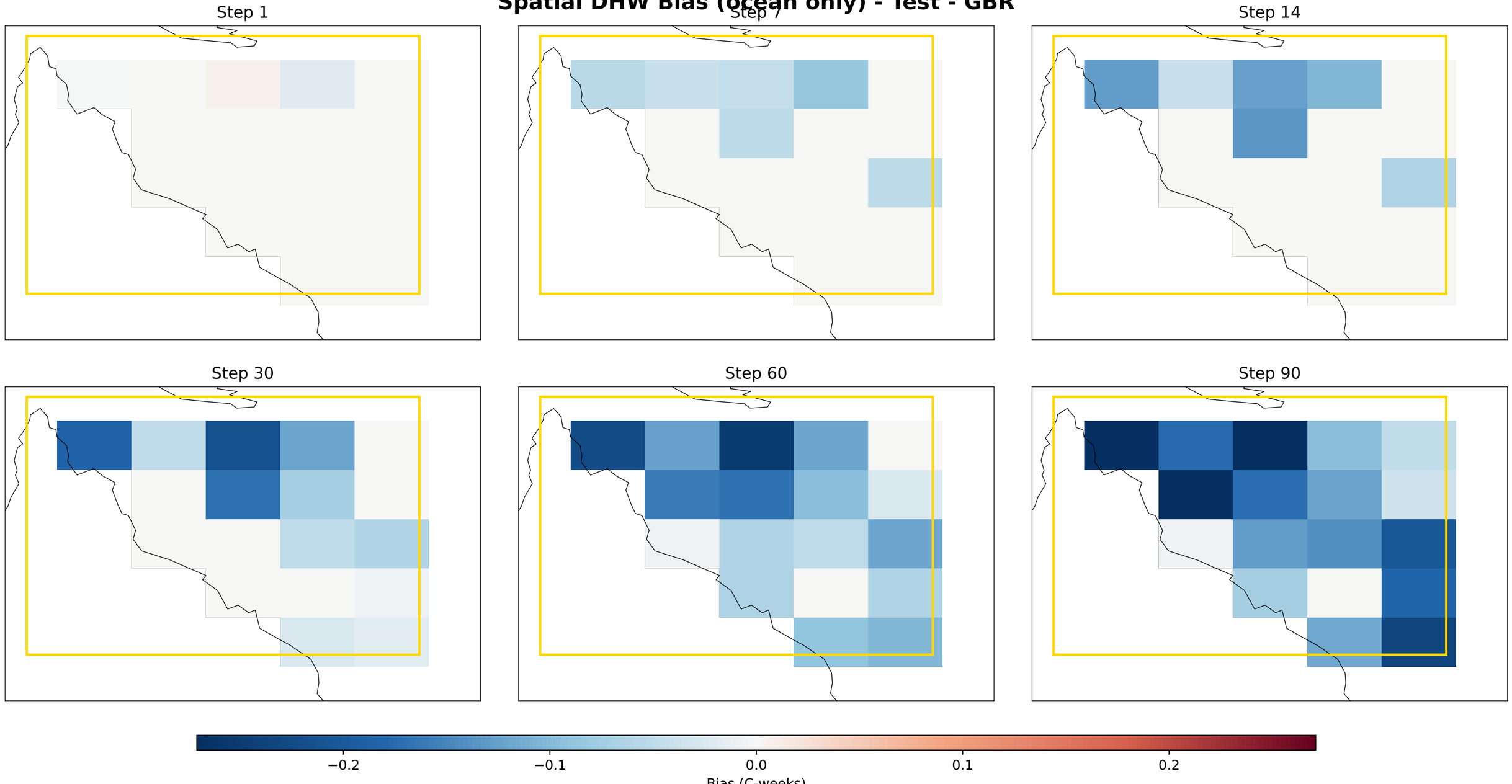





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

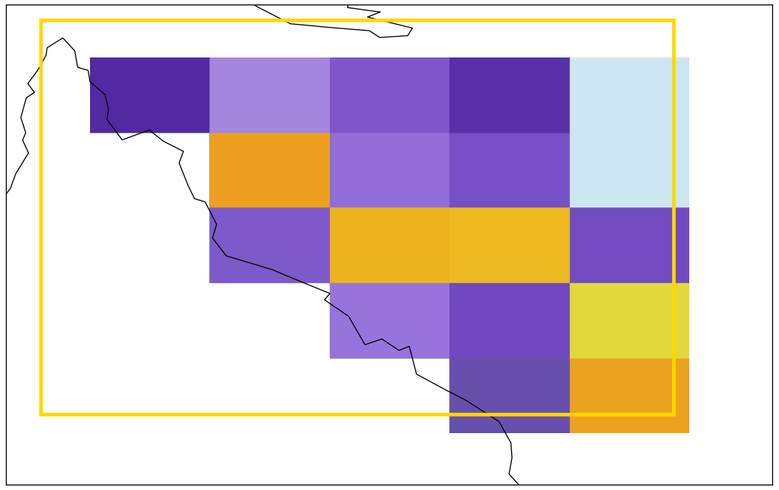

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

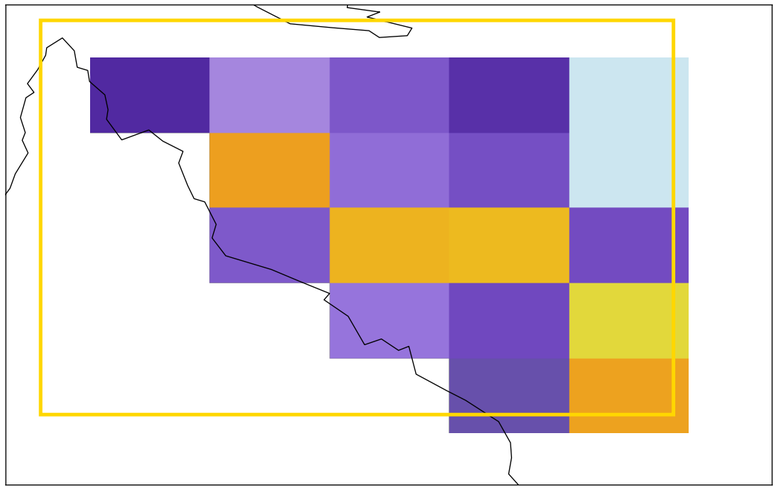

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

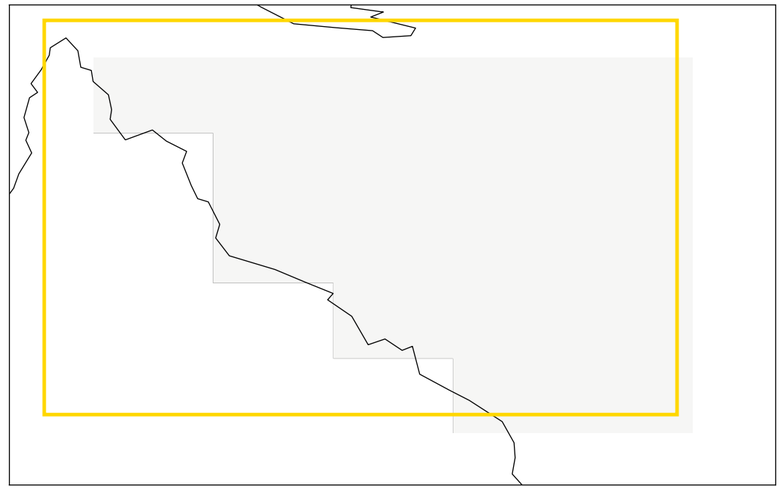
Actual DHW - Step 1



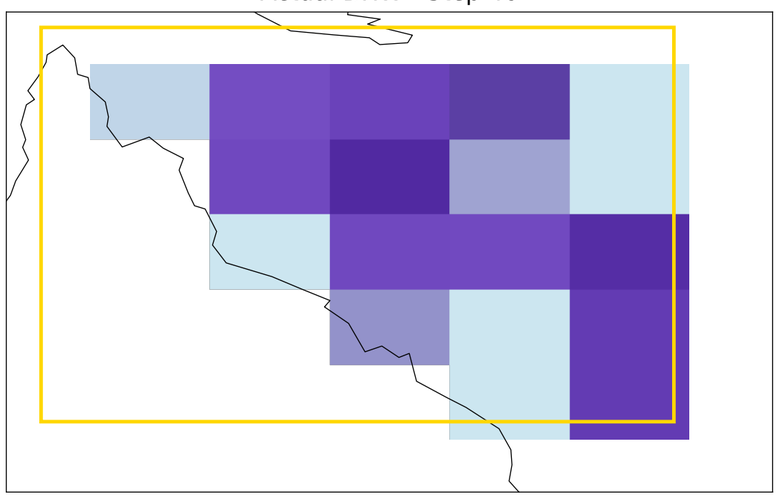
Predicted DHW - Step 1



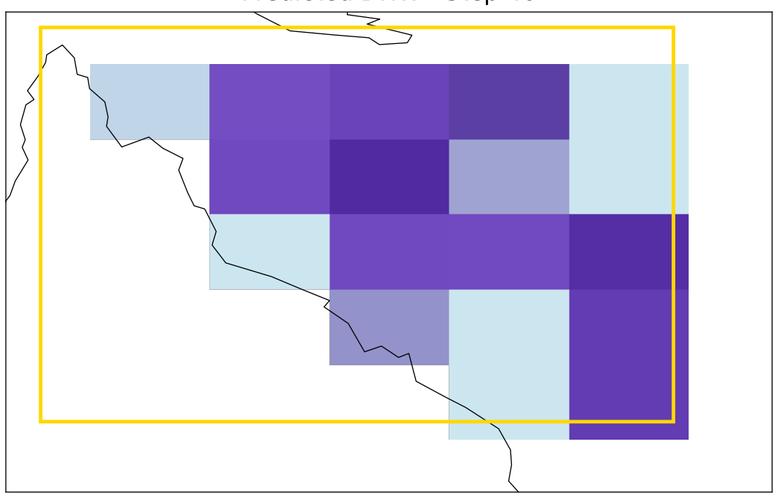
Error (Pred - Actual) - Step 1 [scale ±8]



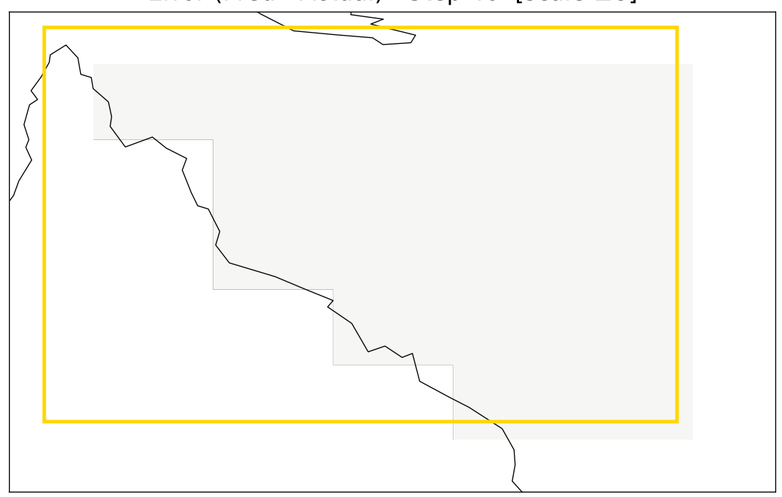
Actual DHW - Step 46



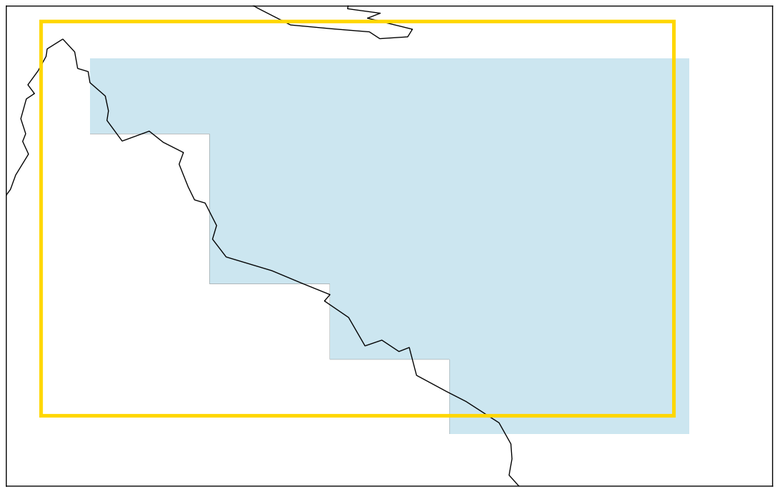
Predicted DHW - Step 46



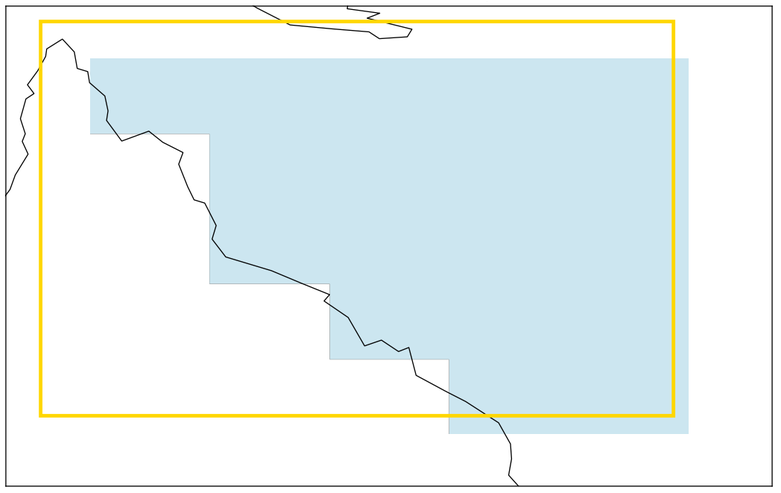
Error (Pred - Actual) - Step 46 [scale ±8]



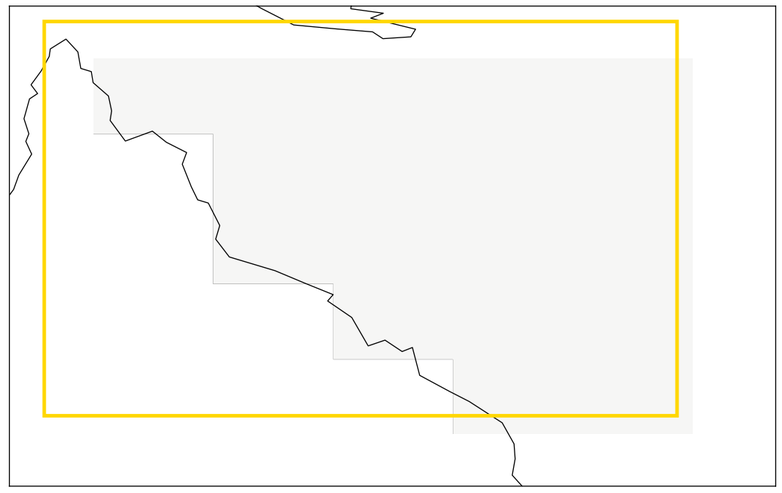
Actual DHW - Step 90

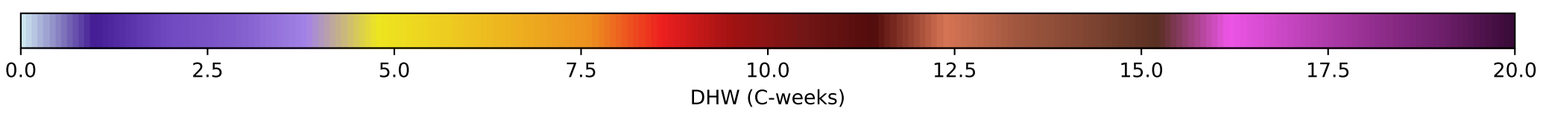


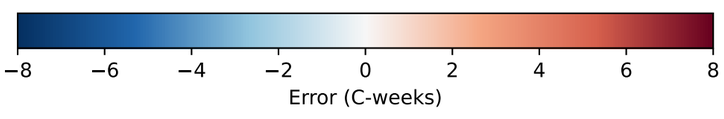
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

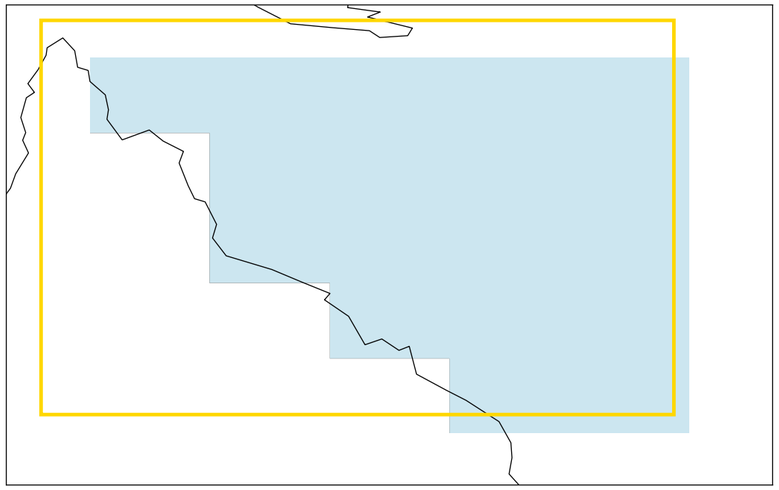




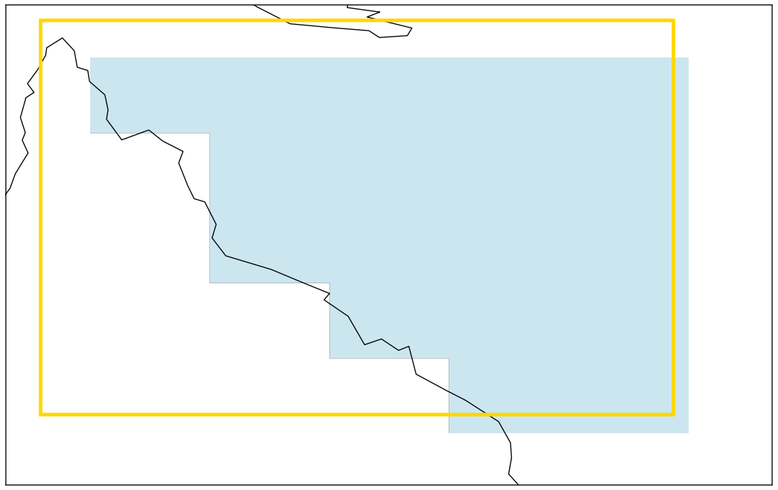


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

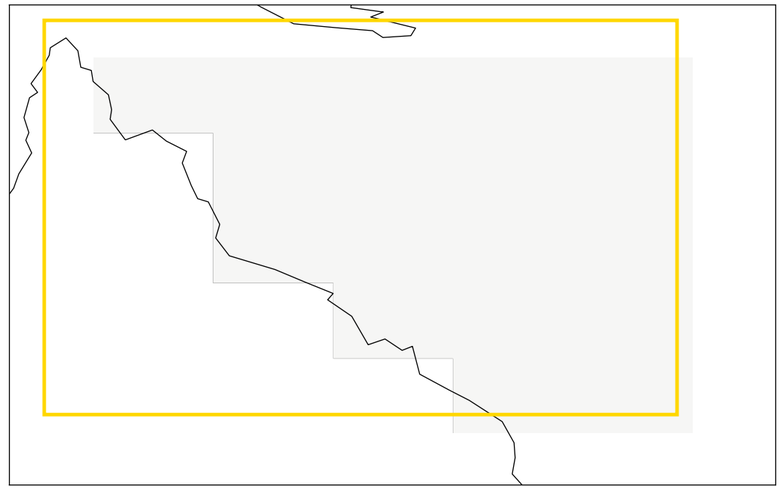
Actual DHW - Step 1



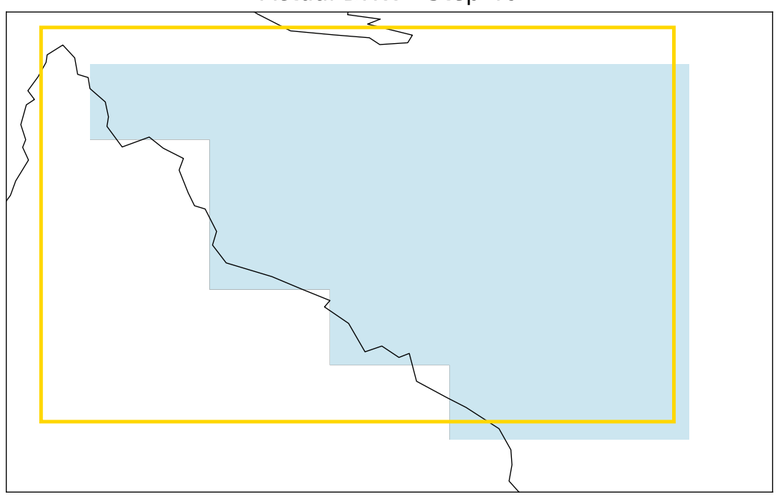
Predicted DHW - Step 1



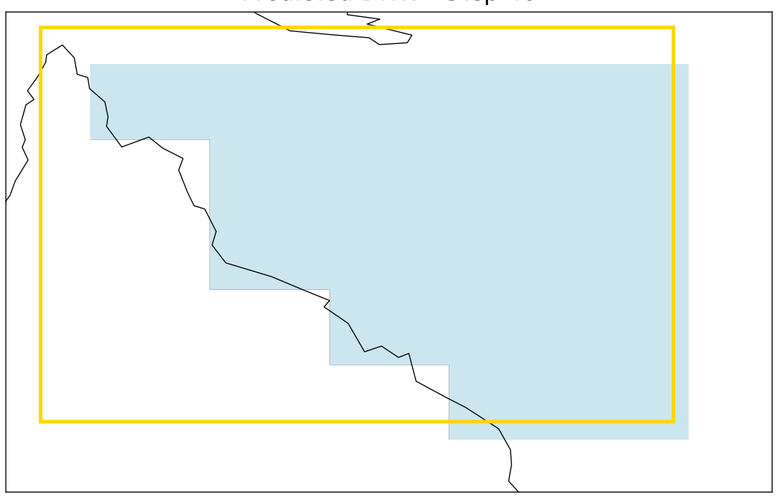
Error (Pred - Actual) - Step 1 [scale ±8]



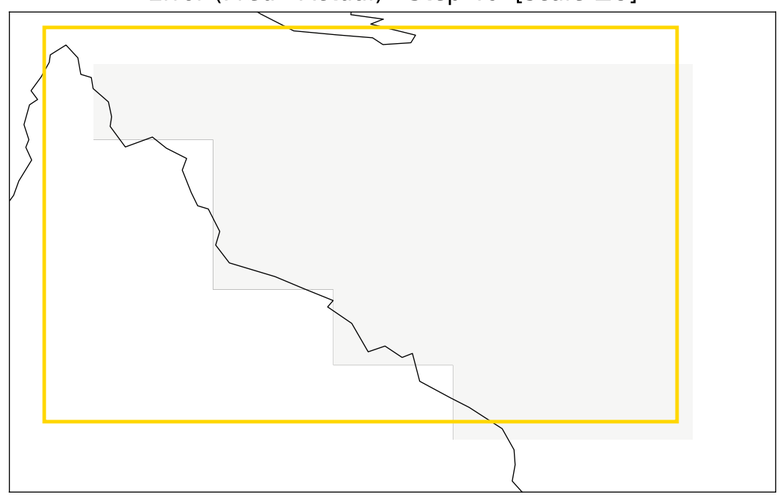
Actual DHW - Step 46



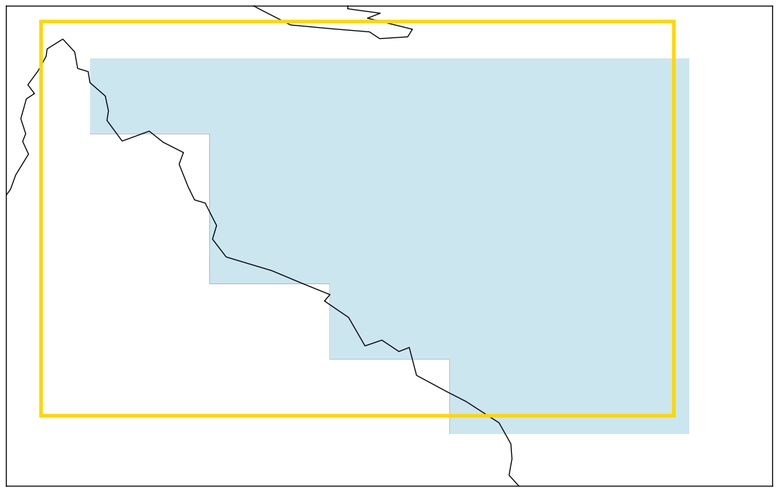
Predicted DHW - Step 46



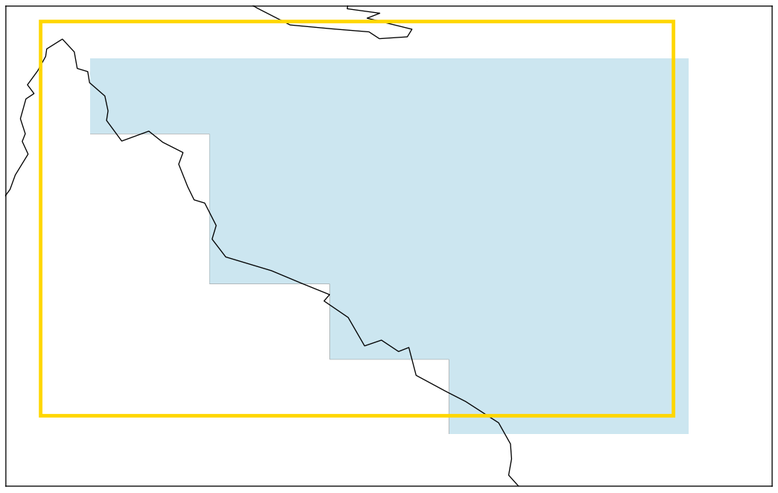
Error (Pred - Actual) - Step 46 [scale ±8]



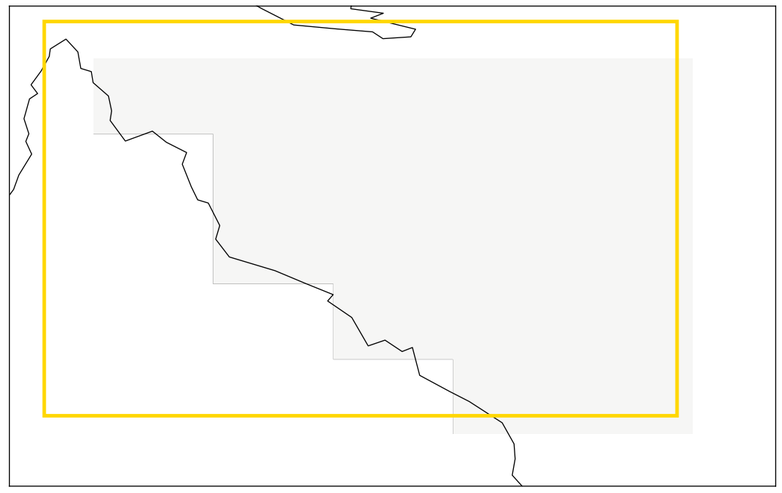
Actual DHW - Step 90

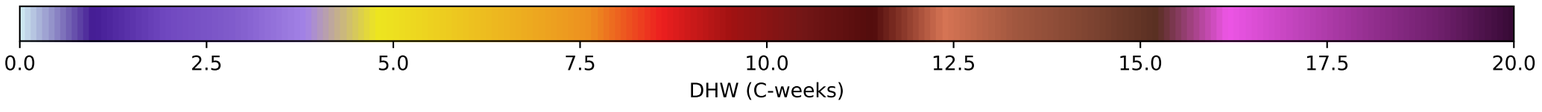


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

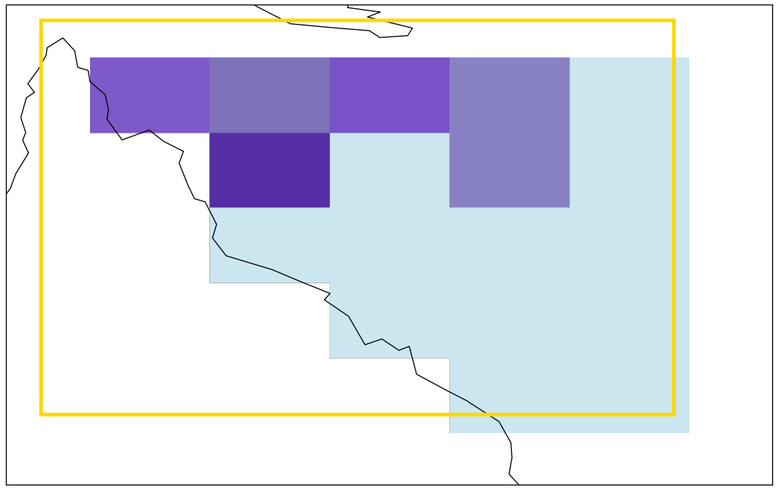




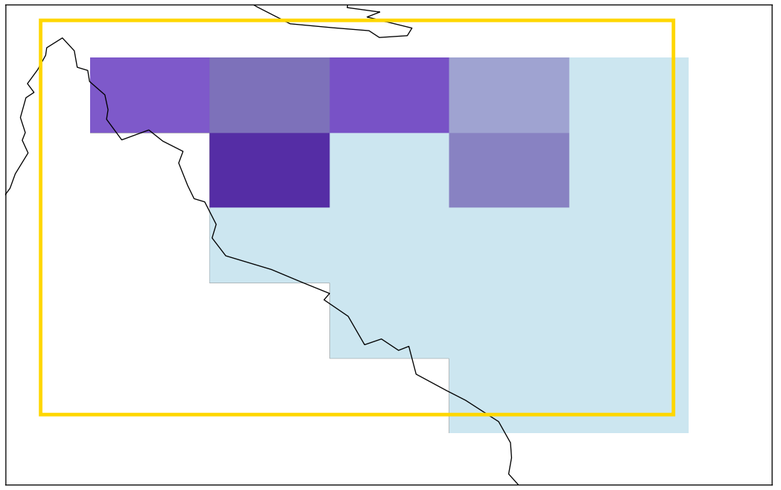


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

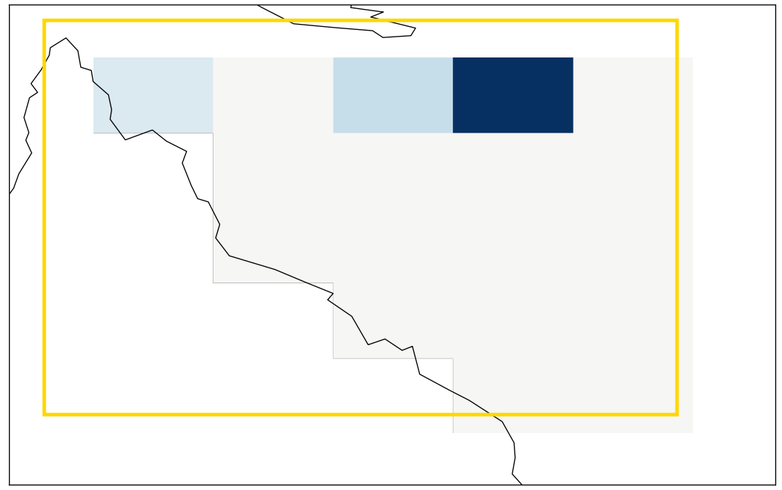
Actual DHW - Step 1



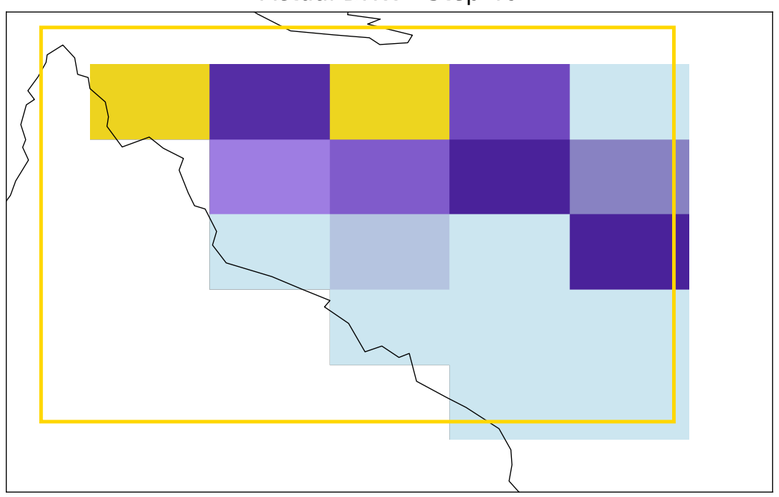
Predicted DHW - Step 1



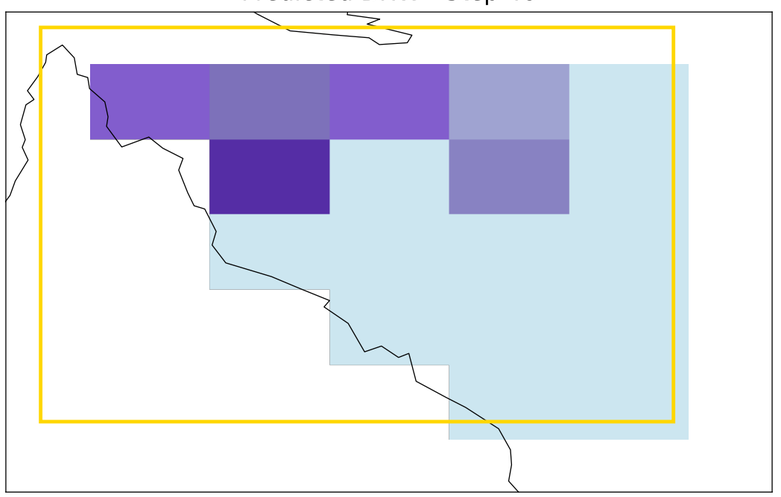
Error (Pred - Actual) - Step 1 [scale ±0.11]



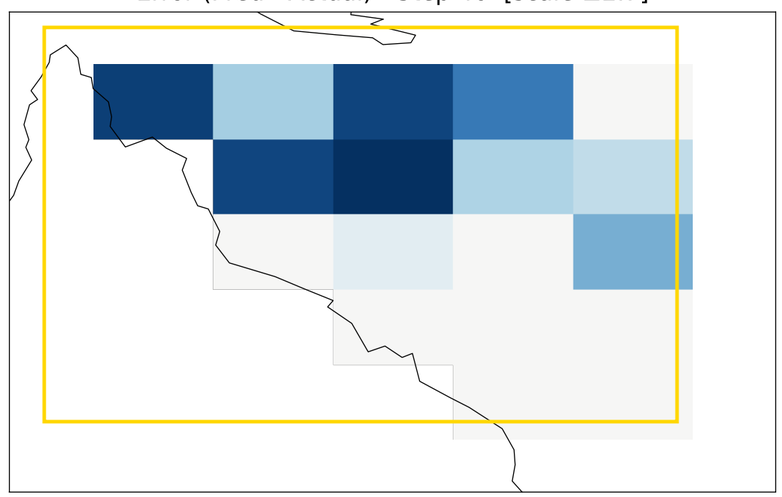
Actual DHW - Step 46



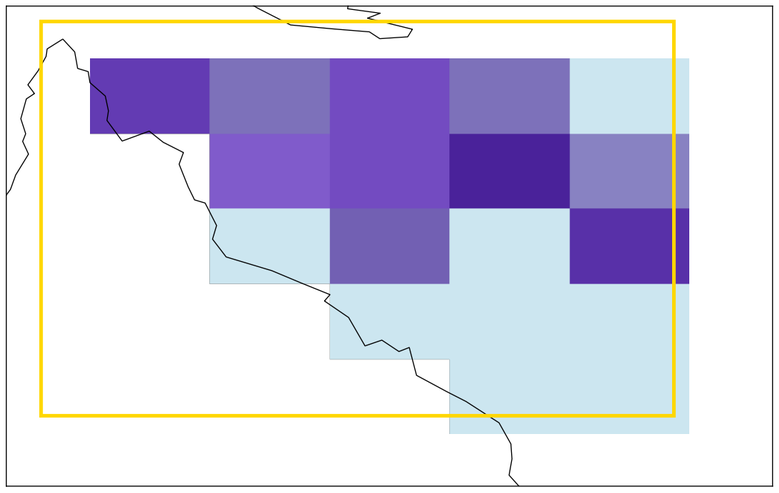
Predicted DHW - Step 46



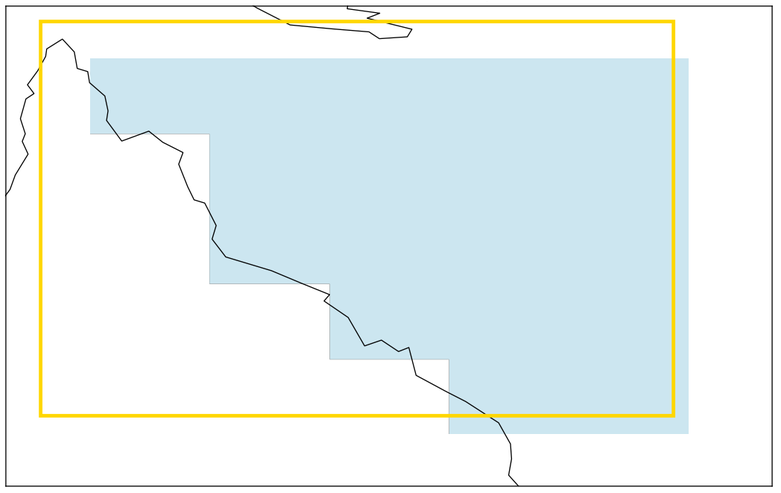
Error (Pred - Actual) - Step 46 [scale ±2.7]



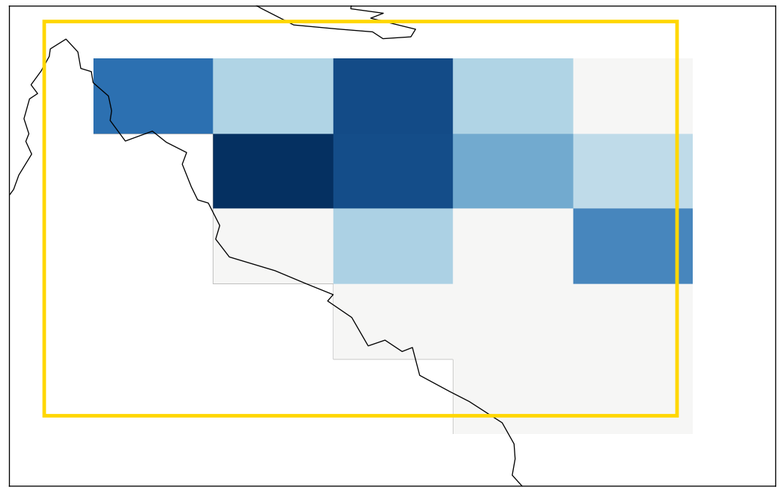
Actual DHW - Step 90

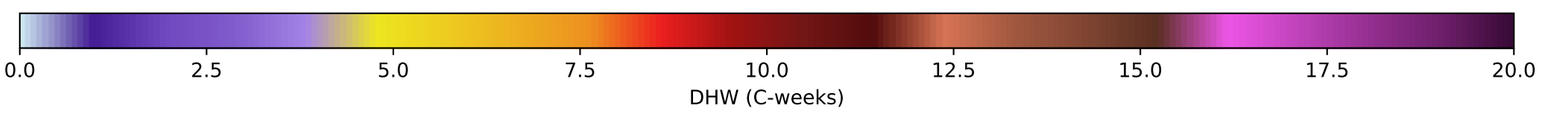


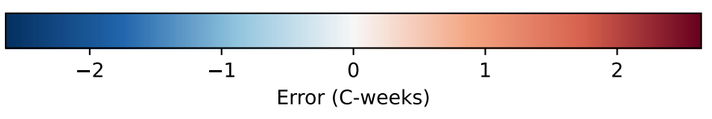
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.6]

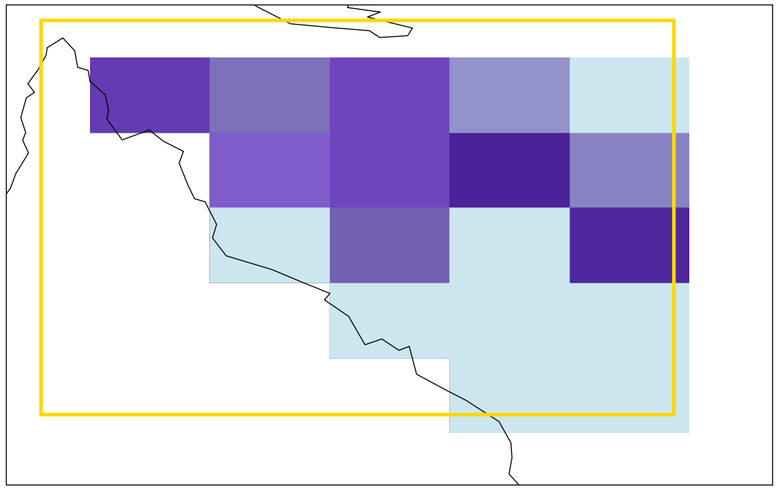




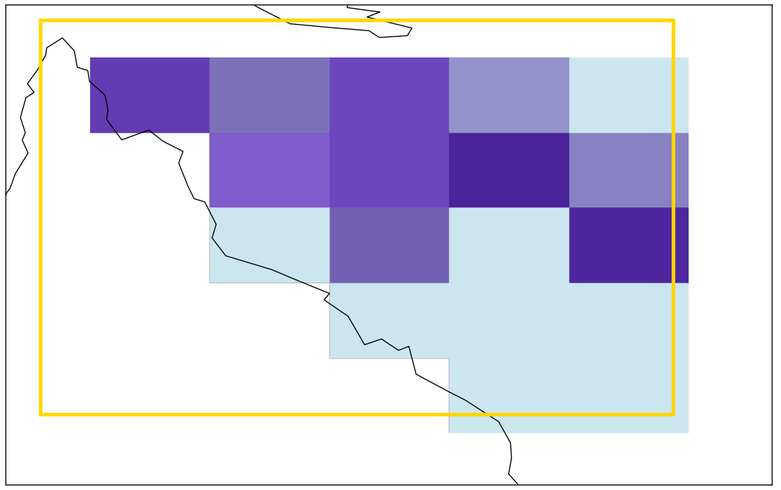


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

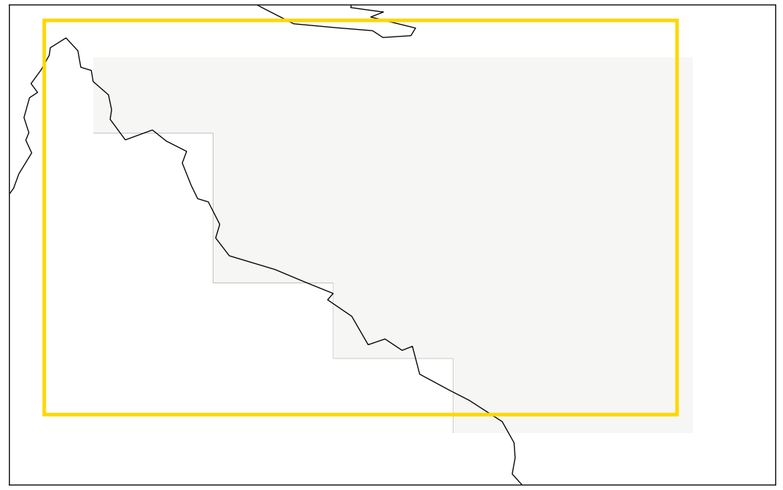
Actual DHW - Step 1



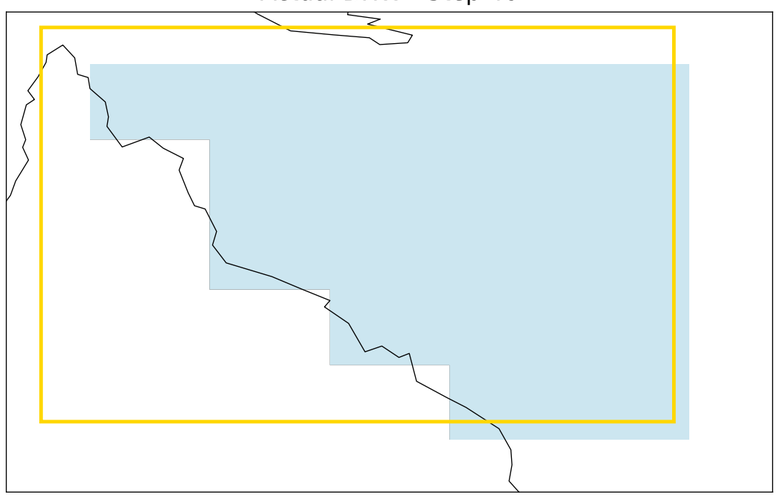
Predicted DHW - Step 1



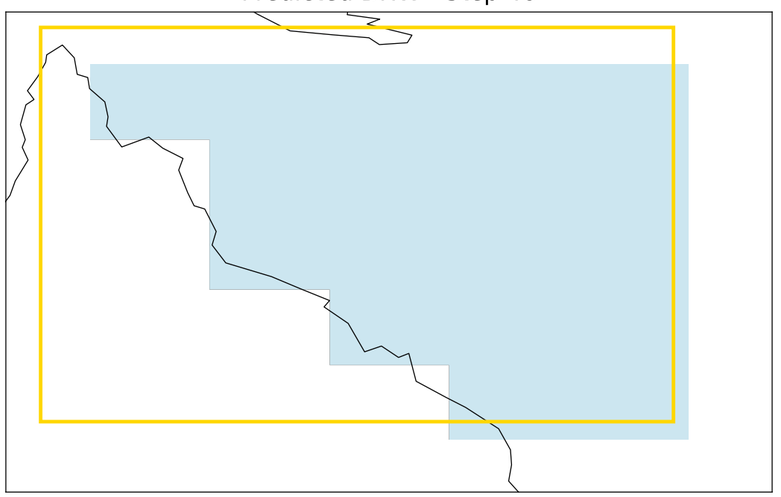
Error (Pred - Actual) - Step 1 [scale ±8]



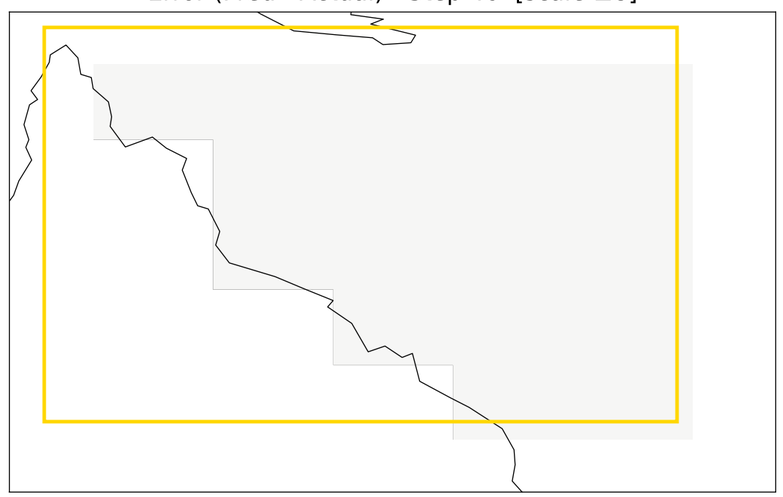
Actual DHW - Step 46



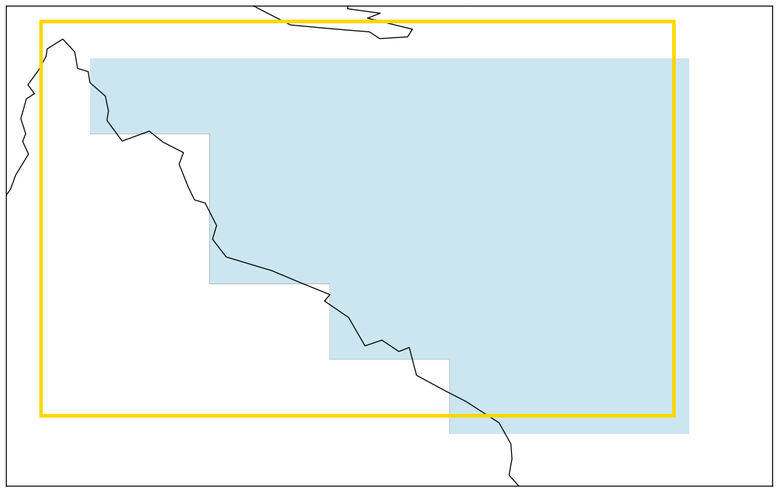
Predicted DHW - Step 46



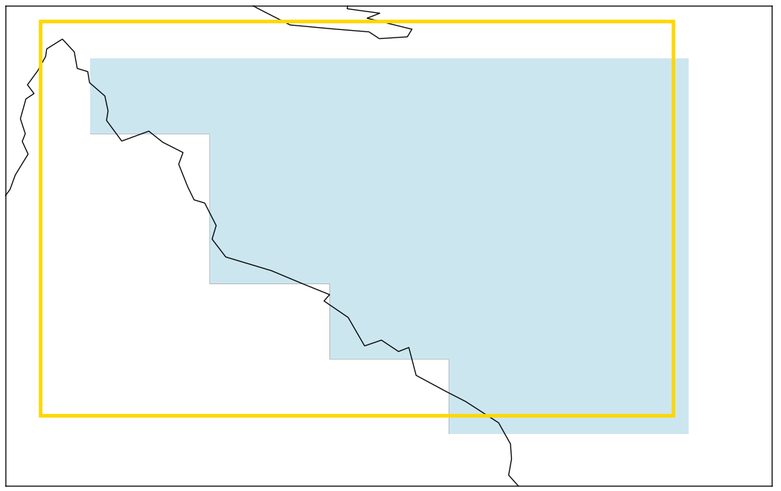
Error (Pred - Actual) - Step 46 [scale ±8]



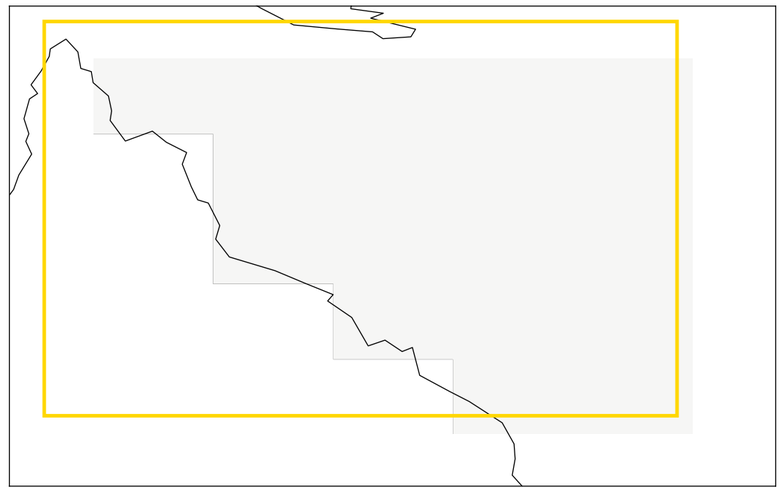
Actual DHW - Step 90

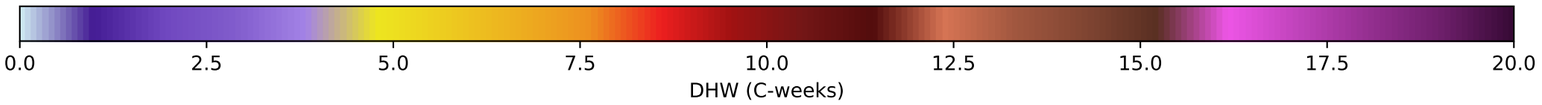


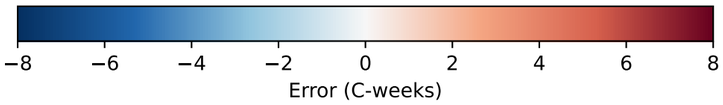
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

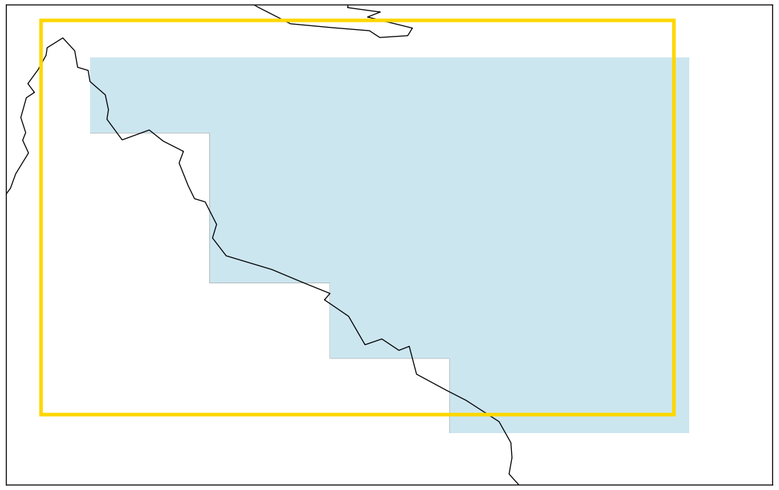




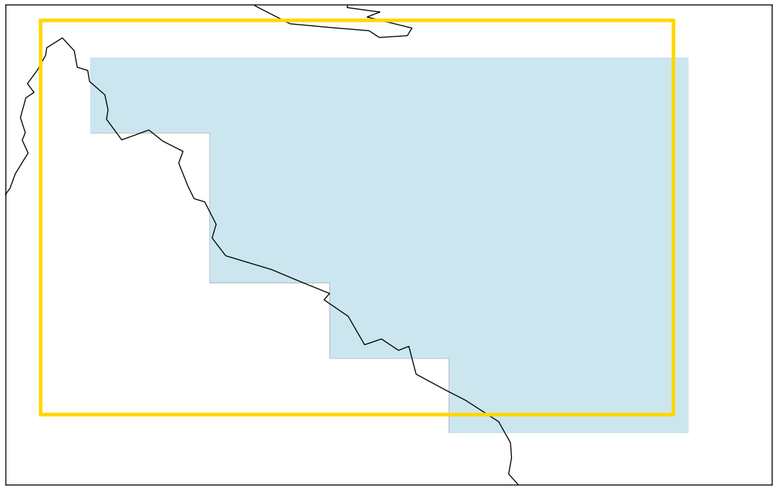


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

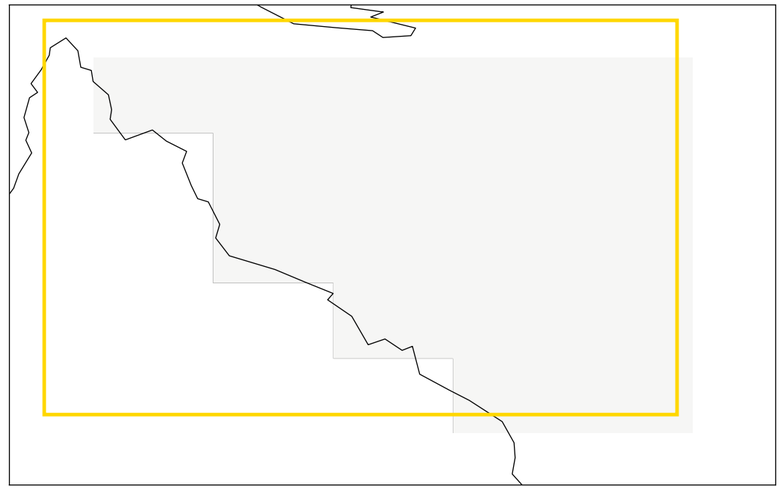
Actual DHW - Step 1



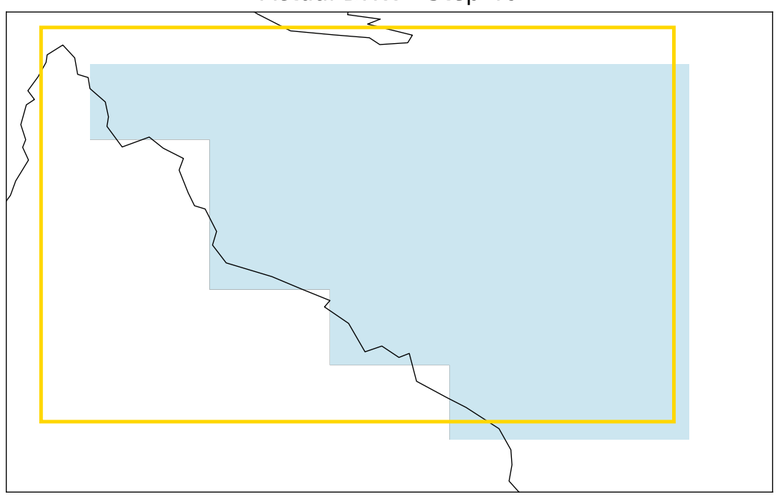
Predicted DHW - Step 1



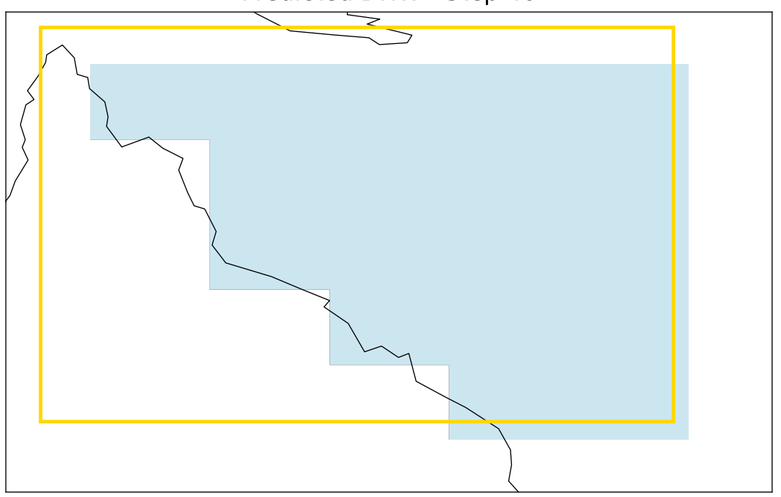
Error (Pred - Actual) - Step 1 [scale ±8]



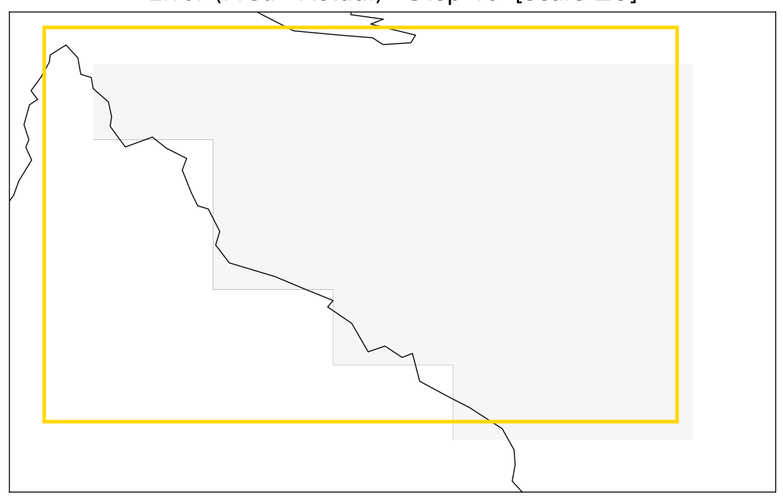
Actual DHW - Step 46



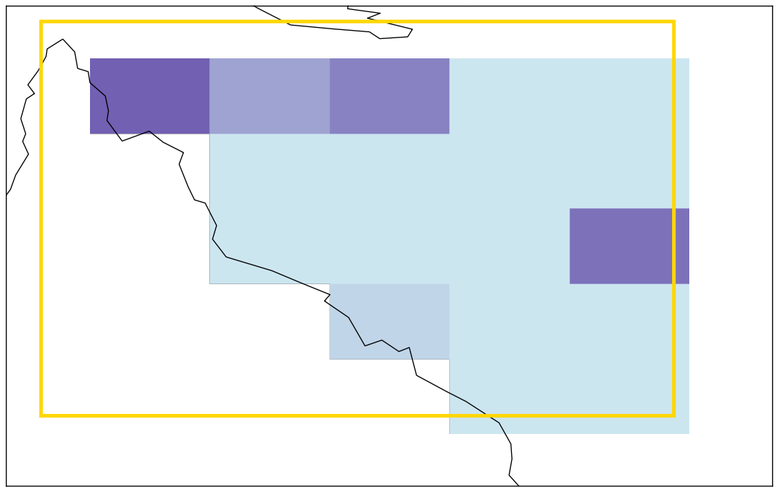
Predicted DHW - Step 46



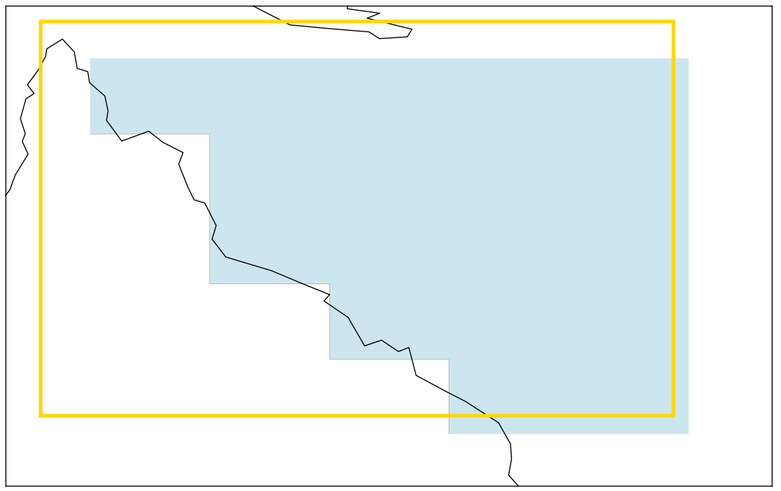
Error (Pred - Actual) - Step 46 [scale ±8]



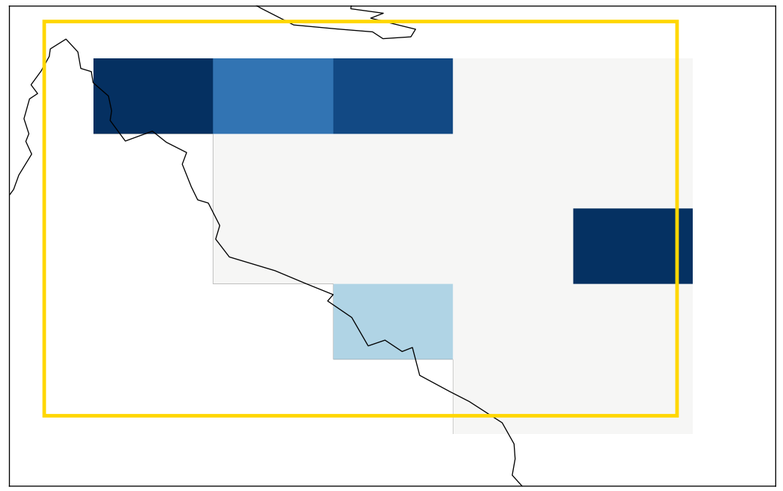
Actual DHW - Step 90

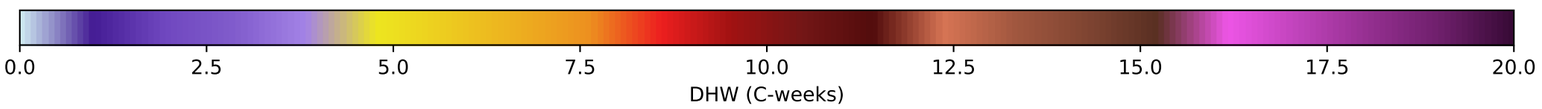


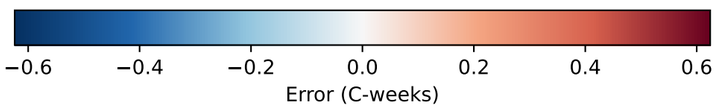
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.62]

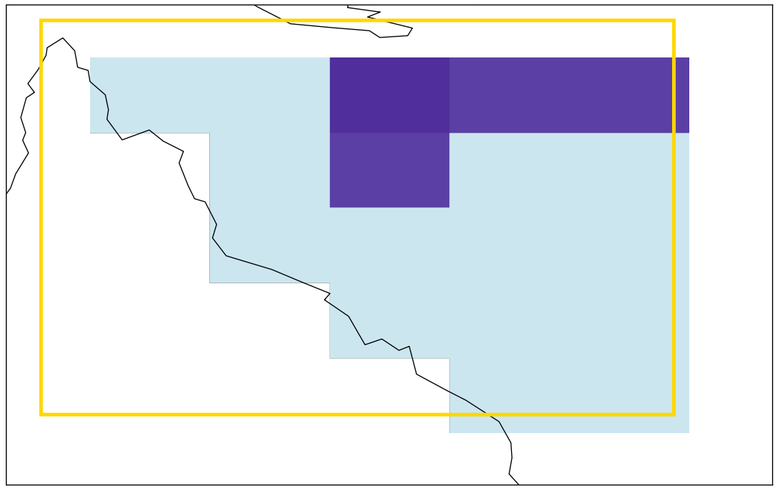




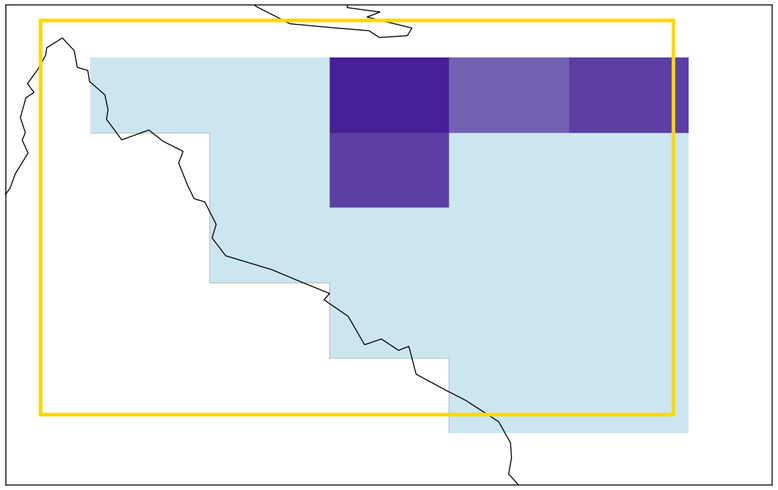


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

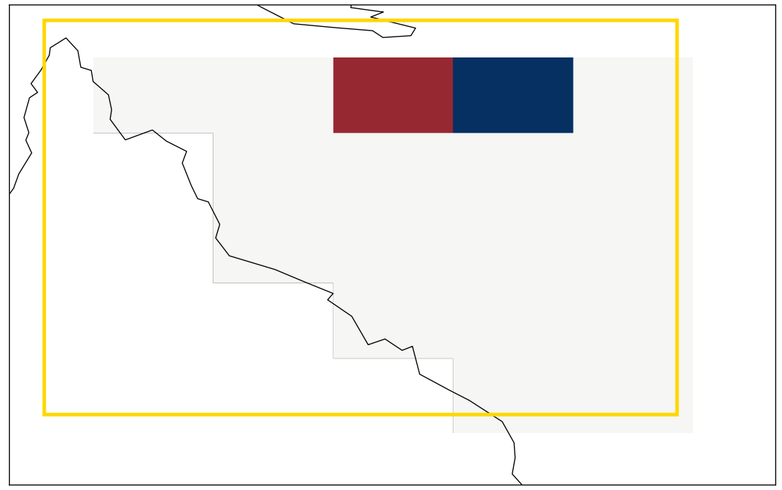
Actual DHW - Step 1



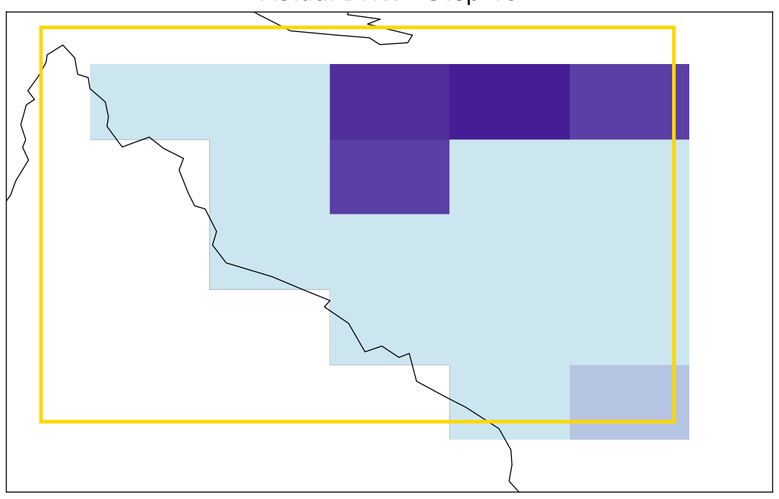
Predicted DHW - Step 1



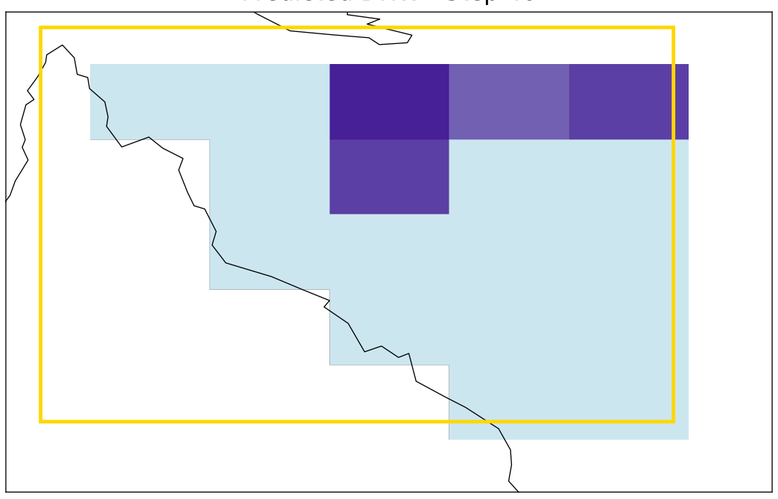
Error (Pred - Actual) - Step 1 [scale ±0.18]



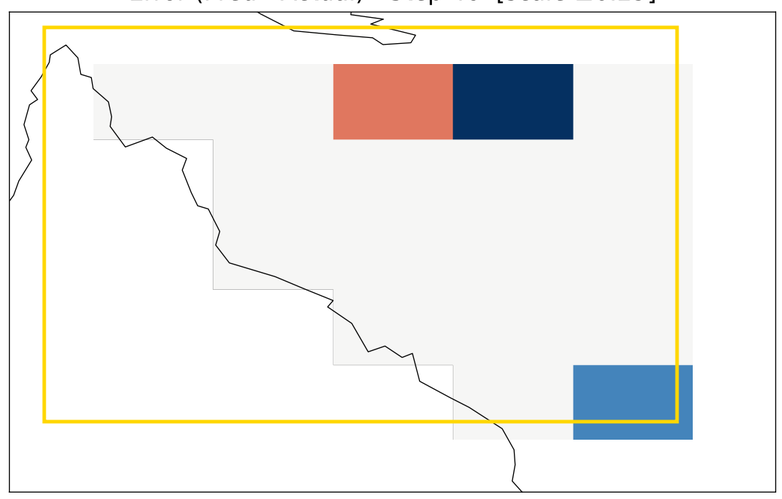
Actual DHW - Step 46



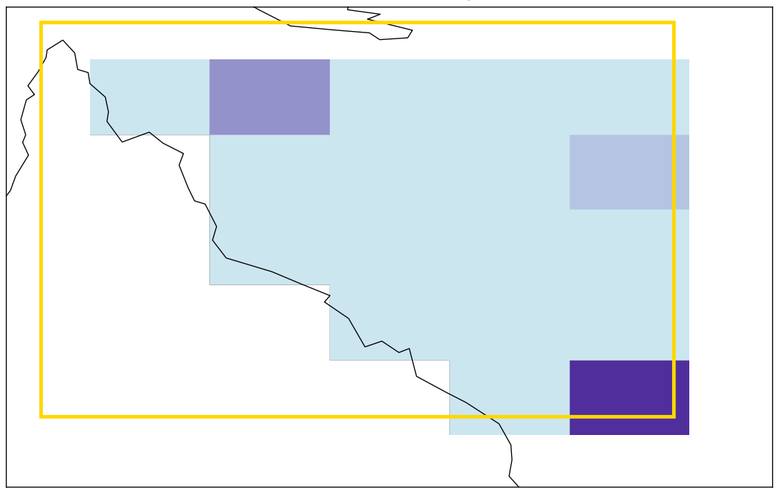
Predicted DHW - Step 46



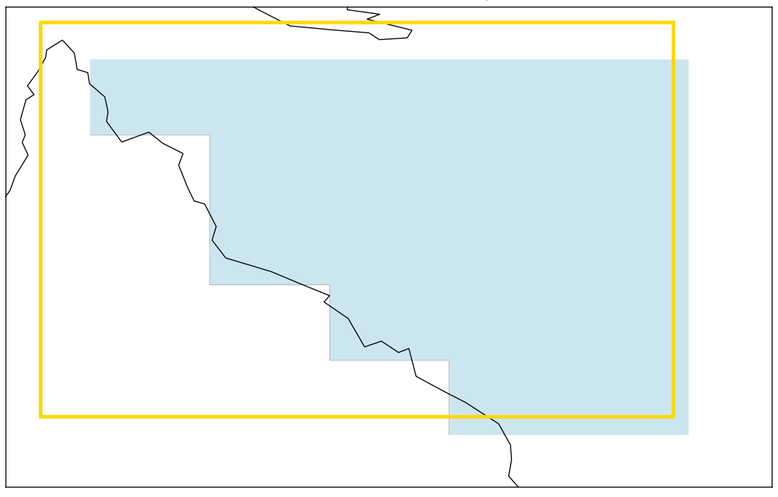
Error (Pred - Actual) - Step 46 [scale ±0.29]



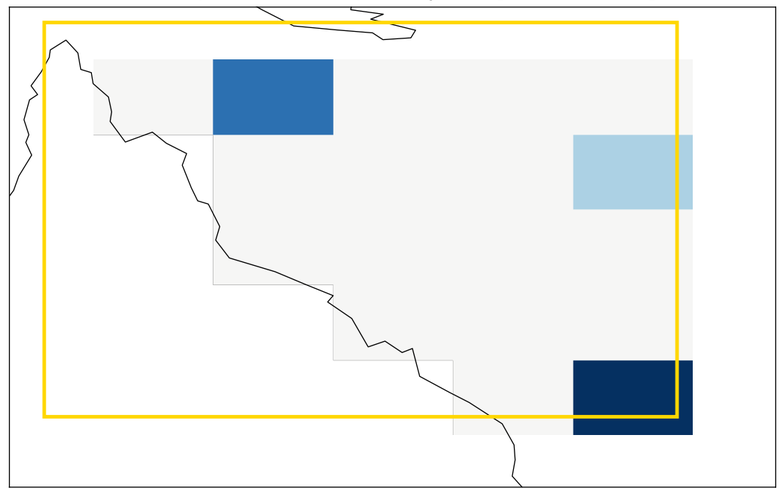
Actual DHW - Step 90

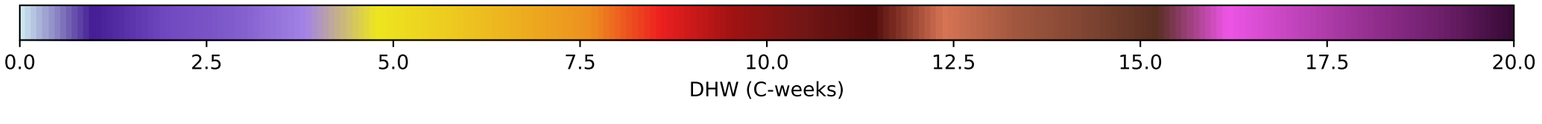


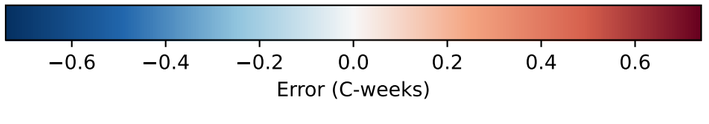
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.74]

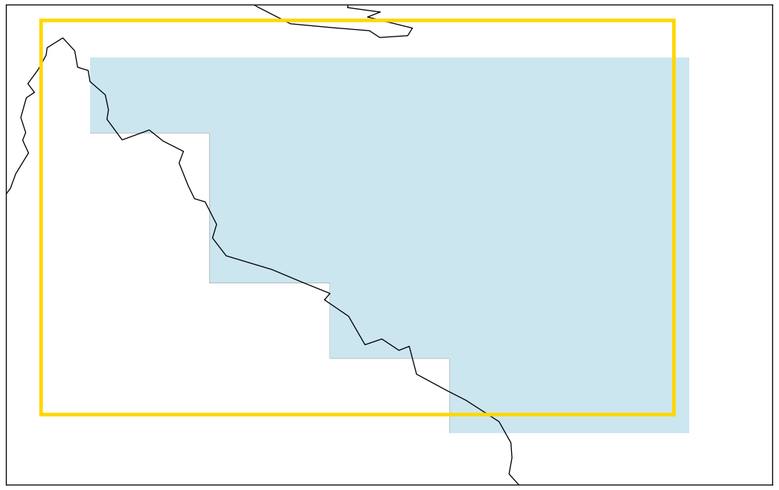




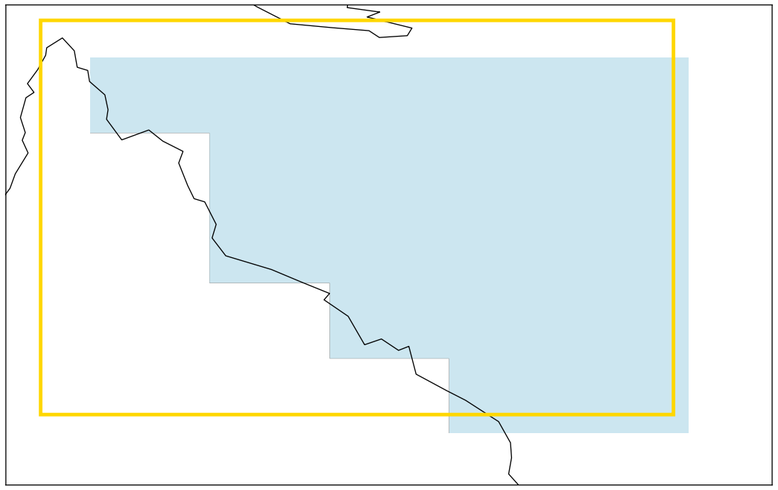


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

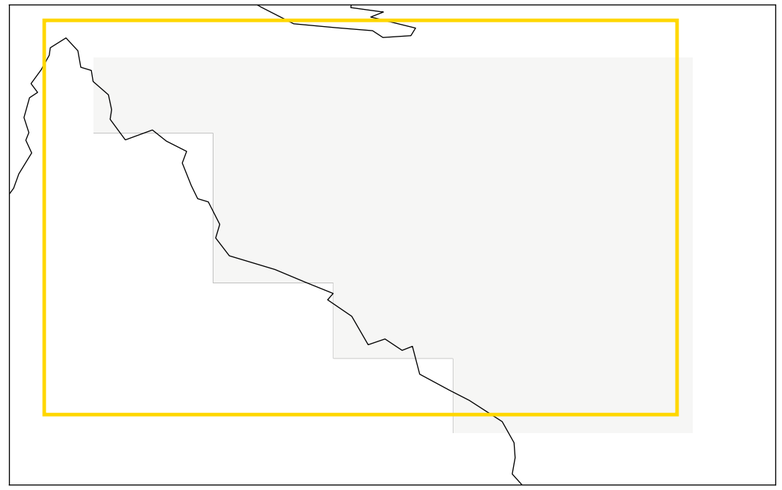
Actual DHW - Step 1



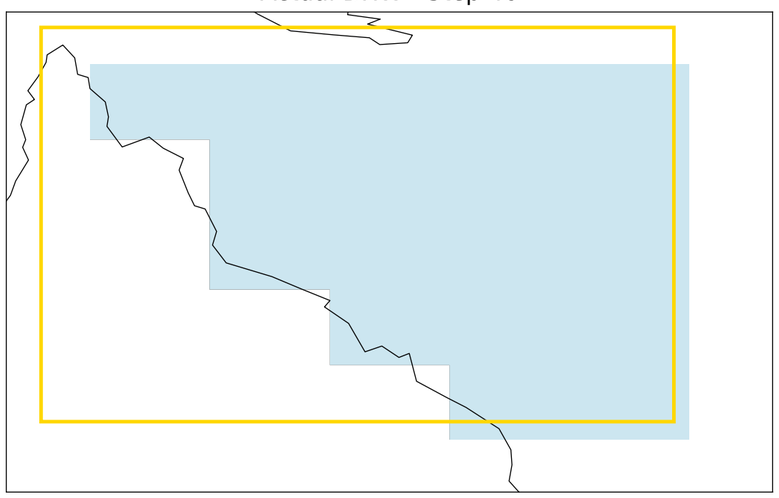
Predicted DHW - Step 1



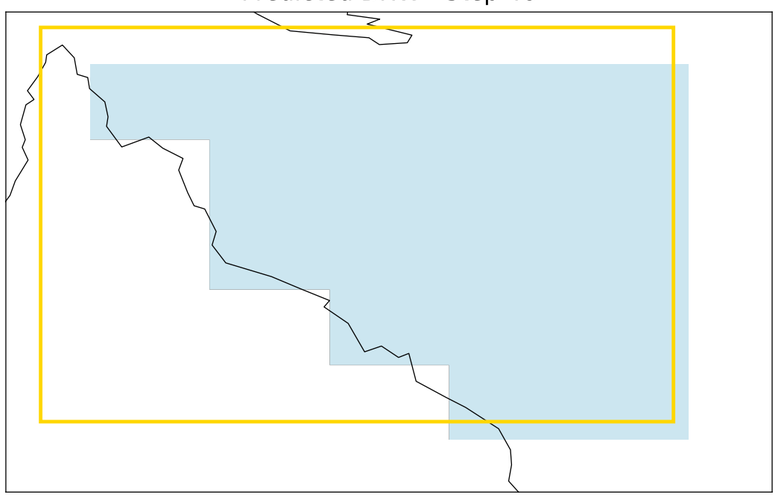
Error (Pred - Actual) - Step 1 [scale ±8]



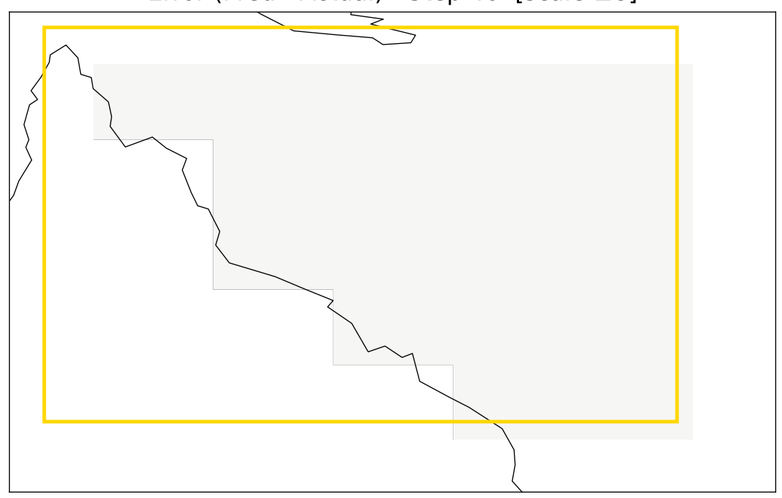
Actual DHW - Step 46



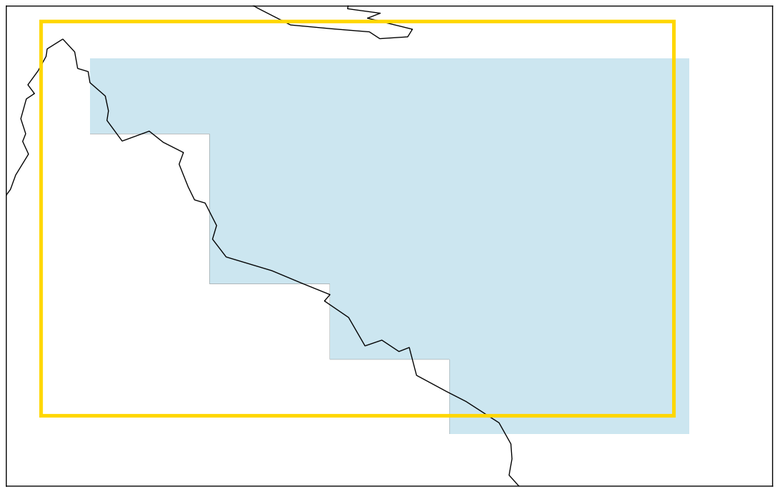
Predicted DHW - Step 46



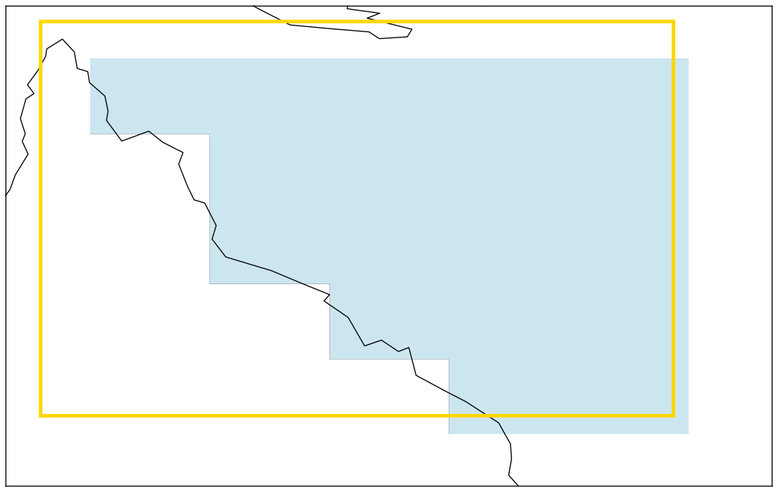
Error (Pred - Actual) - Step 46 [scale ±8]



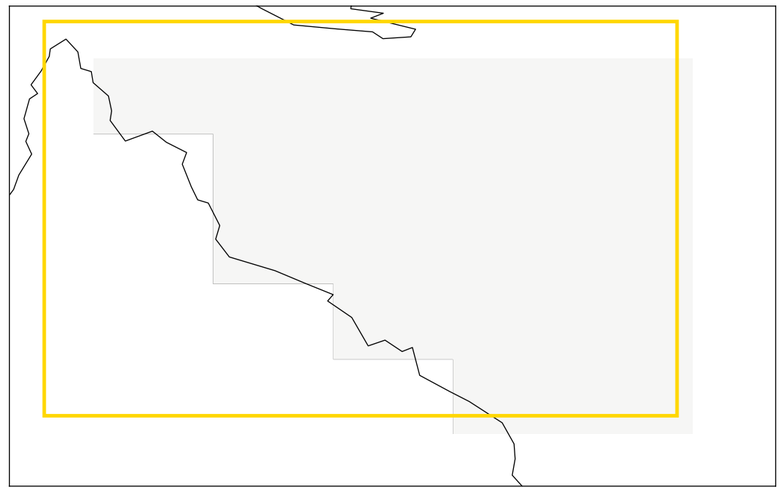
Actual DHW - Step 90

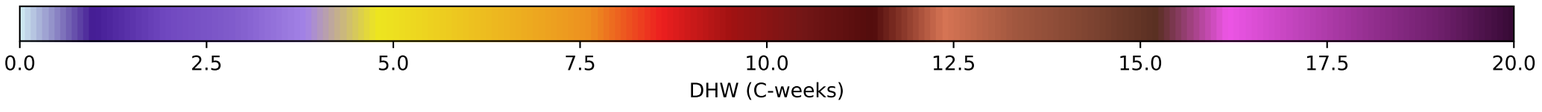


Predicted DHW - Step 90



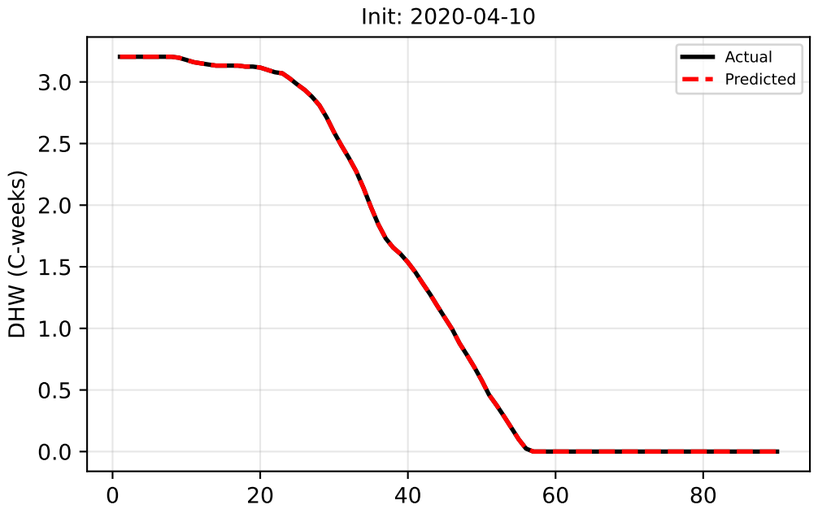
Error (Pred - Actual) - Step 90 [scale ±8]

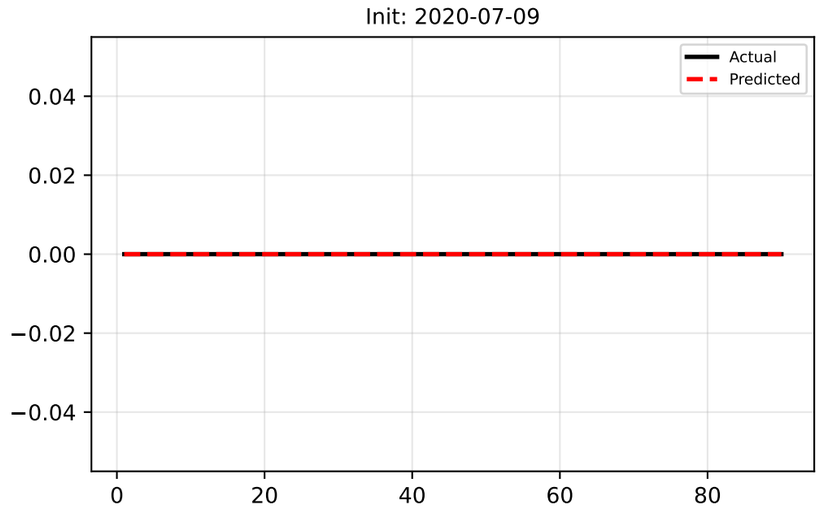


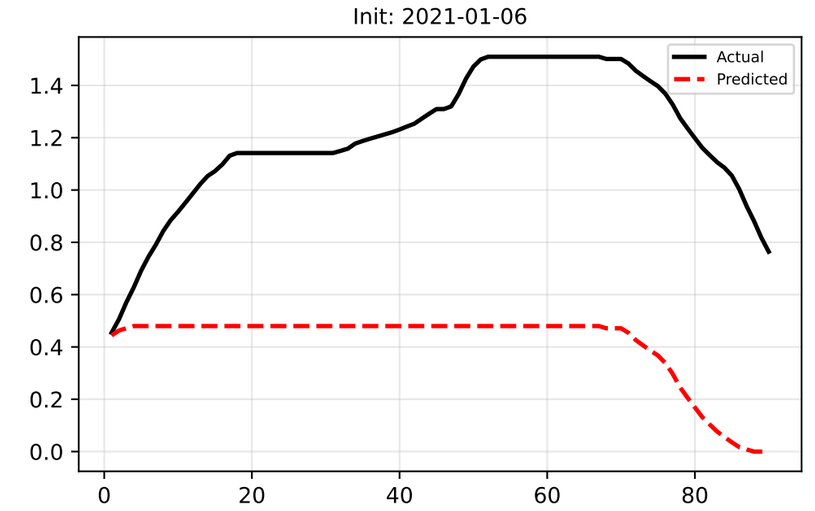


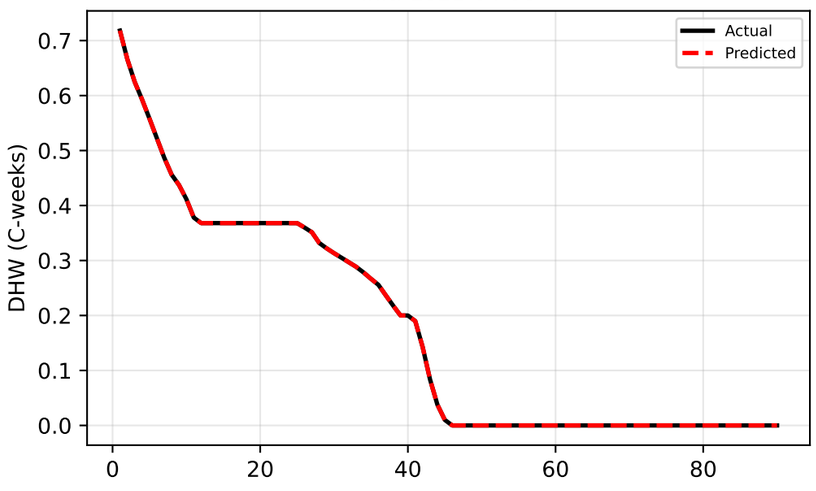


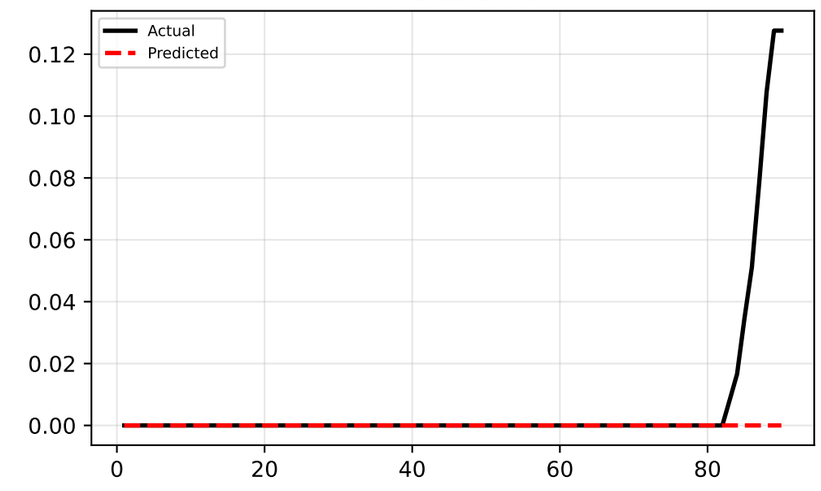
Global-Mean DHW - Test - GBR (C-weeks)

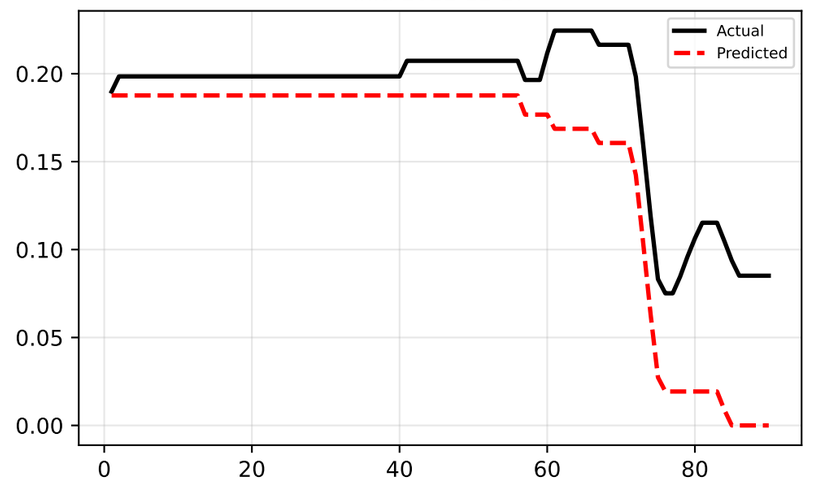


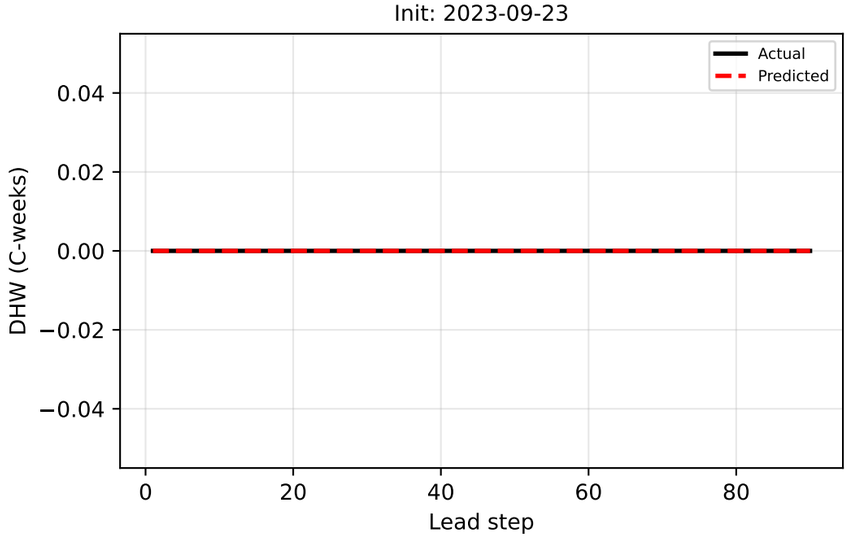


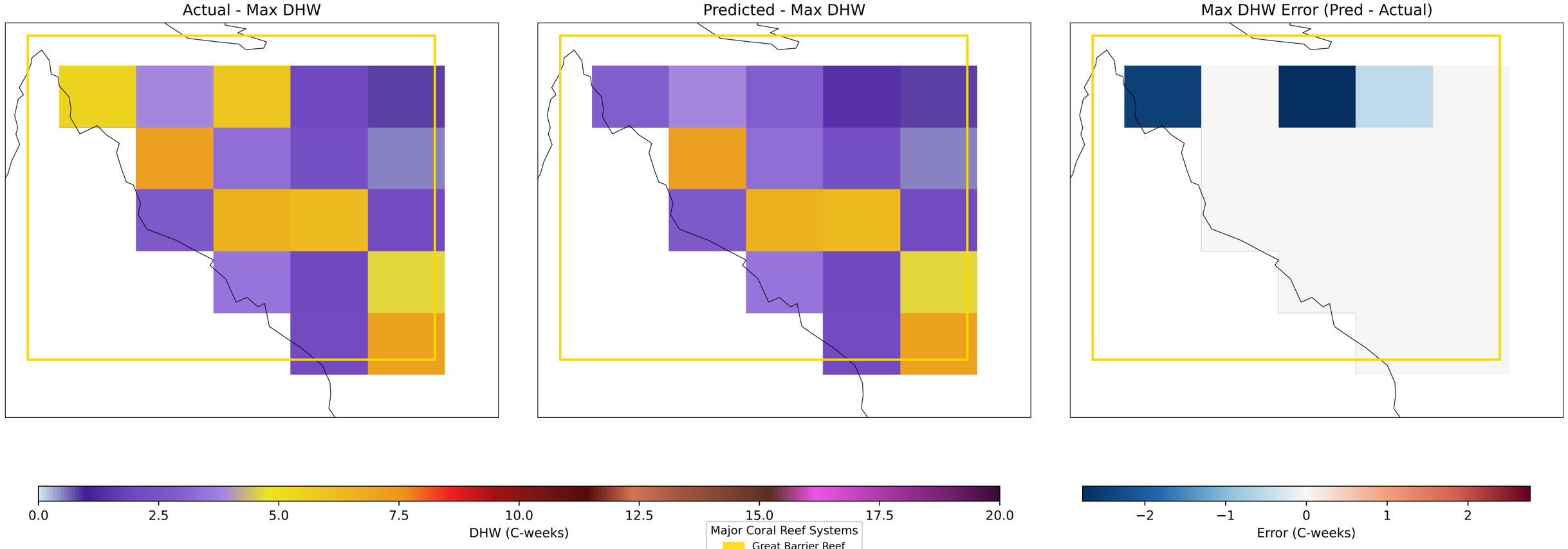


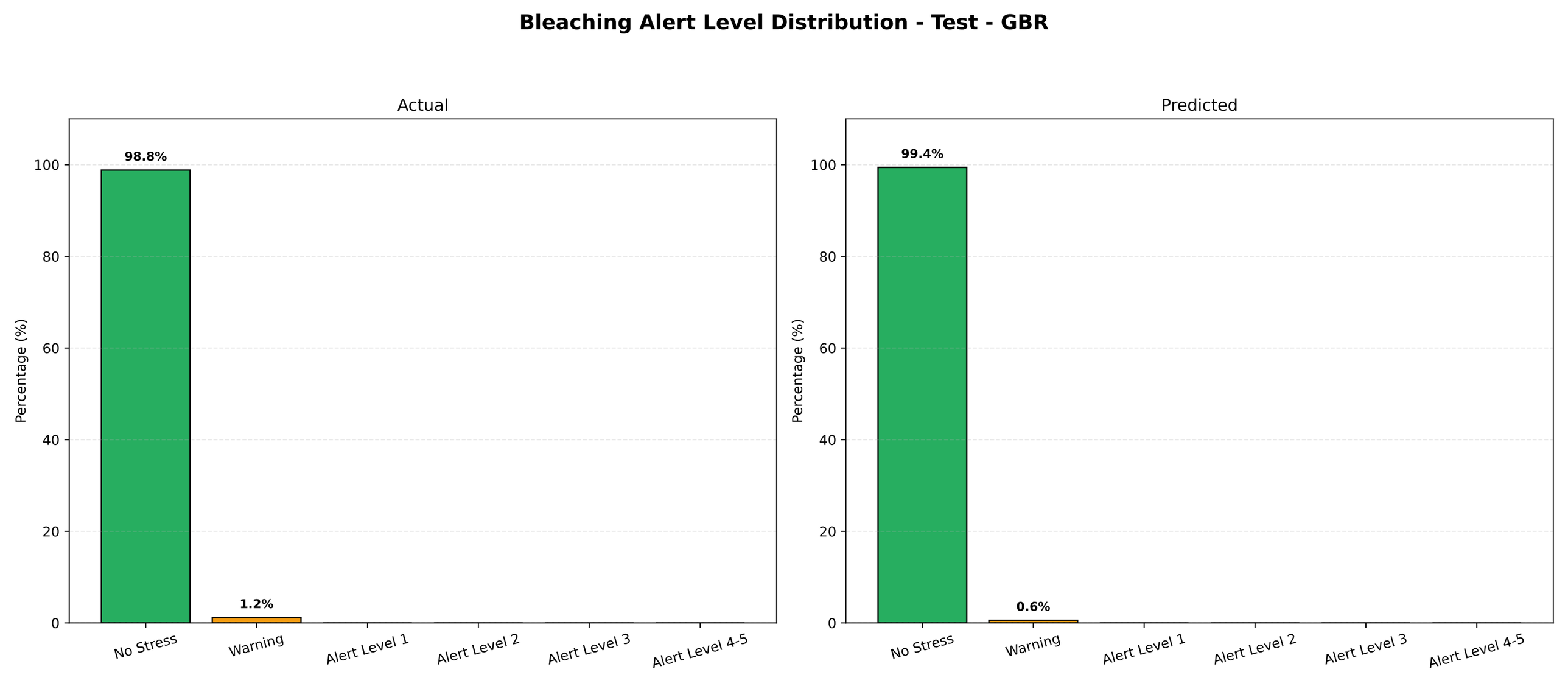




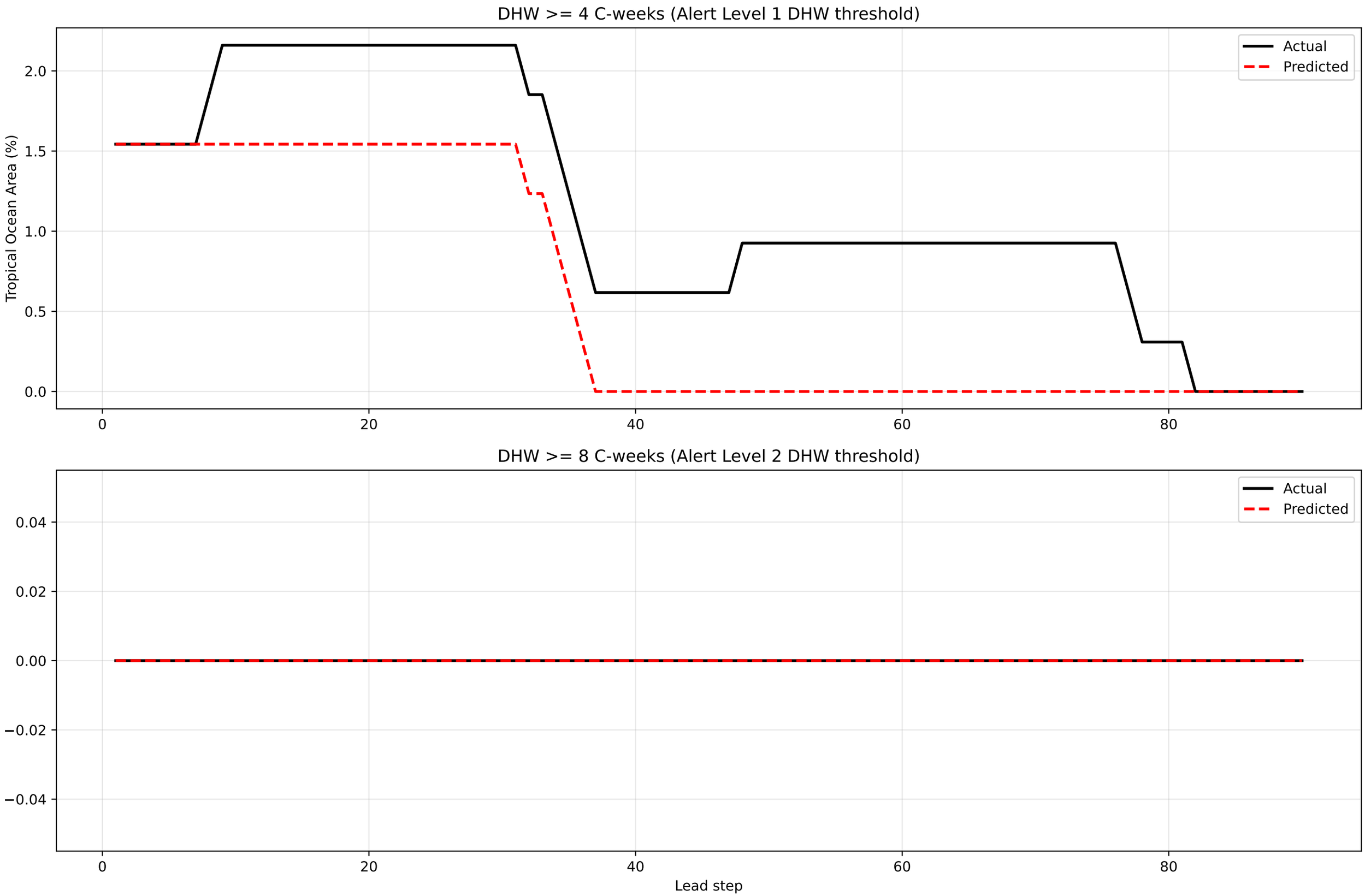





Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−2
−1
0
1
2
Error (C-weeks)


Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.8%
1.2%
99.4%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

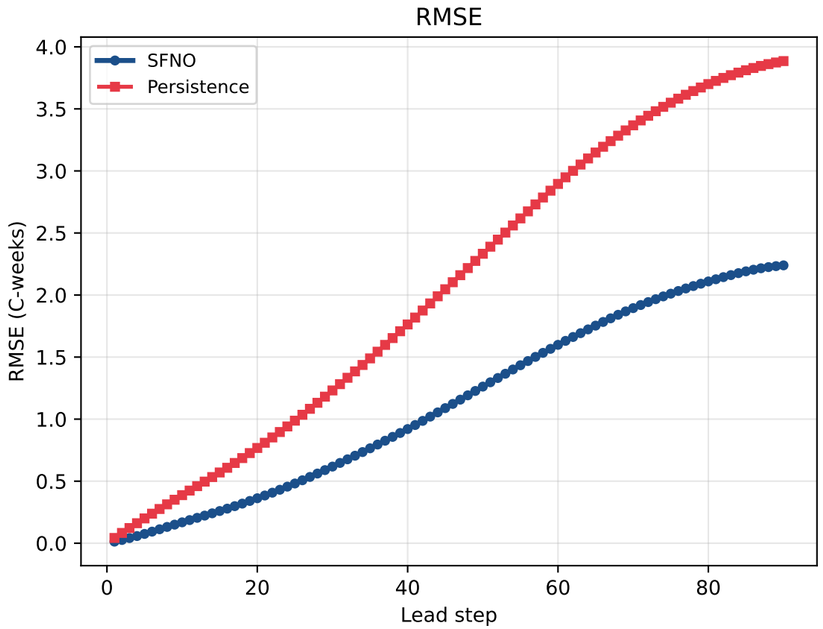
Tropical Ocean Area Under Bleaching Stress - Test - GBR

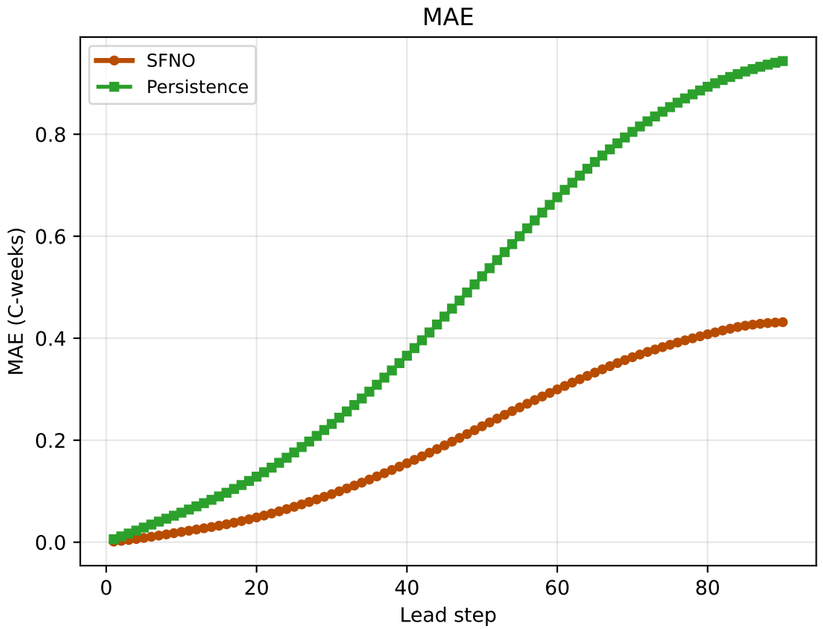


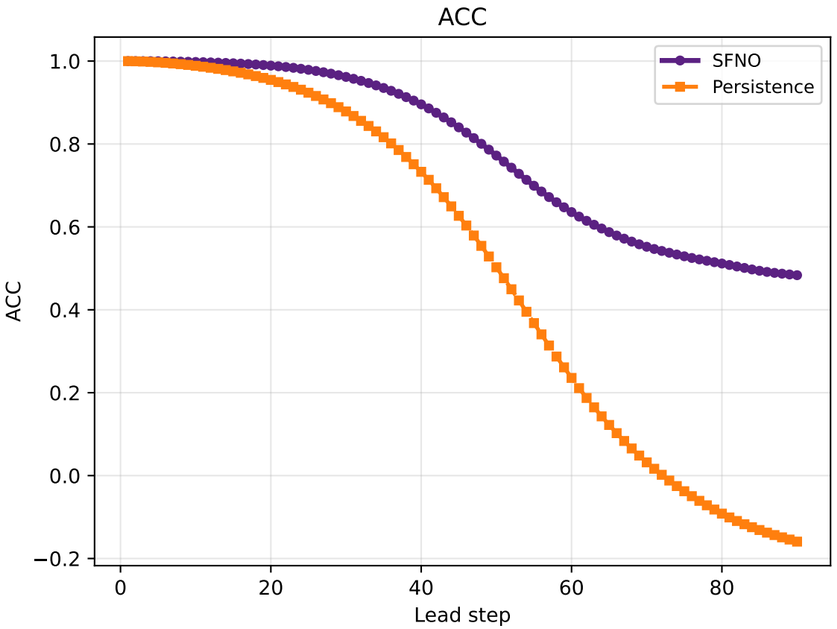
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

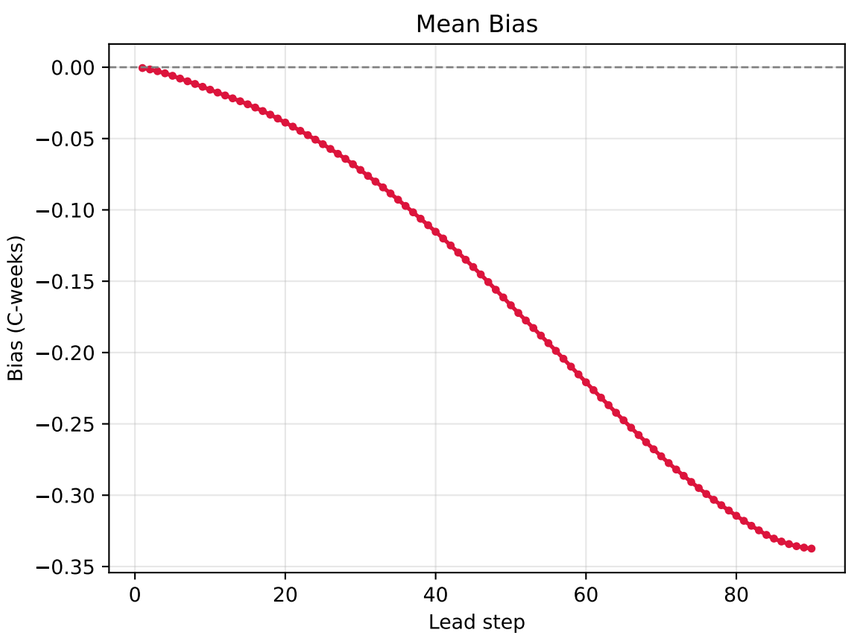
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0129	0.0425	69.7%	0.0011	0.0057	-0.0006	1.0000	0.9999	1.0000
2	0.0263	0.0829	68.3%	0.0026	0.0114	-0.0015	0.9999	0.9995	0.9999
3	0.0417	0.1222	65.9%	0.0044	0.0170	-0.0028	0.9999	0.9989	0.9998
4	0.0578	0.1613	64.2%	0.0063	0.0227	-0.0043	0.9998	0.9980	0.9996
5	0.0750	0.2000	62.5%	0.0084	0.0286	-0.0060	0.9996	0.9970	0.9993
6	0.0934	0.2381	60.8%	0.0107	0.0346	-0.0079	0.9994	0.9957	0.9989
7	0.1125	0.2758	59.2%	0.0130	0.0404	-0.0099	0.9991	0.9943	0.9984
8	0.1312	0.3133	58.1%	0.0153	0.0461	-0.0117	0.9987	0.9926	0.9979
9	0.1497	0.3504	57.3%	0.0177	0.0520	-0.0137	0.9983	0.9907	0.9972
10	0.1683	0.3874	56.6%	0.0201	0.0579	-0.0157	0.9979	0.9886	0.9965
11	0.1865	0.4241	56.0%	0.0225	0.0641	-0.0178	0.9974	0.9863	0.9956
12	0.2045	0.4601	55.5%	0.0249	0.0702	-0.0198	0.9968	0.9839	0.9947
13	0.2226	0.4965	55.2%	0.0273	0.0766	-0.0218	0.9962	0.9812	0.9937
14	0.2410	0.5332	54.8%	0.0297	0.0831	-0.0239	0.9955	0.9783	0.9926
15	0.2595	0.5701	54.5%	0.0324	0.0898	-0.0260	0.9947	0.9752	0.9913
16	0.2786	0.6080	54.2%	0.0352	0.0970	-0.0282	0.9939	0.9717	0.9899
17	0.2988	0.6467	53.8%	0.0383	0.1044	-0.0307	0.9928	0.9680	0.9883
18	0.3192	0.6861	53.5%	0.0415	0.1121	-0.0332	0.9917	0.9639	0.9865
19	0.3406	0.7264	53.1%	0.0449	0.1201	-0.0360	0.9904	0.9594	0.9845
20	0.3624	0.7676	52.8%	0.0485	0.1285	-0.0388	0.9890	0.9546	0.9823
21	0.3845	0.8096	52.5%	0.0523	0.1371	-0.0416	0.9875	0.9494	0.9798
22	0.4072	0.8524	52.2%	0.0563	0.1462	-0.0446	0.9857	0.9437	0.9771
23	0.4308	0.8961	51.9%	0.0605	0.1557	-0.0476	0.9837	0.9376	0.9740
24	0.4559	0.9412	51.6%	0.0649	0.1657	-0.0508	0.9815	0.9309	0.9705
25	0.4813	0.9875	51.3%	0.0694	0.1760	-0.0539	0.9789	0.9237	0.9667
26	0.5073	1.0350	51.0%	0.0741	0.1866	-0.0573	0.9762	0.9158	0.9624
27	0.5342	1.0830	50.7%	0.0789	0.1974	-0.0607	0.9731	0.9074	0.9578
28	0.5615	1.1316	50.4%	0.0839	0.2084	-0.0642	0.9697	0.8984	0.9527
29	0.5892	1.1811	50.1%	0.0890	0.2200	-0.0680	0.9659	0.8888	0.9472
30	0.6176	1.2311	49.8%	0.0943	0.2319	-0.0720	0.9618	0.8785	0.9411
31	0.6470	1.2818	49.5%	0.0998	0.2441	-0.0761	0.9572	0.8675	0.9345
32	0.6762	1.3332	49.3%	0.1054	0.2564	-0.0802	0.9523	0.8558	0.9274
33	0.7052	1.3846	49.1%	0.1110	0.2690	-0.0842	0.9471	0.8434	0.9199
34	0.7346	1.4364	48.9%	0.1168	0.2818	-0.0885	0.9414	0.8304	0.9119
35	0.7648	1.4894	48.6%	0.1229	0.2951	-0.0928	0.9351	0.8164	0.9031
36	0.7953	1.5434	48.5%	0.1290	0.3088	-0.0972	0.9283	0.8014	0.8937
37	0.8260	1.5981	48.3%	0.1353	0.3227	-0.1017	0.9210	0.7854	0.8837
38	0.8570	1.6531	48.2%	0.1416	0.3368	-0.1062	0.9131	0.7687	0.8730
39	0.8882	1.7081	48.0%	0.1481	0.3512	-0.1107	0.9047	0.7511	0.8618
40	0.9200	1.7629	47.8%	0.1547	0.3657	-0.1153	0.8956	0.7329	0.8499
41	0.9525	1.8185	47.6%	0.1614	0.3806	-0.1201	0.8858	0.7136	0.8372
42	0.9860	1.8750	47.4%	0.1684	0.3959	-0.1249	0.8753	0.6932	0.8236
43	1.0202	1.9323	47.2%	0.1755	0.4113	-0.1299	0.8640	0.6717	0.8094
44	1.0544	1.9898	47.0%	0.1826	0.4269	-0.1350	0.8523	0.6495	0.7947
45	1.0886	2.0466	46.8%	0.1898	0.4424	-0.1400	0.8401	0.6267	0.7797
46	1.1226	2.1032	46.6%	0.1971	0.4581	-0.1452	0.8274	0.6033	0.7642
47	1.1570	2.1599	46.4%	0.2045	0.4738	-0.1506	0.8142	0.5792	0.7482
48	1.1919	2.2173	46.2%	0.2120	0.4897	-0.1560	0.8005	0.5542	0.7317
49	1.2270	2.2749	46.1%	0.2196	0.5055	-0.1614	0.7864	0.5286	0.7150
50	1.2622	2.3325	45.9%	0.2273	0.5215	-0.1669	0.7721	0.5026	0.6980
51	1.2969	2.3896	45.7%	0.2349	0.5374	-0.1722	0.7575	0.4762	0.6810
52	1.3314	2.4464	45.6%	0.2423	0.5532	-0.1775	0.7429	0.4495	0.6639
53	1.3657	2.5032	45.4%	0.2498	0.5689	-0.1828	0.7282	0.4226	0.6470
54	1.4001	2.5604	45.3%	0.2571	0.5845	-0.1881	0.7135	0.3952	0.6302
55	1.4342	2.6175	45.2%	0.2643	0.6001	-0.1934	0.6993	0.3679	0.6139
56	1.4681	2.6741	45.1%	0.2715	0.6155	-0.1988	0.6853	0.3407	0.5982
57	1.5015	2.7302	45.0%	0.2787	0.6310	-0.2043	0.6720	0.3138	0.5832
58	1.5340	2.7859	44.9%	0.2857	0.6464	-0.2099	0.6594	0.2872	0.5691
59	1.5664	2.8409	44.9%	0.2927	0.6616	-0.2153	0.6472	0.2611	0.5554
60	1.5986	2.8953	44.8%	0.2996	0.6764	-0.2209	0.6357	0.2357	0.5425
61	1.6307	2.9487	44.7%	0.3064	0.6909	-0.2263	0.6248	0.2110	0.5301
62	1.6619	3.0007	44.6%	0.3130	0.7050	-0.2316	0.6147	0.1873	0.5186
63	1.6927	3.0513	44.5%	0.3196	0.7188	-0.2369	0.6051	0.1647	0.5077
64	1.7231	3.1006	44.4%	0.3261	0.7323	-0.2422	0.5961	0.1429	0.4972
65	1.7534	3.1484	44.3%	0.3326	0.7456	-0.2475	0.5873	0.1222	0.4871
66	1.7831	3.1948	44.2%	0.3390	0.7582	-0.2527	0.5791	0.1024	0.4776
67	1.8123	3.2402	44.1%	0.3452	0.7704	-0.2578	0.5716	0.0835	0.4686
68	1.8408	3.2842	44.0%	0.3512	0.7822	-0.2628	0.5645	0.0655	0.4600
69	1.8682	3.3266	43.8%	0.3571	0.7936	-0.2678	0.5580	0.0483	0.4520
70	1.8945	3.3677	43.7%	0.3627	0.8047	-0.2727	0.5521	0.0319	0.4447
71	1.9194	3.4069	43.7%	0.3681	0.8153	-0.2775	0.5467	0.0165	0.4379
72	1.9433	3.4445	43.6%	0.3731	0.8253	-0.2820	0.5420	0.0020	0.4319
73	1.9665	3.4811	43.5%	0.3779	0.8349	-0.2864	0.5376	-0.0119	0.4263
74	1.9890	3.5164	43.4%	0.3826	0.8443	-0.2907	0.5333	-0.0252	0.4209
75	2.0109	3.5499	43.4%	0.3871	0.8532	-0.2950	0.5291	-0.0377	0.4157
76	2.0324	3.5823	43.3%	0.3915	0.8619	-0.2991	0.5250	-0.0496	0.4106
77	2.0529	3.6139	43.2%	0.3958	0.8703	-0.3032	0.5216	-0.0609	0.4062
78	2.0723	3.6437	43.1%	0.3998	0.8782	-0.3070	0.5182	-0.0716	0.4020
79	2.0912	3.6724	43.1%	0.4038	0.8858	-0.3107	0.5148	-0.0819	0.3978
80	2.1095	3.6998	43.0%	0.4077	0.8931	-0.3144	0.5116	-0.0917	0.3937
81	2.1271	3.7255	42.9%	0.4115	0.9000	-0.3180	0.5081	-0.1009	0.3895
82	2.1442	3.7494	42.8%	0.4152	0.9064	-0.3214	0.5045	-0.1095	0.3852
83	2.1606	3.7716	42.7%	0.4186	0.9123	-0.3246	0.5010	-0.1173	0.3810
84	2.1765	3.7922	42.6%	0.4220	0.9179	-0.3278	0.4973	-0.1245	0.3768
85	2.1909	3.8114	42.5%	0.4245	0.9231	-0.3304	0.4941	-0.1313	0.3732
86	2.2040	3.8295	42.4%	0.4266	0.9278	-0.3325	0.4914	-0.1376	0.3701
87	2.2156	3.8461	42.4%	0.4283	0.9323	-0.3343	0.4891	-0.1434	0.3675
88	2.2251	3.8609	42.4%	0.4297	0.9365	-0.3357	0.4871	-0.1489	0.3652
89	2.2328	3.8738	42.4%	0.4307	0.9402	-0.3367	0.4852	-0.1542	0.3630
90	2.2391	3.8852	42.4%	0.4315	0.9435	-0.3374	0.4835	-0.1593	0.3610

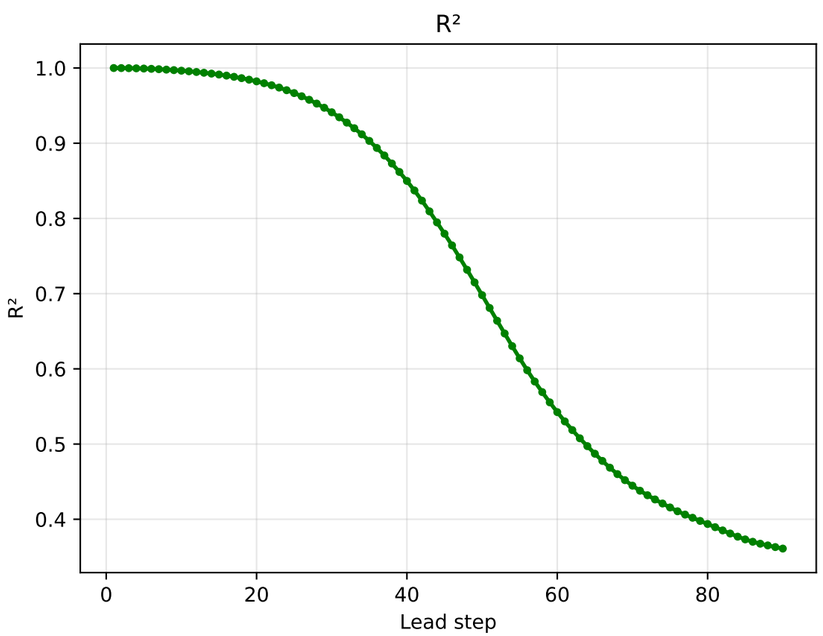
DHW Forecast Skill vs Lead Step - Val

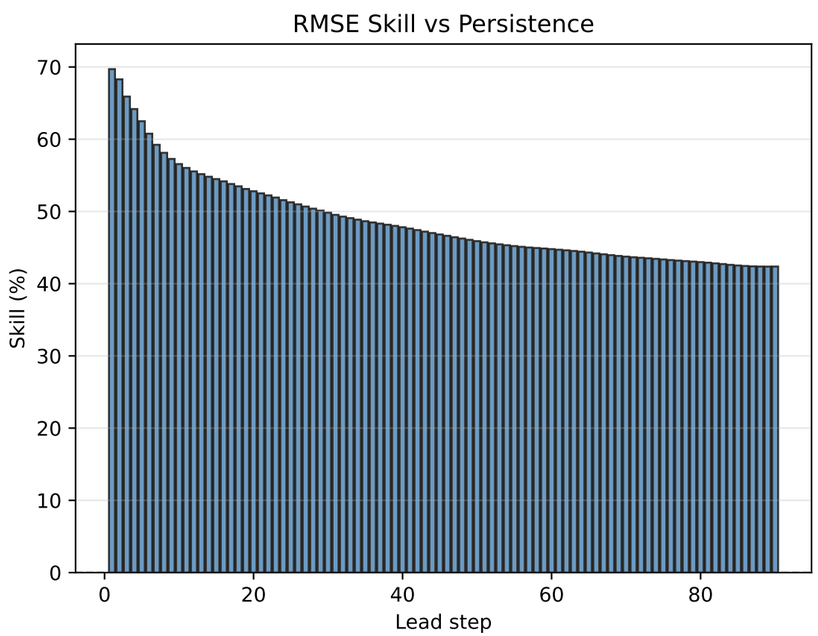


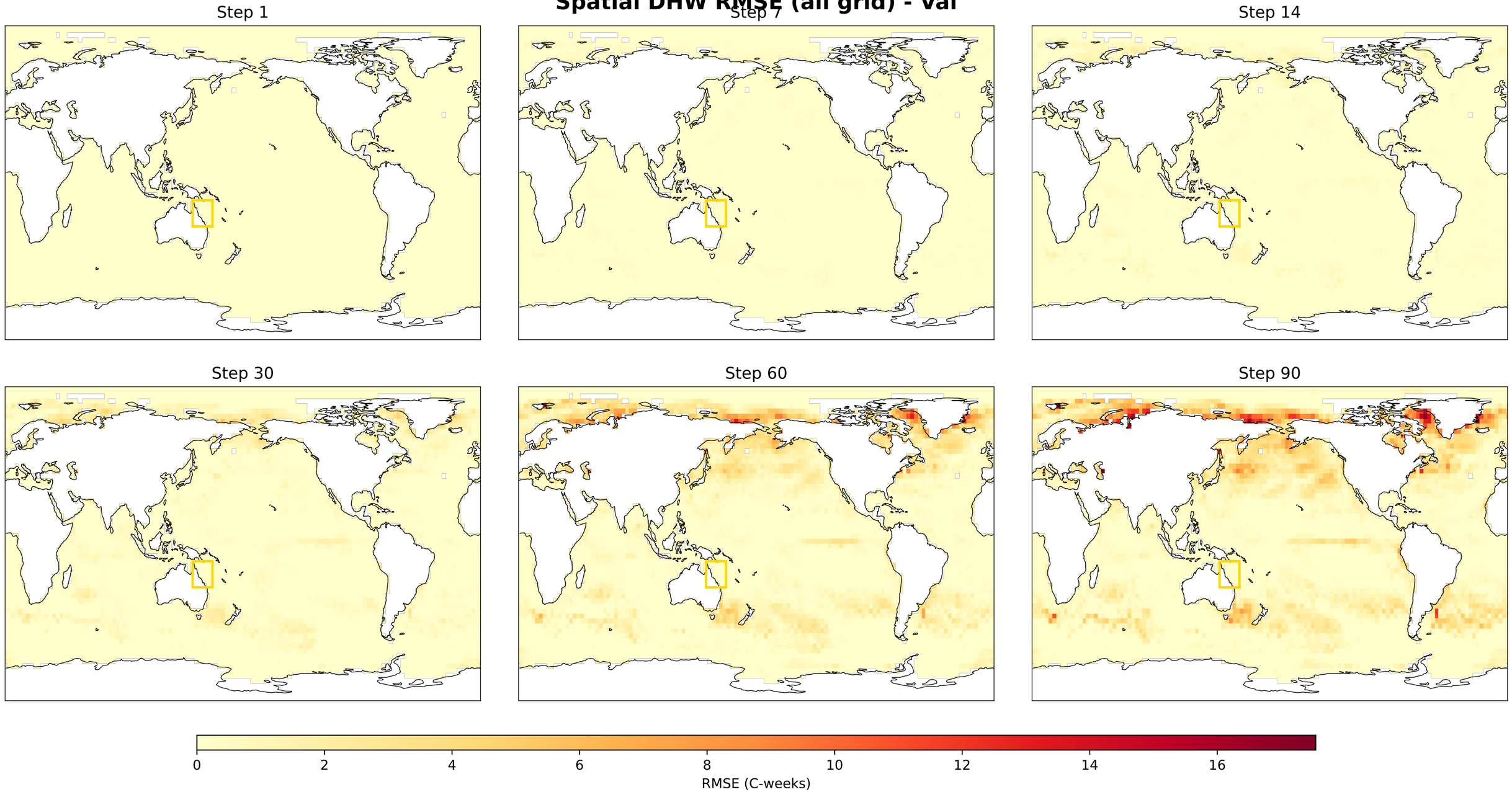


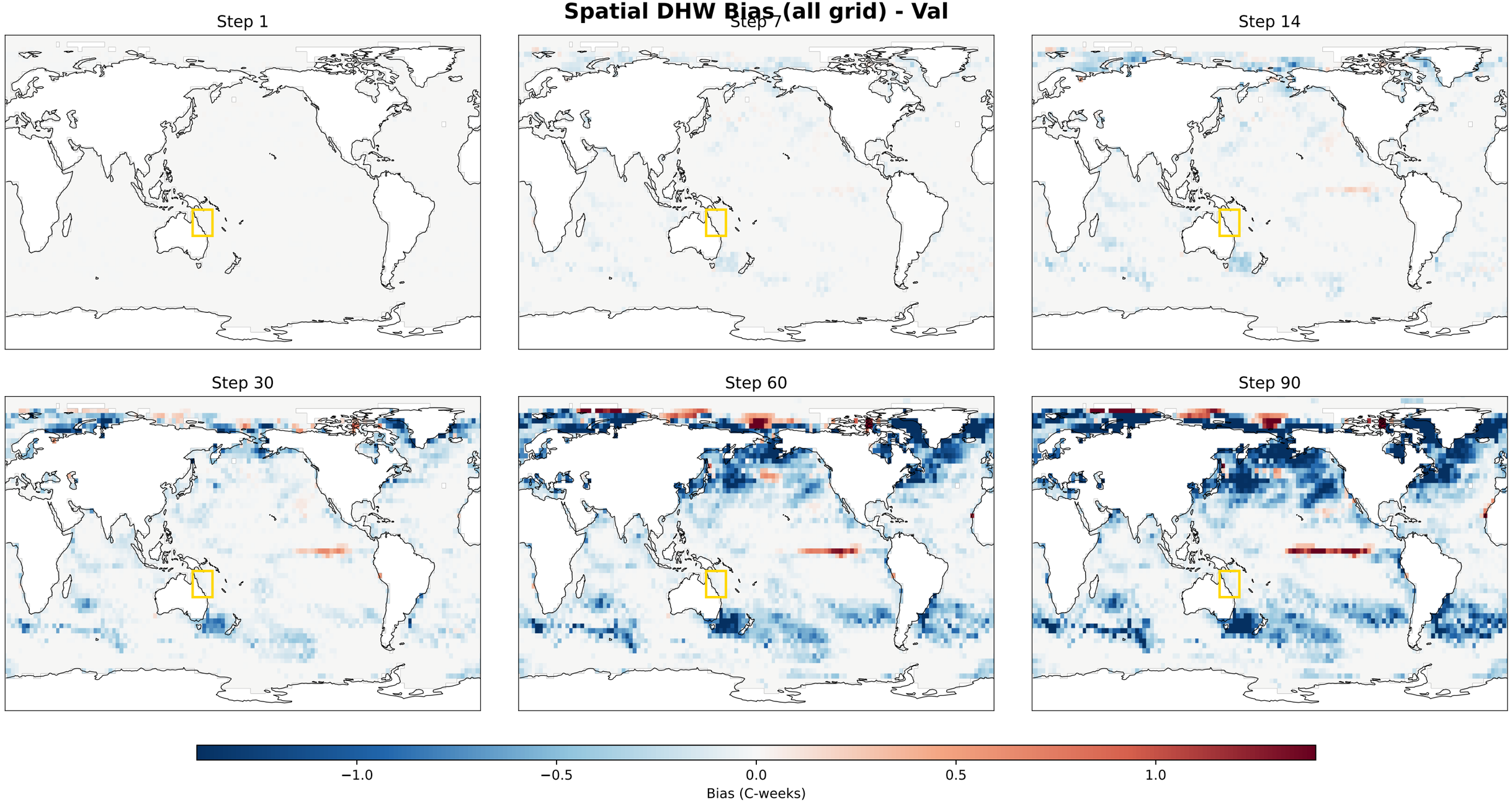


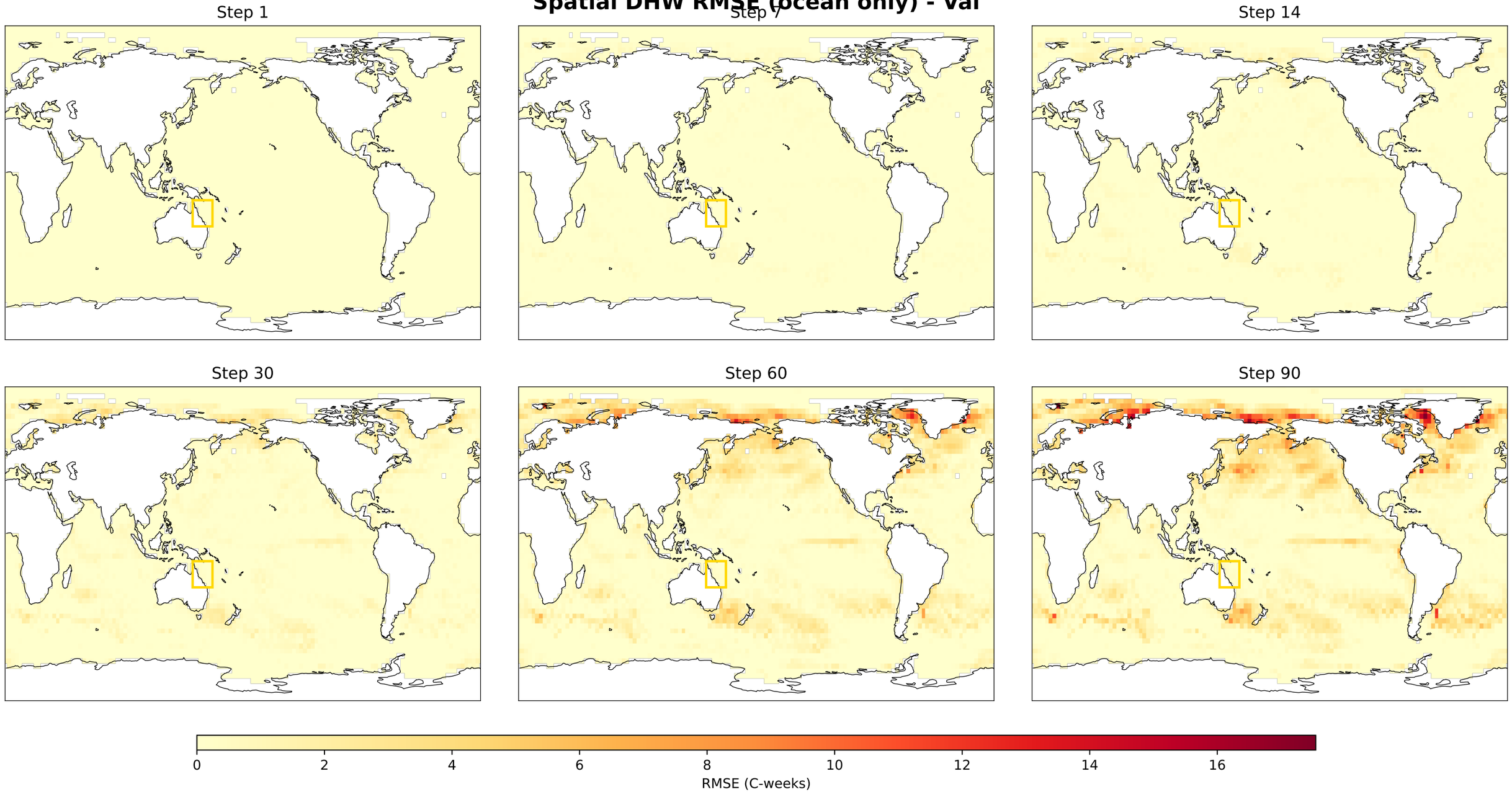


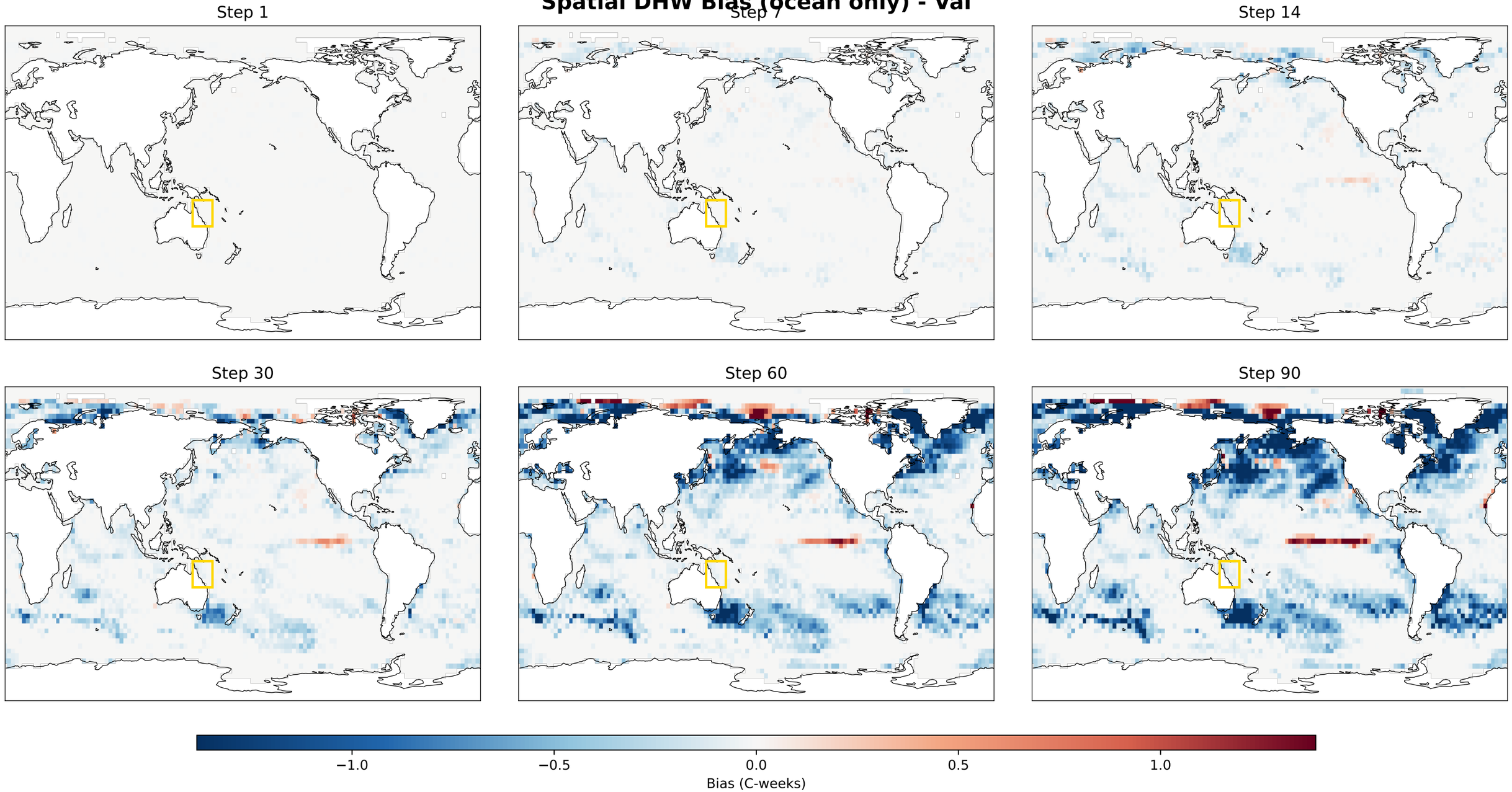





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
16
RMSE (C-weeks)

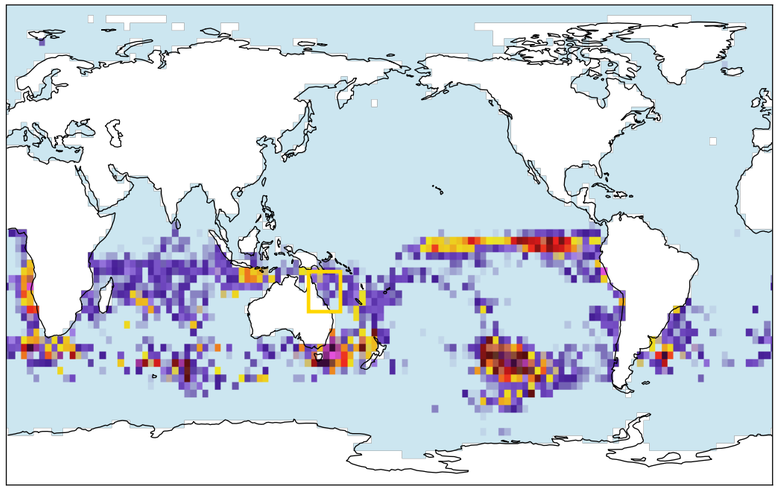

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
16
RMSE (C-weeks)

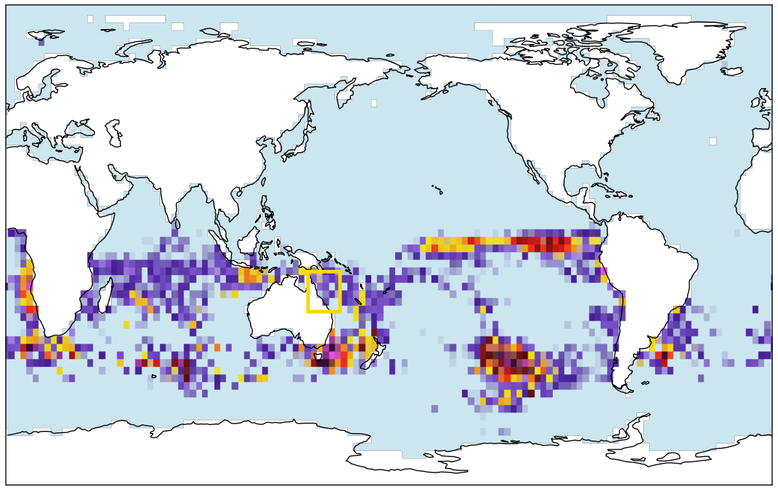

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

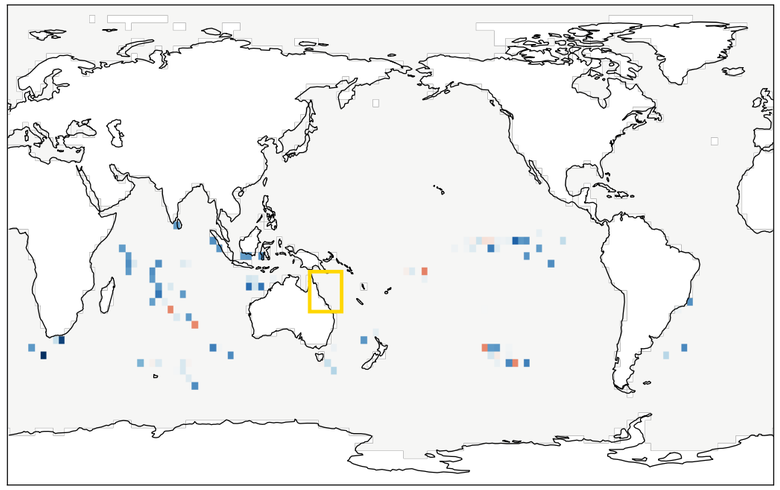
Actual DHW - Step 1



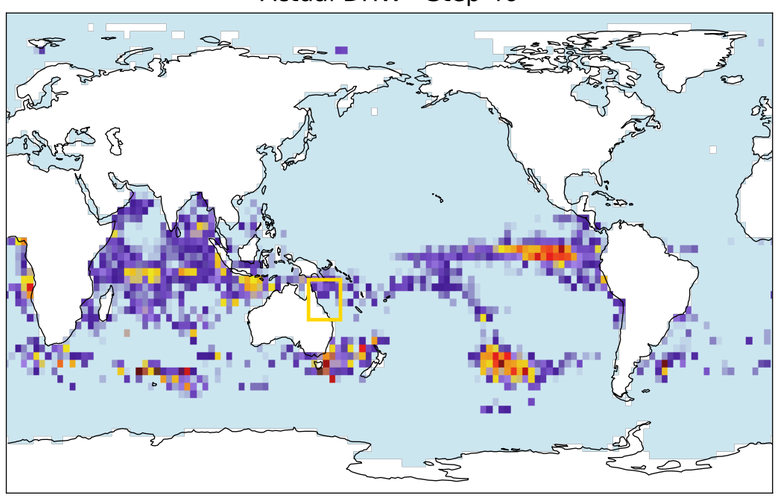
Predicted DHW - Step 1



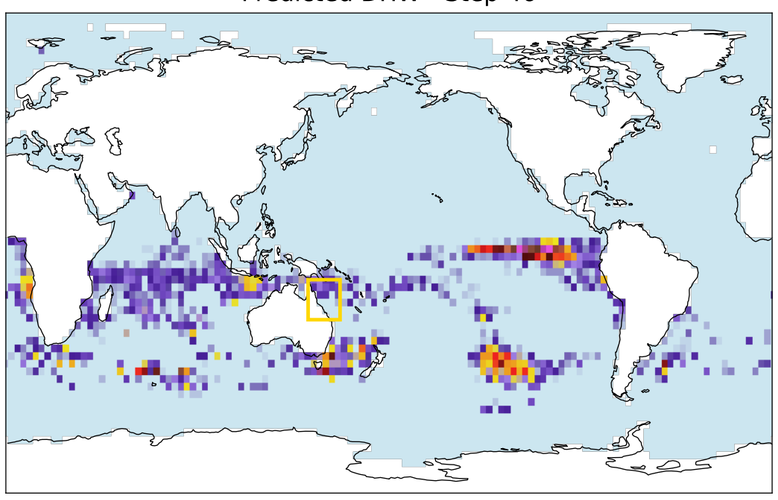
Error (Pred - Actual) - Step 1 [scale ±0.31]



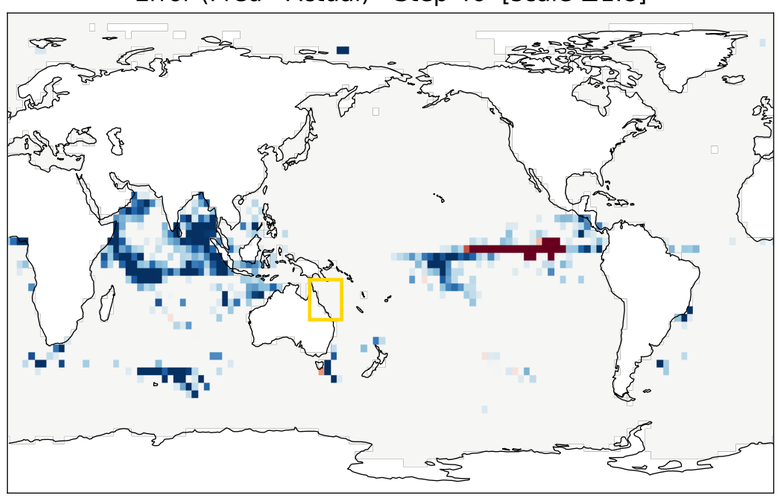
Actual DHW - Step 46



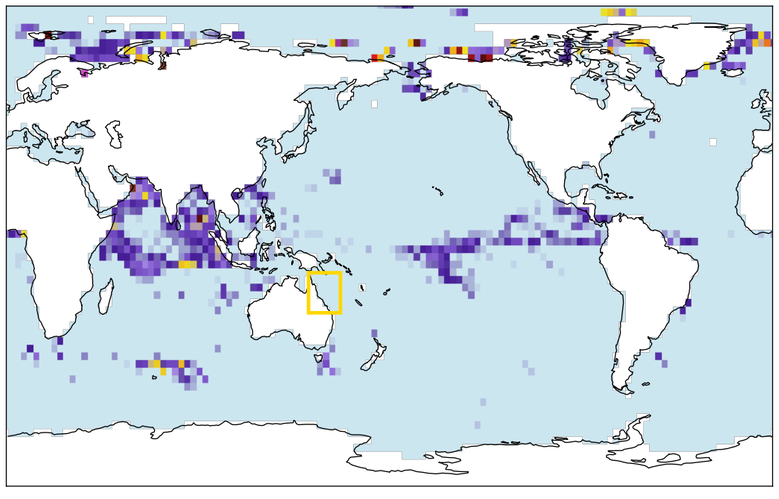
Predicted DHW - Step 46



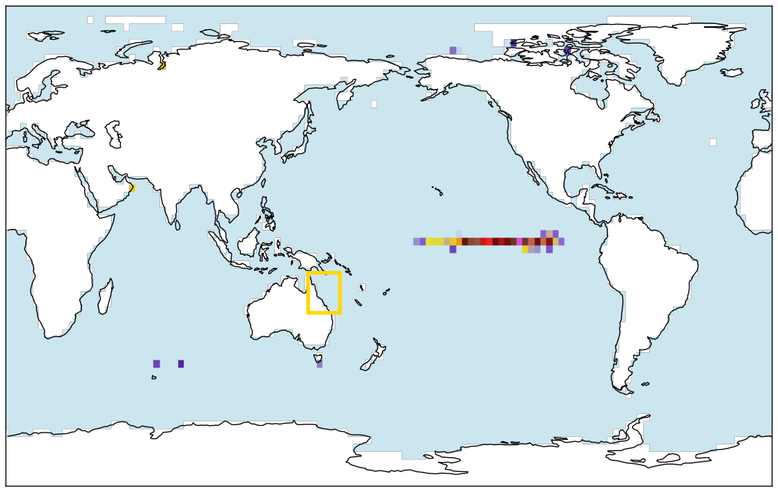
Error (Pred - Actual) - Step 46 [scale ±1.8]



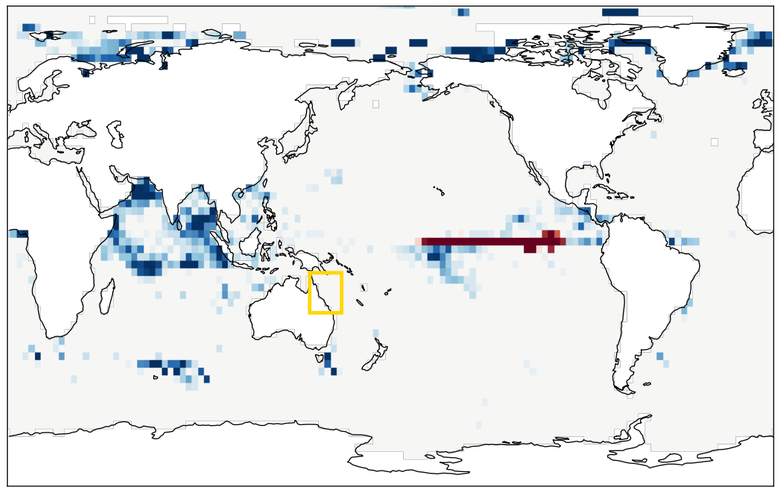
Actual DHW - Step 90

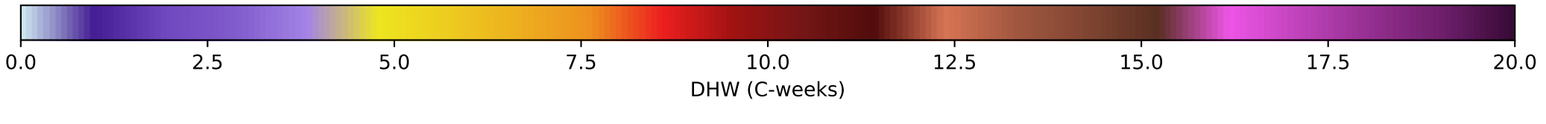


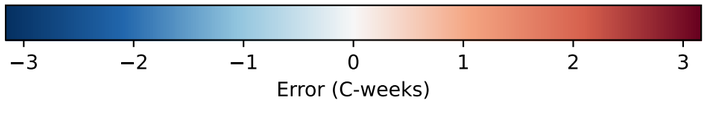
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±3.2]

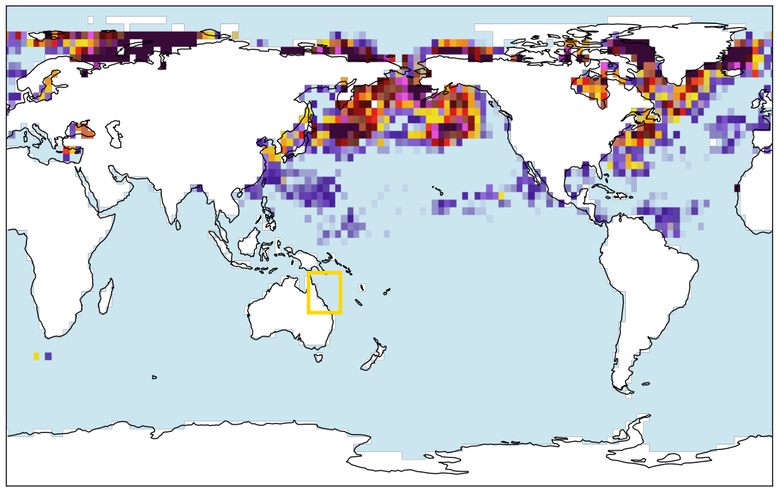




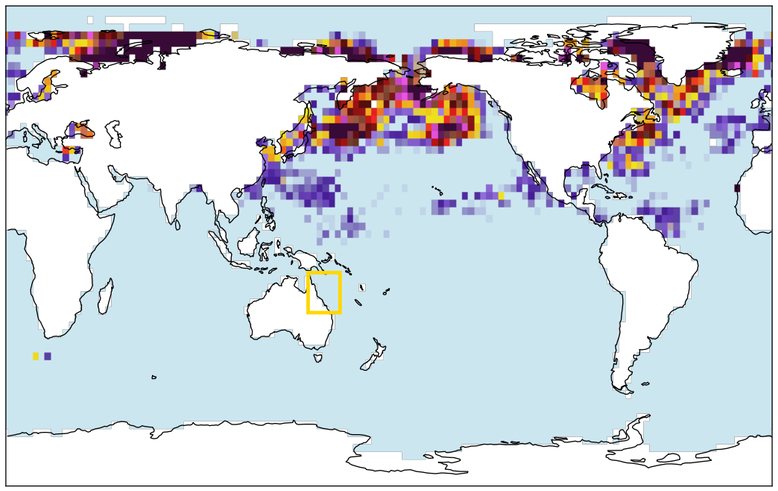


DHW Chain init: 2016-10-07 - Val (C-weeks)

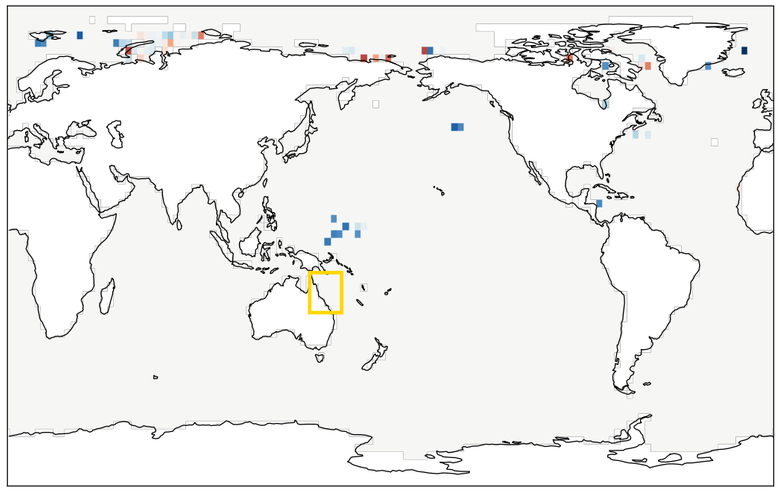
Actual DHW - Step 1



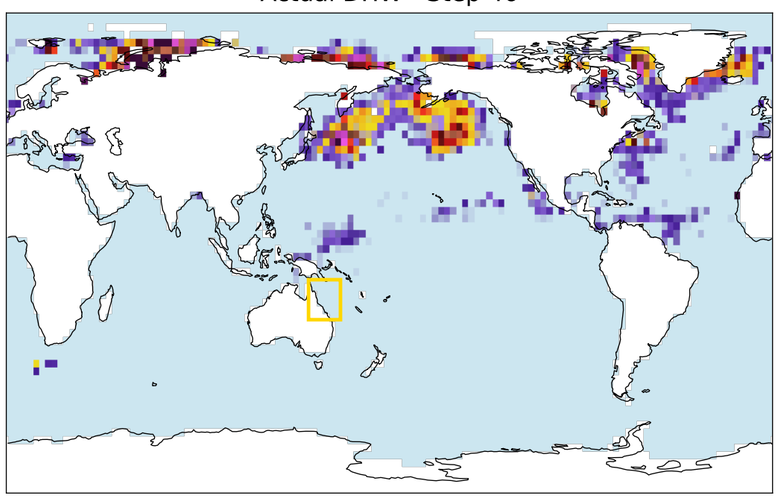
Predicted DHW - Step 1



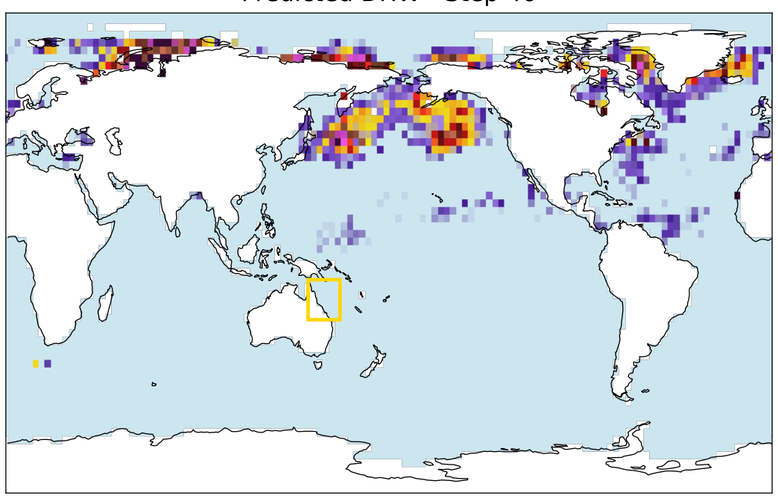
Error (Pred - Actual) - Step 1 [scale ±0.28]



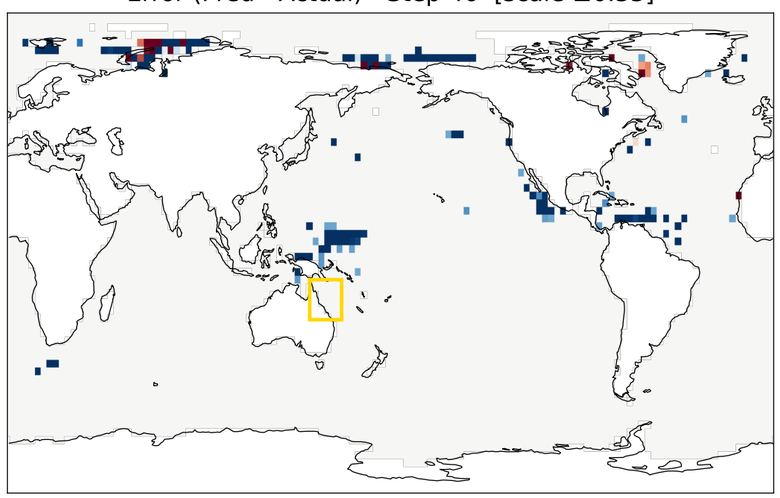
Actual DHW - Step 46



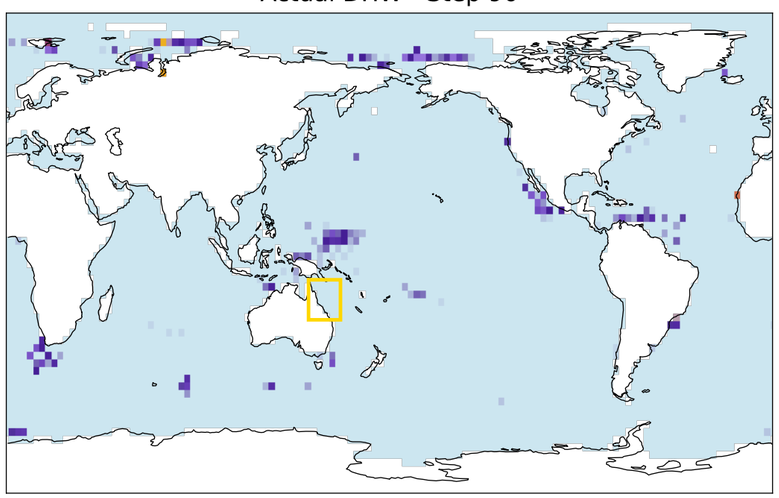
Predicted DHW - Step 46



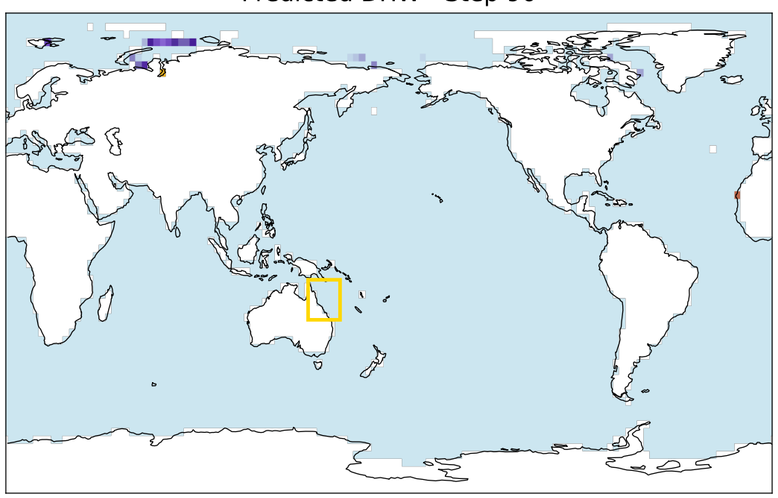
Error (Pred - Actual) - Step 46 [scale ±0.33]



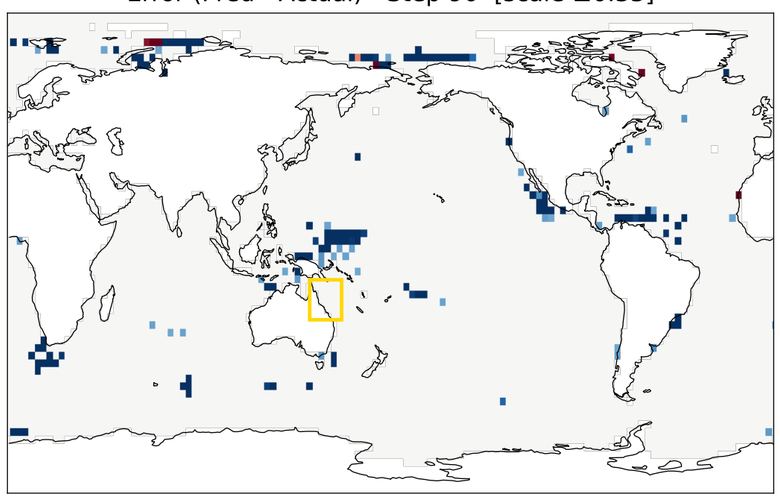
Actual DHW - Step 90

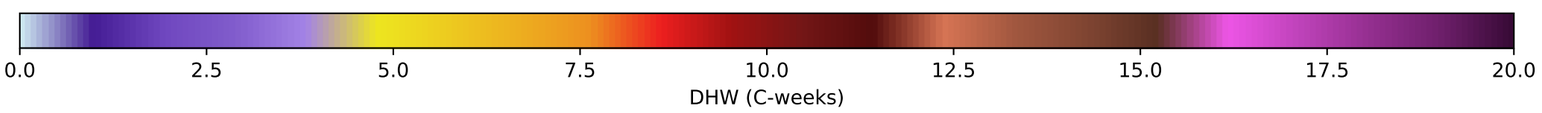


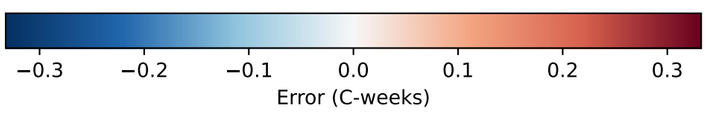
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

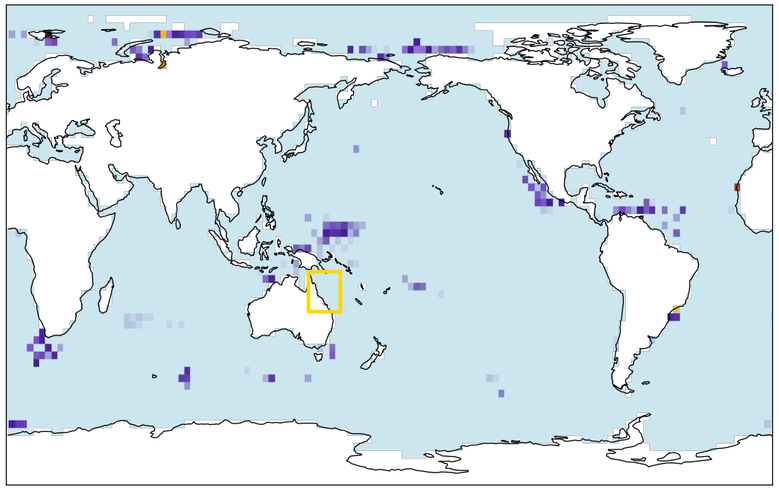




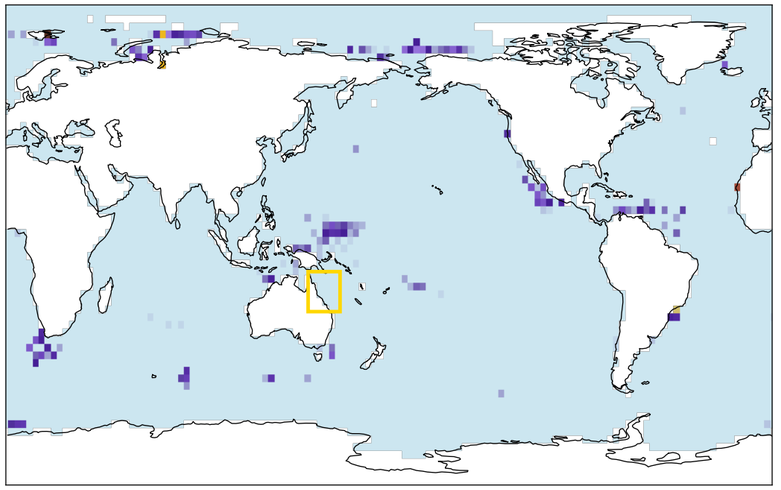


DHW Chain init: 2017-01-05 - Val (C-weeks)

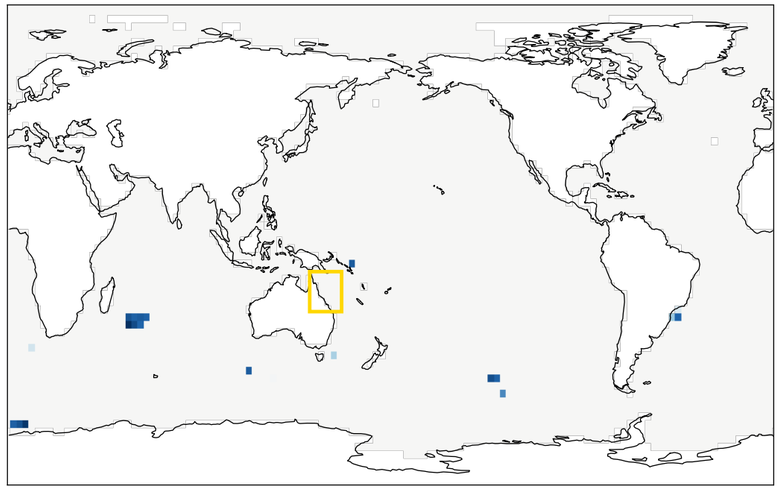
Actual DHW - Step 1



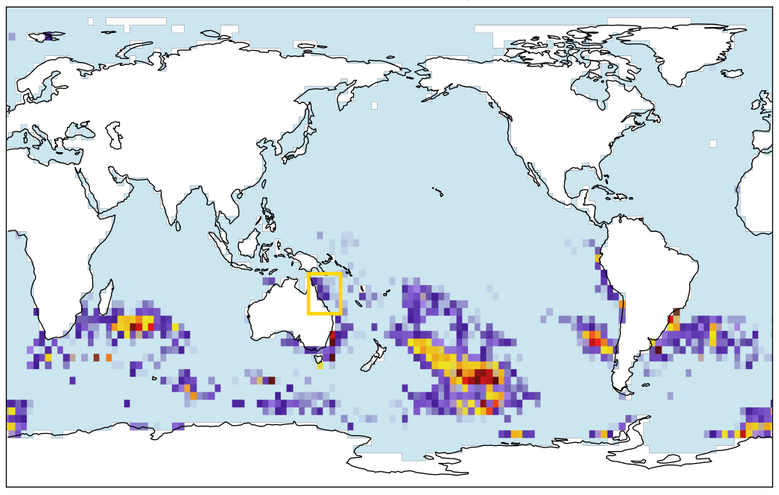
Predicted DHW - Step 1



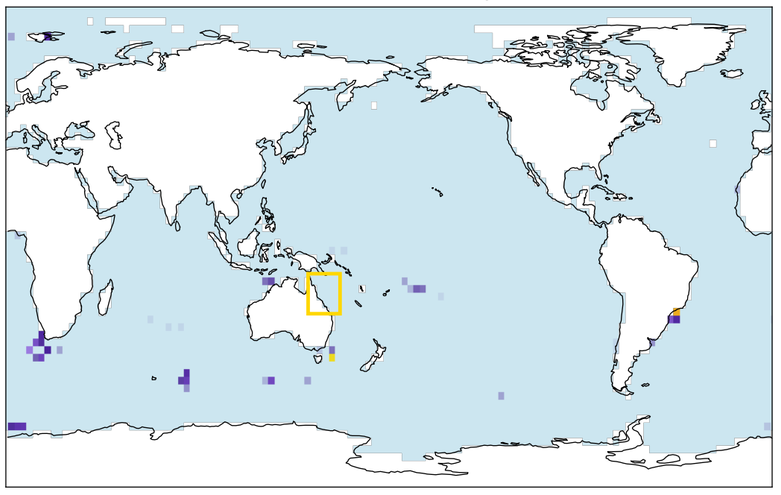
Error (Pred - Actual) - Step 1 [scale ±0.22]



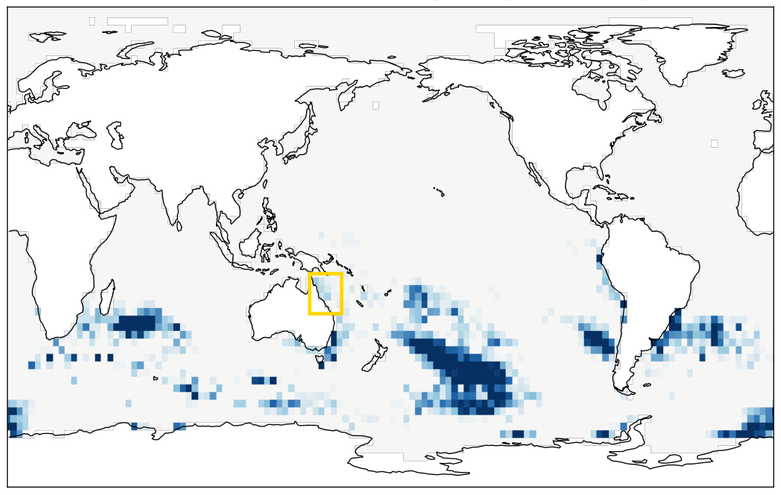
Actual DHW - Step 46



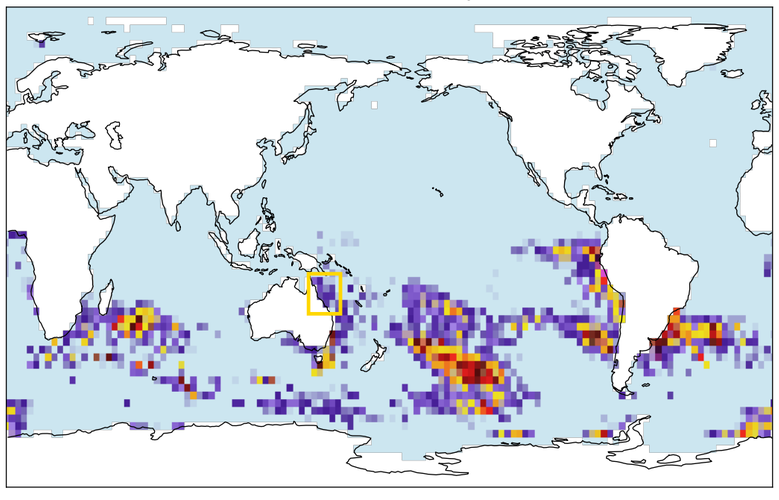
Predicted DHW - Step 46



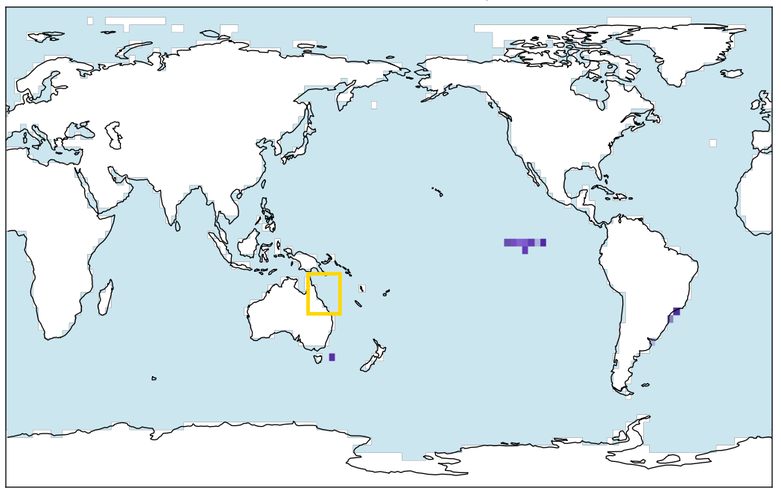
Error (Pred - Actual) - Step 46 [scale ±4.8]



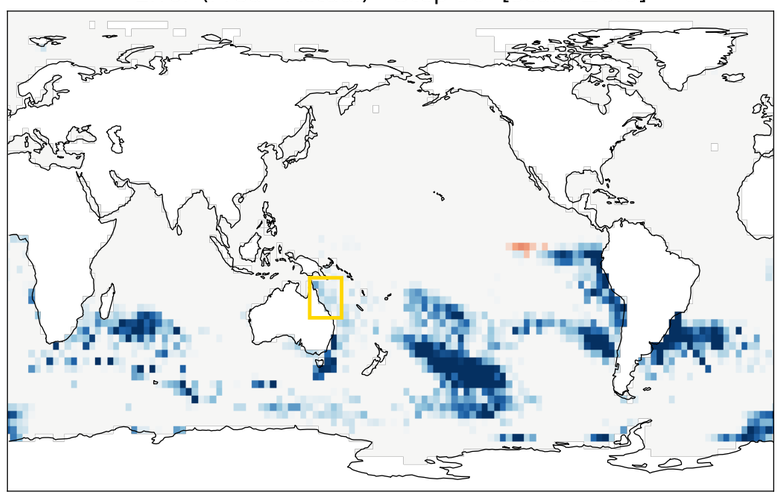
Actual DHW - Step 90

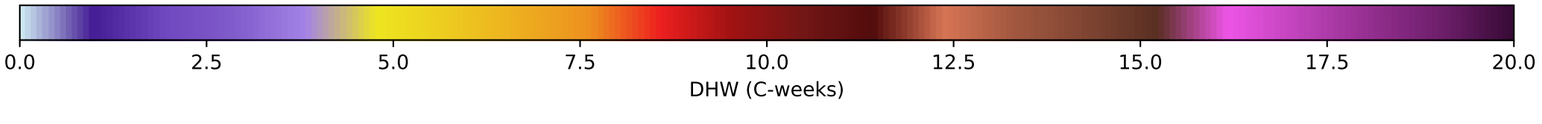


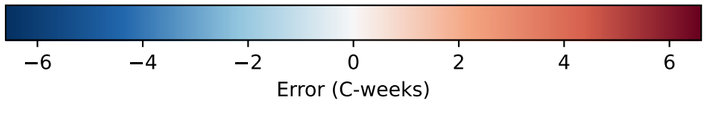
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.6]

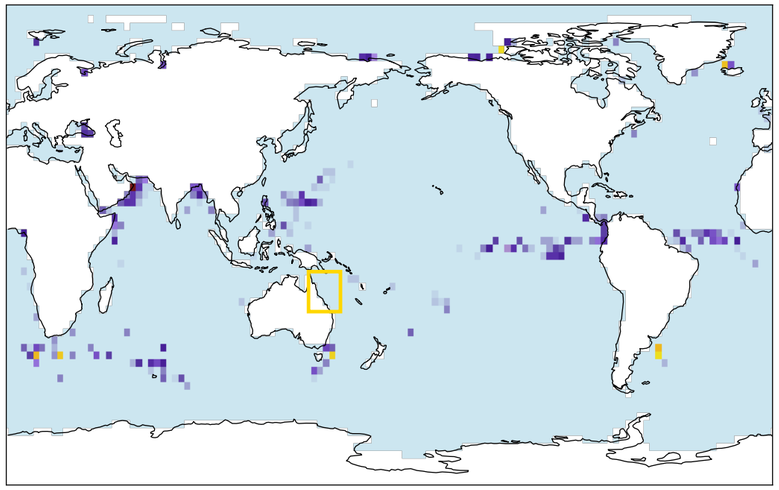




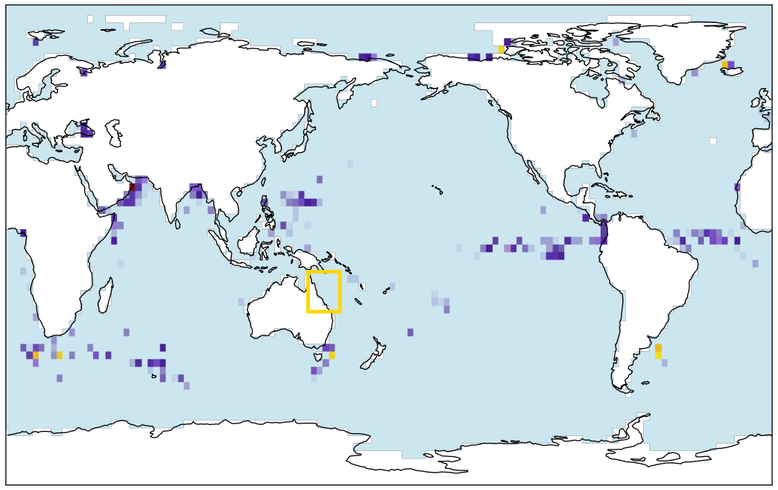


DHW Chain init: 2017-07-04 - Val (C-weeks)

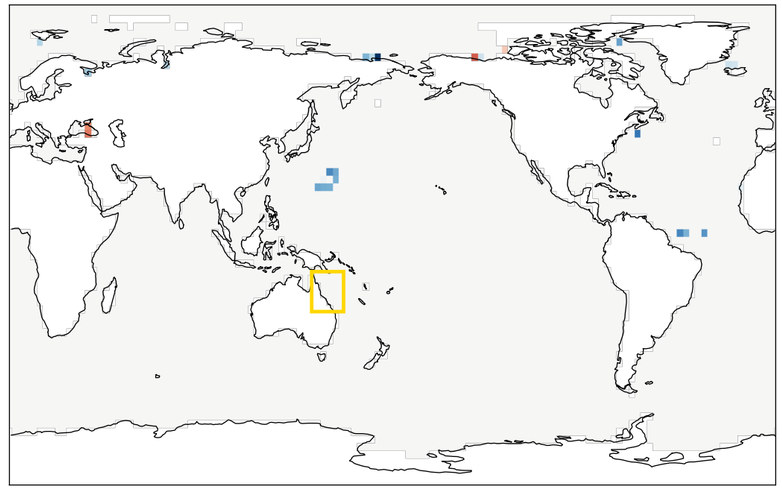
Actual DHW - Step 1



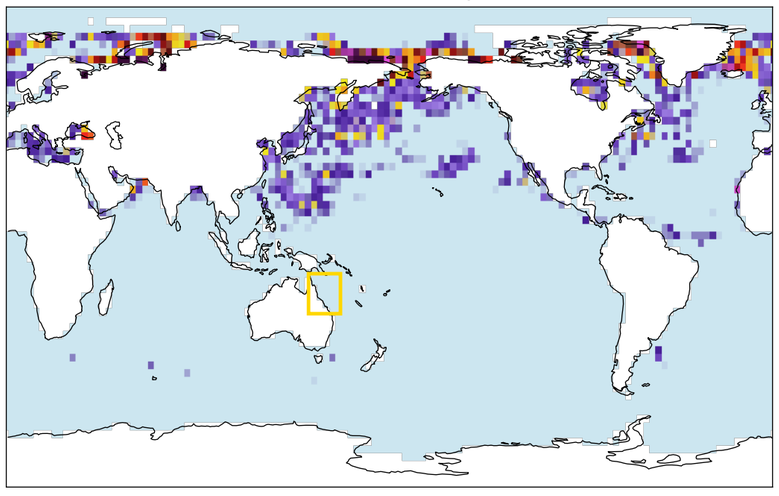
Predicted DHW - Step 1



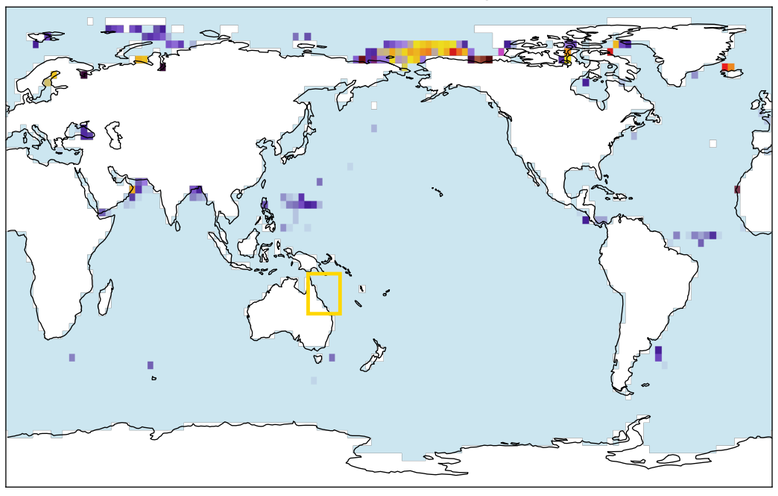
Error (Pred - Actual) - Step 1 [scale ±0.37]



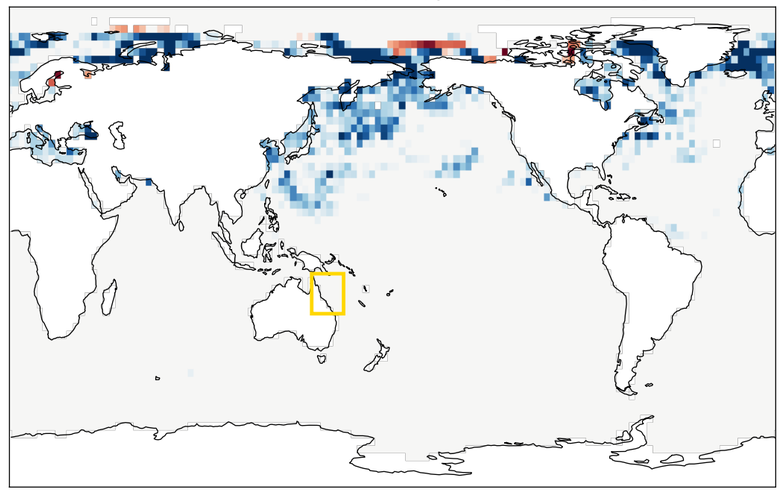
Actual DHW - Step 46



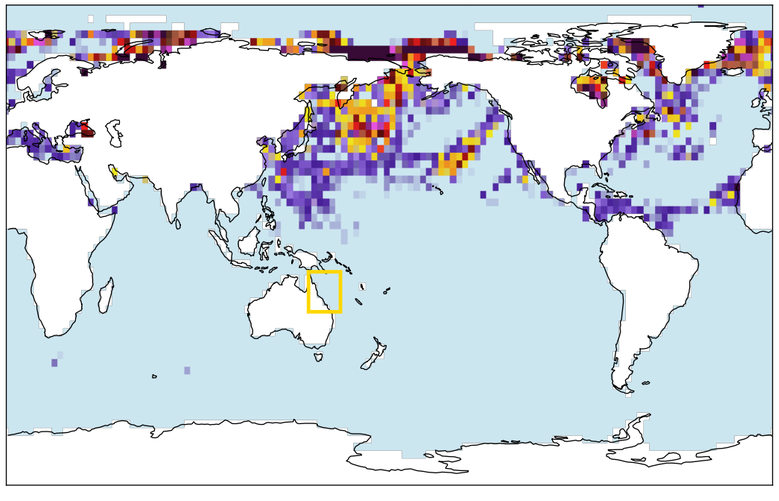
Predicted DHW - Step 46



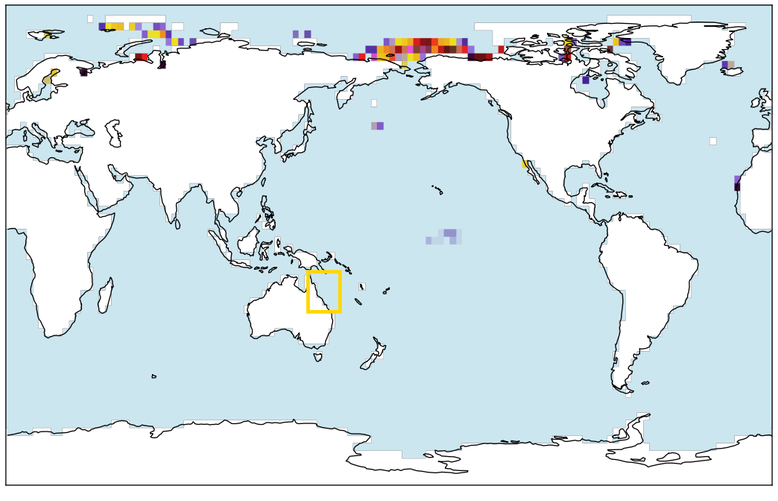
Error (Pred - Actual) - Step 46 [scale ±6.1]



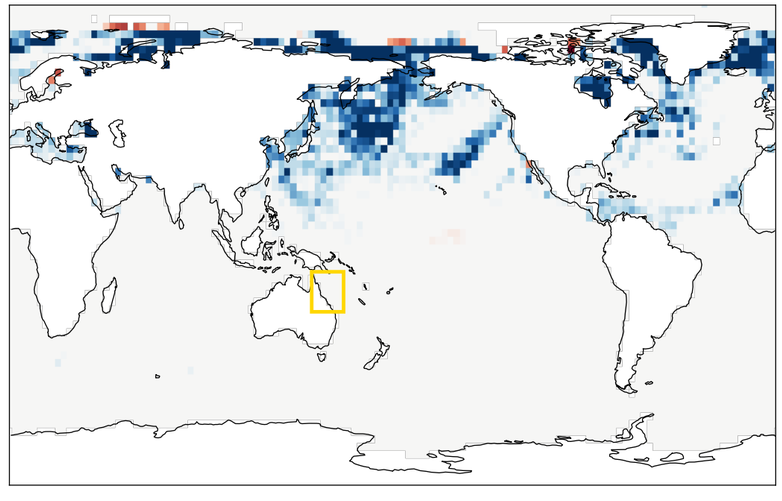
Actual DHW - Step 90

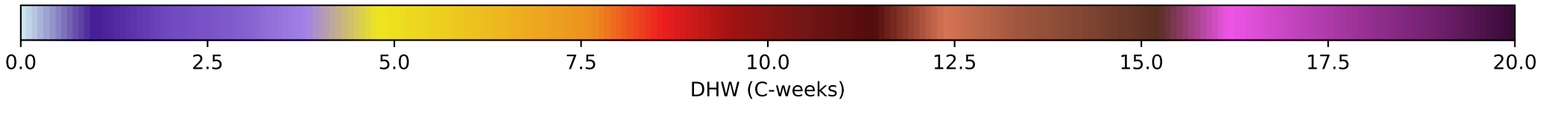


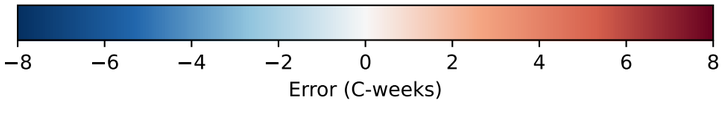
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

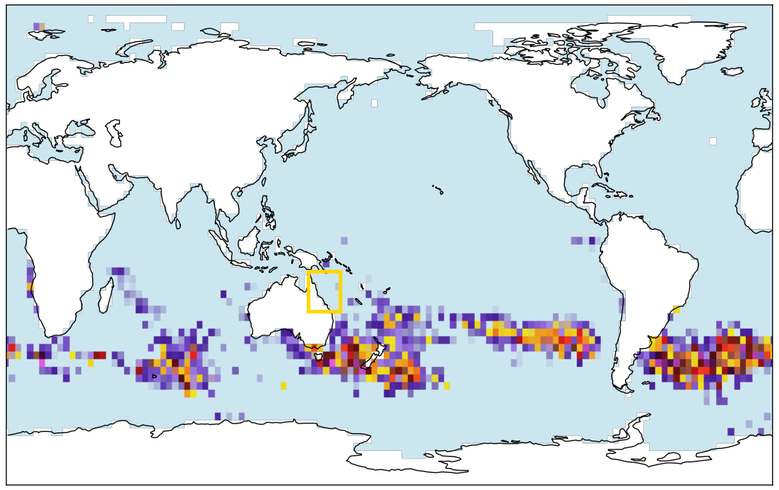




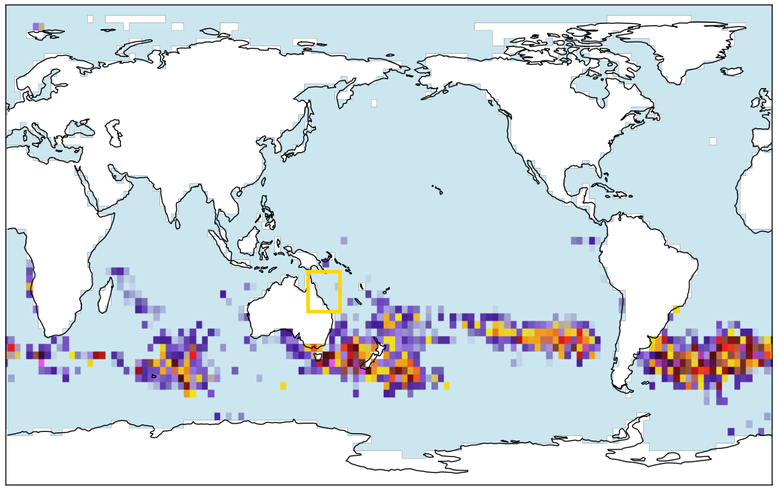


DHW Chain init: 2018-03-31 - Val (C-weeks)

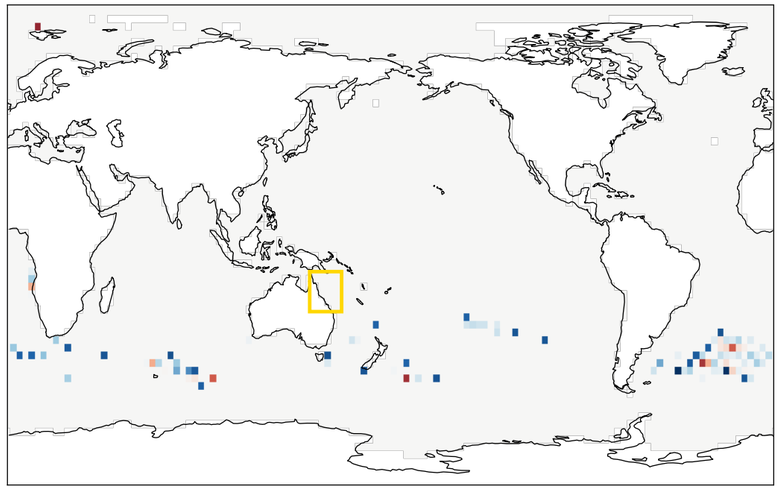
Actual DHW - Step 1



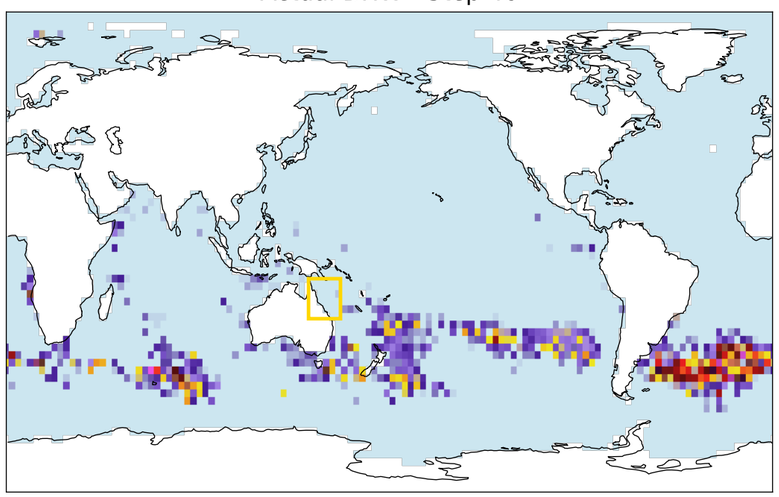
Predicted DHW - Step 1



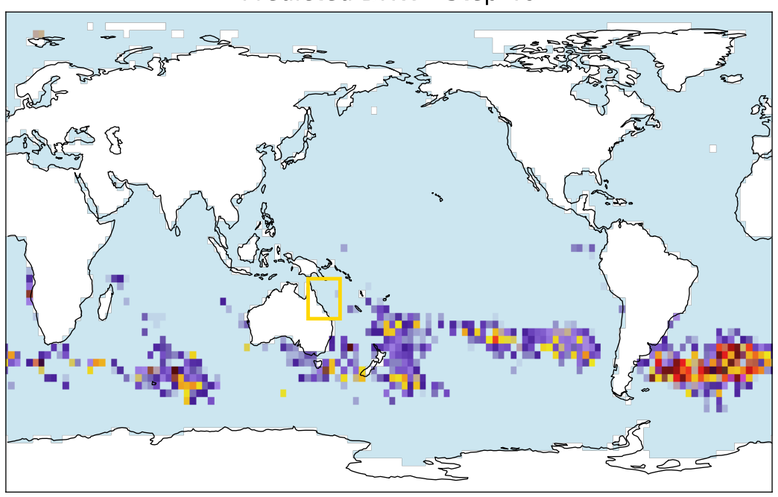
Error (Pred - Actual) - Step 1 [scale ±0.21]



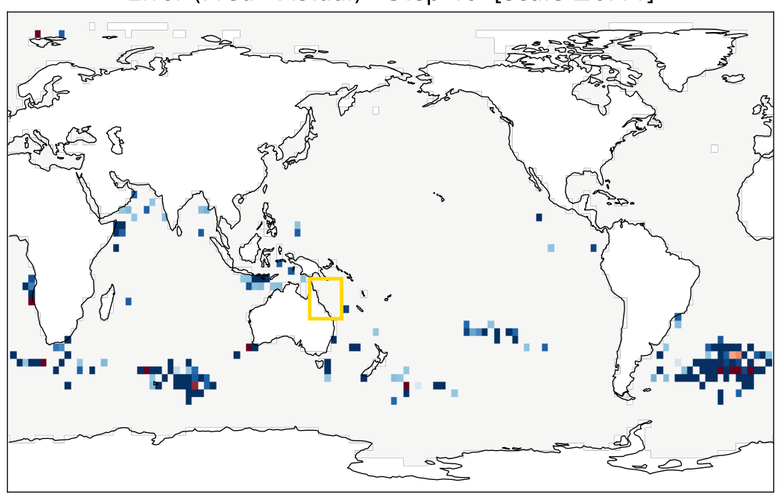
Actual DHW - Step 46



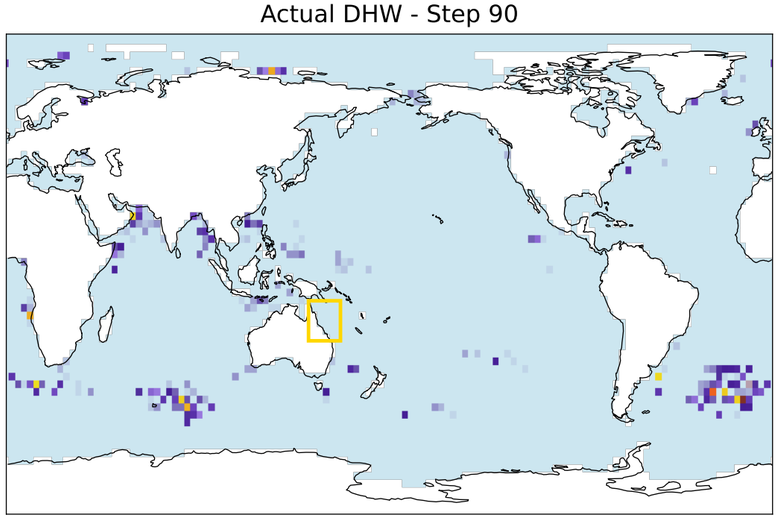
Predicted DHW - Step 46



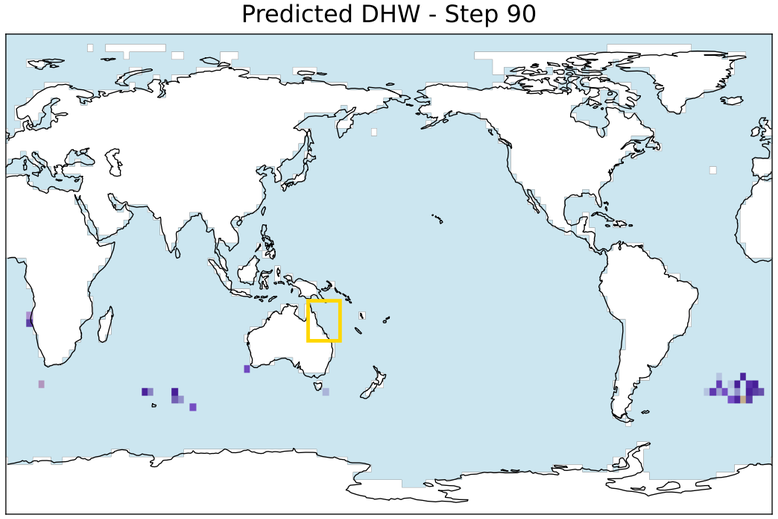
Error (Pred - Actual) - Step 46 [scale ±0.44]



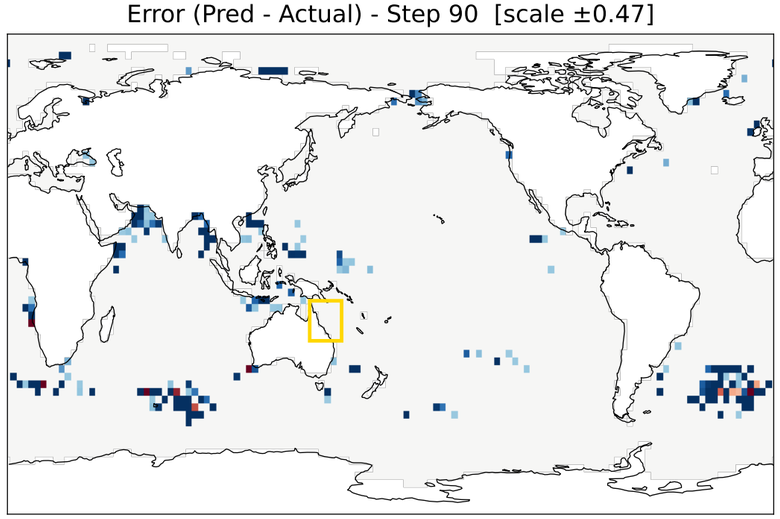
Actual DHW - Step 90

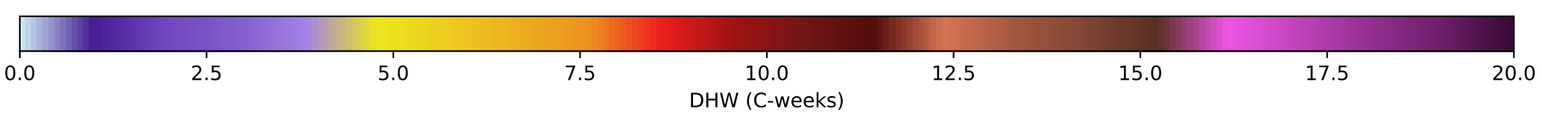


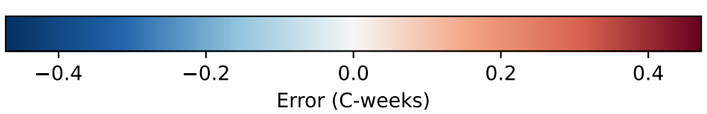
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.47]

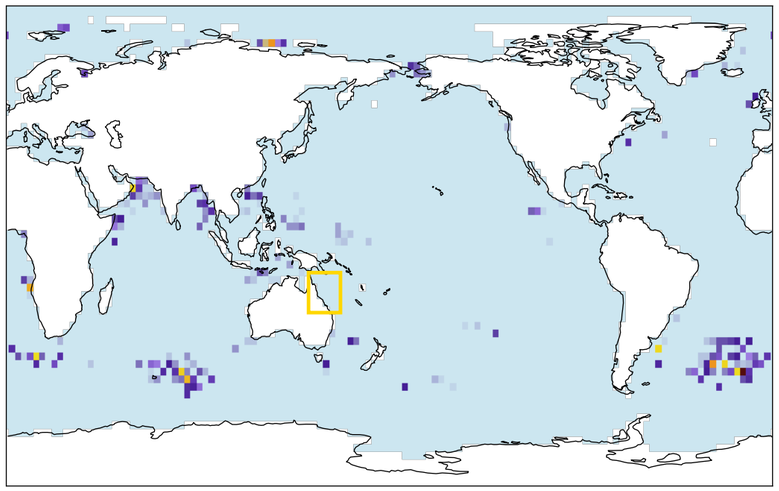




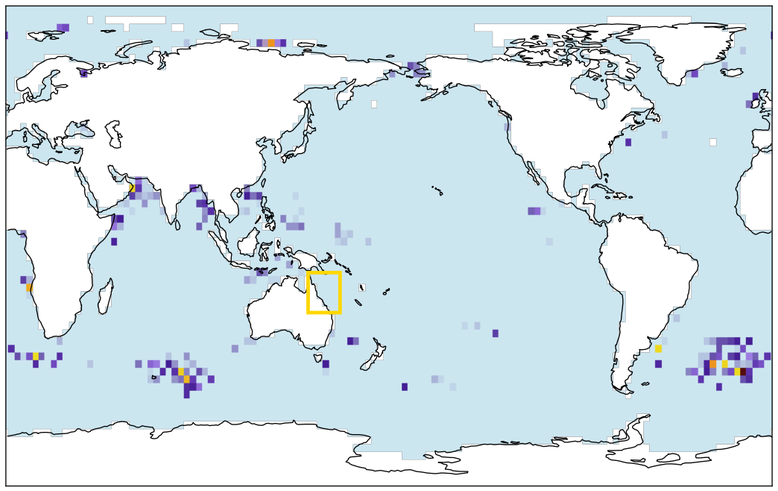


DHW Chain init: 2018-06-29 - Val (C-weeks)

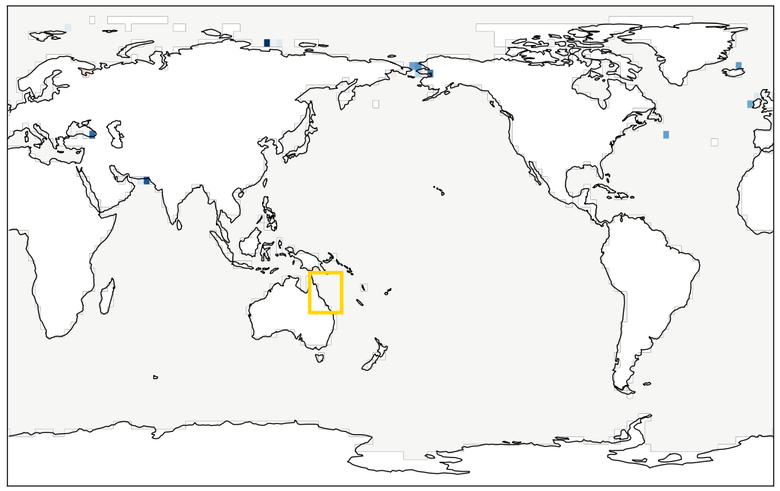
Actual DHW - Step 1



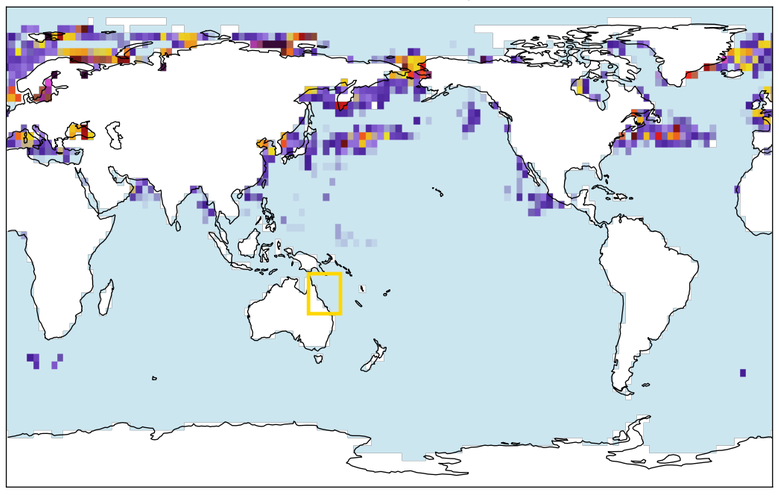
Predicted DHW - Step 1



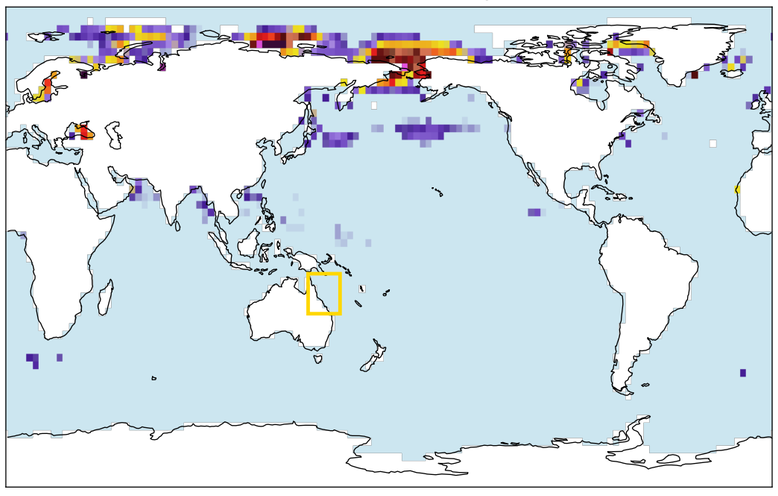
Error (Pred - Actual) - Step 1 [scale ±0.31]



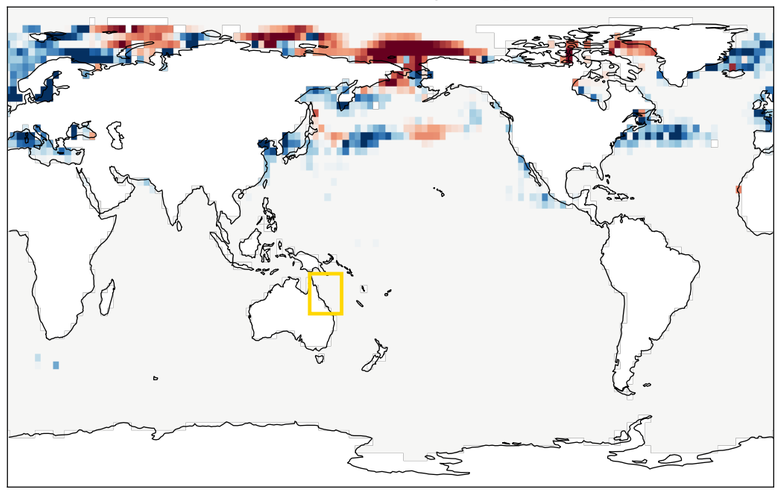
Actual DHW - Step 46



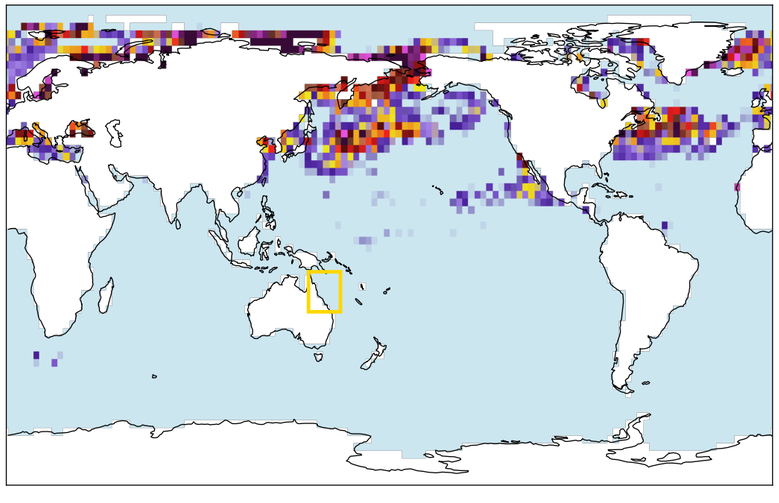
Predicted DHW - Step 46



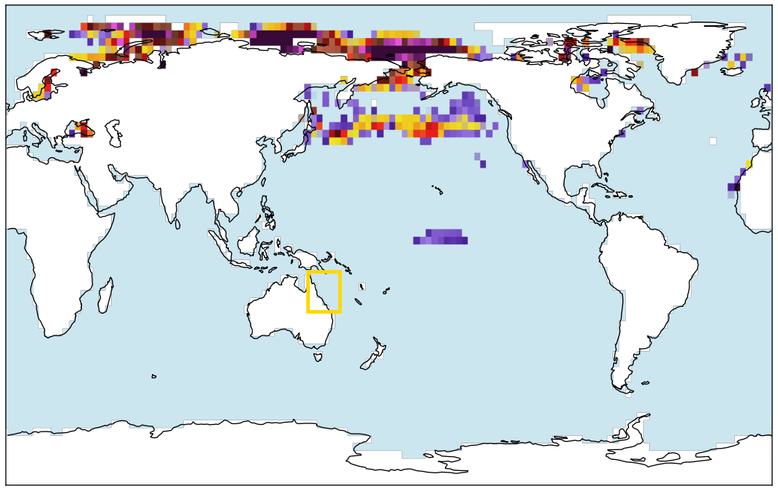
Error (Pred - Actual) - Step 46 [scale ±6.1]



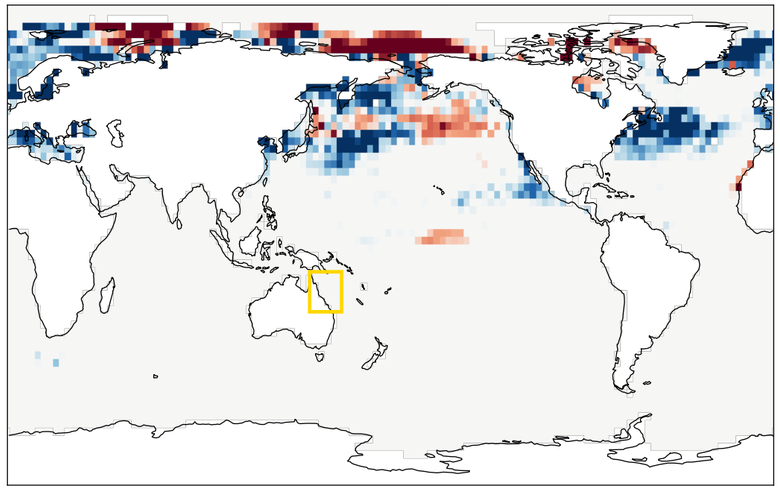
Actual DHW - Step 90

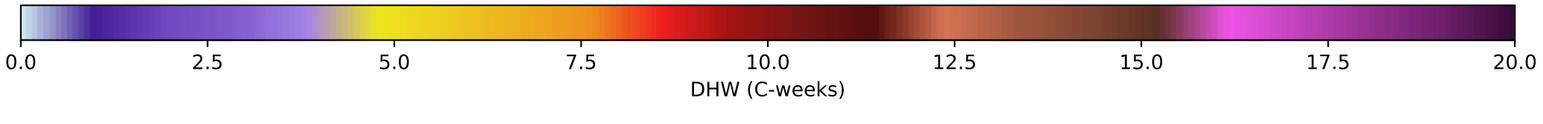


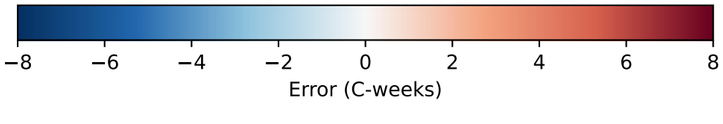
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

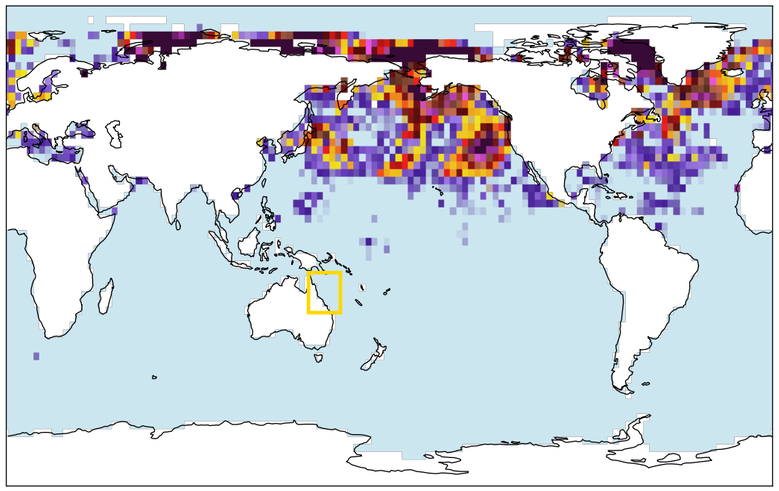




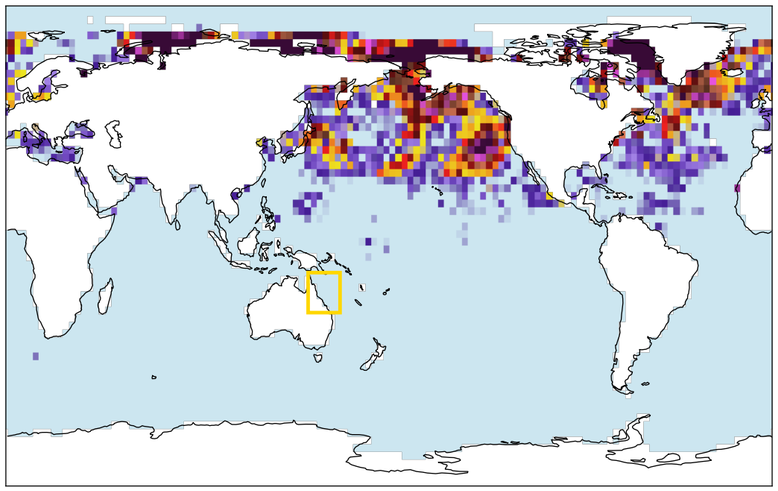


DHW Chain init: 2019-09-22 - Val (C-weeks)

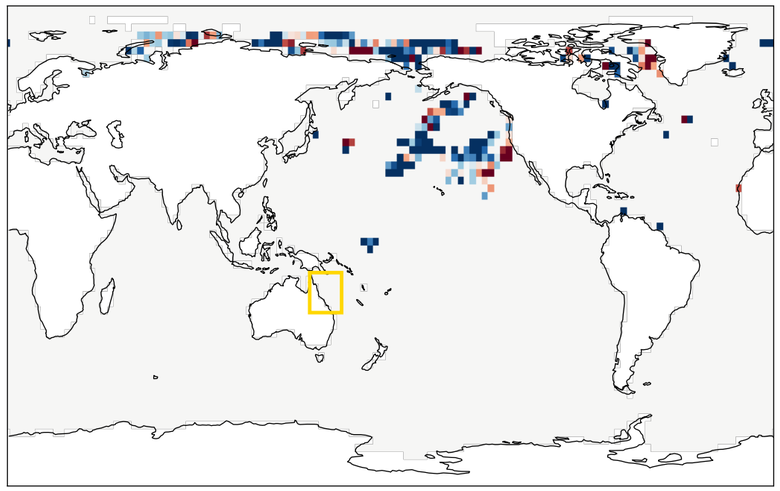
Actual DHW - Step 1



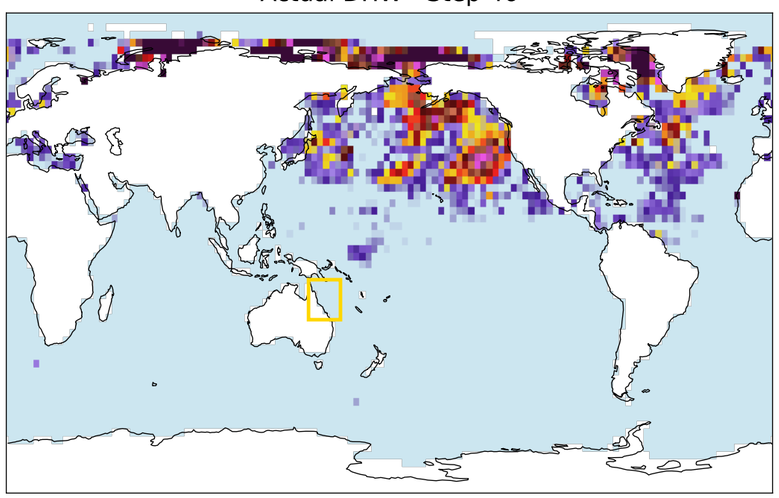
Predicted DHW - Step 1



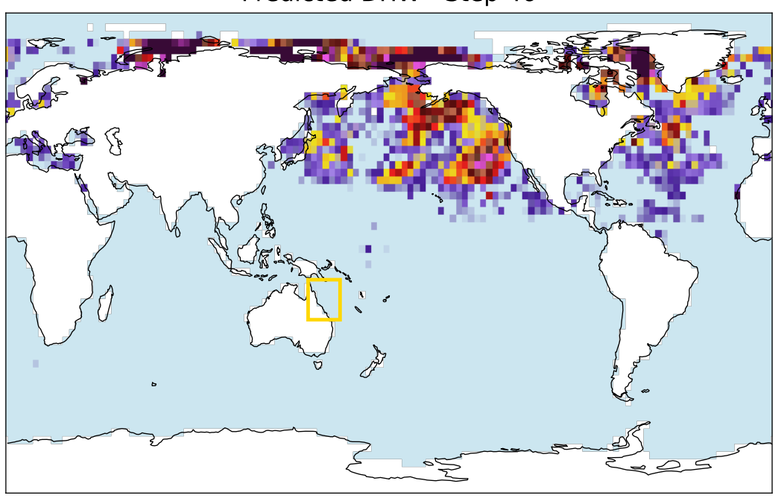
Error (Pred - Actual) - Step 1 [scale ±0.033]



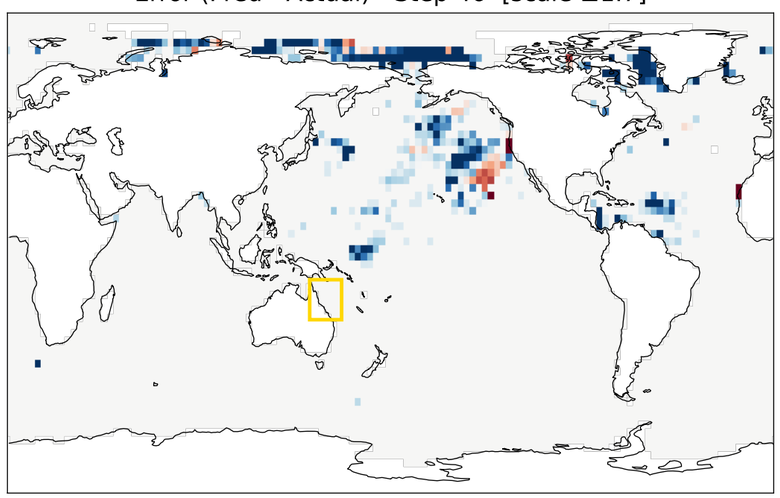
Actual DHW - Step 46



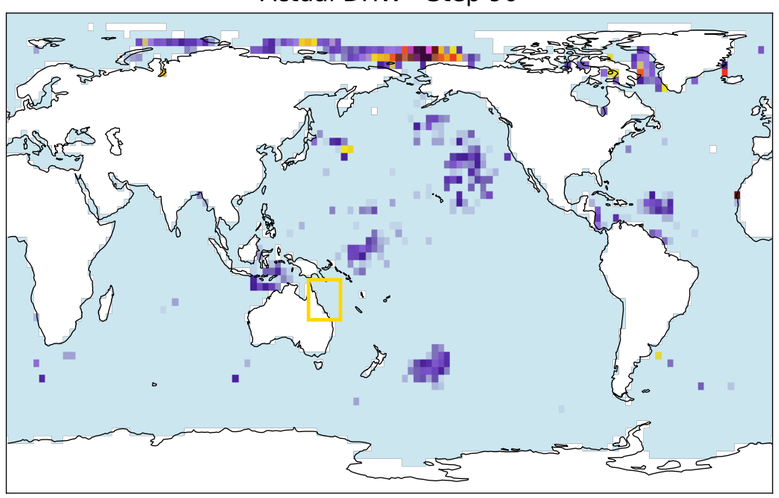
Predicted DHW - Step 46



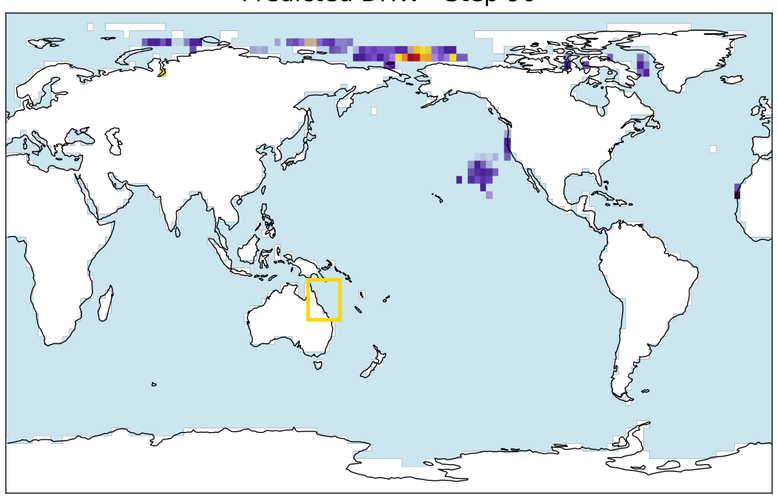
Error (Pred - Actual) - Step 46 [scale ±1.7]



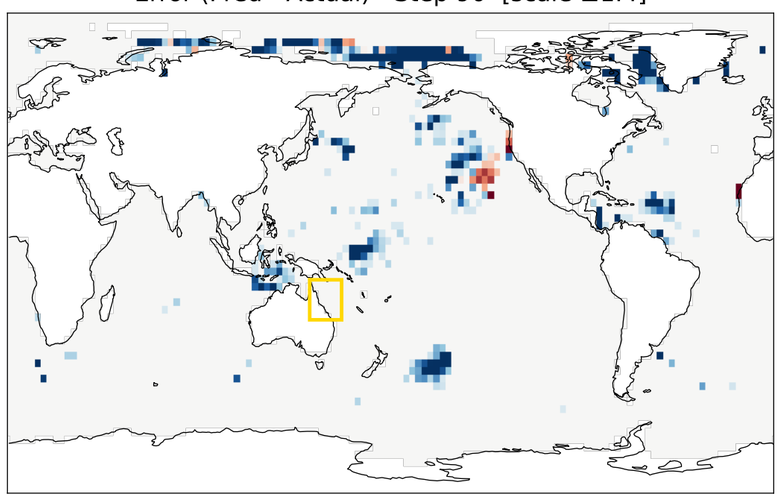
Actual DHW - Step 90

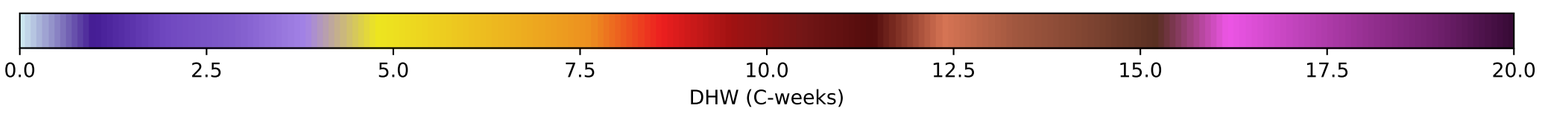


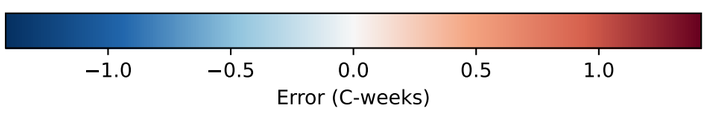
Predicted DHW - Step 90



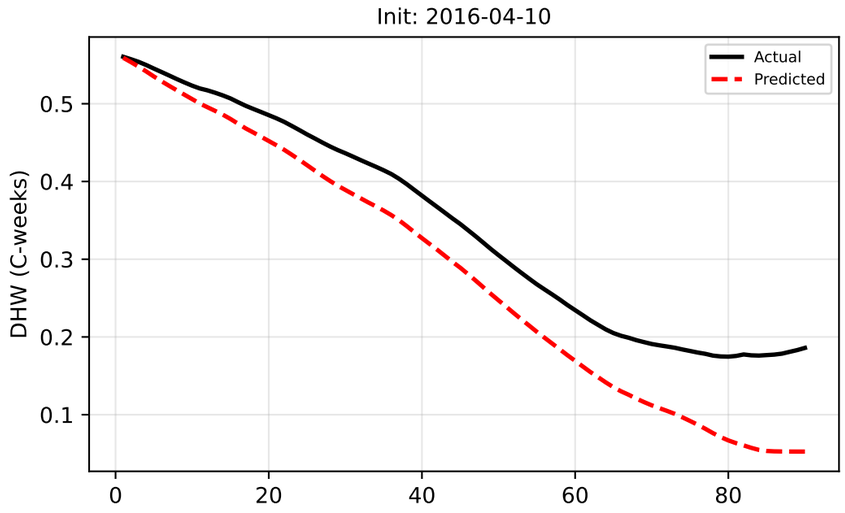
Error (Pred - Actual) - Step 90 [scale ±1.4]

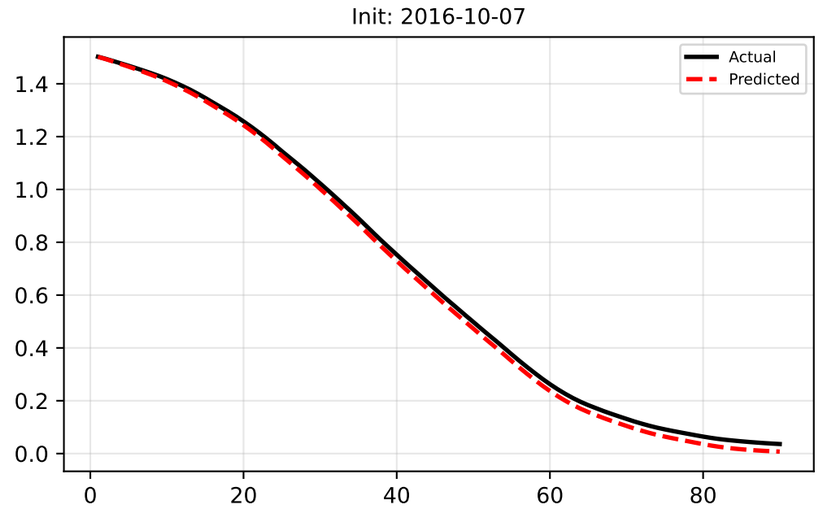


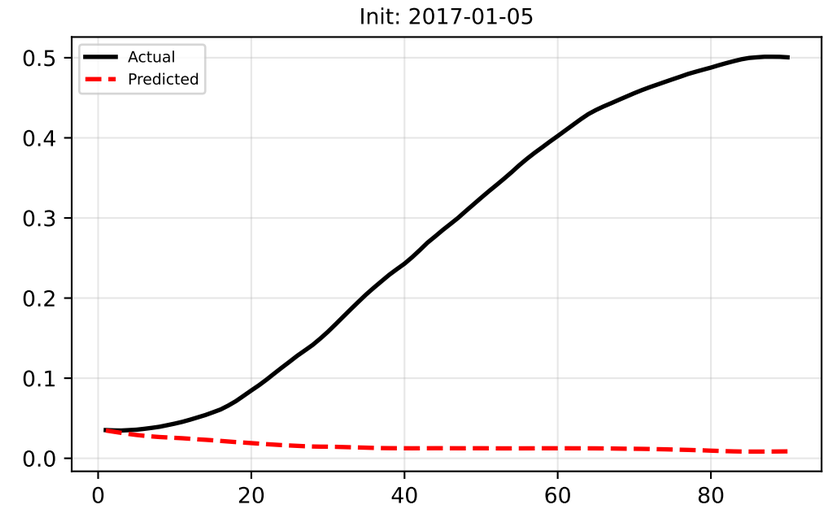


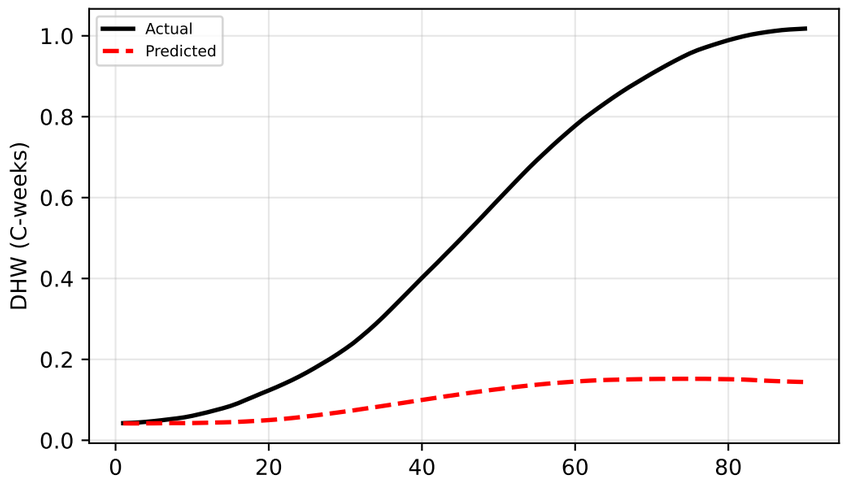


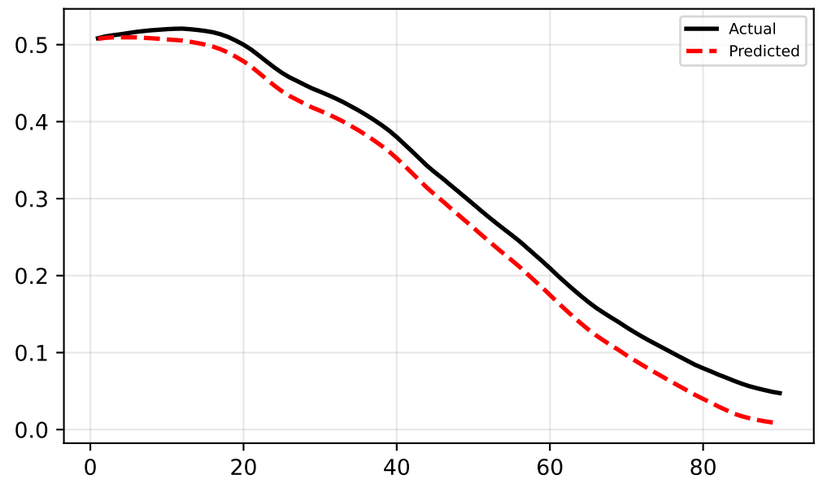
Global-Mean DHW - Val (C-weeks)

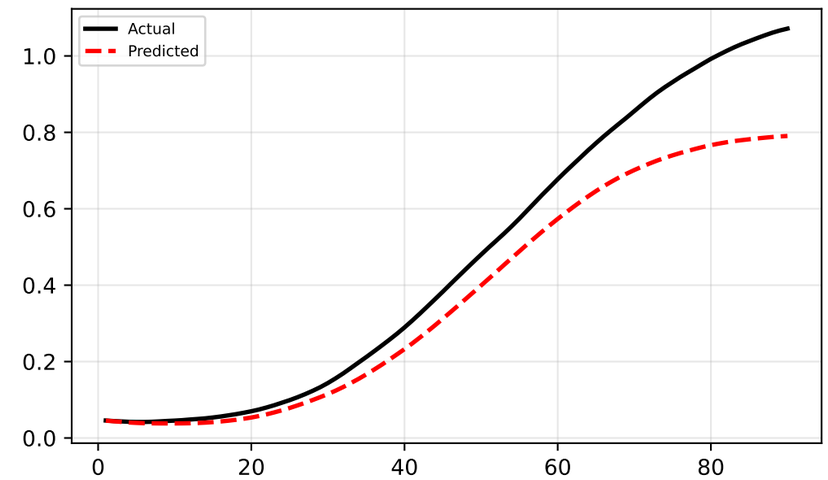


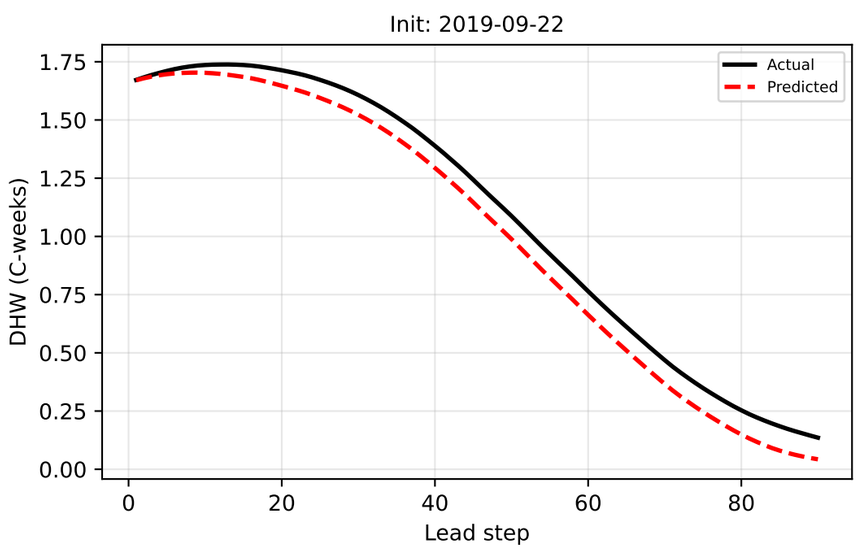




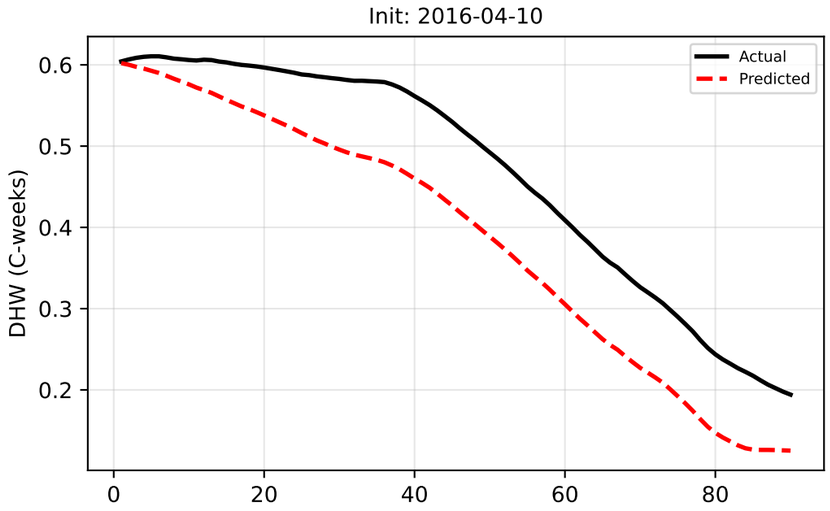


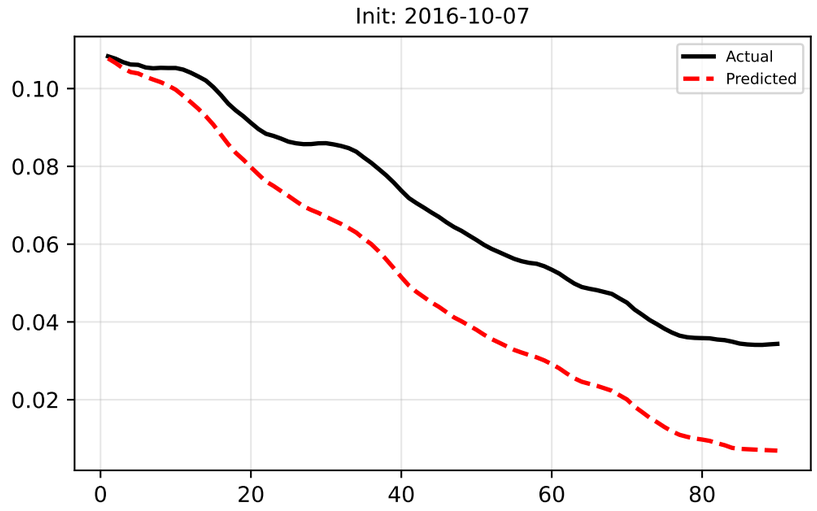


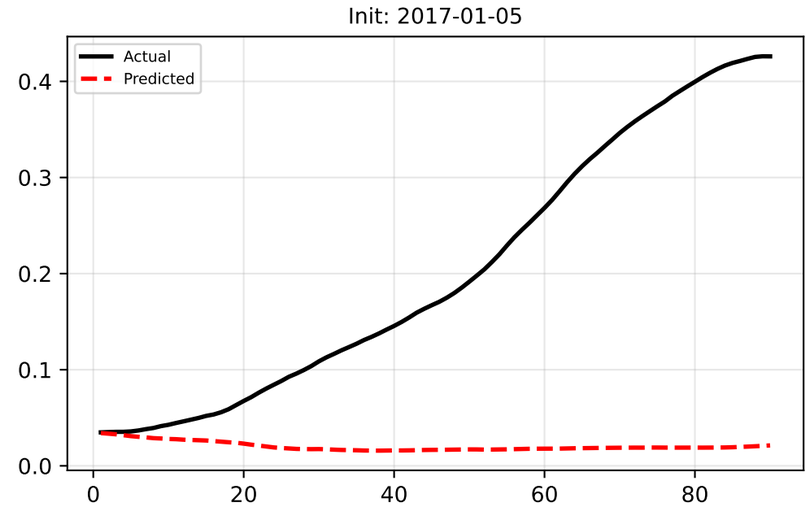


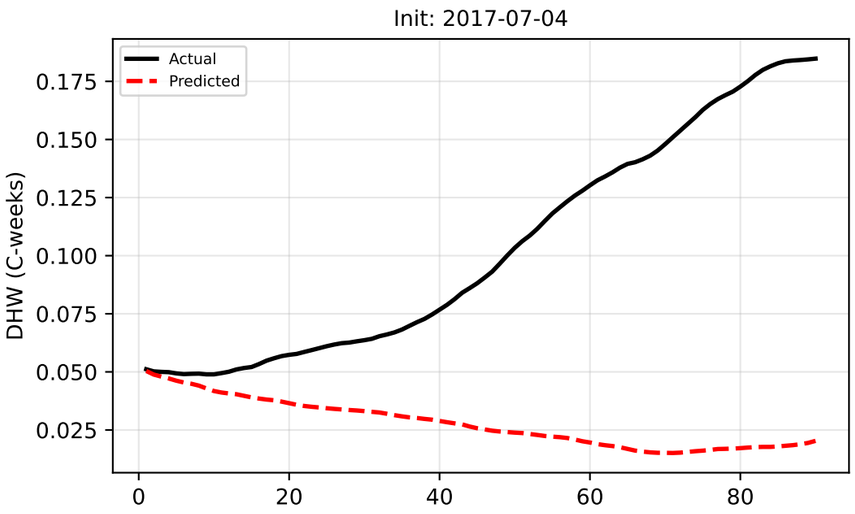


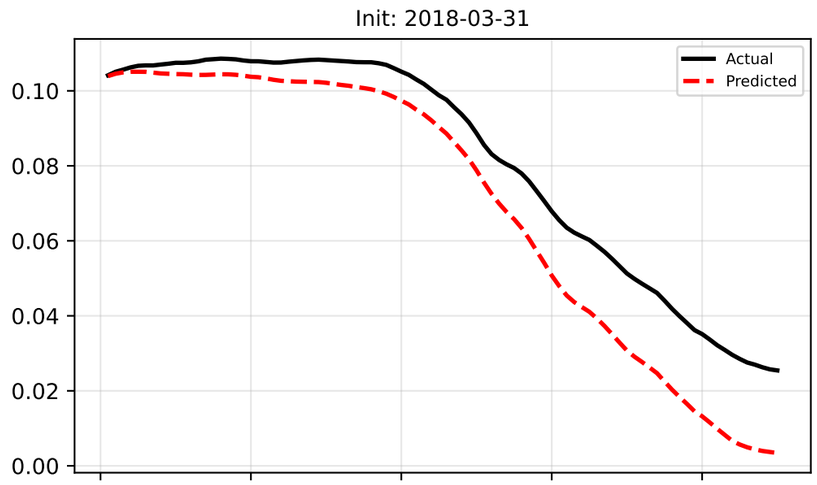
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

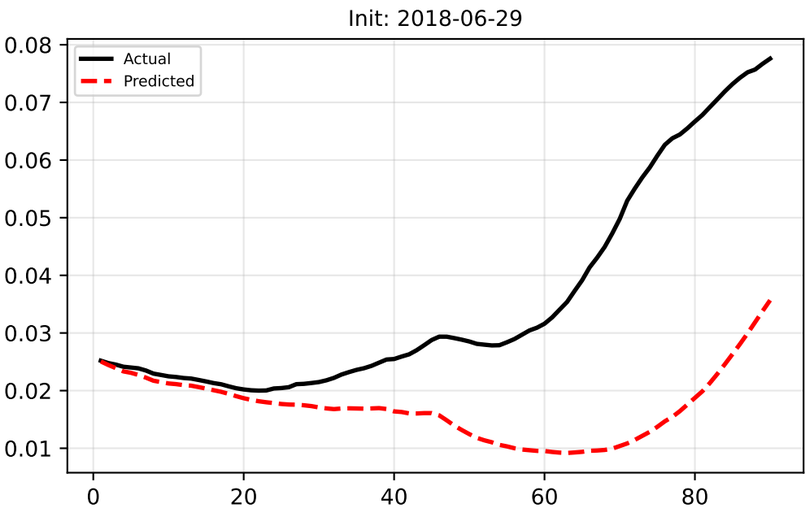


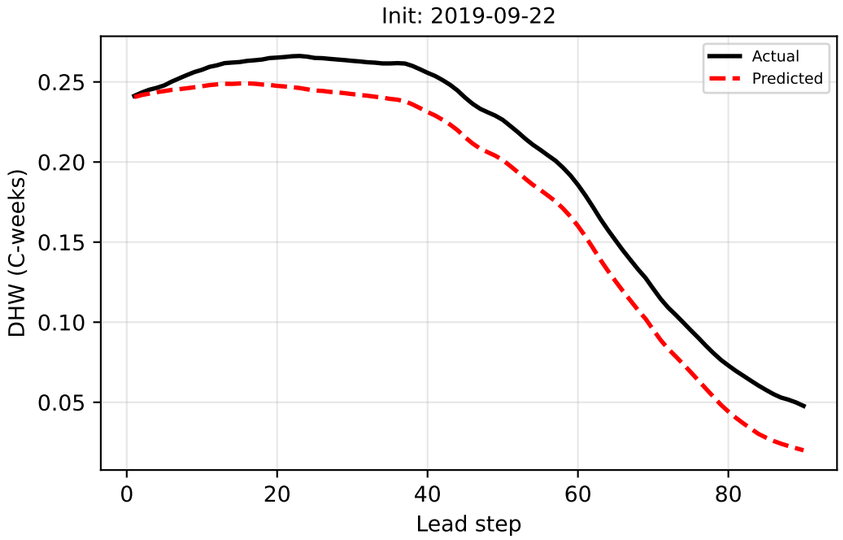


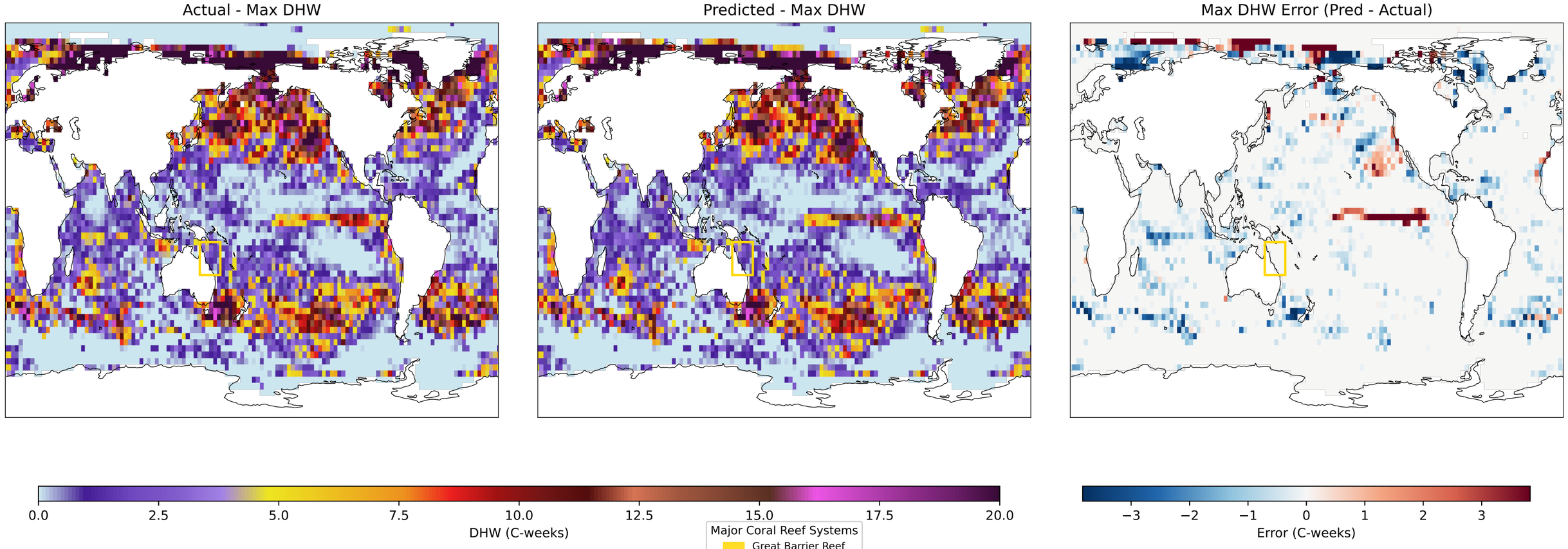


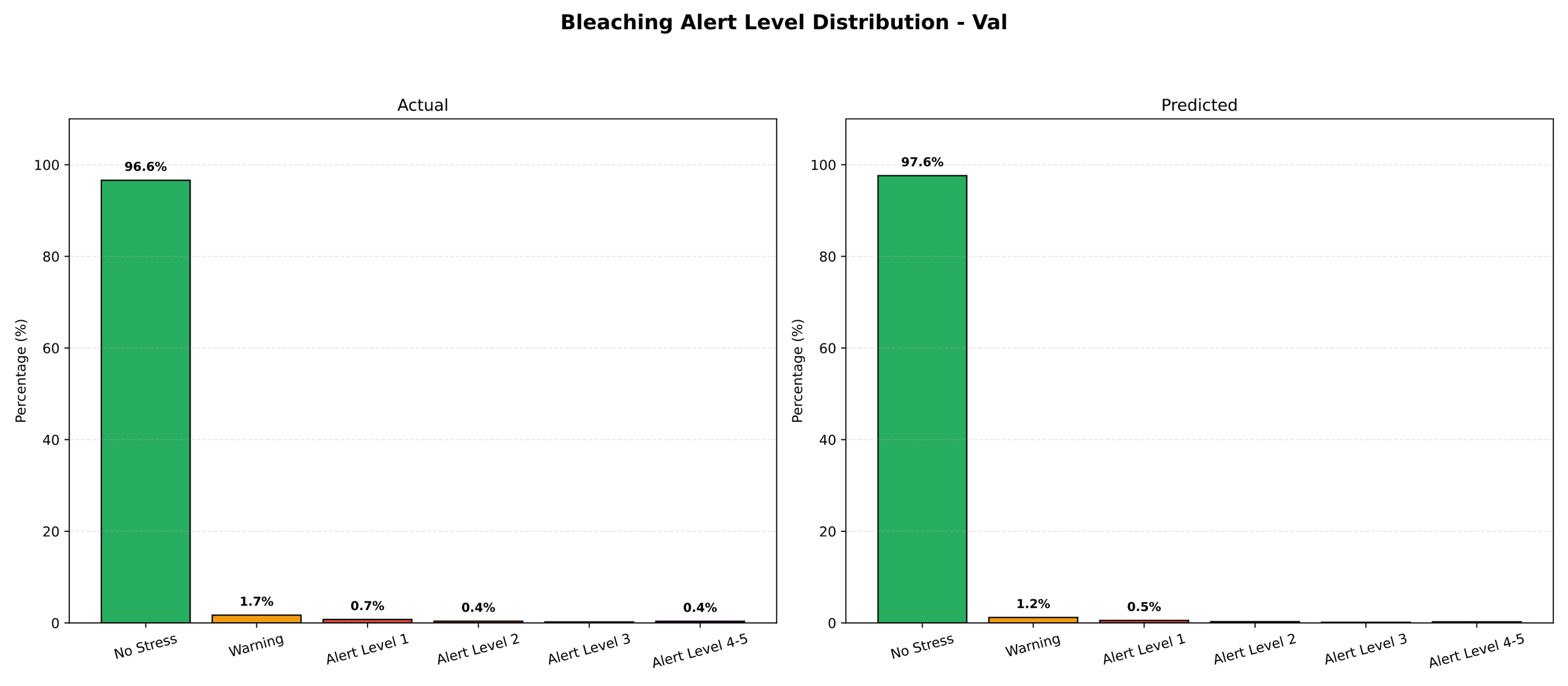






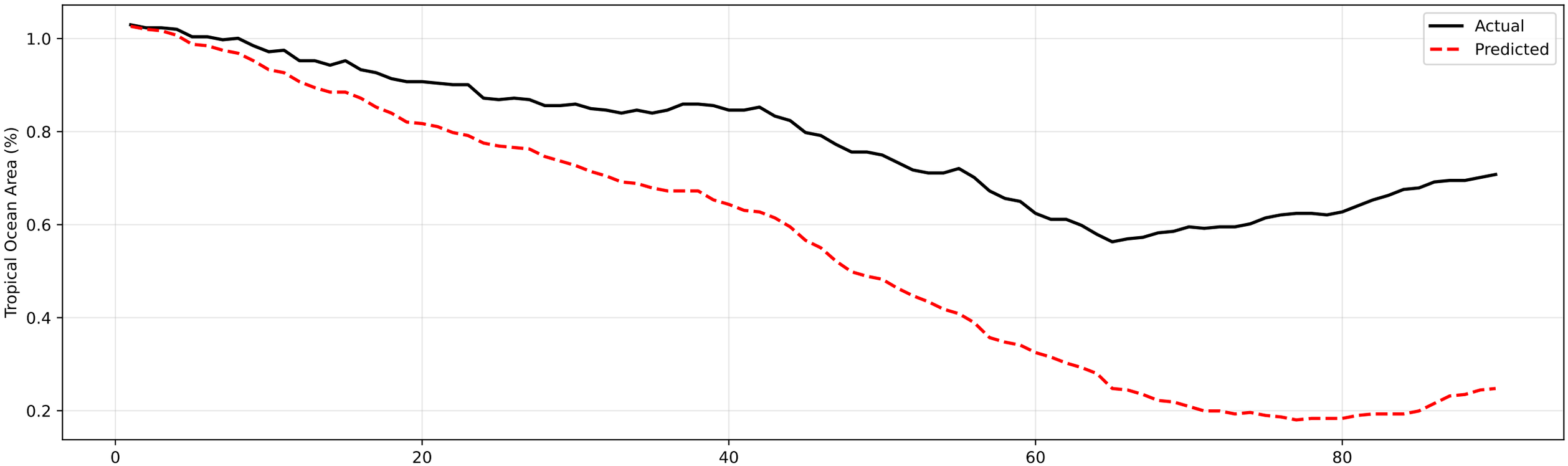



Peak DHW (max over all chains & leads) - Val (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−3
−2
−1
0
1
2
3
Error (C-weeks)

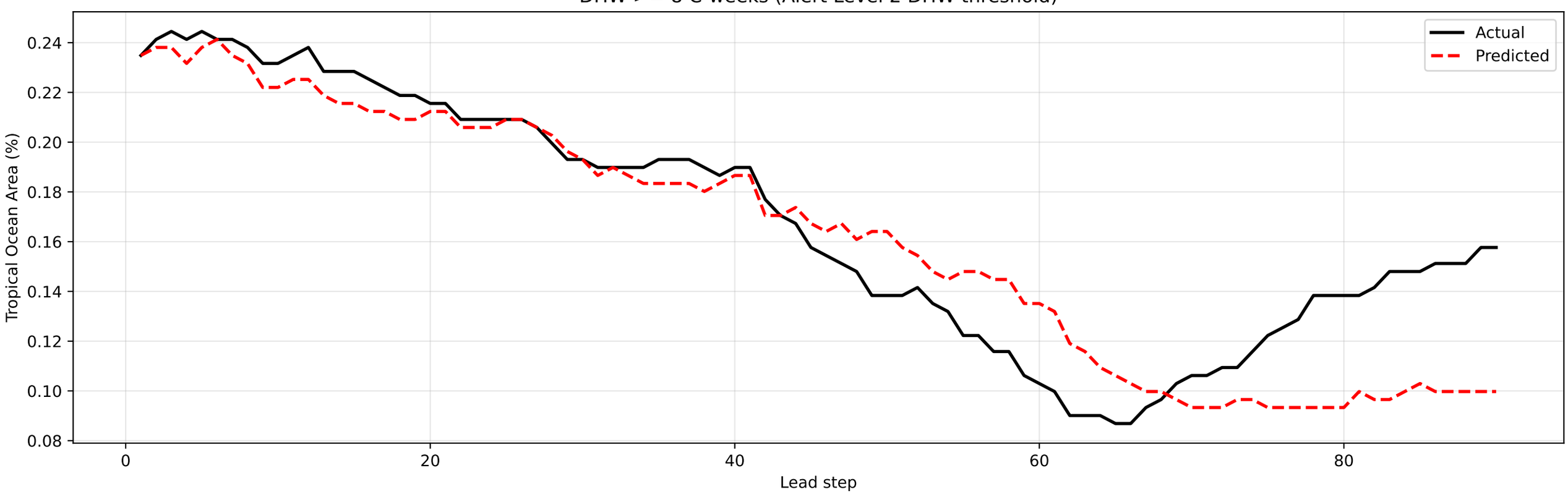

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.7%
0.4%
0.4%
97.6%
1.2%
0.5%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



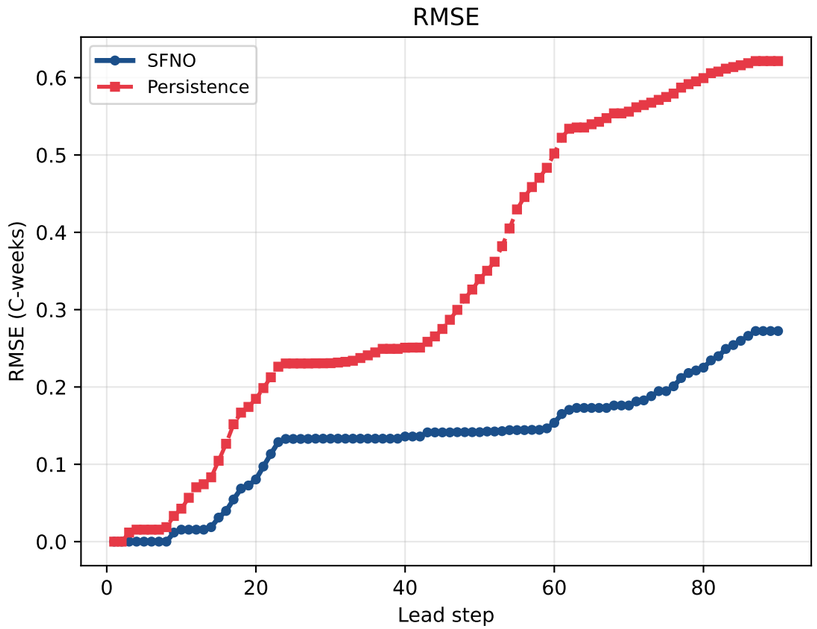
Great Barrier Reef Analysis - Val

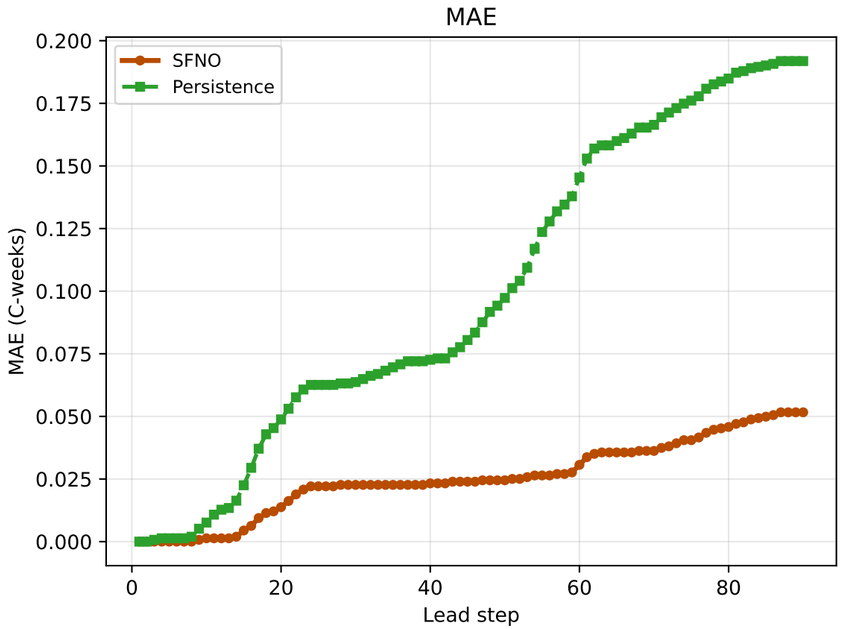
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 5 lat x 5 lon points
```

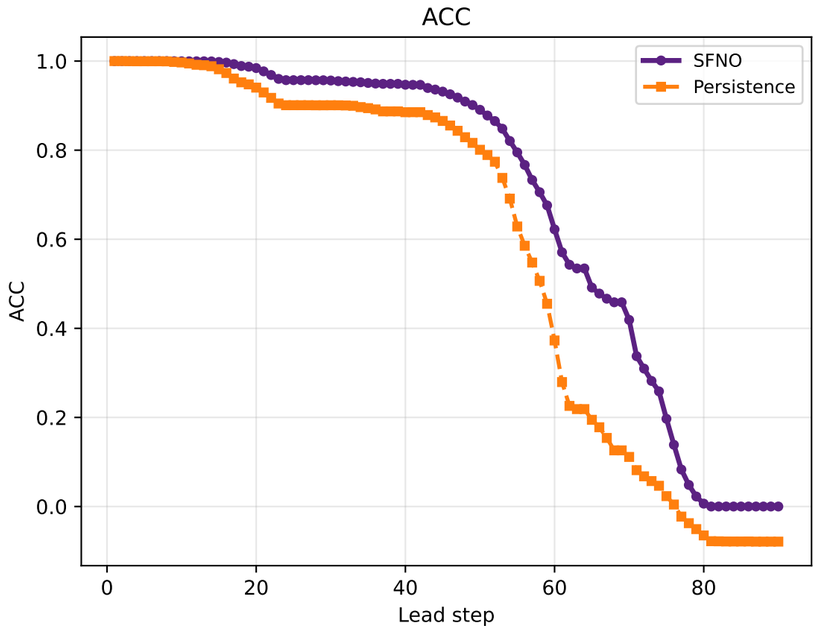
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

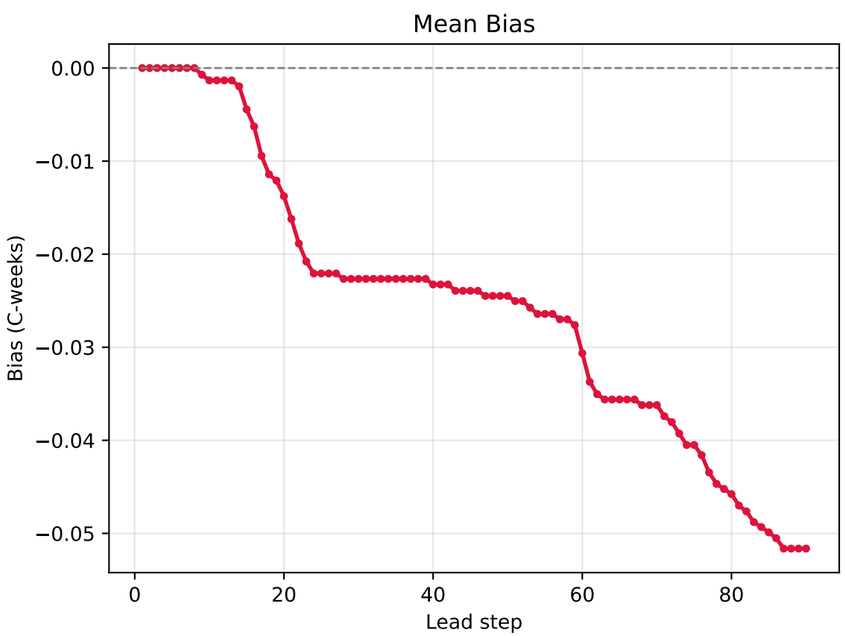
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
2	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
3	0.0000	0.0117	100.0%	0.0000	0.0007	0.0000	1.0000	0.9998	1.0000
4	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
5	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
6	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
7	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
8	0.0000	0.0187	100.0%	0.0000	0.0020	0.0000	1.0000	0.9994	1.0000
9	0.0117	0.0332	64.6%	0.0007	0.0052	-0.0007	0.9997	0.9981	0.9995
10	0.0155	0.0426	63.7%	0.0013	0.0076	-0.0013	0.9996	0.9969	0.9991
11	0.0155	0.0567	72.7%	0.0013	0.0108	-0.0013	0.9995	0.9947	0.9991
12	0.0155	0.0703	78.0%	0.0013	0.0127	-0.0013	0.9995	0.9917	0.9991
13	0.0155	0.0742	79.2%	0.0013	0.0134	-0.0013	0.9995	0.9907	0.9991
14	0.0187	0.0832	77.5%	0.0020	0.0163	-0.0020	0.9993	0.9884	0.9987
15	0.0311	0.1046	70.3%	0.0044	0.0225	-0.0044	0.9980	0.9817	0.9962
16	0.0397	0.1265	68.6%	0.0063	0.0295	-0.0063	0.9966	0.9732	0.9935
17	0.0545	0.1518	64.1%	0.0094	0.0371	-0.0094	0.9932	0.9608	0.9871
18	0.0686	0.1668	58.9%	0.0114	0.0429	-0.0114	0.9889	0.9523	0.9789
19	0.0726	0.1742	58.3%	0.0121	0.0453	-0.0121	0.9874	0.9478	0.9761
20	0.0804	0.1846	56.4%	0.0138	0.0488	-0.0138	0.9843	0.9404	0.9703
21	0.0971	0.1986	51.1%	0.0162	0.0531	-0.0162	0.9770	0.9296	0.9566
22	0.1133	0.2125	46.7%	0.0189	0.0576	-0.0189	0.9688	0.9174	0.9416
23	0.1287	0.2262	43.1%	0.0208	0.0607	-0.0208	0.9600	0.9047	0.9259
24	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
25	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
26	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
27	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
28	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
29	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
30	0.1332	0.2308	42.3%	0.0227	0.0637	-0.0227	0.9564	0.9008	0.9192
31	0.1332	0.2316	42.5%	0.0227	0.0649	-0.0227	0.9550	0.9008	0.9167
32	0.1332	0.2325	42.7%	0.0227	0.0662	-0.0227	0.9542	0.9004	0.9152
33	0.1332	0.2339	43.1%	0.0227	0.0669	-0.0227	0.9533	0.8996	0.9136
34	0.1332	0.2374	43.9%	0.0227	0.0683	-0.0227	0.9525	0.8967	0.9121
35	0.1332	0.2409	44.7%	0.0227	0.0696	-0.0227	0.9509	0.8944	0.9093
36	0.1332	0.2447	45.6%	0.0227	0.0708	-0.0227	0.9496	0.8915	0.9068
37	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
38	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
39	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
40	0.1359	0.2508	45.8%	0.0233	0.0726	-0.0233	0.9469	0.8852	0.9019
41	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
42	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
43	0.1413	0.2584	45.3%	0.0239	0.0756	-0.0239	0.9396	0.8788	0.8887
44	0.1413	0.2654	46.8%	0.0239	0.0776	-0.0239	0.9361	0.8735	0.8822
45	0.1413	0.2750	48.6%	0.0239	0.0805	-0.0239	0.9315	0.8656	0.8740
46	0.1413	0.2869	50.8%	0.0239	0.0834	-0.0239	0.9255	0.8554	0.8631
47	0.1415	0.2997	52.8%	0.0245	0.0876	-0.0245	0.9180	0.8435	0.8499
48	0.1415	0.3142	55.0%	0.0245	0.0917	-0.0245	0.9092	0.8291	0.8344
49	0.1415	0.3260	56.6%	0.0245	0.0942	-0.0245	0.9013	0.8163	0.8207
50	0.1415	0.3395	58.3%	0.0245	0.0973	-0.0245	0.8907	0.8009	0.8025
51	0.1424	0.3503	59.3%	0.0250	0.1012	-0.0250	0.8778	0.7892	0.7809
52	0.1424	0.3620	60.7%	0.0250	0.1041	-0.0250	0.8653	0.7741	0.7605
53	0.1429	0.3821	62.6%	0.0257	0.1093	-0.0257	0.8482	0.7377	0.7322
54	0.1442	0.4049	64.4%	0.0264	0.1169	-0.0264	0.8206	0.6912	0.6890
55	0.1442	0.4296	66.4%	0.0264	0.1236	-0.0264	0.7950	0.6287	0.6501
56	0.1442	0.4457	67.6%	0.0264	0.1278	-0.0264	0.7668	0.5854	0.6088
57	0.1445	0.4585	68.5%	0.0270	0.1318	-0.0270	0.7330	0.5477	0.5614
58	0.1445	0.4705	69.3%	0.0270	0.1346	-0.0270	0.7057	0.5066	0.5248
59	0.1465	0.4834	69.7%	0.0276	0.1379	-0.0276	0.6760	0.4551	0.4861
60	0.1537	0.5019	69.4%	0.0306	0.1454	-0.0306	0.6222	0.3728	0.4186
61	0.1651	0.5223	68.4%	0.0337	0.1529	-0.0337	0.5707	0.2794	0.3594
62	0.1704	0.5339	68.1%	0.0350	0.1569	-0.0350	0.5428	0.2259	0.3295
63	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
64	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
65	0.1729	0.5397	68.0%	0.0356	0.1599	-0.0356	0.4915	0.1947	0.2746
66	0.1729	0.5429	68.2%	0.0356	0.1611	-0.0356	0.4781	0.1779	0.2597
67	0.1729	0.5476	68.4%	0.0356	0.1629	-0.0356	0.4666	0.1539	0.2469
68	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
69	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
70	0.1760	0.5561	68.3%	0.0362	0.1664	-0.0362	0.4189	0.1113	0.2001
71	0.1812	0.5617	67.7%	0.0374	0.1695	-0.0374	0.3375	0.0816	0.1329
72	0.1827	0.5646	67.6%	0.0380	0.1713	-0.0380	0.3096	0.0678	0.1116
73	0.1881	0.5677	66.9%	0.0393	0.1731	-0.0393	0.2819	0.0571	0.0915
74	0.1946	0.5713	65.9%	0.0405	0.1749	-0.0405	0.2585	0.0465	0.0756
75	0.1946	0.5750	66.2%	0.0405	0.1761	-0.0405	0.1969	0.0233	0.0384
76	0.2009	0.5794	65.3%	0.0416	0.1778	-0.0416	0.1385	0.0044	0.0092
77	0.2116	0.5871	64.0%	0.0434	0.1808	-0.0434	0.0831	-0.0226	-0.0142
78	0.2180	0.5916	63.1%	0.0447	0.1826	-0.0447	0.0485	-0.0374	-0.0277
79	0.2214	0.5952	62.8%	0.0452	0.1837	-0.0452	0.0225	-0.0507	-0.0367
80	0.2250	0.5994	62.5%	0.0458	0.1849	-0.0458	0.0064	-0.0649	-0.0418
81	0.2344	0.6058	61.3%	0.0470	0.1872	-0.0470	0.0000	-0.0779	-0.0419
82	0.2398	0.6079	60.6%	0.0476	0.1879	-0.0476	0.0000	-0.0782	-0.0411
83	0.2492	0.6117	59.3%	0.0488	0.1890	-0.0488	0.0000	-0.0785	-0.0399
84	0.2542	0.6137	58.6%	0.0493	0.1896	-0.0493	0.0000	-0.0786	-0.0391
85	0.2597	0.6160	57.8%	0.0499	0.1901	-0.0499	0.0000	-0.0786	-0.0383
86	0.2661	0.6187	57.0%	0.0505	0.1907	-0.0505	0.0000	-0.0785	-0.0374
87	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
88	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
89	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
90	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373

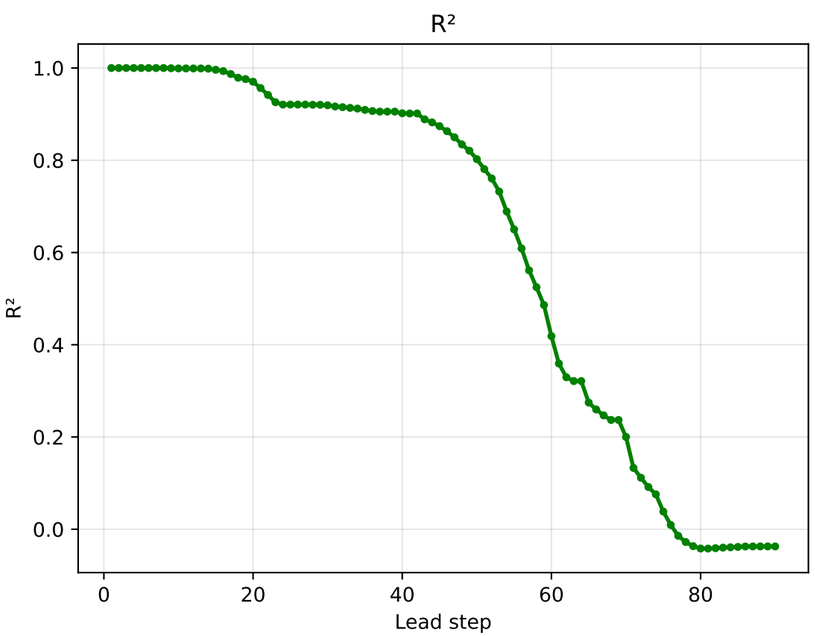
DHW Forecast Skill vs Lead Step - Val - GBR

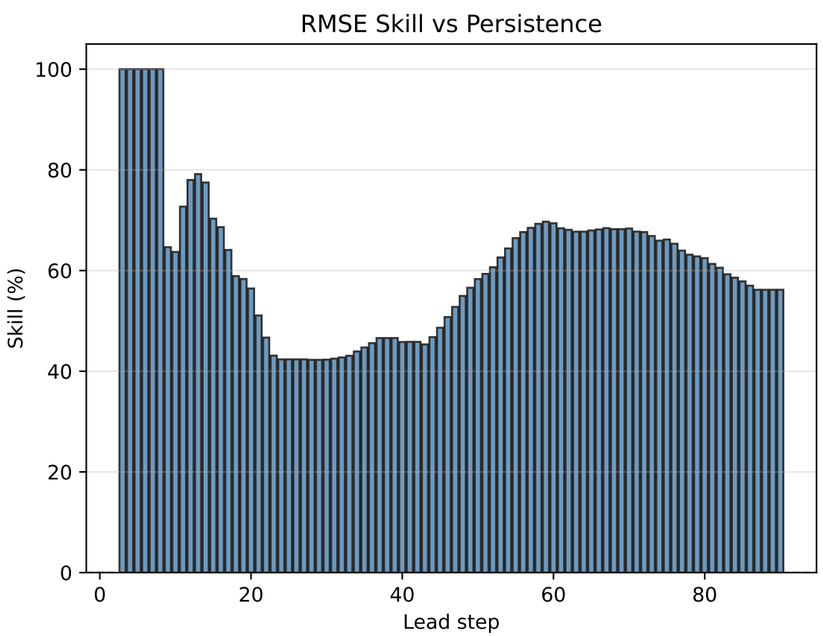


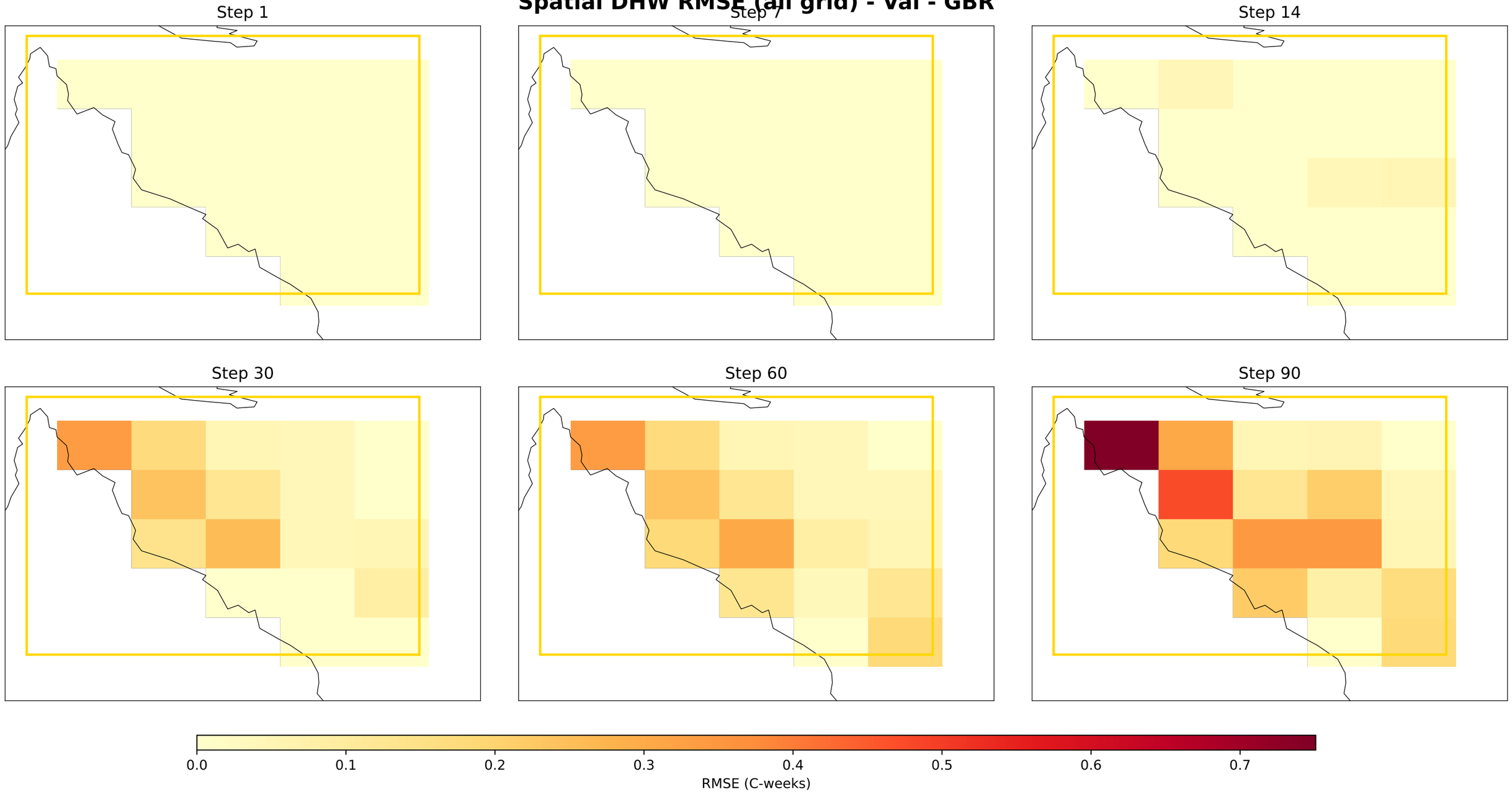


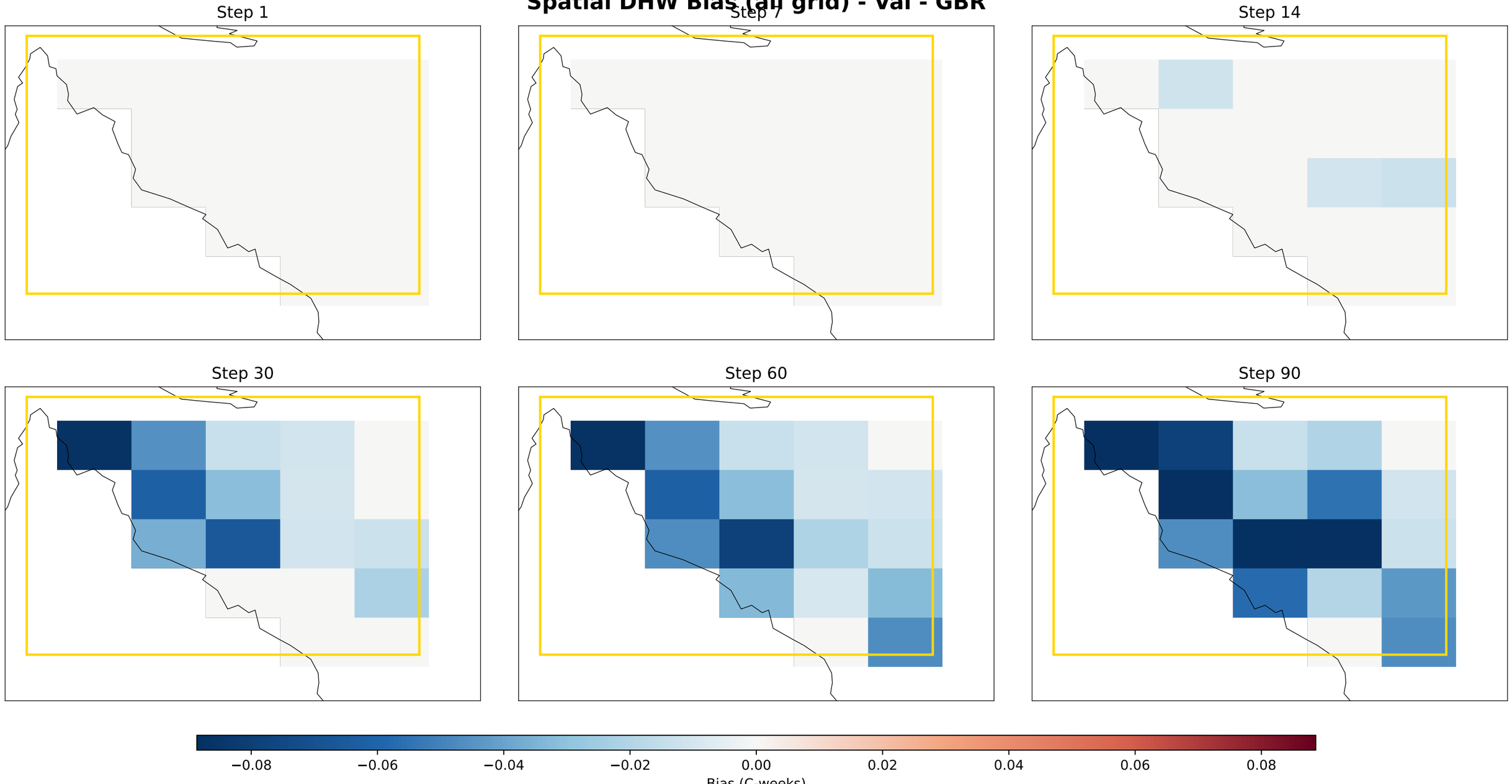


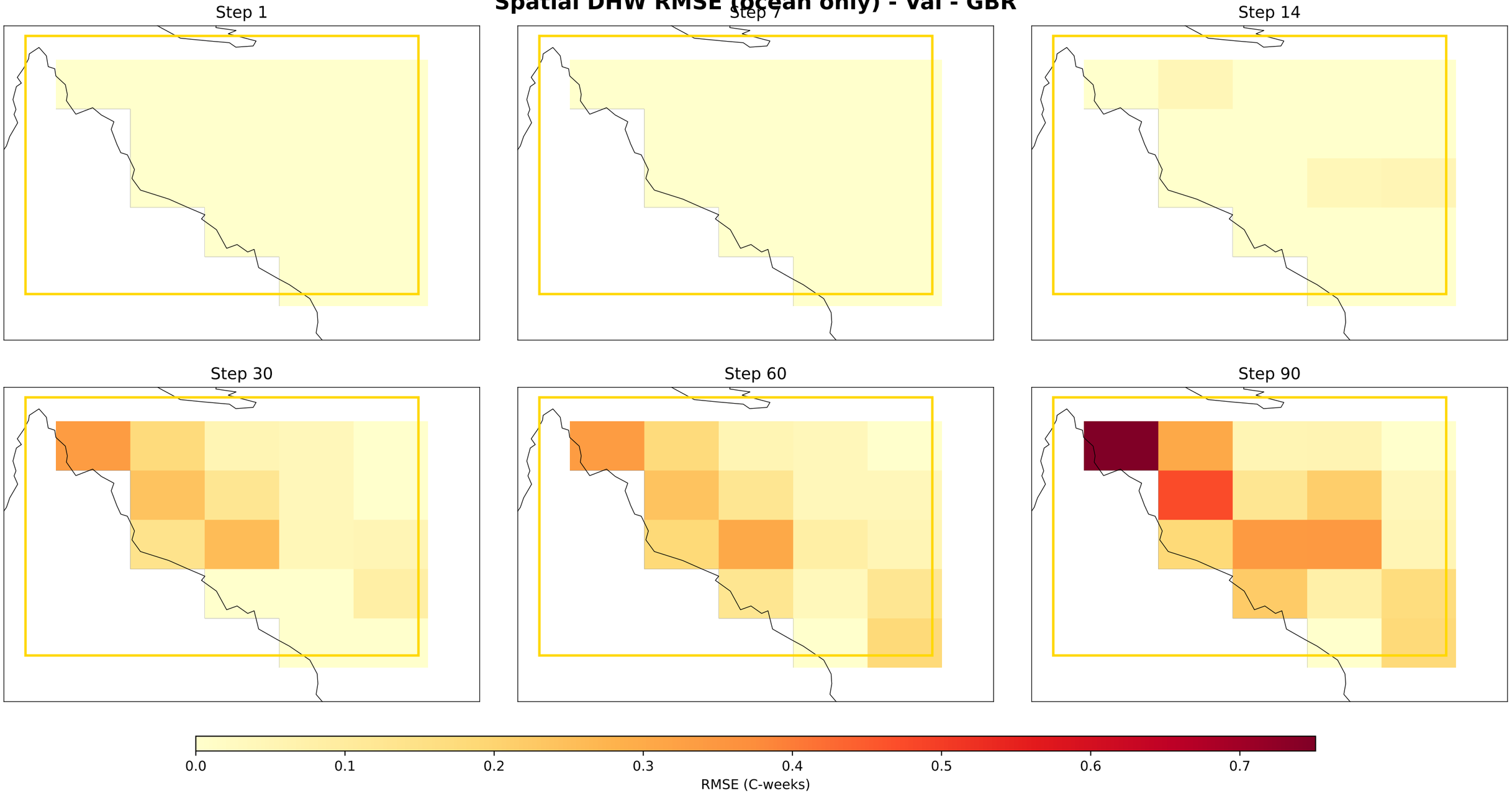


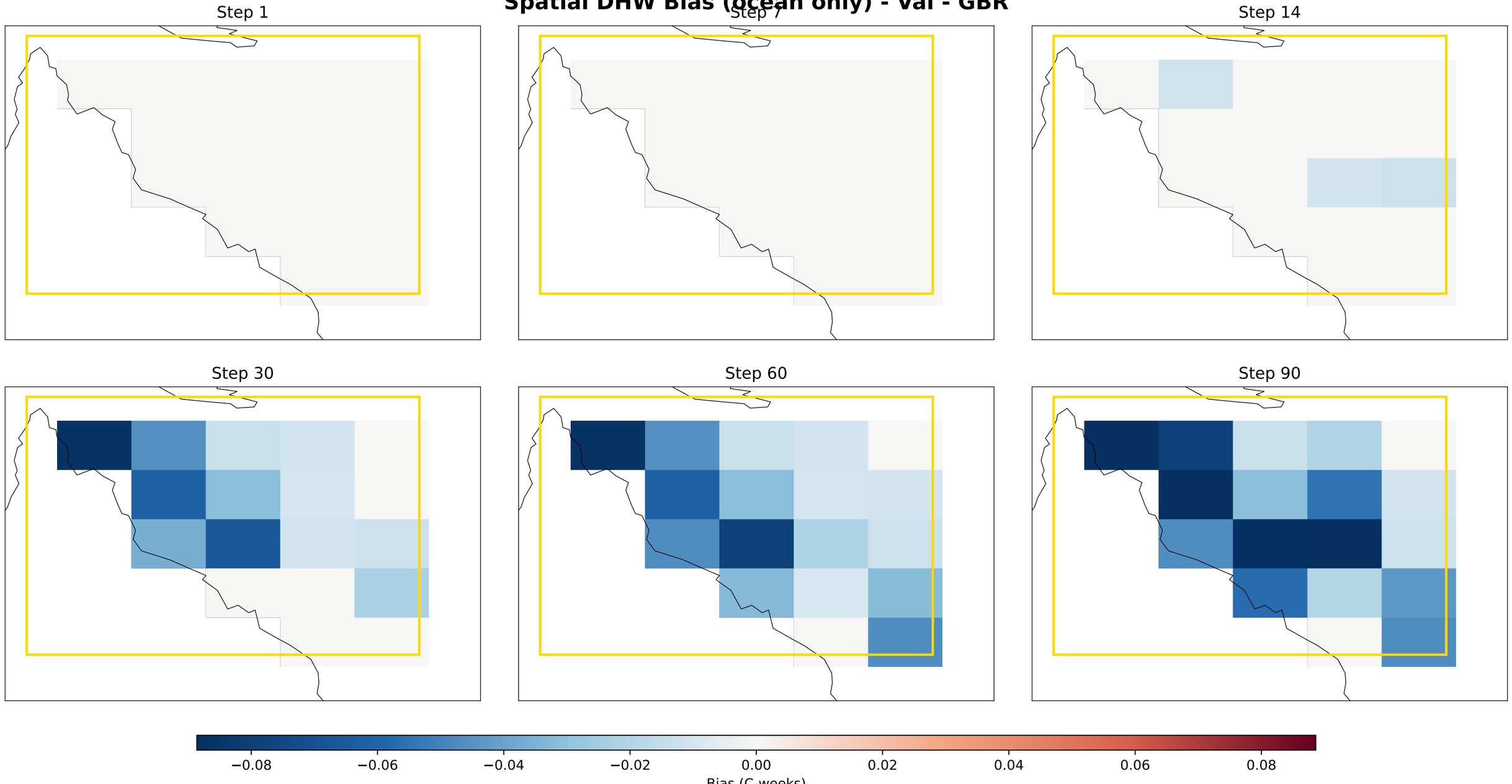





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

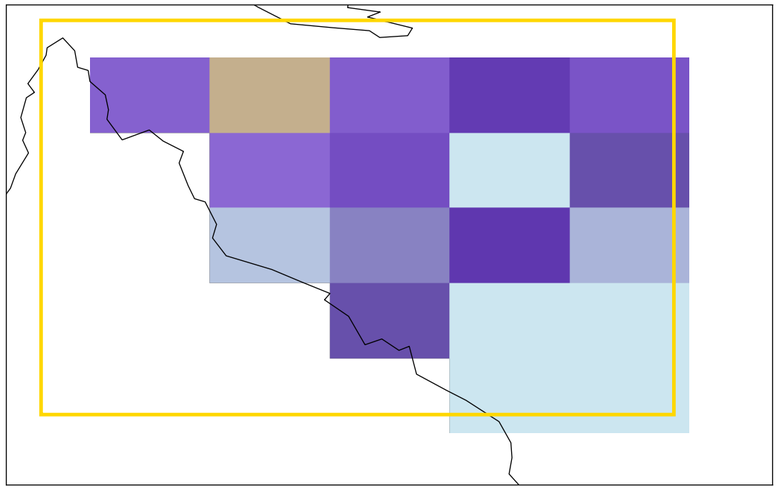

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

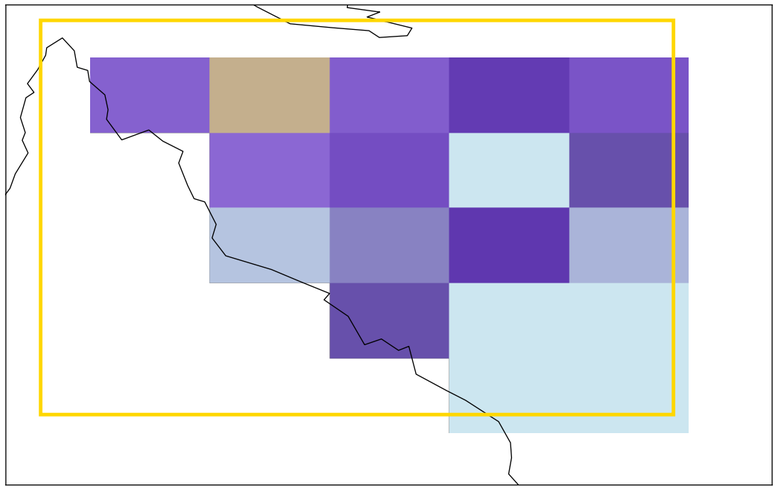

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

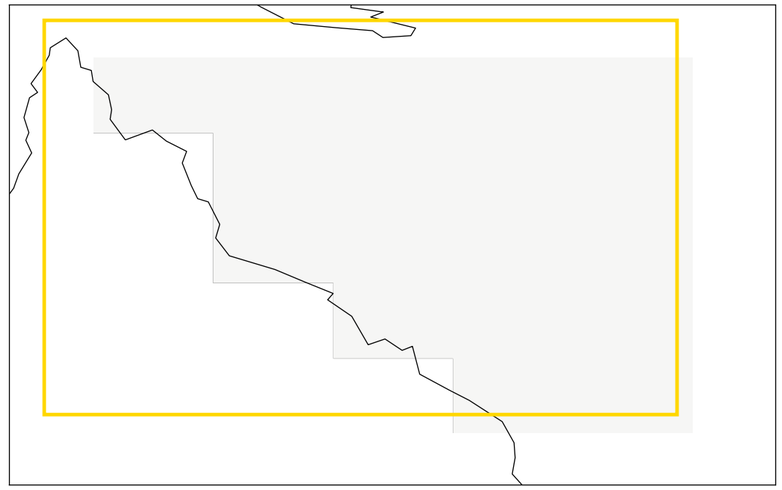
Actual DHW - Step 1



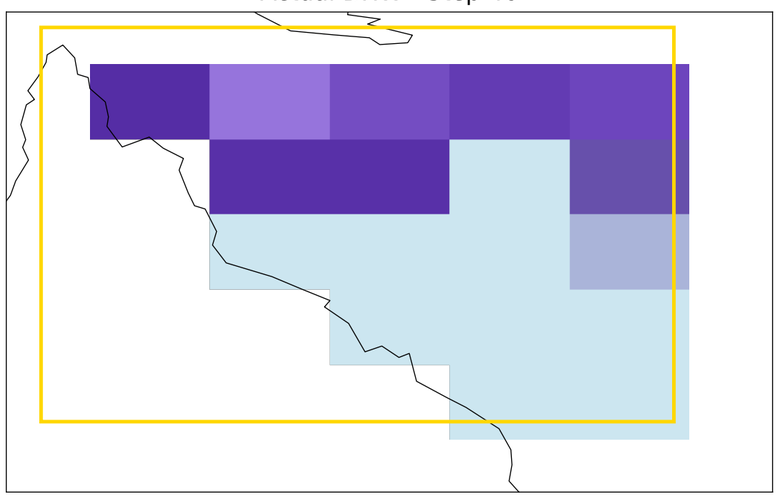
Predicted DHW - Step 1



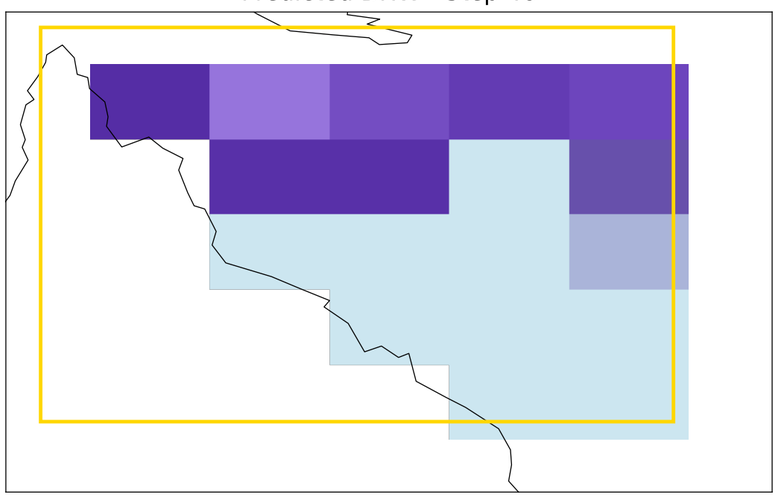
Error (Pred - Actual) - Step 1 [scale ±8]



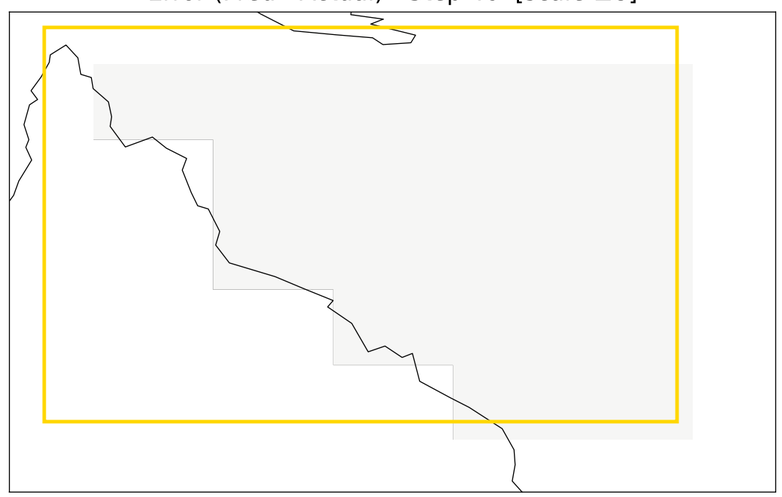
Actual DHW - Step 46



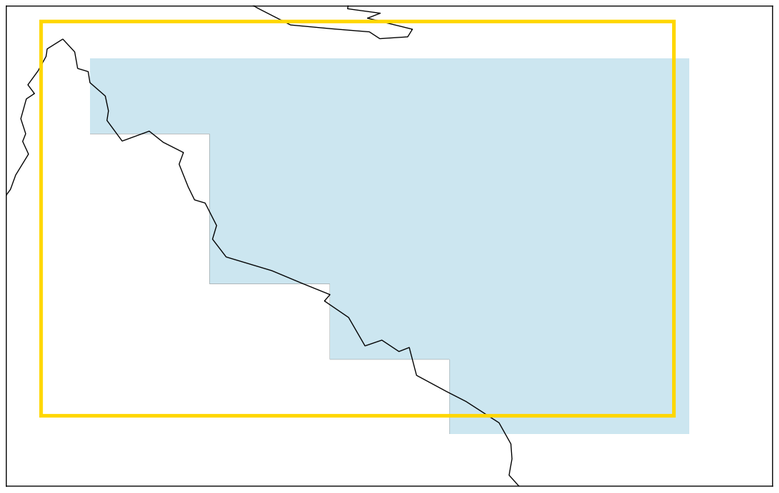
Predicted DHW - Step 46



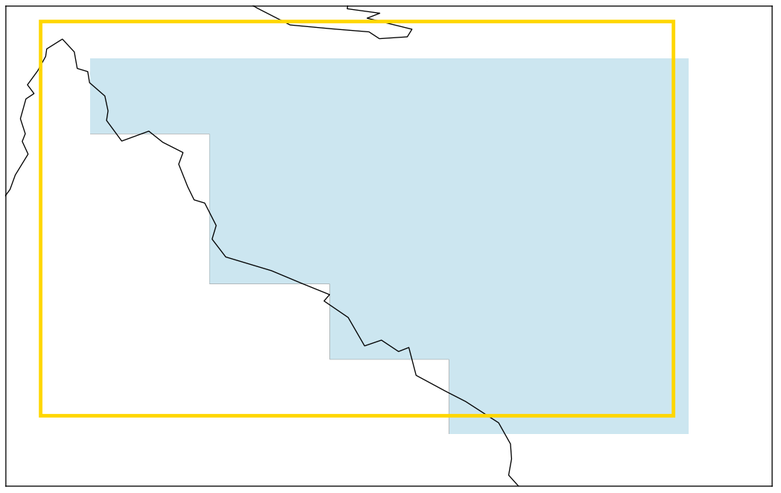
Error (Pred - Actual) - Step 46 [scale ±8]



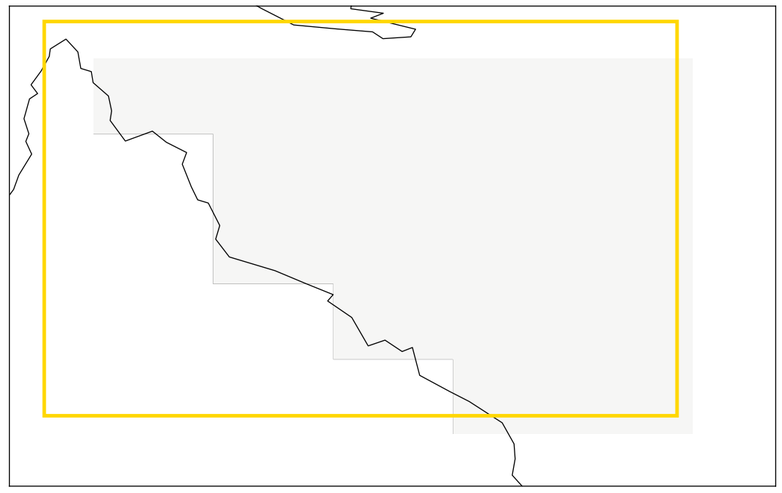
Actual DHW - Step 90

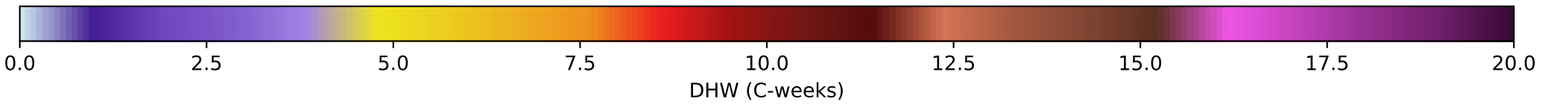


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

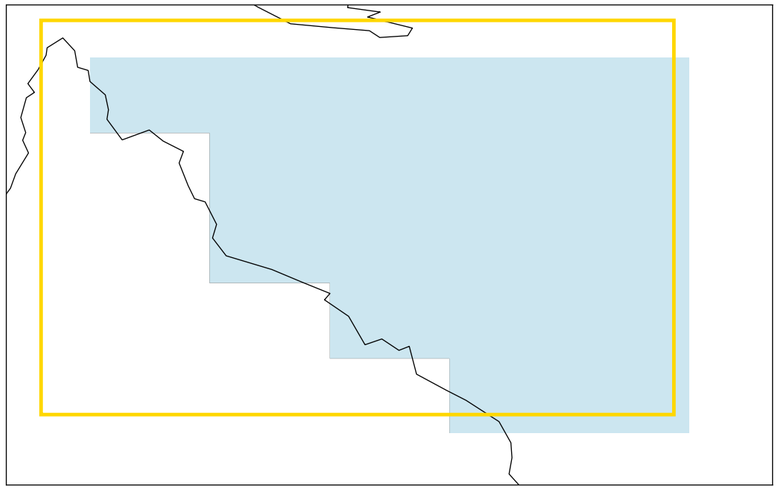




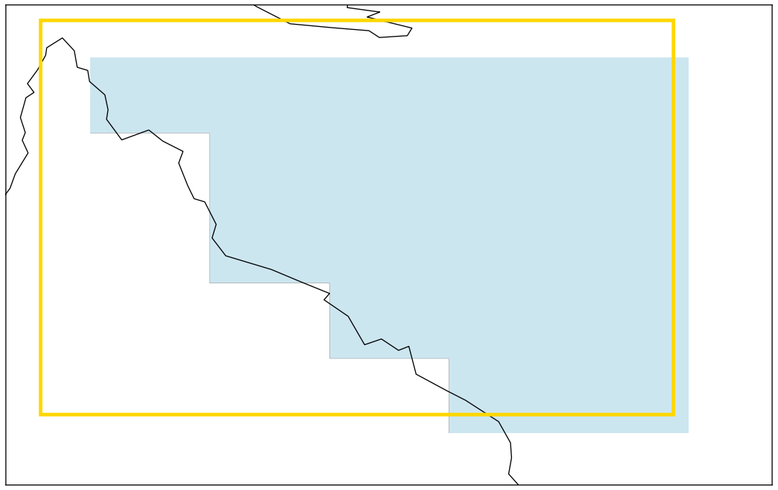


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

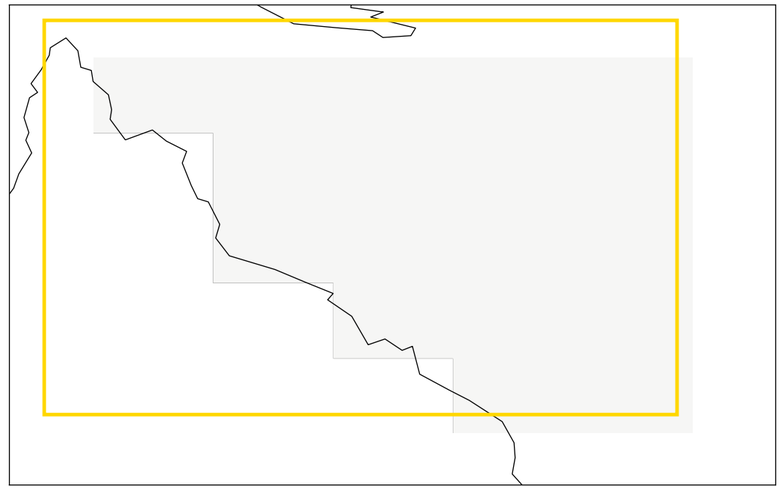
Actual DHW - Step 1



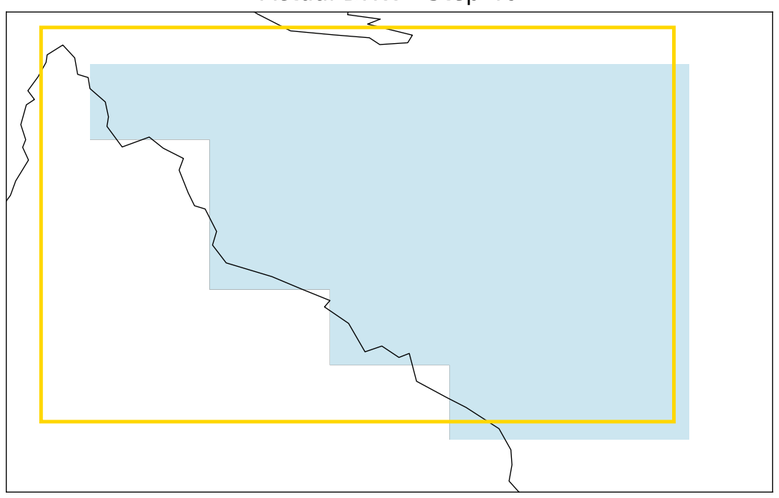
Predicted DHW - Step 1



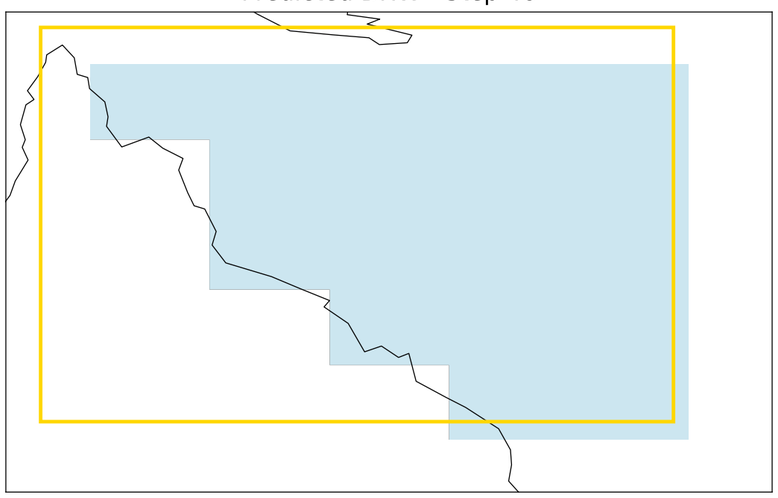
Error (Pred - Actual) - Step 1 [scale ±8]



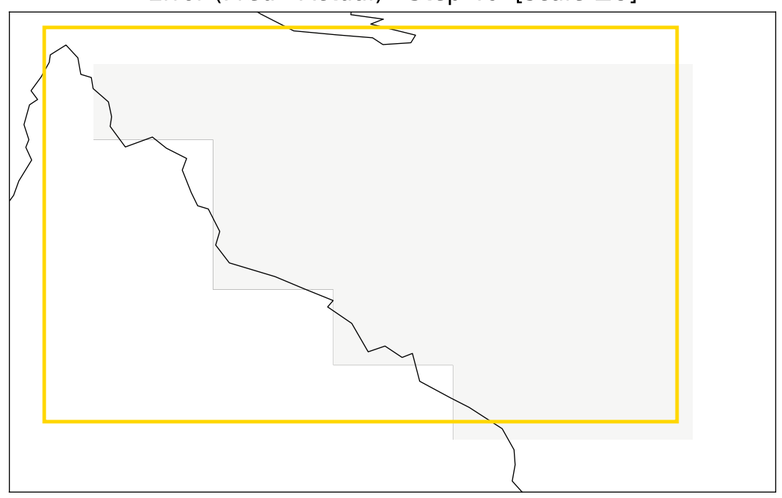
Actual DHW - Step 46



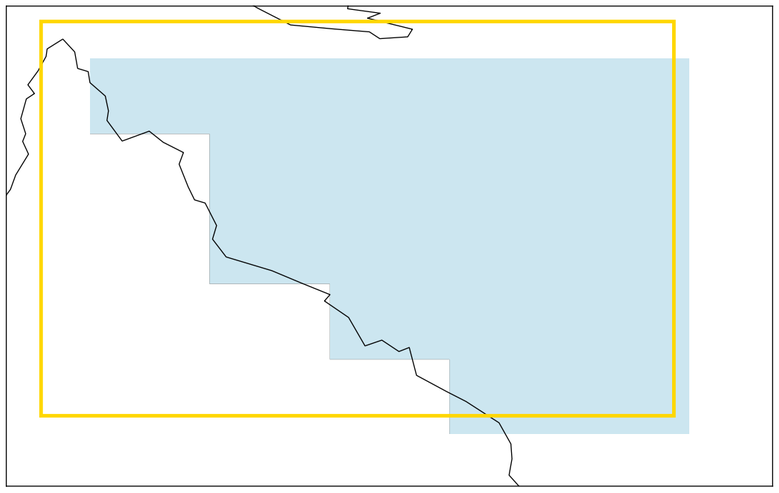
Predicted DHW - Step 46



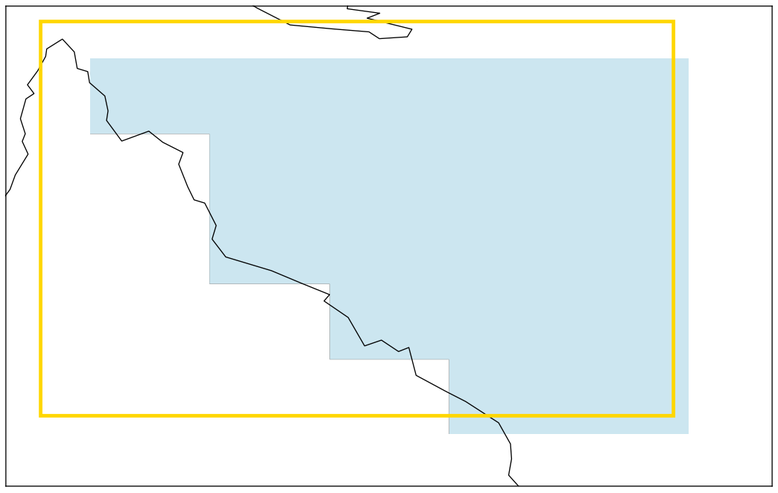
Error (Pred - Actual) - Step 46 [scale ±8]



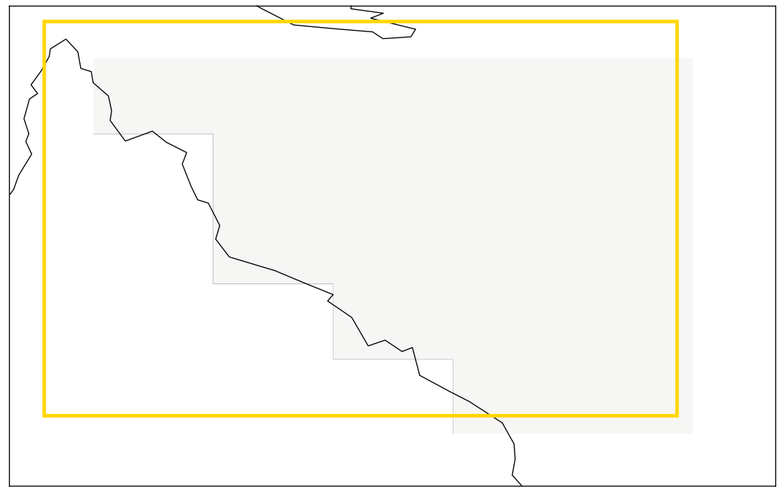
Actual DHW - Step 90

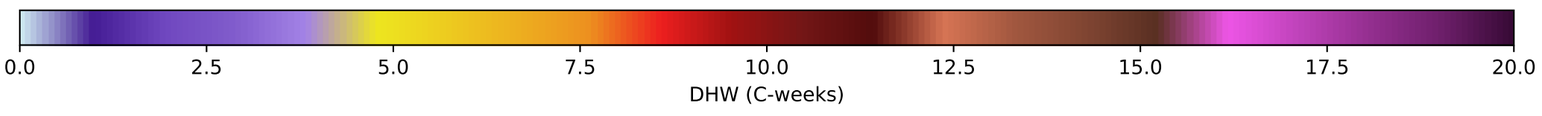


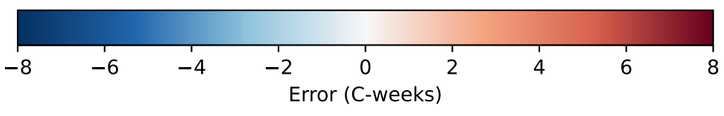
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

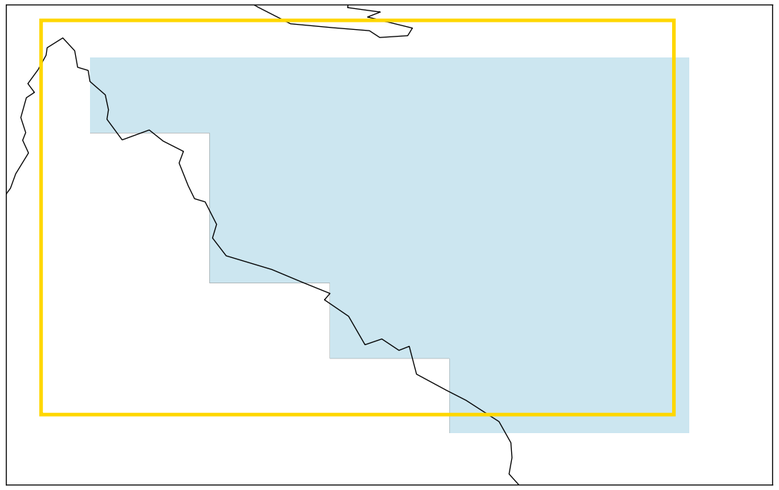




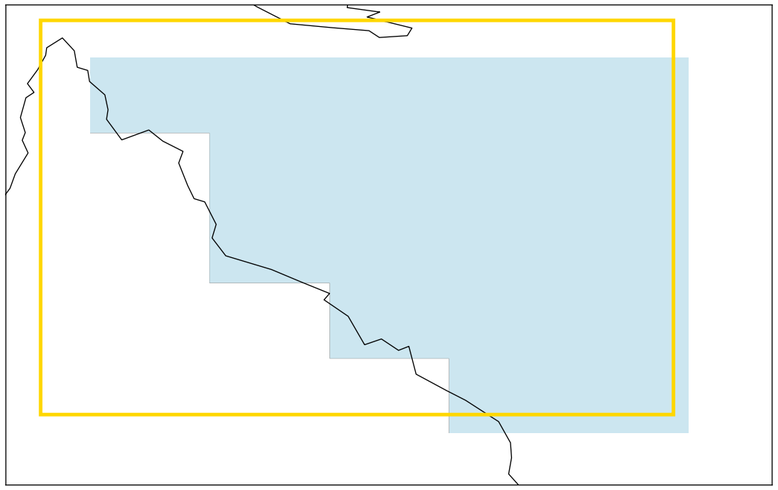


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

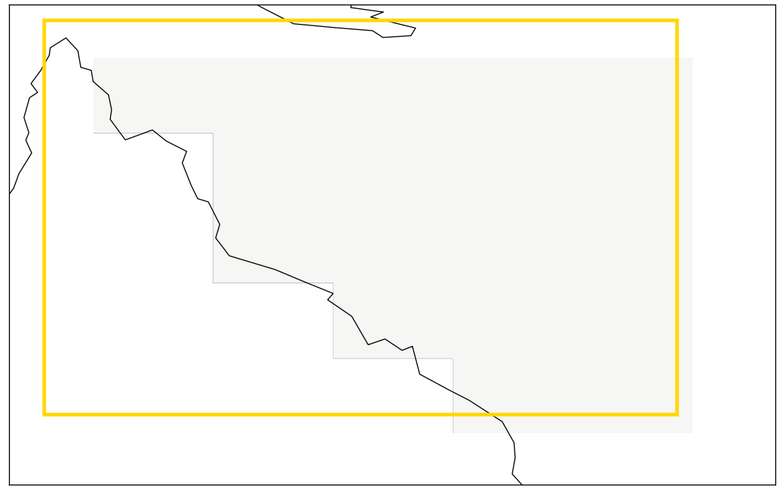
Actual DHW - Step 1



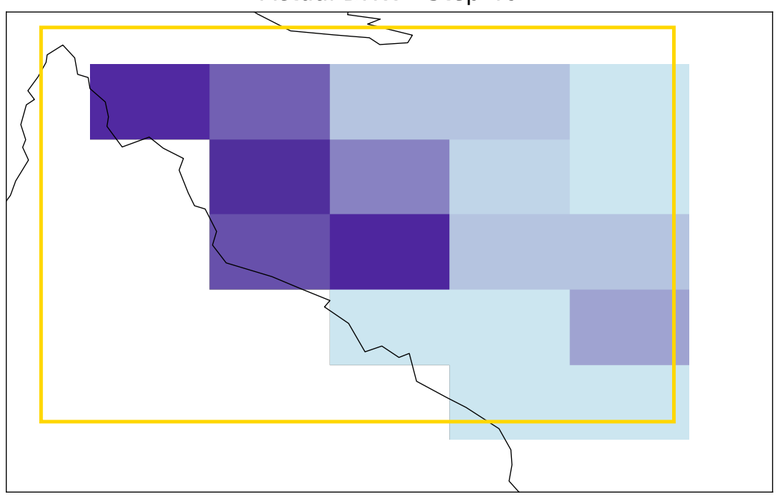
Predicted DHW - Step 1



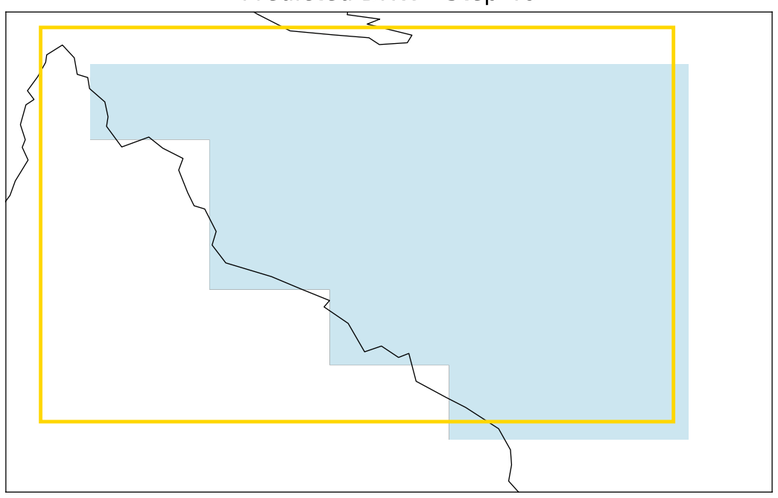
Error (Pred - Actual) - Step 1 [scale ±8]



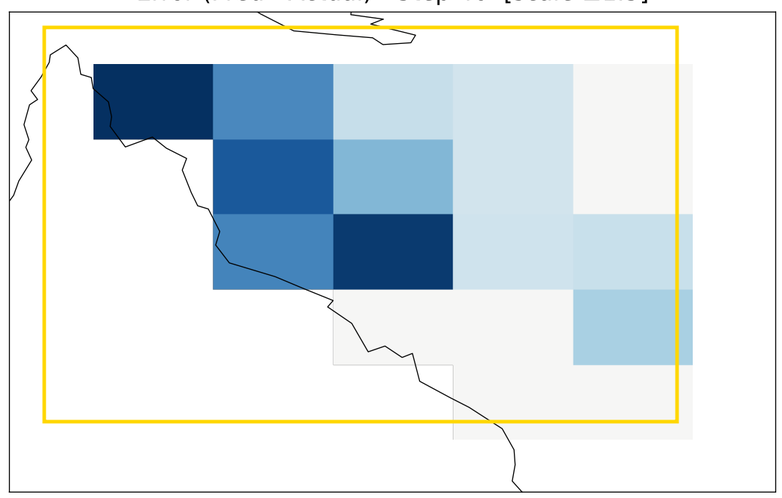
Actual DHW - Step 46



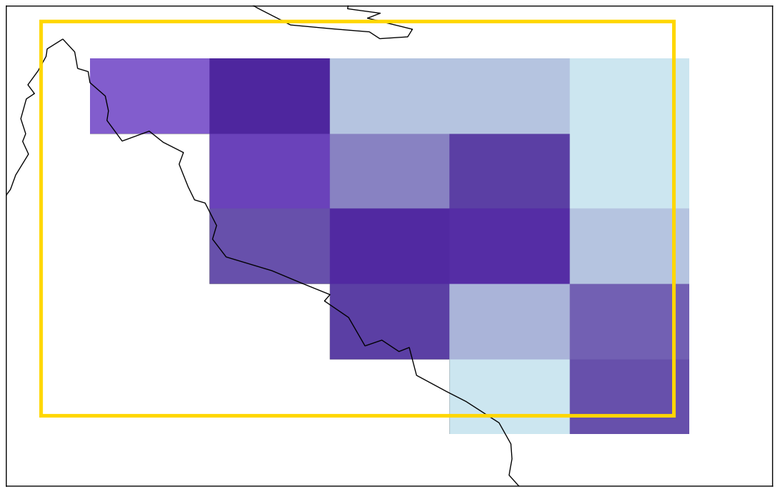
Predicted DHW - Step 46



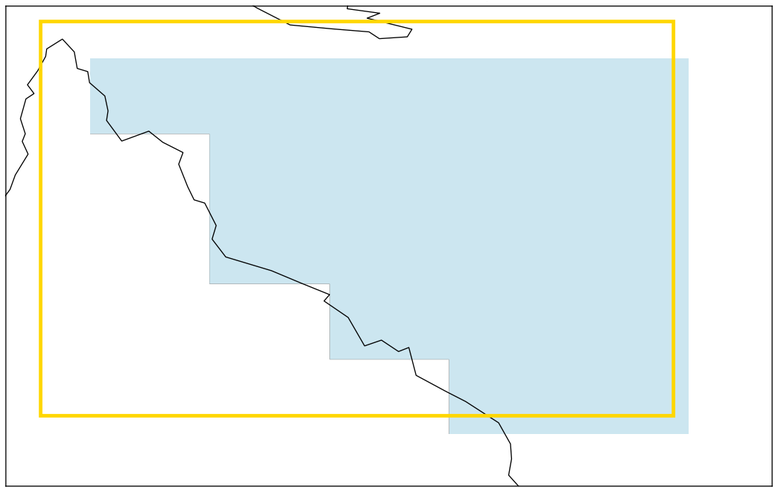
Error (Pred - Actual) - Step 46 [scale ±1.3]



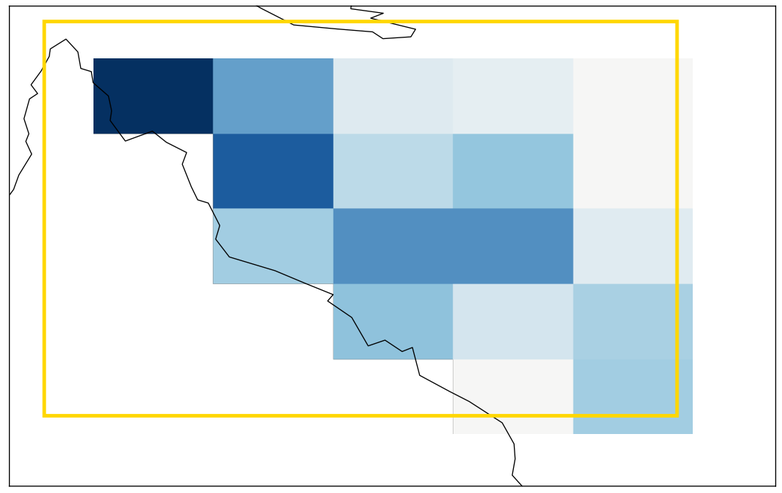
Actual DHW - Step 90

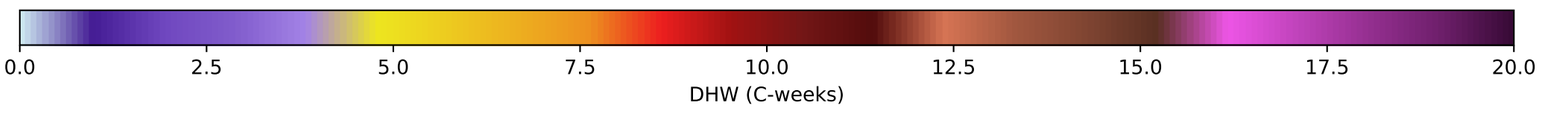


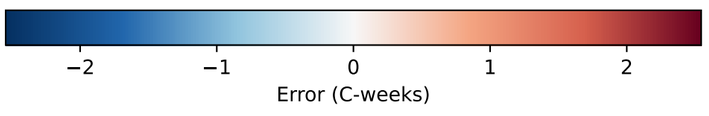
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.5]

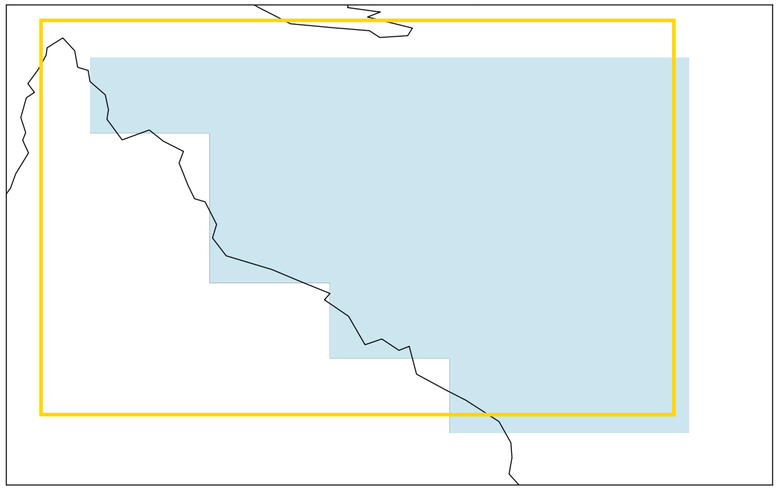




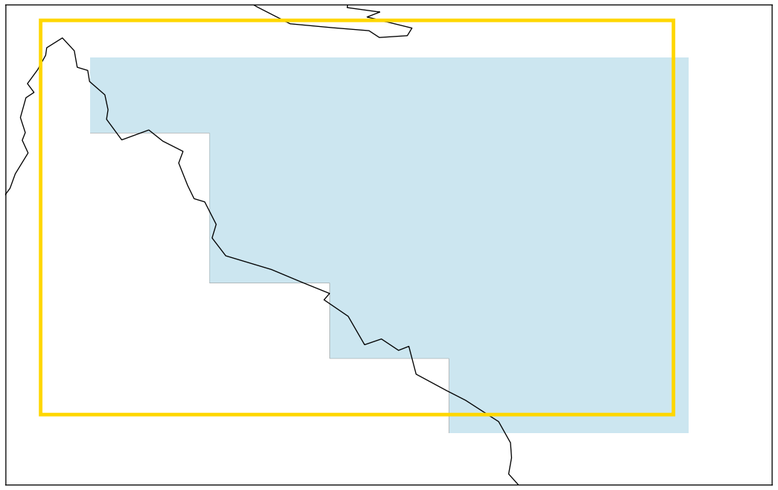


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

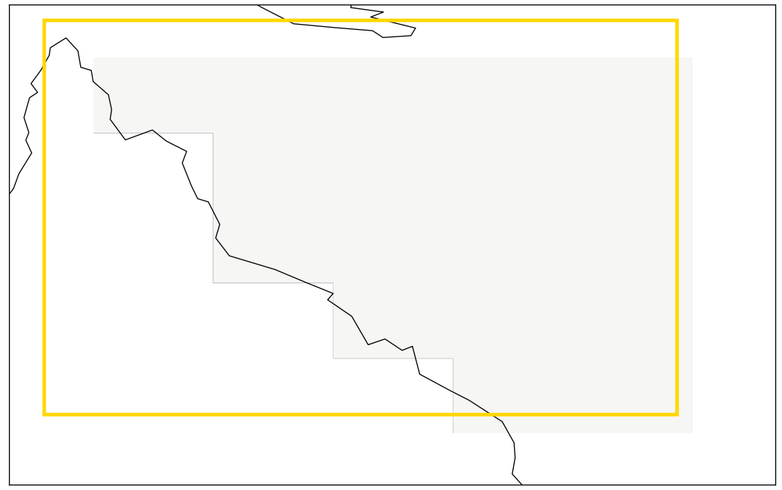
Actual DHW - Step 1



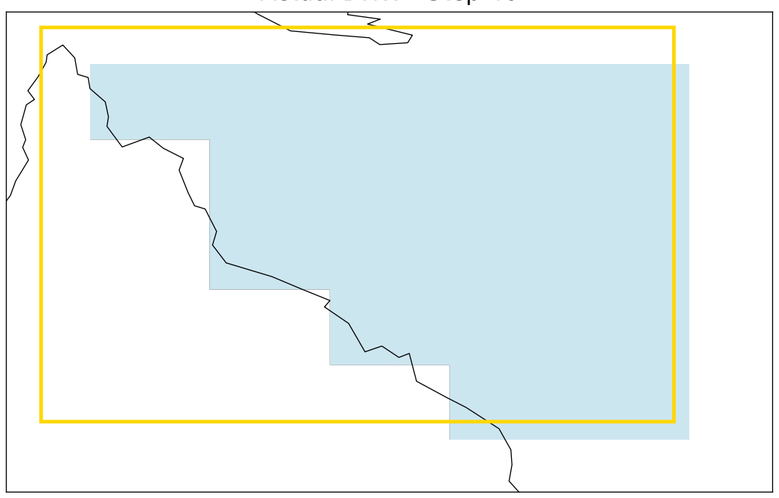
Predicted DHW - Step 1



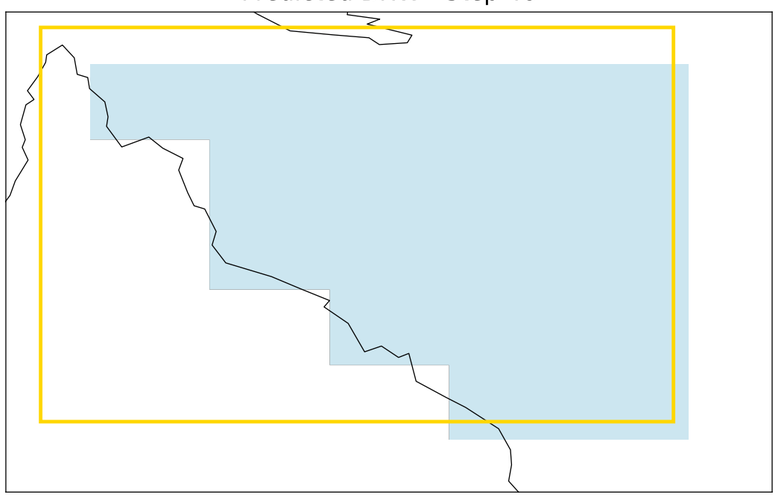
Error (Pred - Actual) - Step 1 [scale ±8]



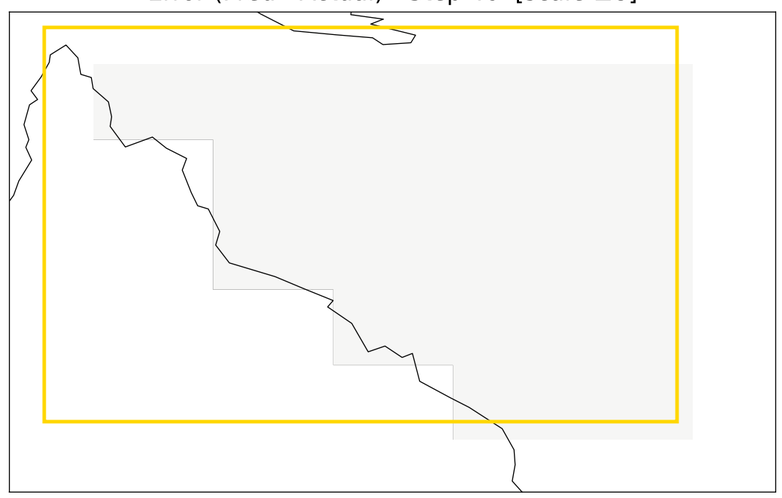
Actual DHW - Step 46



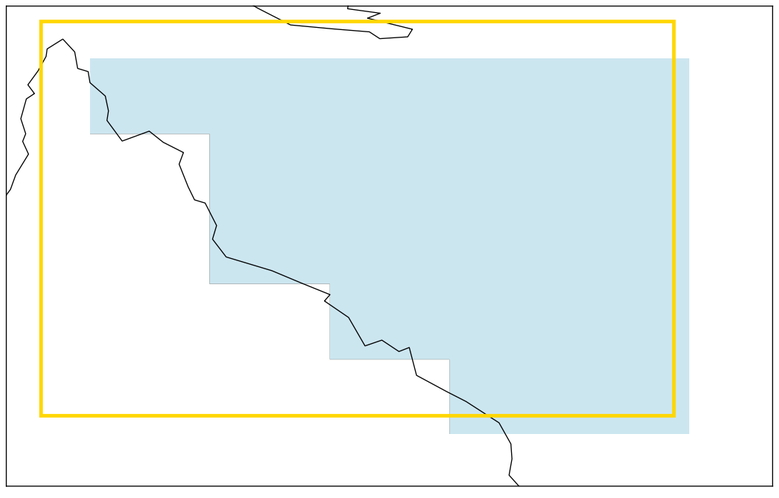
Predicted DHW - Step 46



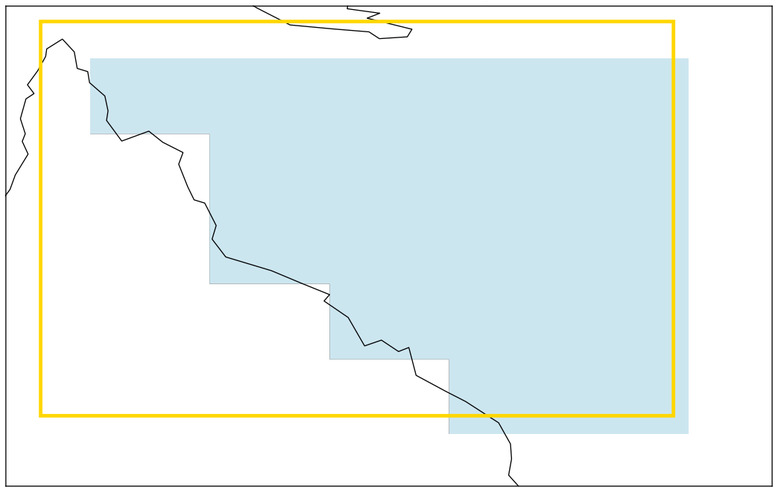
Error (Pred - Actual) - Step 46 [scale ±8]



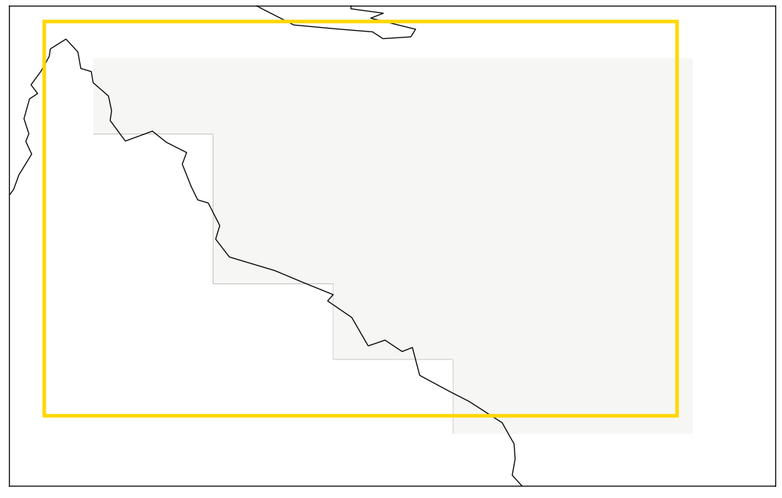
Actual DHW - Step 90

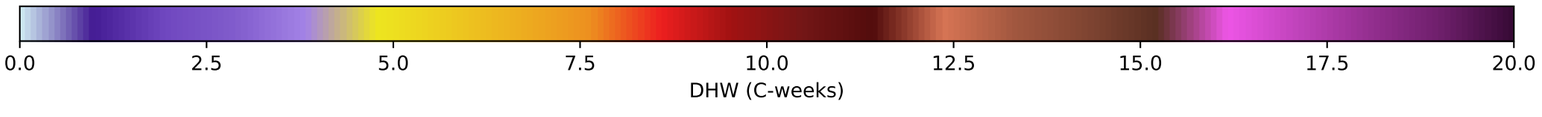


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

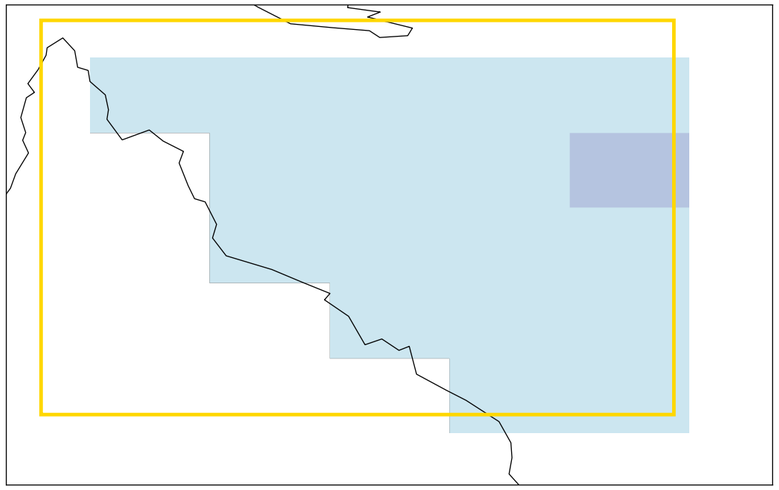




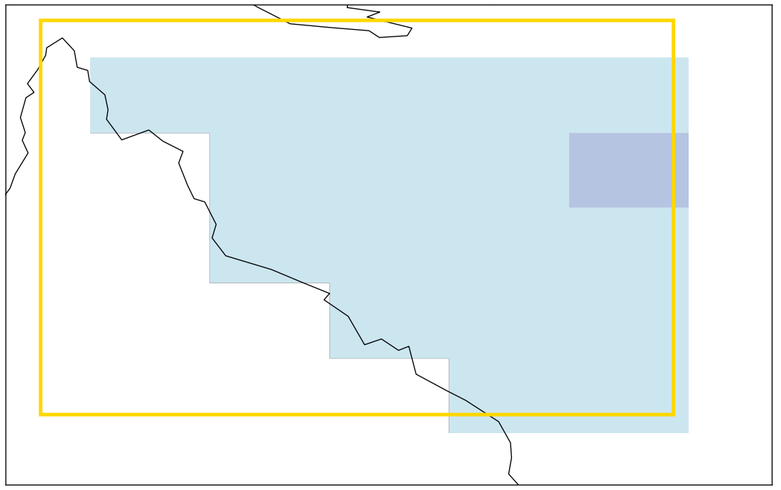


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

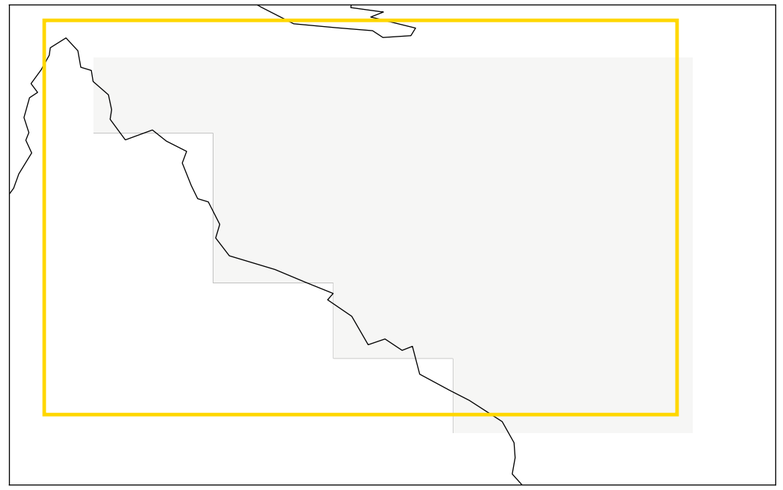
Actual DHW - Step 1



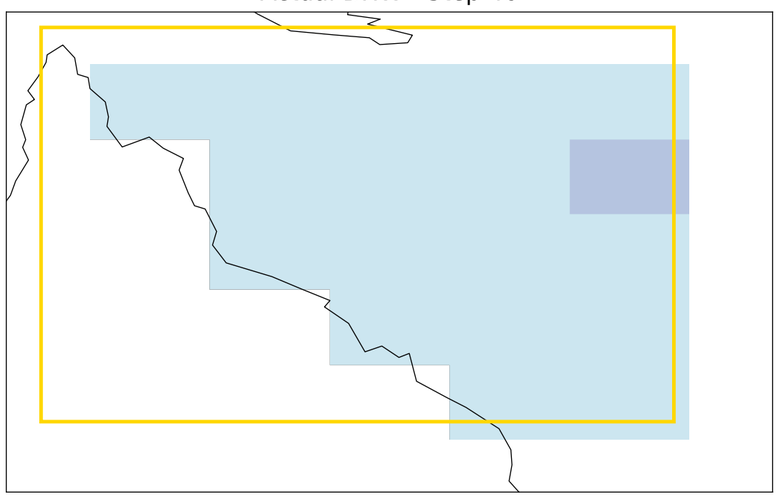
Predicted DHW - Step 1



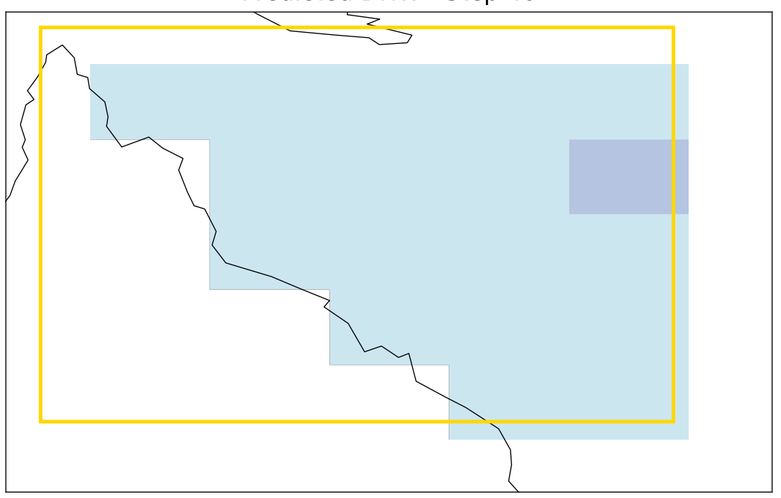
Error (Pred - Actual) - Step 1 [scale ±8]



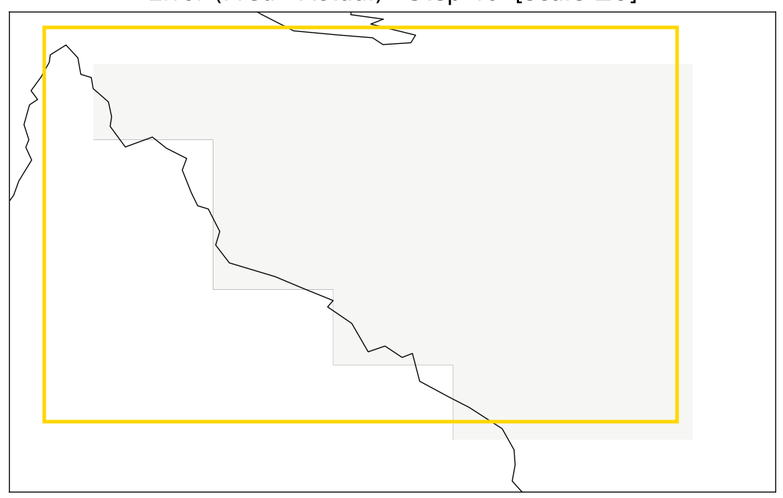
Actual DHW - Step 46



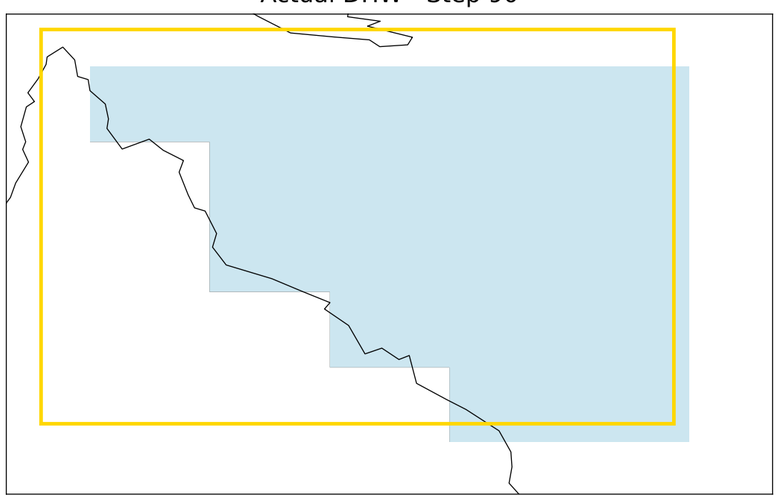
Predicted DHW - Step 46



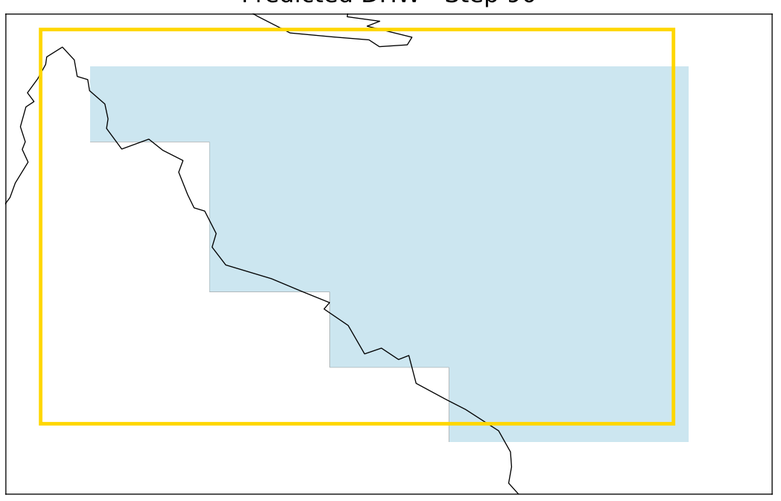
Error (Pred - Actual) - Step 46 [scale ±8]



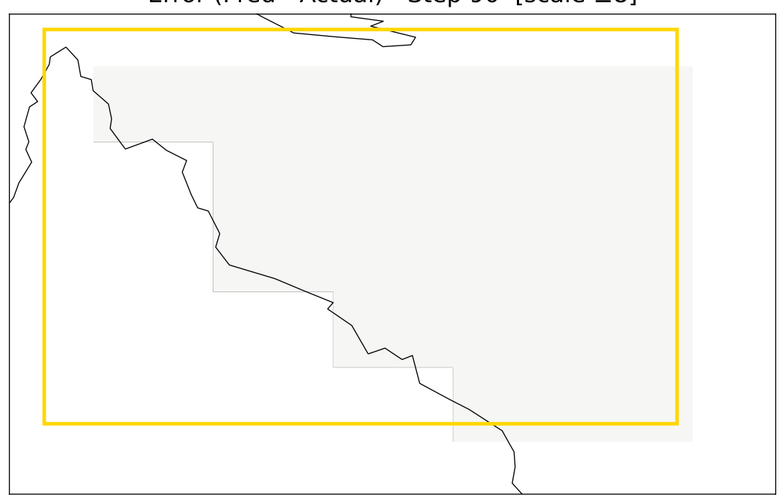
Actual DHW - Step 90

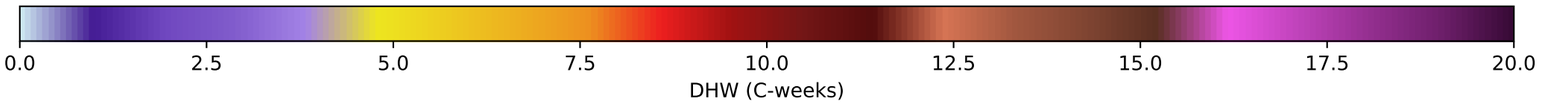


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

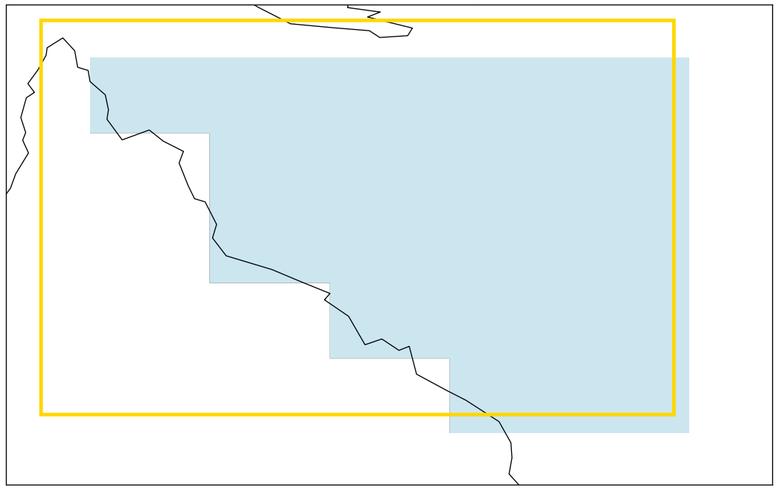




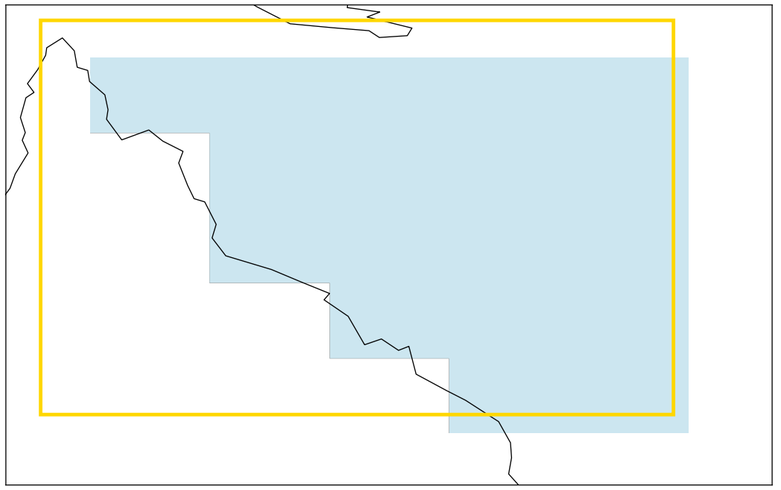


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

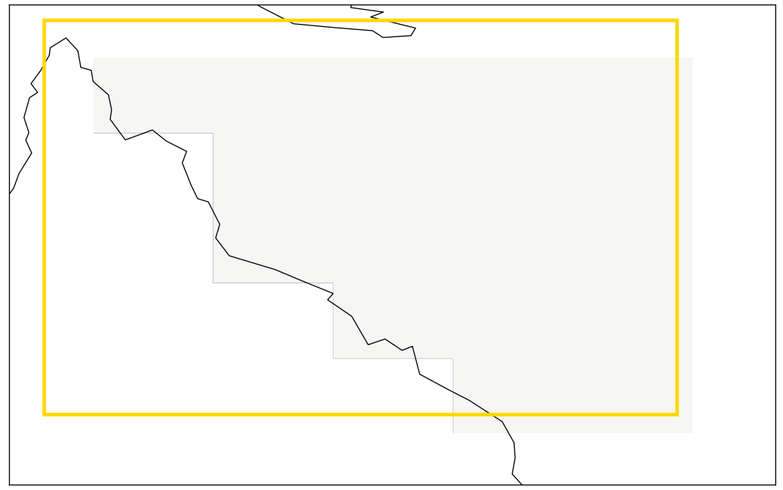
Actual DHW - Step 1



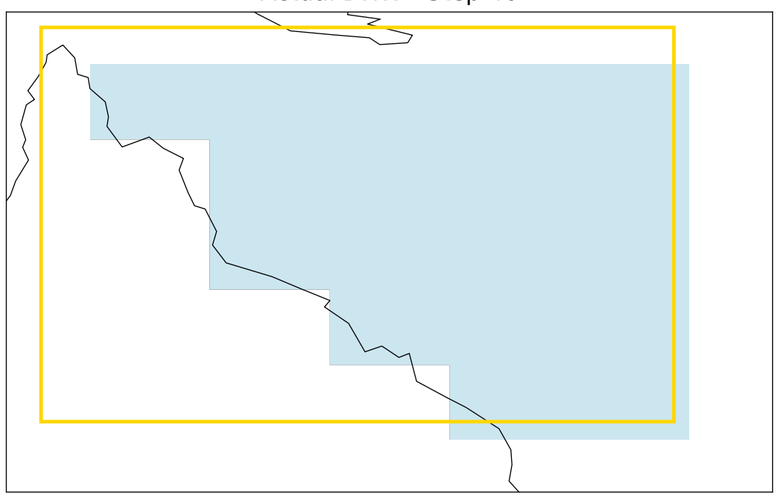
Predicted DHW - Step 1



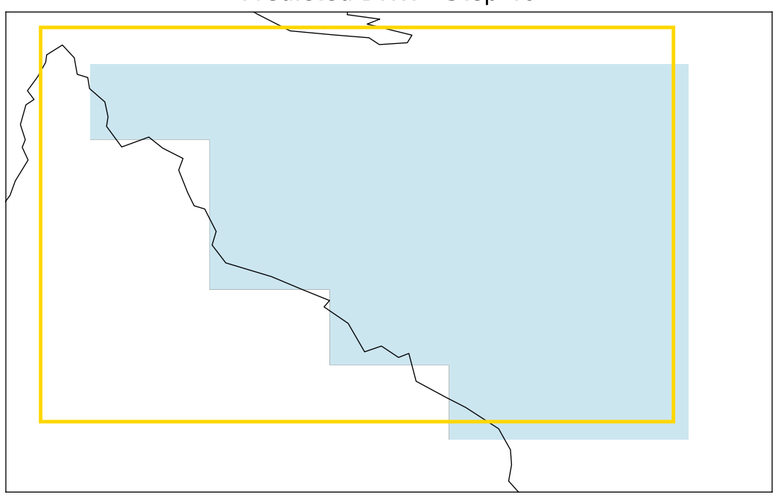
Error (Pred - Actual) - Step 1 [scale ±8]



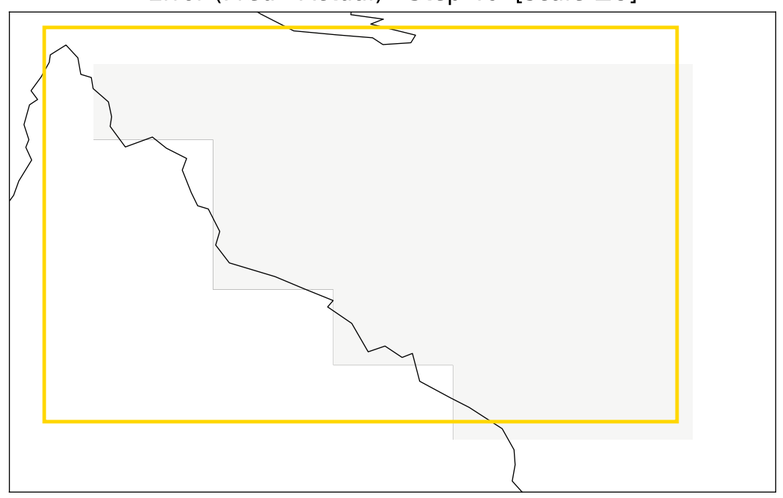
Actual DHW - Step 46



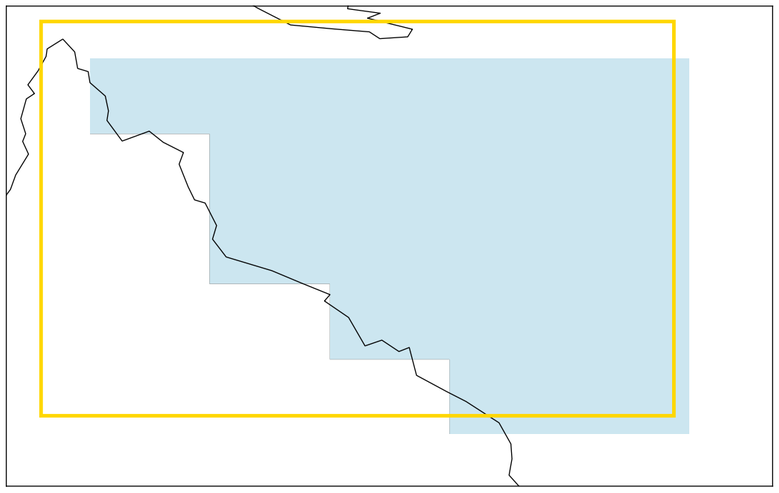
Predicted DHW - Step 46



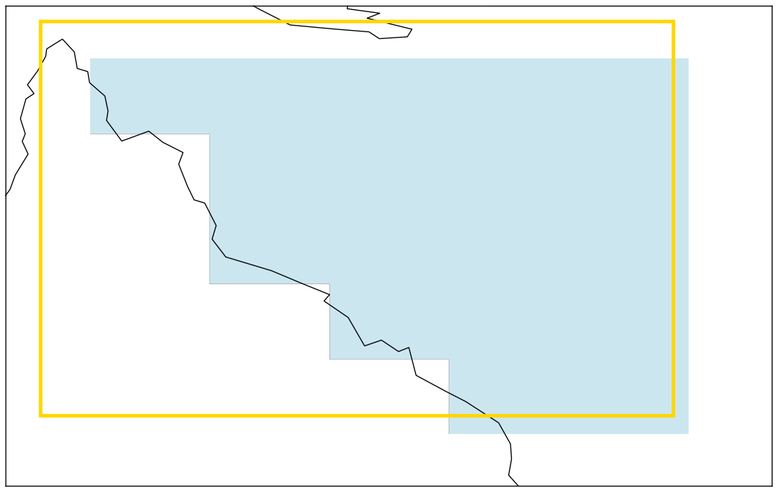
Error (Pred - Actual) - Step 46 [scale ±8]



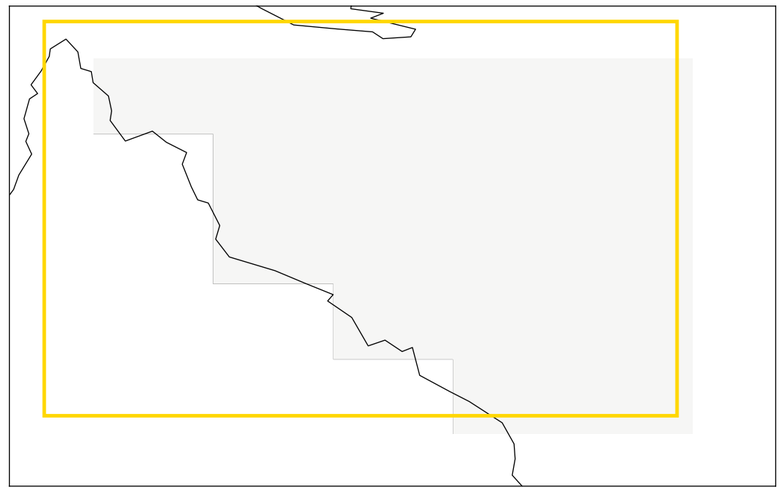
Actual DHW - Step 90

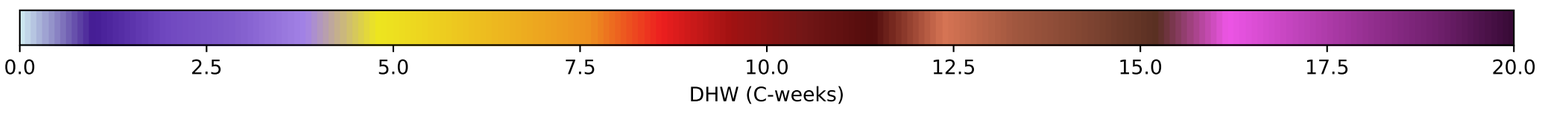


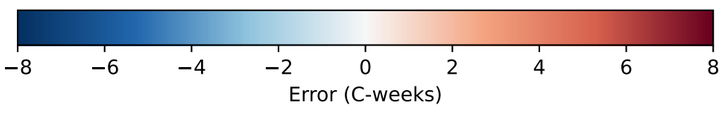
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

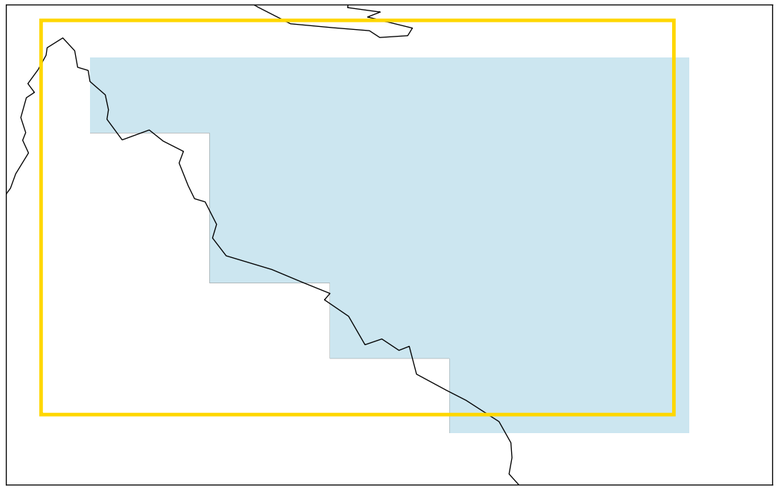




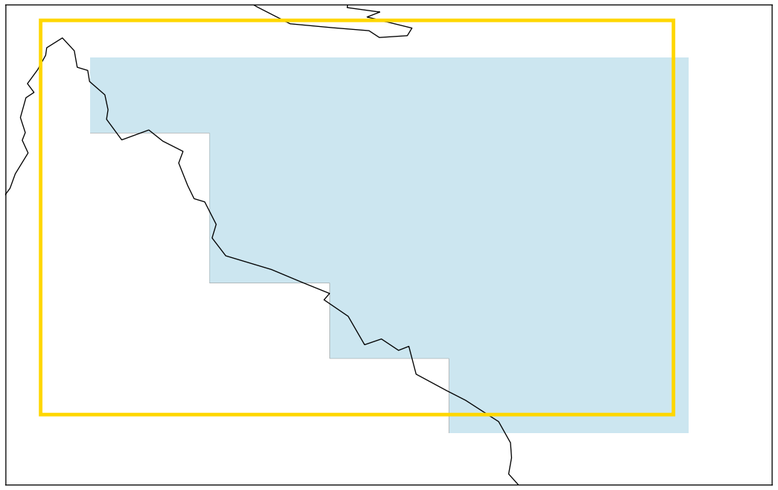


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

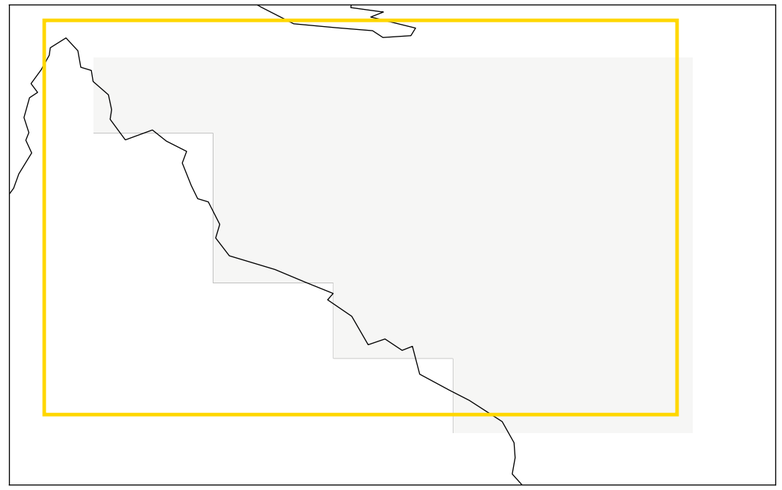
Actual DHW - Step 1



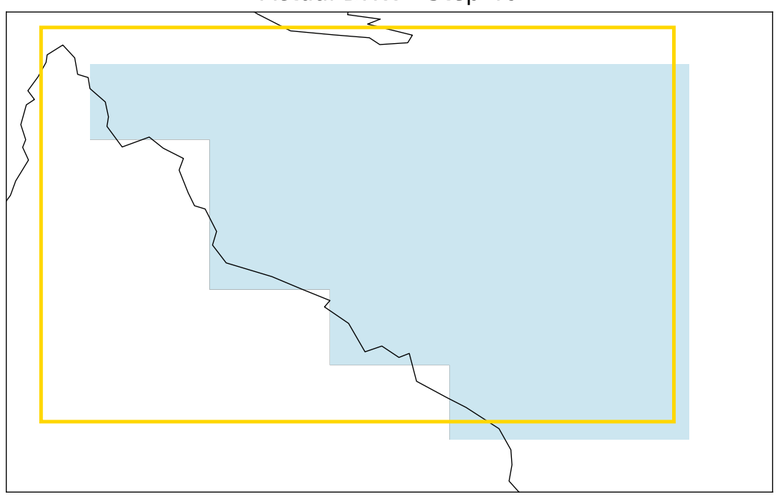
Predicted DHW - Step 1



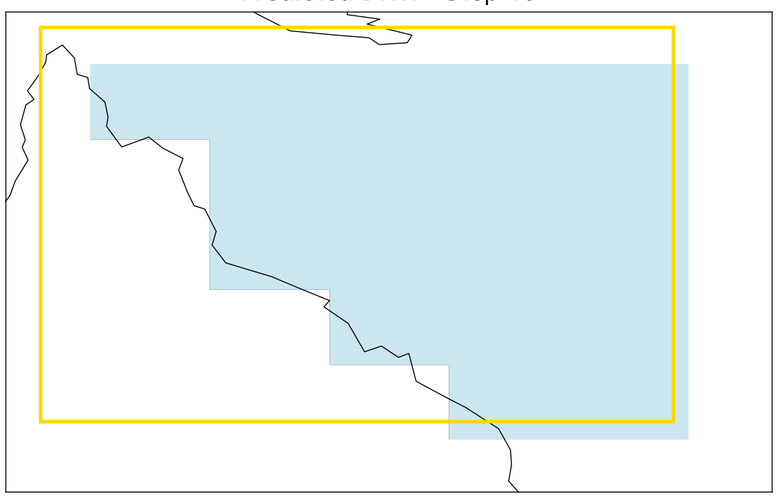
Error (Pred - Actual) - Step 1 [scale ±8]



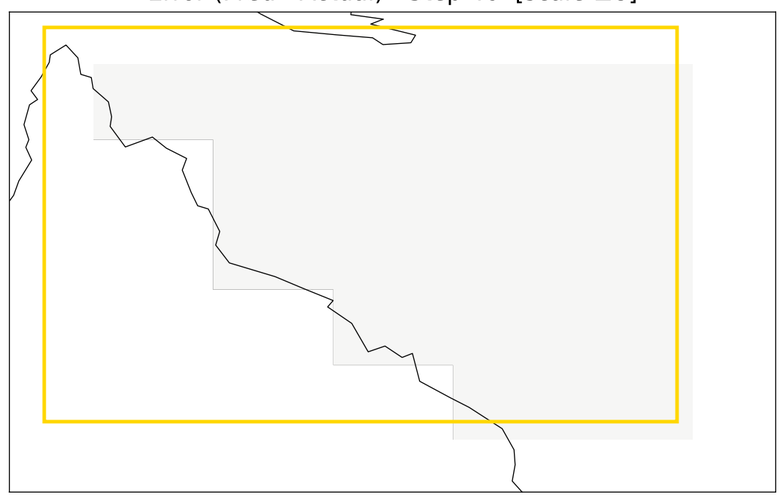
Actual DHW - Step 46



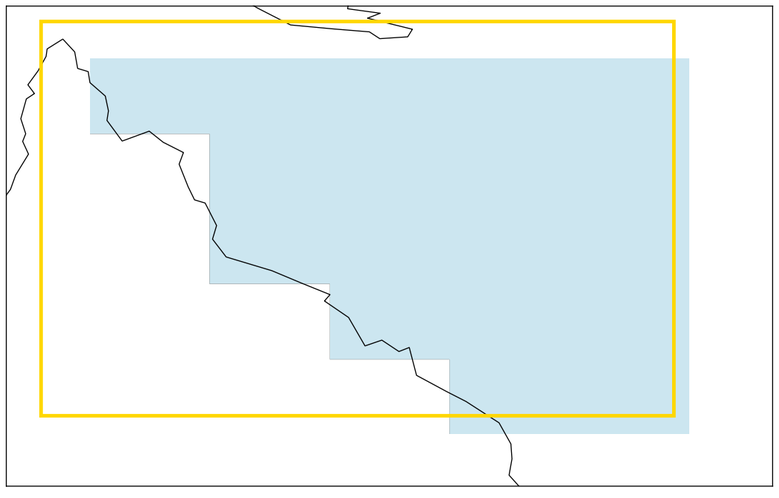
Predicted DHW - Step 46



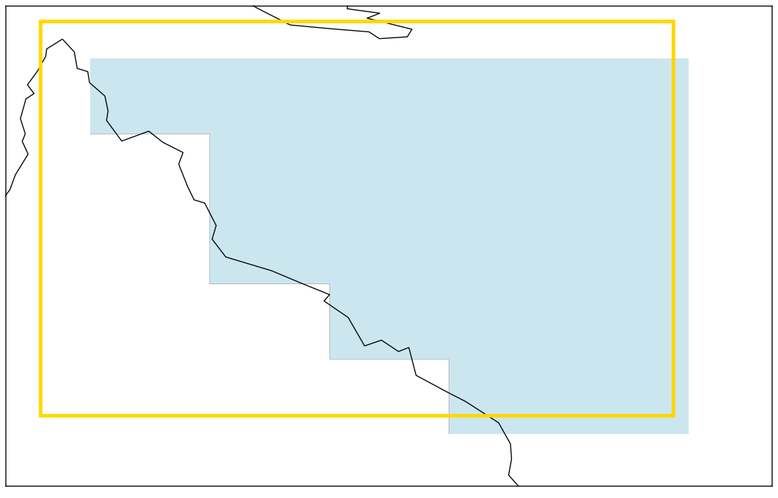
Error (Pred - Actual) - Step 46 [scale ±8]



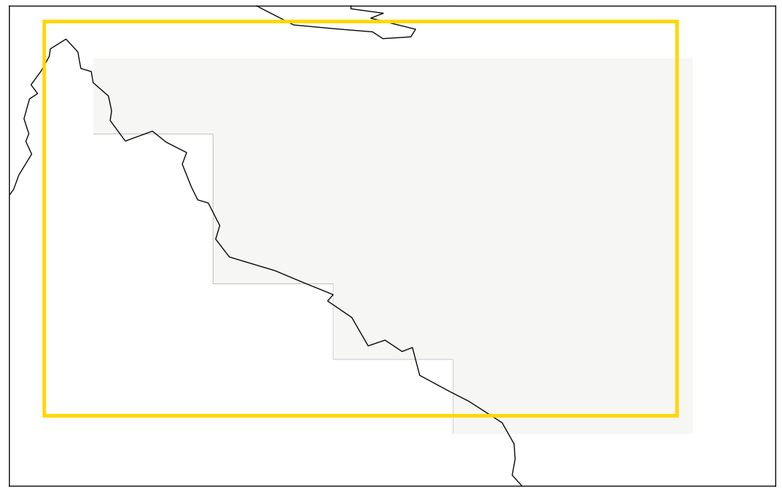
Actual DHW - Step 90

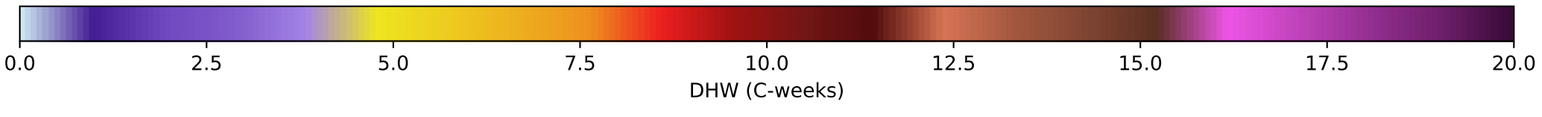


Predicted DHW - Step 90



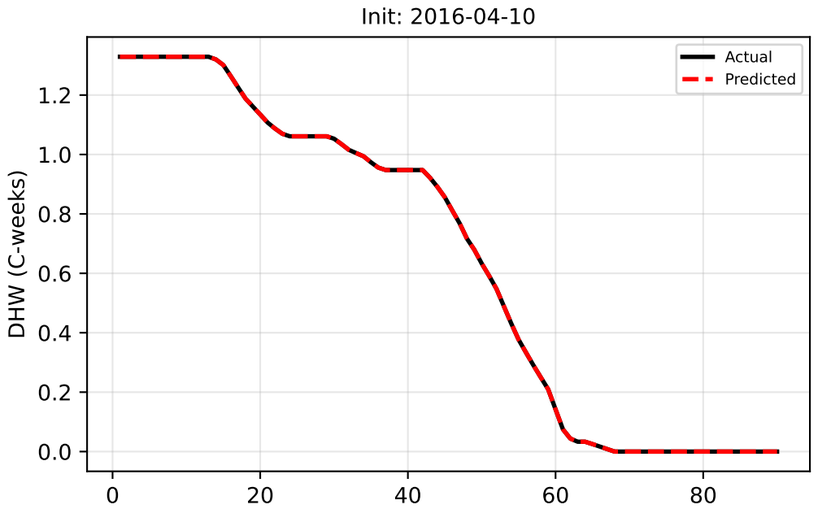
Error (Pred - Actual) - Step 90 [scale ±8]

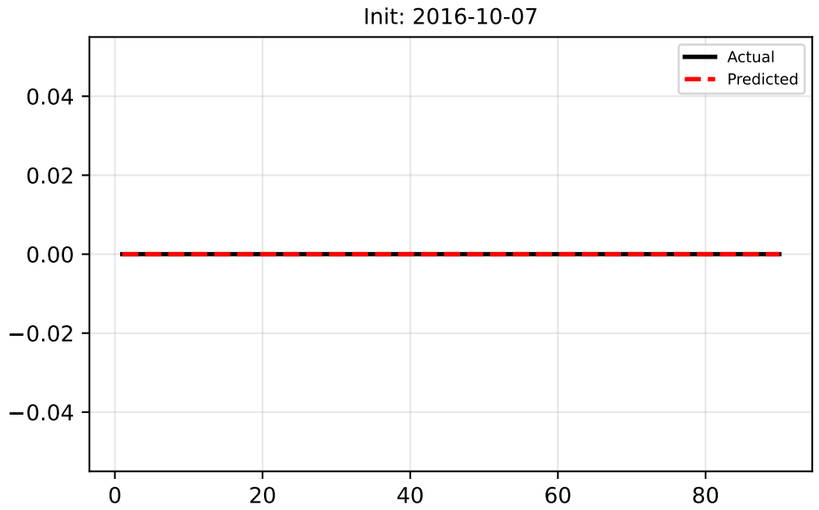


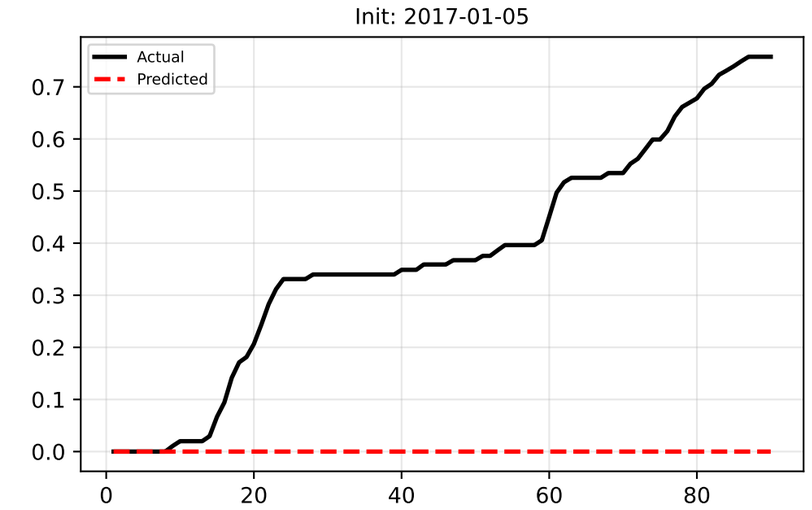


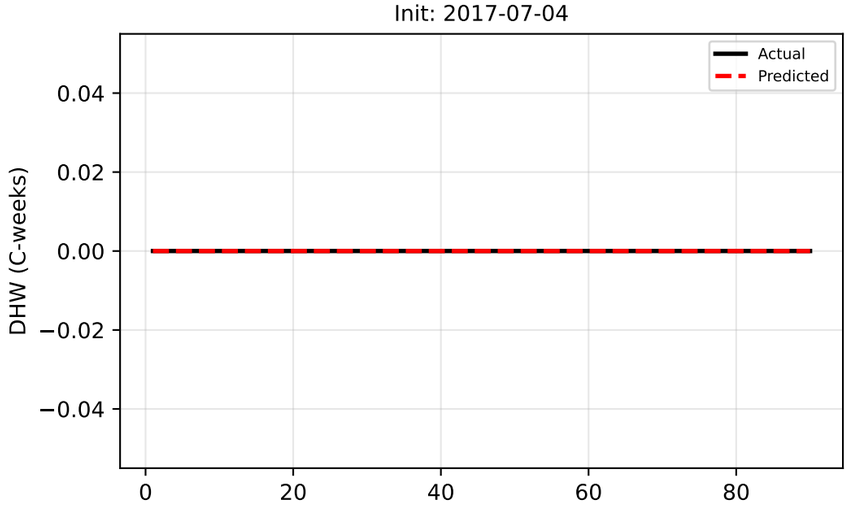


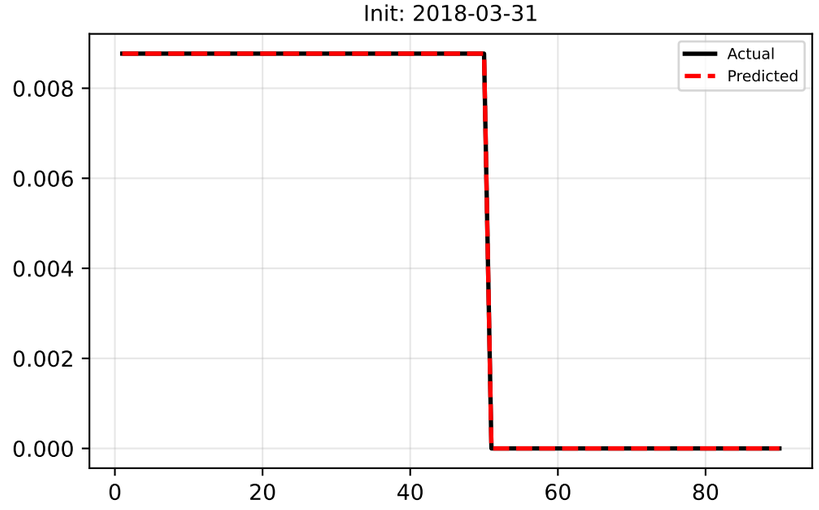
Global-Mean DHW - Val - GBR (C-weeks)

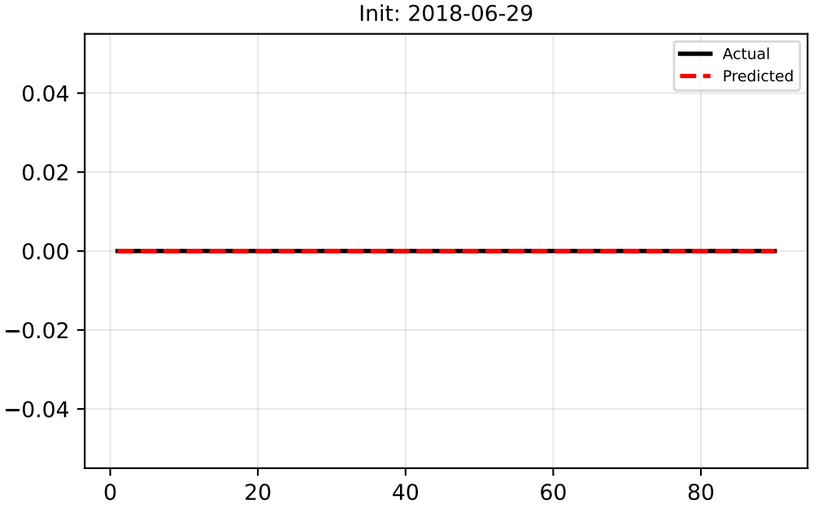


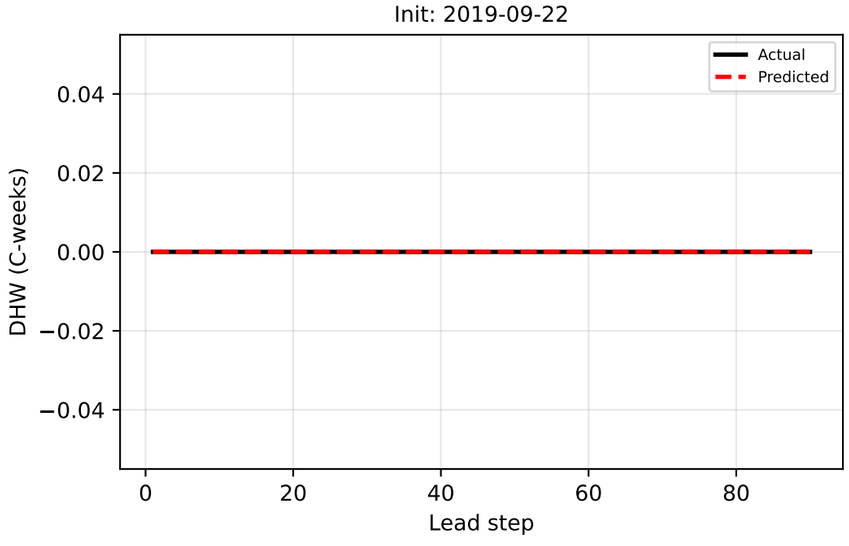


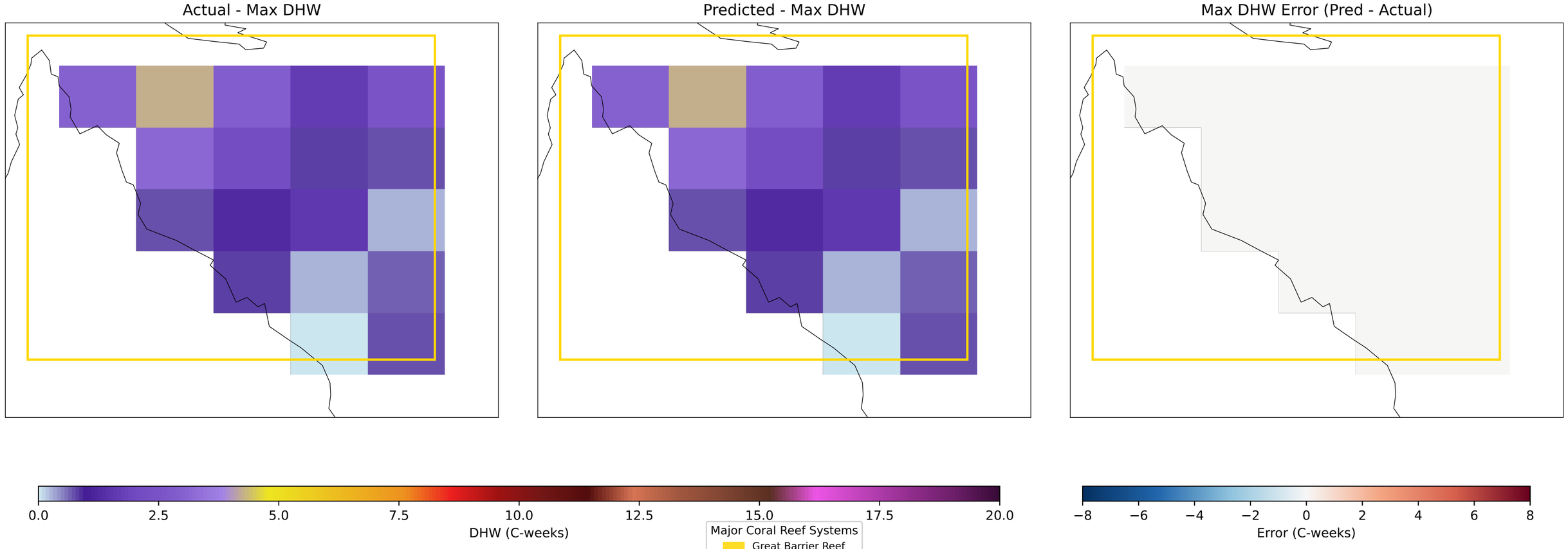


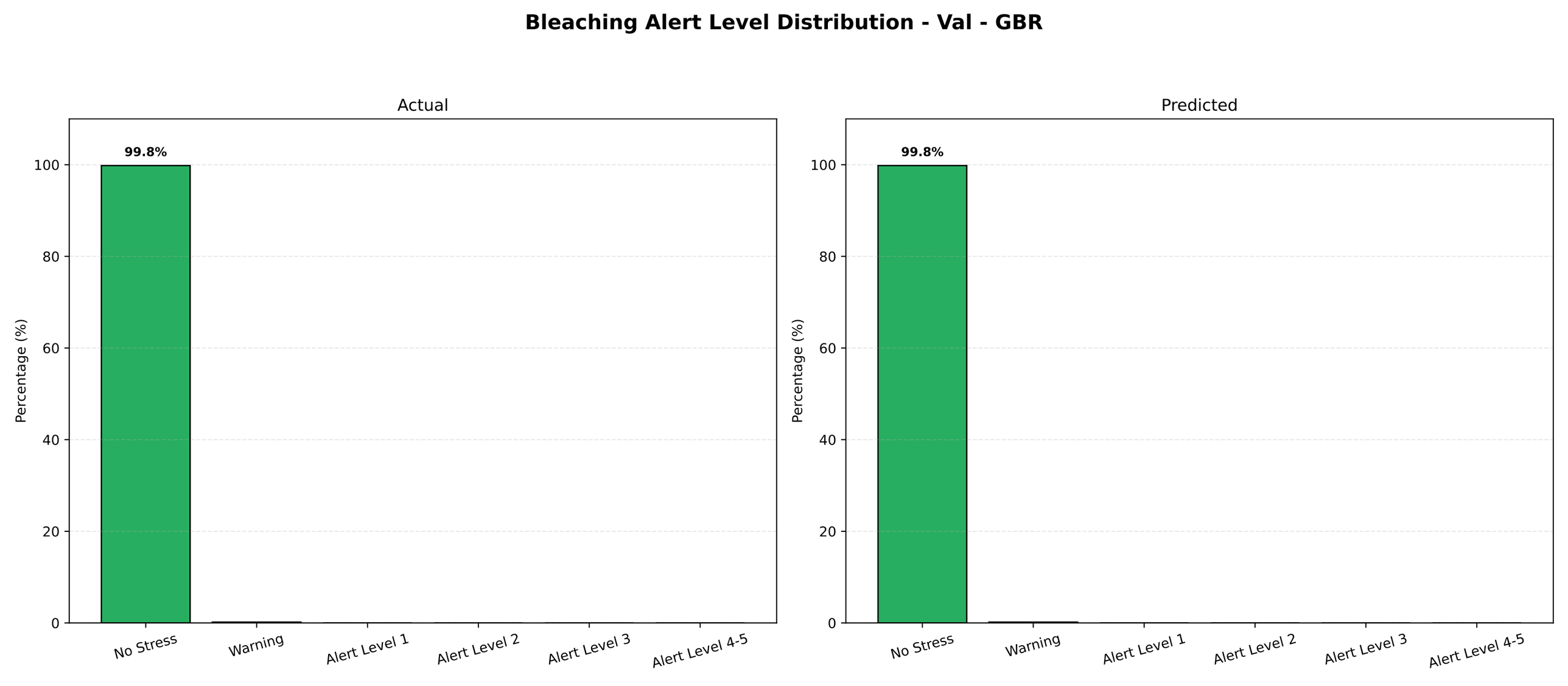






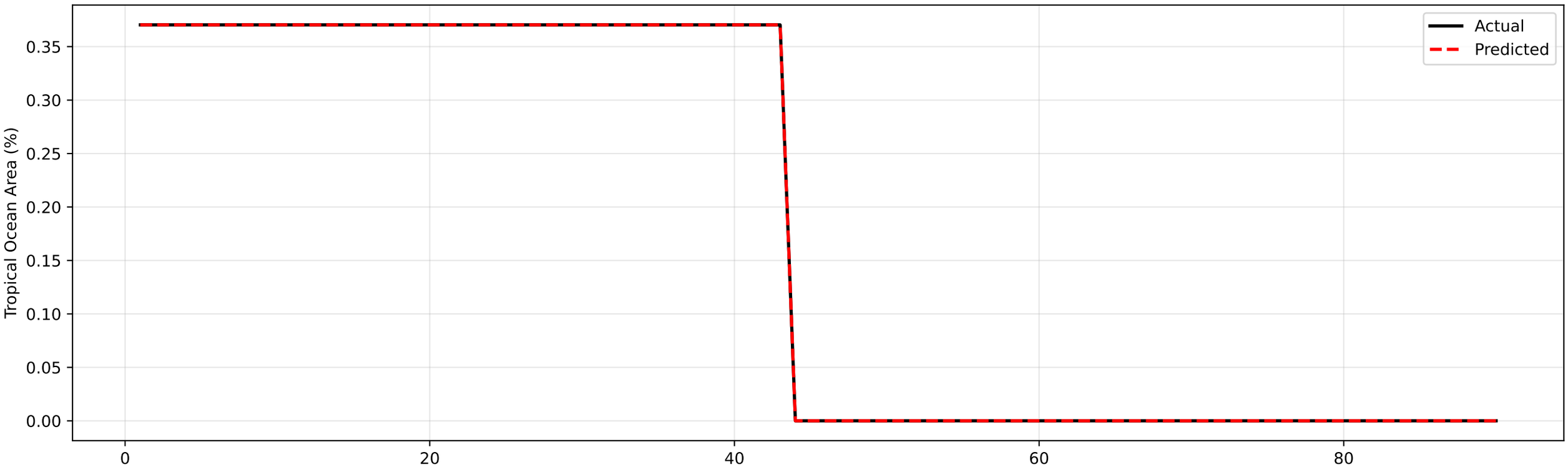



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.8%
99.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

